GPSC CBRT · HOMOEOPATHIC PHYSICIAN · CORE

Core Notes: all 10 topics

The 25-mark core section in one file: every topic with its pictures, memory hooks, rapid revision and practice MCQs.

Topic 1 · Organon of Medicine & Homoeopathic Philosophy
Topic 2 · Homoeopathic Materia Medica
Topic 3 · Homoeopathic Pharmacy
Topic 4 · Repertory & Repertorisation
Topic 5 · General & Applied Human Biology
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Topic 7 · Practice of Medicine & Homoeopathic Therapeutics
Topic 8 · Obstetrics, Gynaecology & Paediatrics
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Topic 10 · Community Medicine, Forensic Medicine & Ethics

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Topic 1: Organon of Medicine & Homoeopathic Philosophy

How to use this guide

- Topic 1 is one of ten core topics in the 25-mark core section, so expect about 2 to 3 questions.
- They ask an aphorism number, a date or edition, who wrote what, one of Kent's 12 observations, or a miasm feature.
- Each section ends with a memory hook; the recall card near the end holds the twenty facts asked most.

Syllabus heading (Advt 06/2026)	What it covers	Section
Fundamental principles and pioneers	Law of similars, simplex, minimum dose, vital force; Hahnemann, Kent, Stuart Close, H. A. Roberts	1.1
Aphoristic teachings	Health, disease, cure, indisposition, drug proving, the unprejudiced observer	1.2
Case taking and posology	Acute and chronic case taking, evaluation of symptoms, second prescription, Kent's 12 observations	1.3
Theory of chronic diseases and miasms	Evolution of chronic disease, fundamental cause, psora, sycosis, syphilis, tubercular diathesis	1.4

Standard books. Hahnemann, *Organon of Medicine*, 6th edition (Boericke's translation); Hahnemann, *The Chronic Diseases*; J. T. Kent, *Lectures on Homoeopathic Philosophy*; Stuart Close, *The Genius of Homoeopathy*; H. A. Roberts, *The Principles and Art of Cure by Homoeopathy*; J. H. Allen, *The Chronic Miasms*. Aphorism numbers follow the 6th edition unless a question says otherwise.

Memory hook: the Organon runs **mission → disease → medicine → application**. Place any aphorism in one of those four blocks first and its number becomes easy to recall.

1.1 Fundamental principles and pioneers

Homoeopathy rests on a few cardinal principles set out by Samuel Hahnemann in six editions of the Organon (1810 to 1842) and developed by Kent, Close and Roberts. Questions ask dates, edition details, aphorism numbers and who wrote which book.

Samuel Hahnemann: dates to know

Year	Event
10 April 1755	Born at Meissen, Saxony; 10 April is World Homoeopathy Day
1779	MD from Erlangen
1790	Cinchona bark experiment on himself while translating Cullen's *Materia Medica*
1796	*Essay on a New Principle for Ascertaining the Curative Powers of Drugs*, in Hufeland's Journal: the formal birth of homoeopathy
1805	*Medicine of Experience*; *Fragmenta de viribus medicamentorum positivis*, the first record of provings (27 drugs)
1810	First edition of the Organon
1811 to 1821	*Materia Medica Pura*, six volumes
1828	*The Chronic Diseases*
1835	Moves to Paris with his second wife, Melanie d'Hervilly
2 July 1843	Dies in Paris; buried at Pere Lachaise

From one experiment to the law of similars

How Cinchona bark started homoeopathy



Drug proving (§§21, 105–145): medicinal powers are learnt only by testing on healthy people.
Self-proving by a healthy, unprejudiced physician is best (§141).
1805: Fragmenta de viribus, the first record of provings (27 drugs).

From one experiment to a law · Cinchona 1790 → provings → similia similibus curentur

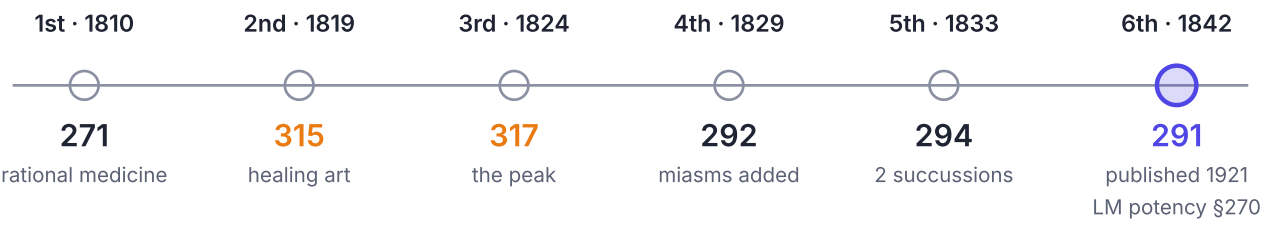
Hahnemann coined the word **allopathy**. His motto was *Aude sapere* (dare to be wise). The word *Organon* was first used by Aristotle.

The six editions of the Organon

Edition	Year	Title	Aphorisms
1st	1810	Organon der rationellen Heilkunde (Organon of Rational Medicine)	271
2nd	1819	Organon der Heilkunst (Organon of the Healing Art)	315
3rd	1824	Organon der Heilkunst	317
4th	1829	Organon der Heilkunst	292
5th	1833	Organon der Heilkunst	294
6th	written 1842, published 1921	Organon der Heilkunst; in English *Organon of Medicine*	291

Six editions: 271 aphorisms rising to 317, then falling to 291

Organon of Medicine, year of each edition and its number of aphorisms



6th edition: published by Richard Haehl, translated by William Boericke, introduction by James Krauss.

Organon editions · year and aphorism count

Read it left to right: the count climbs to 317 in the 3rd edition, then settles in the 290s.

- The 6th edition was published by **Richard Haehl** in 1921 and translated by **William Boericke** (1922); **James Krauss** wrote the introduction.
- The 6th edition introduced the **50 millesimal (LM) potency** (§270) and calls the vital principle the *dynamis*.
- The 4th edition (1829) first carried the theory of chronic diseases into the Organon.

Memory hook: aphorism counts **271 → 315, 317 → 292, 294, 291**: one start, two 'three-hundreds', then three 'two-nineties' falling to the last.

Cardinal principles

Principle	Meaning	Where
Law of similars (*similia similibus curentur*)	A medicine cures the disease whose symptoms it can produce in the healthy	§26 (law of nature), §§24 to 27
Individualisation	Each patient is treated by his own totality, not by the disease name	§§82 to 104
Totality of symptoms	The sum of all signs and symptoms is the only indication for the remedy	§§7, 18
Drug proving	Medicinal powers are learnt only by testing on healthy people	§§21, 105 to 145
Simplex (single remedy)	One single, simple medicine at a time	§§272 to 274
Minimum dose	The smallest dose that cures gently	§§275 to 283
Potentisation	Trituration and succussion develop the dynamic power	§§269 to 271

Principle	Meaning	Where
Vital force	Spirit-like force that keeps the organism in harmony; disease is its dynamic derangement	§§9 to 16
Chronic miasms	Chronic disease arises from psora, syphilis and sycosis	§§78 to 81

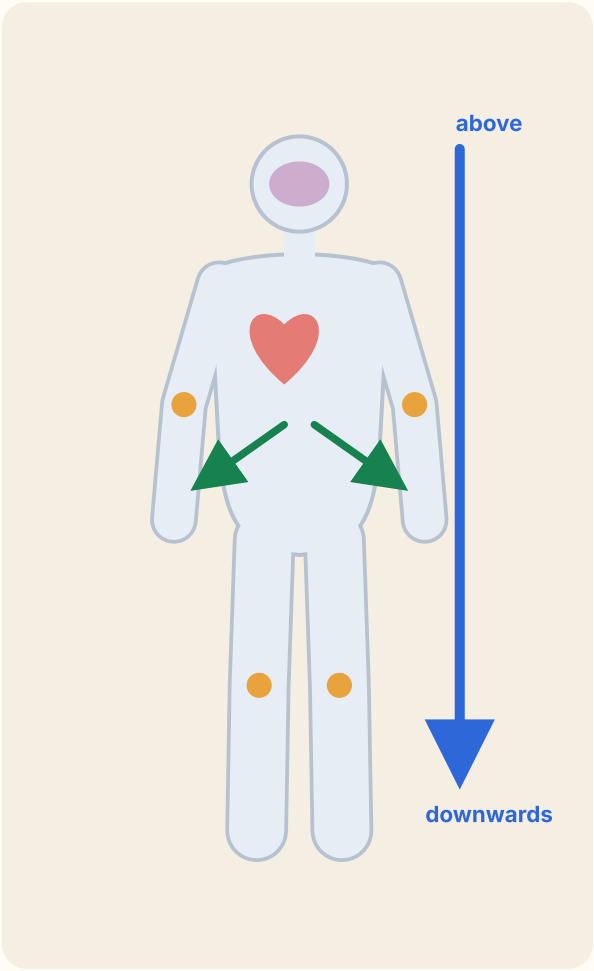
The **Arndt-Schulz law** is quoted in support of the minimum dose: weak stimuli excite vital activity, moderate ones favour it, strong ones retard it and very strong ones arrest it.

Later pioneers

Pioneer	Main work	Ideas examiners ask about
J. T. Kent (USA, 1849 to 1916)	*Lectures on Homoeopathic Philosophy* (1900); *Repertory* (1897); *Lectures on Materia Medica*	Mental and general symptoms first; potency series 30, 200, 1M, 10M, 50M, CM; 12 observations; second prescription; influenced by Swedenborg
Stuart Close (USA, 1860 to 1929)	*The Genius of Homoeopathy* (1924)	Susceptibility as the basis of cure and of dose; homoeopathy as a general therapeutic law
H. A. Roberts (USA, 1868 to 1950)	*The Principles and Art of Cure by Homoeopathy* (1936); *Sensations As If*	Vital energy and the direction of cure
Constantine Hering (1800 to 1880)	*Guiding Symptoms of our Materia Medica*	Hering's law: above downwards, within outwards, more to less important organs, reverse order of appearance
C. von Boenninghausen (1785 to 1864)	*Therapeutic Pocket Book* (1846)	The complete symptom: location, sensation, modality, concomitant

Hering's law: the direction of cure

Four arrows that tell you the case is healing



1 Above downwards

Symptoms clear from the head towards the feet

2 Within outwards

From the inner organs towards the skin

3 More to less important organs

Vital organs (heart, brain) clear before joints and skin

4 Reverse order of appearance

The newest symptoms go first, the oldest go last

Constantine Hering (1800–1880), author of the Guiding Symptoms. Use it to judge whether the cure is going the right way.

Hering's law · above downwards, within outwards, vital organs first, reverse order

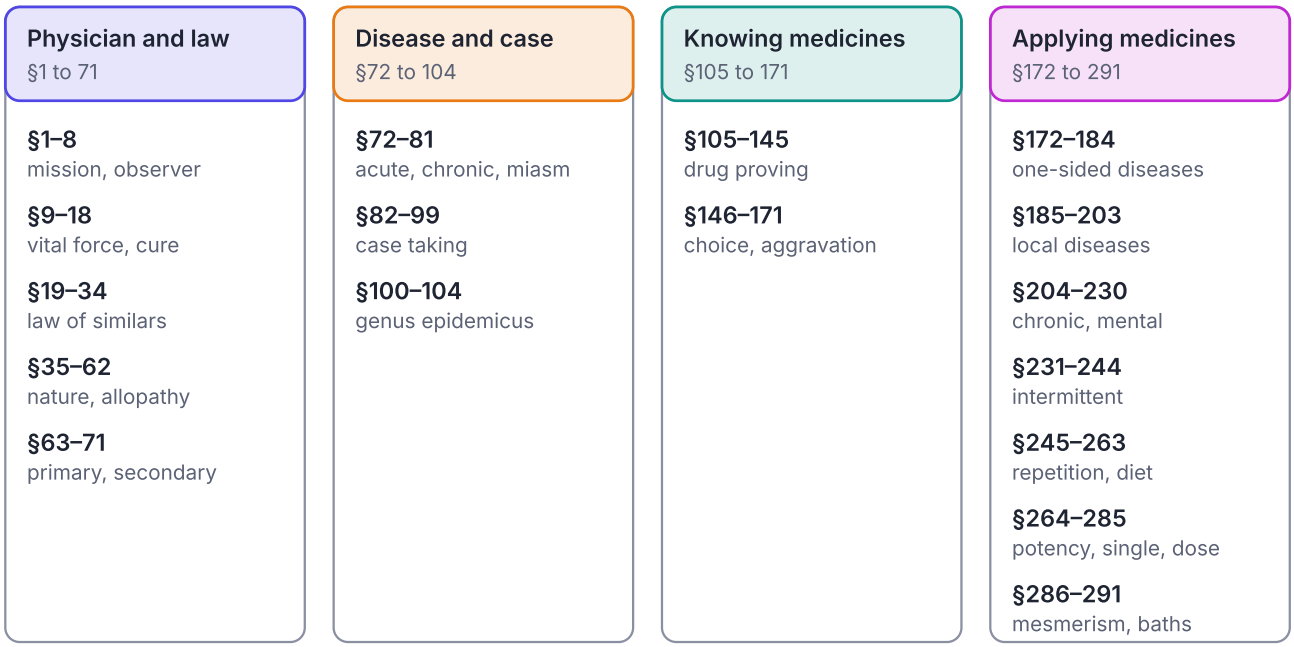
Memory hook: **K**ent = **K**ey symptoms of mind first; **C**lose = **C**apacity to react (susceptibility); **R**oberts = **R**oad of cure (direction).

1.2 Aphoristic teachings

The 291 aphorisms of the 6th edition follow one plan: the physician's mission, then disease, then medicines, then how medicines are applied. The map below places each block; the table after it holds the single aphorisms asked most.

The Organon in four blocks: law, disease, medicines, application

6th edition, 291 aphorisms; place the number in its block first



Map of the Organon · 4 blocks, 291 aphorisms

Any aphorism above 172 is about applying medicines; anything below 72 is about the physician and the law.

Key aphorisms

§	What it says
1	The physician's high and only mission is to restore the sick to health, to cure
2	Highest ideal of cure: rapid, gentle and permanent restoration of health, on easily understood principles
3	The physician must know the disease, the medicinal powers, how to apply them, and the obstacles to recovery
4	The physician is also a preserver of health
5	Aids to cure: exciting cause, fundamental cause (chronic miasm), constitution, character, occupation, habits, age, sexual function
6	The unprejudiced observer sees only the changes in body and mind: the signs and symptoms
7	The totality of symptoms is the outwardly reflected image of the inner disease and the only guide to the remedy
9	In health the spirit-like vital force rules the body with unbounded sway

§	What it says
11	In disease only the vital force is primarily deranged
16	The vital force is deranged, and cured, only by dynamic influences
17	Removing the totality of symptoms removes the disease: that is cure
21	Curative power of medicines is known only from changes they produce in the healthy
26	Law of nature: a weaker dynamic affection is extinguished by a stronger one similar to it
29	The remedy sets up a similar, slightly stronger artificial disease
33	The organism is more susceptible to medicines than to natural disease
63, 64	Primary action belongs to the medicine; secondary action (counteraction) belongs to the vital force
73	Acute diseases: individual, sporadic, epidemic, acute miasms
74	Drug diseases from allopathic treatment are the most incurable
77	Pseudo-chronic diseases from avoidable causes
78, 80	True chronic disease arises from a chronic miasm; psora is the oldest and mother of chronic disease
83, 84, 87	Case taking: freedom from prejudice; patient speaks first; no leading questions
100 to 102	Genus epidemicus
110	First rudiment of pure materia medica
127, 128	Prove on both sexes; prove in the 30th potency
141	Provings made by the healthy, unprejudiced physician on himself are the best
150	Indisposition: trivial complaint from an obvious cause; remove the cause
153	Choose by the striking, singular, uncommon and peculiar symptoms
157 to 160	Homoeopathic aggravation
172, 173	One-sided diseases
210	Mental diseases are bodily diseases with the mental symptom predominant

§	What it says
253	The earliest sign of improvement is in mood and mind
270	50 millesimal (LM) method
273	Never more than one single, simple medicine at a time

Memory hook: **6** eyes see (unprejudiced observer), **7** symptoms make the picture (totality), **26** is the law, **153** is the peculiar one, **273** is the single one.

The six concepts named in the syllabus

- **Health (§9):** the vital force keeps all parts in harmonious operation so the mind can use the body for the higher purposes of existence.
- **Disease (§§11 to 12):** a dynamic derangement of the vital force, shown outwardly only through signs and symptoms.
- **Cure (§§2, 17):** removal of the whole totality of symptoms, rapidly, gently and permanently. Kent separates cure from **recovery**, the return to health once a cause is removed.
- **Indisposition (§150):** a slight, passing complaint from an obvious cause such as a dietary error. Remove the cause; medicine is not needed.
- **Drug proving (§§105 to 145):** one pure drug at a time (§122), on healthy people of both sexes (§127), in the 30th potency (§128), with a controlled diet (§125), on reliable provers (§126). Many provers are needed for a complete picture (§135). In India these are now called homoeopathic pathogenetic trials and are run by the CCRH.
- **Unprejudiced observer (§6):** sees without theory or bias, noting only what the senses perceive; the first quality §83 asks of a case taker.

Memory hook: health is **harmony** (§9), disease is **derangement** (§11), cure is **removal of the totality** (§17), indisposition is **removal of the cause** (§150).

1.3 Case taking, evaluation, second prescription and posology

The exam asks how a case is taken, which symptoms rank highest, how to read the response to the first remedy (Kent's 12 observations) and how dose and repetition are decided.

Hahnemann's rules for case taking (§§83 to 104)

1. Bring freedom from prejudice, sound senses, attention in observing and fidelity in tracing the picture (§83).
2. Let the patient tell the story without interruption; then hear relatives and attendants (§84).
3. Write in the patient's own words, each symptom on a separate line, leaving space to add details.
4. Complete each symptom: where, what sensation, what makes it better or worse, what accompanies it.
5. Never ask a question that suggests the answer or can be answered with a bare yes or no (§87).
6. Add what the physician observes: appearance, behaviour, expression (§90).
7. In chronic cases ask about small complaints the patient no longer mentions (§95).
8. Acute cases are easier: the changes are recent and striking (§99).
9. In an epidemic, combine several patients' symptoms to find the genus epidemicus (§§100 to 102).

	Acute case	Chronic case
Focus	Recent change from the normal state	The whole life of the patient
Key questions	Cause, onset, modalities, concomitants, change in mood, thirst, temperature	Past illness, family history (miasm), appetite, thirst, desires and aversions, sleep, dreams, thermal reaction, menses, mental make-up
Remedy found from	The striking acute symptoms	The totality, mental and general symptoms first
Time	Short	Long; may need several sittings

Evaluation of symptoms (highest to lowest)

1. Strange, rare and peculiar symptoms (§153).
2. Mental generals, in Kent's order: will and emotions, then understanding, then memory.
3. Physical generals: heat and cold, appetite, thirst, desires and aversions, sleep, sweat, sexual and menstrual function.
4. Particulars: symptoms of one part, best when complete.
5. Common symptoms of the disease diagnosis: lowest value.

- A **general** symptom is said of 'I' (I am chilly); a **particular** of a part (my knee aches).
- Boenninghausen's **complete symptom** = location + sensation + modality + concomitant.
- Guernsey's **keynote** = one prominent feature pointing to a remedy.
- **Causation** ('ailments from' grief, a fall) ranks high when clear.

Memory hook: S-M-P-P-C: **S**trange first, then **M**ind, **P**hysical generals, **P**articulars, **C**ommon last.

Kent's 12 observations after the first prescription

#	What you see	What it means
1	Prolonged aggravation and final decline	Incurable; remedy too deep or potency too high for advanced organic disease
2	Long aggravation, then slow improvement	Marked structural change but still curable
3	Quick, short, strong aggravation, then rapid improvement	Good prognosis; no serious organic change
4	Cure without any aggravation	No organic disease; remedy and potency exactly right
5	Improvement first, aggravation afterwards	Remedy only superficially similar, or incurable
6	Relief that lasts too short a time	Structural change, or something interfering
7	Full relief of symptoms but no real relief of the patient	Incurable; the remedy only palliates
8	Some patients prove every remedy	Oversensitive, idiosyncratic patients
9	Action of medicines on provers	Healthy provers often left in better health
10	New symptoms appear	Many new symptoms suggest the wrong remedy
11	Old symptoms reappear	Curative: disease leaving in reverse order; wait
12	Symptoms take the wrong direction	Unfavourable; antidote the remedy

Kent's 12 observations: 3, 4 and 11 are good; 1, 7 and 12 are bad

What follows the first prescription, in four groups

Aggravation 1 to 4, worst to best	**1 · incurable** long aggravation, then decline	**2 · curable** long aggravation, slow recovery	**3 · best sign** short, sharp, then rapid recovery	**4 · just right** cure with no aggravation
Quality of relief 5 to 7	**5 · superficial** better first, worse later	**6 · obstacle** relief lasts too short	**7 · palliation** symptoms better, patient not	
Sensitive people 8 and 9	**8 · provers** prove every remedy given	**9 · provings** healthy provers end up better		
Which symptoms 10 to 12	**10 · new ones** new symptoms: wrong remedy	**11 · old return** curing in reverse order: wait	**12 · wrong way** outer to inner: antidote	

Green = good sign · red = bad sign · outlined = depends on the case

Kent's 12 observations · four groups, good and bad signs

Learn the colours first: the three green boxes and the three red ones answer most questions on this table.

Memory hook: 1 to 4 = aggravation (bad to best), **5 to 7 = quality of relief**, **8 to 9 = sensitive people**, **10 to 12 = which symptoms appear**. Old returning = good (11); wrong direction = antidote (12).

The second prescription (Kent)

- **Wait** while improvement continues; repeating too soon spoils the cure.
- **Repeat** the same remedy when improvement has stopped and the same symptoms return.
- **Antidote** when symptoms take the wrong direction or a violent, needless aggravation occurs.
- **Complementary** remedy when the first has done all it can and the picture has shifted to a related remedy.
- **Change** the remedy only when the totality has clearly changed or the first prescription was wrong.
- An **intercurrent** remedy (often a nosode) is given when reaction is deficient or a miasm blocks progress.

Posology

- **Minimum dose (§§275 to 279):** just enough for a gentle cure; too large a dose causes needless aggravation. The dose can hardly be too small (§279).
- **Potency choice:** greater susceptibility calls for higher potencies (Close). Mental and functional states and sensitive patients take higher potencies; advanced pathology, weak vitality and phlegmatic types take lower ones.
- **Kent's series:** 30, 200, 1M, 10M, 50M, CM, then DM and MM.
- **Repetition:** acute cases may need frequent doses. In the 5th edition a chronic case gets a single dose and the physician waits; the 6th edition allows repeated LM doses in water, succussed before each dose.
- **LM scale:** 1 in 50,000 per step, 100 succussions; one drop medicates 500 poppy-seed globules. The 5th edition used 2 succussions per potency.
- **Homoeopathic aggravation** is the primary action of a well-chosen remedy; a strong one means the dose was too large.
- **12C (24X)** is roughly where Avogadro's number is passed and no molecule of the original substance is likely to remain.

Memory hook: W-R-A-C-C: **W**ait, **R**epeat, **A**ntidote, **C**omplement, **C**hange: the five choices of the second prescription.

1.4 Chronic diseases and miasms

Hahnemann traced chronic disease to three infectious miasms (psora, syphilis and sycosis), with psora behind about seven-eighths of all chronic disease. Questions test how the theory developed and the features that separate the miasms.

How the theory developed

- Hahnemann saw chronic diseases treated homoeopathically keep returning in new forms. He studied the problem from about 1816, explained it to his pupils Stapf and Gross in 1827, and published *The Chronic Diseases* in 1828.
- **Acute miasms** run a course and end. Some, such as smallpox, measles and whooping cough, occur once in a lifetime and are **fixed miasms**. Kent divides an acute miasm into prodromal, progressive and declining periods.
- **Chronic miasms** never end on their own; the vital force cannot throw them off.
- **Three stages of infection:** the moment of infection; the whole organism becomes diseased internally; then the outward local sign (itch eruption, chancre or fig-wart), which quiets the internal disease.

- **Suppression** of the local sign drives the disease inwards to more vital organs; this is why Hahnemann opposed external applications.
- **Psora** (from suppressed itch) is the oldest, most universal and most destructive miasm, the 'mother' of chronic disease (§80). It can stay latent for years.
- **Pseudo-psora** is J. H. Allen's name for the tubercular diathesis, a mix of psora and syphilis.
- One-word keys (P. S. Ortega): psora = **deficiency**, sycosis = **excess**, syphilis = **destruction**.

Psora too little, sycosis too much, syphilis destroys

One picture per miasm, its key word, origin, aggravation and leading remedies



The four miasms at a glance · key word, origin, aggravation, remedies

Picture the gauge: psora is the near-empty one, sycosis overflows, syphilis is broken, and the tubercular arrows never settle.

The miasms compared

Feature	Psora	Sycosis	Syphilis	Tubercular
Key word	Deficiency, functional disorder	Excess, overgrowth	Destruction, ulceration	Change, wasting
Origin	Suppressed itch eruption	Gonorrhoea and fig-warts	Venereal chancre	Psora with syphilis
Mind	Anxious, fearful, quick but tires soon; lacks confidence	Secretive, suspicious, jealous, fixed ideas, hurried	Depressed, dull, morose, suicidal, destructive	Restless, changeable, discontented, longs to travel

Feature	Psora	Sycosis	Syphilis	Tubercular
Skin	Dry, rough, itching, burning	Warts, moles, thickening, oily	Ulcers, copper-coloured eruptions, deep fissures	Ringworm, eczema, easy bleeding
Discharges	Scanty	Thick, greenish-yellow, fishy	Thin, acrid, offensive, bloody	Profuse, offensive, blood-streaked
Worse	Cold, standing, suppression	Damp, cold, rainy weather	Night, sunset to sunrise	Before storms, closed rooms
Better	Natural discharges and eruptions	Unusual discharges, motion	Daytime	Open air
Leading remedies	Sulphur (king of antipsorics), Psorinum, Calcarea carb., Lycopodium	Thuja, Medorrhinum, Natrum sulph., Nitric acid	Mercurius, Syphilinum, Aurum met., Kali iod.	Tuberculinum, Bacillinum, Calcarea phos., Phosphorus

Memory hook: Psora = Poverty (too little), **Sycosis = Swelling** (too much), **Syphilis = Smashing** (destruction), **Tubercular = Travel** (restless change). Night aggravation means syphilis; damp weather means sycosis.

Rapid revision: one-line facts

Area	Fact
Hahnemann	Born 10 April 1755, Meissen; died 2 July 1843, Paris; Cinchona experiment 1790; Essay 1796
Organon	1810, 1819, 1824, 1829, 1833, 1842 (published 1921); 271, 315, 317, 292, 294, 291 aphorisms
6th edition	Published by Richard Haehl; translated by William Boericke; introduction by James Krauss; LM potency §270
First edition	Organon der rationellen Heilkunde (rational medicine); 'Heilkunst' from the 2nd
Aphorisms	§1 mission; §2 ideal cure; §3 knowledge; §4 preserver of health; §5 causes; §6 unprejudiced observer; §7 totality
Aphorisms	§9 health; §11 disease; §16 dynamic influences; §17 cure; §26 law of similars; §29 artificial disease; §33 susceptibility

Area	Fact
Aphorisms	§63 primary action (drug); §64 secondary action (vital force); §74 drug diseases most incurable; §80 psora
Aphorisms	§§84, 87 own words, no leading questions; §§100 to 102 genus epidemicus; §128 prove in 30th; §141 self-proving best
Aphorisms	§150 indisposition; §153 peculiar symptoms; §173 one-sided; §210 mental disease; §253 mood improves first; §273 single remedy
Pioneers	Close = Genius of Homoeopathy (susceptibility); Roberts = Principles and Art of Cure; Kent = Lectures on Philosophy
Hering's law	Above downwards, within outwards, more to less important organs, reverse order of appearance
Kent's observations	3 = best prognosis; 4 = no organic disease; 11 = old symptoms return (good); 12 = wrong direction (antidote)
Posology	Kent's series 30, 200, 1M, 10M, 50M, CM; LM 1:50,000, 100 succussions, 500 globules per drop
Miasms	Psora deficiency, sycosis excess, syphilis destruction; pseudo-psora = tubercular (J. H. Allen)
Remedies	Sulphur king of antipsorics; Thuja and Medorrhinum for sycosis; Mercurius and Syphilinum for syphilis
Traps	Primary action belongs to the drug; recovery is not cure; an indisposition needs no medicine

Memory hook: read this table aloud twice the evening before the exam; say each aphorism number before you read its meaning.

One-glance recall card

Use it as a self-test: cover the small lines and say each fact from its cue.

Topic 1 recall card: the 15 facts asked most

Cover the small lines, say the fact from the big cue, then check

291 aphorisms 6th edition, written 1842, published 1921 by Haehl	**271 → 291** counts: 271, 315, 317, 292, 294, 291 (1810 to 1842)	**§6 · §7** unprejudiced observer; totality is the only guide
§9 · §11 · §17 health = harmony; disease = derangement; cure = removal	**§26** law of similars stated as a law of nature	**§63 · §64** primary action = drug; secondary action = vital force
§128 · §141 prove in the 30th potency; self-proving is best	**§150** indisposition: remove the cause, no medicine needed	**§153** strange, rare, peculiar symptoms choose the remedy
§270 · §273 LM potency (1:50,000); one single remedy at a time	**Kent 3 · 4 · 11** good signs: short sharp agg, no agg, old symptoms return	**Kent 1 · 7 · 12** bad signs: decline, palliation, wrong direction (antidote)
Hering's law above down, within out, vital to less vital, reverse order	**Close · Roberts** Genius of Homoeopathy (1924); Principles and Art of Cure (1936)	**P · Sy · Syph** deficiency, excess, destruction; pseudo-psora = TB

Topic 1 recall card · 15 facts

Practice MCQs

Twenty questions grouped by syllabus heading; the answer key follows.

Fundamental principles and pioneers

1. The 6th edition of the Organon contains: (a) 271 aphorisms (b) 294 aphorisms (c) 291 aphorisms (d) 317 aphorisms
2. The first edition of the Organon (1810) was titled: (a) Organon der Heilkunst (b) Organon der rationellen Heilkunde (c) Organon of Medicine (d) Materia Medica Pura
3. The 50 millesimal potency is described in: (a) §128 (b) §153 (c) §270 (d) §273
4. 'The Genius of Homoeopathy' was written by: (a) J. T. Kent (b) H. A. Roberts (c) Stuart Close (d) C. Hering
5. The law of similars is stated as a law of nature in: (a) §6 (b) §26 (c) §63 (d) §150

Aphoristic teachings

6. The unprejudiced observer is described in: (a) §6 (b) §9 (c) §83 (d) §153
7. A trivial complaint from an obvious cause that needs no medicine is called: (a) a one-sided disease (b) an indisposition (c) a pseudo-chronic disease (d) a fixed miasm
8. Hahnemann advised proving medicines in the: (a) mother tincture (b) 6th potency (c) 30th potency (d) 200th potency
9. Secondary action (counteraction) belongs to: (a) the medicine (b) the vital force (c) the miasm (d) the physician
10. According to §253, the earliest sign of improvement appears in the: (a) skin (b) appetite (c) mood and mind (d) sleep

Case taking and posology

11. Asking questions that suggest the answer is forbidden in: (a) §84 (b) §87 (c) §100 (d) §141
12. In Kent's 12 observations, a quick, short, strong aggravation followed by rapid improvement means: (a) incurable case (b) good prognosis (c) wrong remedy (d) palliation only
13. When symptoms take the wrong direction after a remedy, Kent advises to: (a) wait (b) repeat (c) antidote (d) raise the potency
14. The highest-ranking symptoms in evaluation are: (a) common symptoms (b) particulars (c) strange, rare and peculiar symptoms (d) pathological symptoms
15. In the LM scale, the drug-to-vehicle ratio at each step is: (a) 1:10 (b) 1:100 (c) 1:1000 (d) 1:50,000

Chronic diseases and miasms

16. Hahnemann published 'The Chronic Diseases' in: (a) 1810 (b) 1821 (c) 1828 (d) 1835
17. The miasm marked by excess and overgrowth is: (a) psora (b) sycosis (c) syphilis (d) tubercular
18. J. H. Allen called the tubercular diathesis: (a) latent psora (b) pseudo-psora (c) fixed miasm (d) acute miasm
19. Aggravation from sunset to sunrise is typical of: (a) psora (b) sycosis (c) syphilis (d) tubercular
20. The king of antipsoric remedies is: (a) Thuja (b) Mercurius (c) Sulphur (d) Tuberculinum

Answer key

Q	Answer	Why
1	c	291 aphorisms in the 6th edition

Q	Answer	Why
2	b	'Rational medicine'; 'Heilkunst' from the 2nd edition
3	c	§270 describes the LM method
4	c	Stuart Close, 1924
5	b	§26 states the therapeutic law of nature
6	a	§6: only the perceptible changes in health
7	b	§150: remove the cause
8	c	§128: the 30th potency
9	b	§64: counteraction of the vital force
10	c	§253: mood and mental state first
11	b	§87: no leading questions
12	b	Observation 3: no serious organic change
13	c	Observation 12: antidote
14	c	§153 characteristic symptoms
15	d	1 in 50,000 per step
16	c	1828
17	b	Sycosis = excess
18	b	Pseudo-psora
19	c	Syphilis: night aggravation
20	c	Sulphur

Topic 2: Homoeopathic Materia Medica

How to use this guide

- Materia Medica is one of the ten Core topics (the Core section carries 25 of the 60 CBRT marks), so expect about 2 to 3 questions.
- Most of them name a keynote and ask for the remedy, or name a remedy and ask its modality, source or relationship.
- The GPSC syllabus has four headings, and this guide follows them:

Syllabus heading	What it covers	Section
Nature, scope and drug relationships	Sources, construction, comparative MM, modalities, complementary, inimical and antidotal relationships	2.1
Plant remedies (41)	Aconite to Veratrum viride	2.2
Mineral, chemical and acid remedies (39)	Alumina to Zincum	2.3
Nosodes, sarcodes and special therapeutics	Bacillinum, Medorrhinum, Psorinum, Pyrogenium, Thuja, Glonoine; organ remedies	2.4

- Each remedy is set as one row: **source**, **keynote**, **modality** and one **exam hook**.
- Learn the keynote column first; it answers most questions.
- Standard books: Boericke, *Pocket Manual of Homoeopathic Materia Medica*; Allen, *Keynotes*; Kent, *Lectures on Materia Medica*; Clarke, *Dictionary of Practical Materia Medica*; Phatak, *Materia Medica of Homoeopathic Medicines*.

Memory hook: picture each remedy as a person with **one sentence**: Aconite is "sudden fear in cold dry wind", Bryonia is "lie still, worse any motion". If you can say the sentence, you have the remedy.

2.1 Nature, scope and drug relationships

Homoeopathic Materia Medica is the record of the pathogenetic effects of drugs, found by proving on healthy people, together with toxicological data and clinical confirmations. Hahnemann's *Materia Medica Pura* (1811 to 1821, six volumes) is the first pure MM; *Fragmenta de viribus* (1805) recorded 27 drugs.

Sources of the Materia Medica

1. **Drug provings** on healthy people: the primary source (Organon §§105 to 145).
2. **Toxicology**: accidental and intentional poisonings, which show the gross effects.
3. **Clinical observations**: symptoms cured though never proved, marked as clinical.
4. **Animal experiments**, for tissue changes only.

Construction: how MMs are arranged

Type	Plan	Examples
Pure / schematic	Proving symptoms in Hahnemann's head-to-foot schema	Hahnemann's *MM Pura*; T. F. Allen's *Encyclopedia* (10 vols); Hering's *Guiding Symptoms* (10 vols)
Clinical	Arranged by disease or clinical use	Boericke (with repertory); Clarke's *Dictionary*
Keynote	Only the leading features	H. C. Allen's *Keynotes*; Nash's *Leaders*
Comparative	Remedies compared side by side	Farrington's *Comparative MM*; Gibson-Miller *Relationship of Remedies*
Lecture / drug picture	The remedy as a person	Kent's *Lectures*; Tyler's *Drug Pictures*
Physiological / group	By family or chemical group	Farrington's *Clinical MM*; Phatak

Modalities

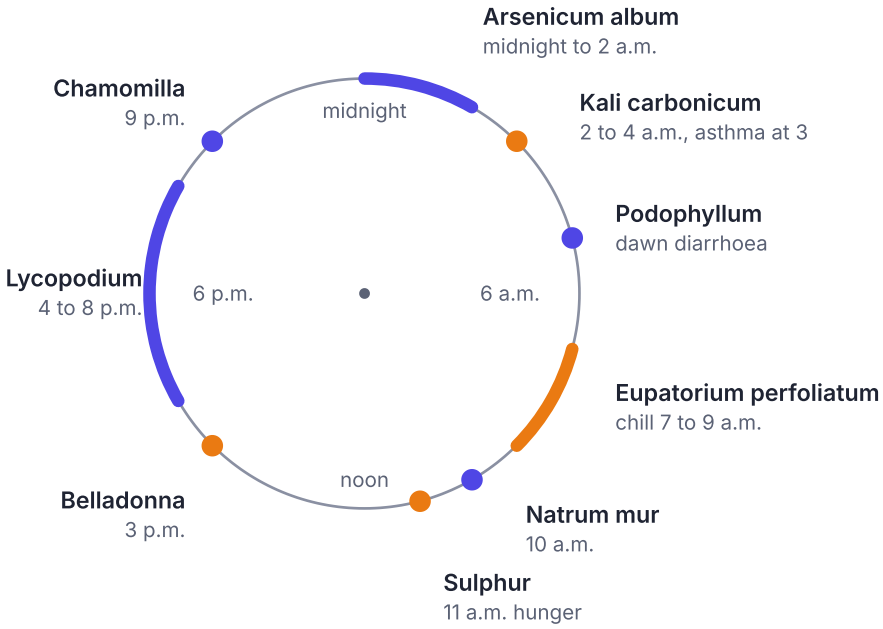
A **modality** is a condition that makes a symptom better (amel., >) or worse (agg., <): time, temperature, weather, position, motion, touch, eating, emotions. Modalities turn a common symptom into a characteristic one, and Boenninghausen generalised them to the whole patient.

Classic modality	Remedy
Worse from slightest motion, better lying on painful side	Bryonia
Better from continued motion, worse first motion, worse wet weather	Rhus tox
Worse 4 to 8 p.m.	Lycopodium
Worse 1 to 2 a.m. (after midnight)	Arsenicum album

Classic modality	Remedy
Worse 3 a.m.	Kali carb
Worse after sleep	Lachesis
Worse in warm room, better open air	Pulsatilla
Worse cold dry wind	Aconite, Hepar sulph, Causticum
Better from hard pressure, bending double	Colocynth
Worse touch, jar, light, noise	Belladonna

The aggravation clock: when each remedy is worst

24 hours, midnight at the top; arcs mark a span of hours, dots a single hour



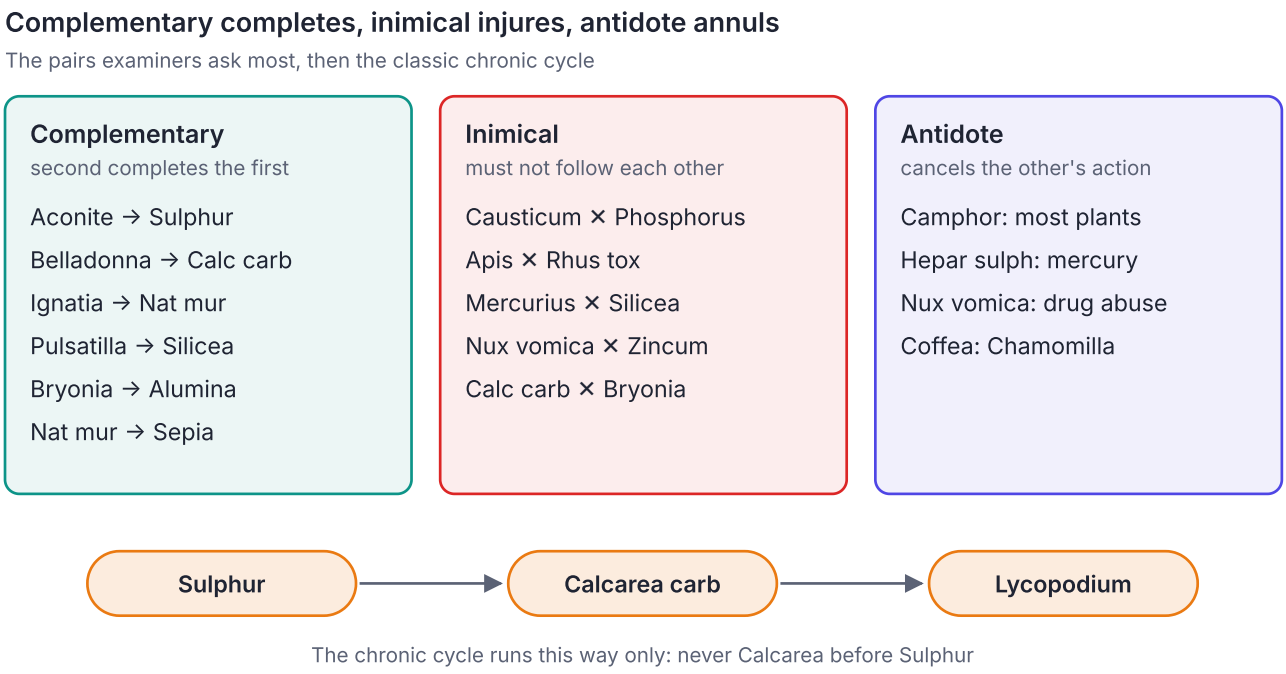
aggravation clock · 9 remedies around 24 hours

Walk round the clock from midnight: Arsenicum, Kali carb, Podophyllum, Eupatorium, Nat mur, Sulphur, Belladonna, Lycopodium, Chamomilla.

Drug relationships

Relationship	Meaning	Pairs to remember
Complementary	Completes the action of the first remedy	Aconite → Sulphur; Belladonna → Calc carb; Nat mur → Sepia; Pulsatilla → Silicea; Bryonia → Alumina; Sulphur → Calc carb → Lycopodium (the trio)

Relationship	Meaning	Pairs to remember
Inimical	Should not follow each other	Causticum ↔ Phosphorus; Apis ↔ Rhus tox; Mercurius ↔ Silicea; Nux vomica ↔ Zincum; Calc carb not after Bryonia or Nitric acid
Antidote	Removes the effects of another remedy	Camphor antidotes most plant remedies; Nux vomica antidotes drug abuse; Hepar sulph antidotes mercury; Coffea antidotes Chamomilla
Follows well	A good next remedy	Arnica → Rhus tox; Belladonna → Mercurius; Ipecac → Arsenicum
Acute and chronic	Acute remedy and its chronic complement	Bryonia / Alumina; Belladonna / Calc carb; Ignatia / Nat mur; Rhus tox / Calc fluor



drug relationships · complementary, inimical, antidote

Green pairs work together, red pairs clash, blue ones cancel: recall the colour first, then the pair.

The common teaching cycle **Sulphur → Calcarea carb → Lycopodium** (and back to Sulphur) is asked often; the order is not reversed: Calcarea is not given before Sulphur.

Memory hook: C-I-A — **C**omplementary completes, **I**nimical injures, **A**ntidote annuls. For inimicals chant "**Cause Phos**-trouble, **Apis Rhus**-hes in, **Merc Sil**-ences": Causticum–Phosphorus, Apis–Rhus tox, Mercurius–Silicea.

Differentiating allied remedies

Comparative Materia Medica (Farrington, Gibson-Miller) sets look-alike remedies side by side. The exam gives one telling feature and expects the right twin.

Look-alikes	First remedy	Second remedy
Aconite vs Belladonna (sudden, violent fevers)	**Aconite**: after exposure to cold dry wind; fear of death, anxious restlessness; great thirst	**Belladonna**: heat, redness, throbbing; dilated pupils; delirium; worse touch, jar, light, noise
Bryonia vs Rhus tox (pains and motion)	**Bryonia**: worse from the slightest motion; better rest, pressure, lying on the painful side; thirst for large quantities	**Rhus tox**: worse on first motion, better from continued motion; restless; worse cold damp and rest
Ignatia vs Natrum mur (grief)	**Ignatia**: recent, acute grief; sighing; changeable, contradictory symptoms; lump in the throat	**Natrum mur**: old, silent grief; consolation aggravates; craves salt; headache from sunrise to sunset
Pulsatilla vs Sepia (women's remedies)	**Pulsatilla**: mild, weepy, wants consolation; thirstless; worse warm room, better open air; changeable symptoms	**Sepia**: indifferent to loved ones; bearing-down sensation; better from vigorous exercise
Arsenicum vs Phosphorus (anxious, burning)	**Arsenicum album**: chilly, restless anxiety; burning pains better by heat; thirst for small sips; worse after midnight	**Phosphorus**: craves cold water, vomited once warm in the stomach; bright red bleeding; fears being alone and thunder
Nux vomica vs Lycopodium (digestive)	**Nux vomica**: irritable, chilly, sedentary; ineffectual urging for stool; worse in the morning	**Lycopodium**: full after a few mouthfuls, bloating; right-sided; worse 4 to 8 p.m.
Apis vs Cantharis (burning, urinary)	**Apis**: oedema; stinging pains; thirstless; better from cold	**Cantharis**: violent burning and cutting; constant urging, urine drop by drop; burns and scalds

Memory hook: pick the twin by the **modality or the mind**, never by the common symptom: both twins share the fever, the pain or the grief; only one has the wind, the motion, the consolation or the time.

2.2 Plant remedies

Forty-one plant remedies are named in the syllabus. Each row gives the common name and family, the keynote, the modality and the hook examiners use. Read across a row as one sentence.

Aconite to Eupatorium

Remedy (source)	Keynote	Worse / better	Exam hook
Aconitum napellus (monkshood, Ranunculaceae)	Sudden violent onset after cold dry wind; great fear of death, predicts the hour; restless anxiety	< evening, night, warm room; > open air	First stage of inflammation, before exudation
Aethusa cynapium (fool's parsley, Umbelliferae)	Infant cannot digest milk: vomits curds forcibly soon after nursing, then drowsy	< 3 to 4 a.m., summer	Milk intolerance; "inability to think"
Aloe socotrina (aloes, Liliaceae)	Insecure rectum: stool escapes with flatus; jelly-like lumpy stools; drives out of bed in the morning	< early morning, heat; > cold	Piles like a bunch of grapes, > cold water
Arnica montana (leopard's bane, Compositae)	Bruised sore feeling, bed feels too hard; says he is well when very ill; fears being touched	< touch, motion; > lying, head low	First remedy for injury and shock
Baptisia tinctoria (wild indigo, Leguminosae)	Typhoid state: feels body scattered, tries to gather the pieces; falls asleep mid-answer; offensive discharges	< humid heat	Painless throat: swallows only liquids
Belladonna (deadly nightshade, Solanaceae)	Sudden heat, redness, throbbing; dilated pupils; violent delirium, wants to bite or escape	< touch, jar, light, noise, 3 p.m.; > semi-erect	Hot, red, dry, sudden
Berberis vulgaris (barberry, Berberidaceae)	Radiating stitching pains from the kidney region; bubbling sensation; urine changes often	< motion, jar, standing	Renal and biliary colic, pains radiate
Bryonia alba (white bryony, Cucurbitaceae)	Stitching pains; dry membranes; thirst for large drinks at long intervals; irritable, wants to go home	< slightest motion; > pressure, lying on painful side, rest	Dry hard stool as if burnt
Cannabis indica (hashish, Cannabinaceae)	Exaggerated time and space: minutes seem ages; uncontrollable laughter	< morning, coffee	Time distortion
Capsicum annuum (cayenne, Solanaceae)	Homesickness with red cheeks; burning smarting pains; fat, lazy, lax fibre	< cold air, draught; > heat, eating	Mastoiditis; homesickness

Remedy (source)	Keynote	Worse / better	Exam hook
Chamomilla (German chamomile, Compositae)	Cannot bear pain; one cheek red, one pale; child wants to be carried; green stools like chopped eggs	< anger, heat, 9 p.m., night; > being carried	Teething; ailments from anger
Cinchona officinalis (China) (Peruvian bark, Rubiaceae)	Weakness from loss of vital fluids; periodicity; bloating not relieved by belching	< slightest touch, draught, alternate days; > hard pressure	First drug Hahnemann proved (1790)
Cocculus indicus (Indian cockle, Menispermaceae)	Sick from riding in a vehicle; ailments from loss of sleep and nursing the sick; hollow feeling	< vehicle motion, loss of sleep	Travel sickness
Coffea cruda (unroasted coffee, Rubiaceae)	Over-sensitive; sleepless from rush of ideas or joy; toothache > ice-cold water	< excess joy, noise, open air	Sleeplessness from good news
Colchicum autumnale (meadow saffron, Liliaceae)	Gout; nausea and faintness from the smell of cooking food; autumnal dysentery	< motion, evening, smell of food	Nausea from the smell of food
Colocynthis (bitter cucumber, Cucurbitaceae)	Violent cutting colic; ailments from anger with indignation	< anger; > bending double, hard pressure	Colic better bending double
Digitalis purpurea (foxglove, Scrophulariaceae)	Very slow, weak, irregular pulse; feels the heart would stop if he moves; faint sinking at stomach	< sitting erect, motion	Slow pulse; cardiac jaundice
Dulcamara (bittersweet, Solanaceae)	Complaints from damp cold, getting wet, hot days and cold nights; flat smooth warts	< cold damp, night; > warmth	Ailments from wet weather
Eupatorium perfoliatum (boneset, Compositae)	Bone-breaking pains in fever; chill 7 to 9 a.m. with thirst; vomiting of bile	< 7 to 9 a.m., periodically	Influenza, dengue: "break-bone"

Gelsemium to Veratrum viride

Remedy (source)	Keynote	Worse / better	Exam hook
Gelsemium sempervirens (yellow jasmine, Loganiaceae)	Dull, dizzy, drowsy, drooping; trembling; thirstless fever; anticipatory anxiety, diarrhoea from bad news	< emotion, anticipation, damp; > profuse urination	Heavy eyelids (ptosis); exam fright
Hyoscyamus niger (henbane, Solanaceae)	Jealousy, obscenity, wants to uncover; foolish laughter; dry cough < lying, > sitting	< lying, night, touch	Jealous, shameless delirium
Ignatia amara (St Ignatius bean, Loganiaceae)	Grief and disappointment; sighing; lump in throat; contradictory symptoms	< grief, consolation, coffee, tobacco; > eating, change of position	Remedy of contradictions
Ipecacuanha (ipecac root, Rubiaceae)	Constant nausea not relieved by vomiting; clean tongue; bright red bleeding; rattling cough	< vomiting, lying, periodically	Nausea with a clean tongue
Ledum palustre (marsh tea, Ericaceae)	Punctured wounds and stings; the part is cold to touch yet > cold; gout ascends	< warmth of bed; > cold applications	Prevents tetanus; black eye
Lilium tigrinum (tiger lily, Liliaceae)	Bearing down as if pelvic organs would fall out; hurried; religious gloom alternating with sexual excitement	< consolation, warm room	Uterine prolapse with heart pain
Mezereum (spurge olive, Thymelaeaceae)	Eczema with thick crusts and pus beneath; intolerable itching; pain after herpes zoster	< cold air, night, warm bed, touch	Crusts with pus under them
Nux vomica (poison nut, Loganiaceae)	Irritable, fastidious, over-sensitive; sedentary life, stimulants; ineffectual urging for stool; chilly	< morning, mental strain, after eating, cold; > evening, rest, pressure	Ineffectual urging; "businessman"
Opium (poppy, Papaveraceae)	Painlessness, stupor, complaints from fright; constipation with no urge, round black balls; contracted pupils	< heat, during and after sleep	Lack of reaction

Remedy (source)	Keynote	Worse / better	Exam hook
Podophyllum peltatum (may apple, Berberidaceae)	Profuse gushing painless morning diarrhoea, then weakness; diarrhoea alternates with headache	< early morning, hot weather, teething	Gushing diarrhoea at dawn
Pulsatilla nigricans (wind flower, Ranunculaceae)	Mild, yielding, weeps easily, wants consolation; changeable symptoms; thirstless; bland yellow-green discharges	< warm room, evening, rich food; > open air, consolation	Weepy, thirstless, wants open air
Rhododendron chrysanthum (snow rose, Ericaceae)	Rheumatic pains worse before a thunderstorm; orchitis, testis drawn up	< before a storm, wind; > after the storm breaks	Dread of thunder
Rhus toxicodendron (poison ivy, Anacardiaceae)	Restless; stiffness < first motion, > continued motion; red triangular tip of tongue; vesicles	< cold damp, rest, overlifting; > warmth, continued motion	The "rusty gate"
Rumex crispus (yellow dock, Polygonaceae)	Dry tickling cough from cold air, covers the mouth; morning diarrhoea	< inhaling cold air, uncovering, pressure on throat-pit	Cough < cold air
Ruta graveolens (rue, Rutaceae)	Bruised bones, periosteum and tendons; eyestrain from fine work; rectal prolapse	< lying, cold, wet, using the eyes	Tendons and eyestrain
Sabina (savine, Cupressaceae)	Bright red uterine bleeding with clots; pain from sacrum to pubis; threatened abortion in the 3rd month	< heat, least motion; > cool fresh air	Pain sacrum to pubis
Sanguinaria canadensis (blood root, Papaveraceae)	Right-sided periodic headache from occiput to over the right eye; burning palms and soles; flushes	< sun, right side; > sleep, vomiting	Right-sided sun headache
Spigelia anthelmia (pinkroot, Loganiaceae)	Left-sided neuralgia of face and eye rising and setting with the sun; violent palpitation; fear of pins	< touch, motion, noise; > lying on right side	Left-sided neuralgia; worms
Staphisagria (stavesacre, Ranunculaceae)	Suppressed anger and indignation; effects of insult; clean cuts and surgical wounds; styes	< anger, touch	Incised wounds; "honeymoon" cystitis

Remedy (source)	Keynote	Worse / better	Exam hook
Stramonium (thorn apple, Solanaceae)	Violent delirium; fear of the dark, wants light and company; stammering; fear of water	< dark, alone, shining objects	Fear of the dark
Veratrum album (white hellebore, Melanthiaceae)	Collapse with cold sweat on the forehead; profuse vomiting and purging; thirst for cold water	< least motion, drinking	Cold sweat on the forehead (cholera)
Veratrum viride (American hellebore, Melanthiaceae)	Violent congestion, high blood pressure; red streak down the middle of the tongue; sunstroke	< rising; > lying	Red stripe on the tongue

Families worth knowing

- **Solanaceae:** Belladonna, Hyoscyamus, Stramonium (the three delirium remedies), Capsicum, Dulcamara.
- **Loganiaceae:** Nux vomica, Ignatia, Gelsemium, Spigelia.
- **Ranunculaceae:** Aconite, Pulsatilla, Staphisagria.
- **Rubiaceae:** Cinchona, Ipecac, Coffea.
- **Compositae:** Arnica, Chamomilla, Eupatorium.

Memory hook: the delirium trio by temperature of mind: **Bell burns** (hot, violent, red), **Hyos hides nothing** (jealous, shameless, uncovers), **Stram screams for light** (terror of the dark). And for motion: **Bryonia freezes, Rhus tox keeps walking.**

2.3 Mineral, chemical and acid remedies

The syllabus puts 39 remedies under this heading. A few are not minerals (Anacardium is a nut, Apis a bee, Bovista a puff-ball, Cantharis a beetle, Spongia a sponge), and the source column says so, since source questions are common.

Alumina to Fluoric acid

Remedy (source)	Keynote	Worse / better	Exam hook
Alumina (aluminium oxide)	Extreme dryness; constipation with no urge, even a soft stool needs straining; craves chalk and charcoal; time passes too slowly	< morning, potatoes, alternate days; > evening, open air	Chronic of Bryonia
Ammonium carbonicum (ammonium carbonate)	Fat, sedentary women; nosebleed on washing the face in the morning; dyspnoea, weak heart	< 3 a.m., cold wet; > lying on the stomach	Nosebleed on washing the face
Anacardium orientale (marking nut, a plant)	Two wills, contradiction; feeling of a plug in rectum and elsewhere; memory loss; cursing	< hot water applications; > eating	Better while eating
Antimonium crudum (black sulphide of antimony)	Thick milky-white tongue; cross, cannot bear to be touched or looked at; horny callosities; overeating	< sun's heat, cold bathing; > rest, open air	White-coated tongue
Antimonium tartaricum (tartar emetic)	Coarse rattling of mucus with little expectoration; drowsy, weak; nausea	< lying, warmth, damp cold; > sitting up, expectorating	Rattling chest, young and old
Apis mellifica (honey bee)	Oedema; stinging, burning pains; thirstless; jealous and busy; scanty urine	< heat, touch, after sleep; > cold, open air	Stinging, thirstless, puffy
Argentum nitricum (silver nitrate)	Anticipatory anxiety with diarrhoea; hurried, impulsive; craves sweets that disagree; flatulence; fear of heights	< warmth, sweets, emotion; > cold, fresh air	Splinter pains; fear of heights
Arsenicum album (arsenious oxide)	Anxiety, restlessness and prostration out of proportion; burning > heat; thirst for small sips; fastidious; fear of death	< midnight to 2 a.m., cold, cold drinks; > heat, head raised	Burning better by heat
Aurum metallicum (gold)	Profound depression, suicidal, feels worthless; bone pains at night; high blood pressure	< night, cold, winter	Suicidal depression
Baryta carbonica (barium carbonate)	Dwarfish in mind and body; slow, shy, hides behind furniture; big tonsils; children and the old	< thinking of complaints, cold; > walking in open air	Slow, shy child

Remedy (source)	Keynote	Worse / better	Exam hook
Borax (sodium borate)	Fear of downward motion; aphthae in the mouth; infant cries when laid down	< downward motion, sudden noise; > pressure	Fear of downward motion
Bovista (puff-ball fungus)	Puffy, awkward, drops things; blunt instruments leave deep marks; menses only at night	< full moon, heat	Deep marks from blunt edges
Calcarea carbonica (middle layer of oyster shell)	Fat, fair, flabby, chilly; sweat on the head; craves eggs and chalk; slow; open fontanelles	< cold wet, exertion, milk; > dry weather	Sweaty head, craves eggs
Calcarea phosphorica (calcium phosphate)	Delayed dentition and late-closing fontanelles; tall, thin; craves smoked meat; growing pains	< cold damp, weather change; > summer, lying	Craves bacon
Cantharis (Spanish fly, a beetle)	Burning, raw, cutting pains; constant urging, urine drop by drop with burning; burns and scalds	< touch, urinating, cold water; > warmth, rubbing	Burning cystitis; burns
Carbo vegetabilis (vegetable charcoal)	Collapse, wants to be fanned; cold breath; flatulence > belching; venous stasis	< evening, fat food, warm damp; > fanning, belching	The "corpse reviver"
Causticum (Hahnemann's tinctura acris sine kali)	Gradual paralysis (vocal cords, bladder, right face); burning rawness; sympathy for others; urine escapes on coughing	< dry cold wind, clear weather; > damp wet weather	Better in wet weather
Ferrum metallicum (iron)	Anaemia with false plethora, face flushes easily; vomits food after eating	< rest, midnight, eggs; > walking slowly	Pseudo-plethora
Ferrum phosphoricum (iron phosphate)	First stage of inflammation, less violent than Aconite; flushing; nosebleed	< night, 4 to 6 a.m., right side; > cold	Tissue salt for early fever
Fluoricum acidum (hydrofluoric acid)	Varicose veins, bone caries; a hot patient; more able to exert than before	< warmth; > cold, walking	Hot patient, bone decay

Graphites to Zincum

Remedy (source)	Keynote	Worse / better	Exam hook
Graphites (black lead, plumbago)	Fat, chilly, constipated; eczema with honey-like sticky discharge, cracks behind the ears; timid; music makes her weep	< warmth (skin), night, menses; > dark, wrapping up	Honey-like discharge
Hepar sulphuris calcareum (Hahnemann's calcium sulphide)	Over-sensitive to pain, cold and touch; splinter pains; tends to suppurate; very chilly, irritable	< cold dry air, uncovering, touch; > warmth, damp weather	Suppuration: low potency ripens, high aborts
Iodium (iodine)	Thin despite a ravenous appetite, > eating; restless, hot; hard enlarged glands, goitre	< warmth, rest, fasting; > eating, cold air	Eats well, yet wastes
Kali carbonicum (potassium carbonate)	Stitching pains; weak back; bag-like swelling of the upper lids; asthma at 3 a.m. > bending forward; rigid, dutiful	< 2 to 4 a.m., cold, lying on left side; > warm weather, sitting forward	Puffy upper lids
Magnesia carbonica (magnesium carbonate)	Sour-smelling child; green frothy stools like frog-pond scum; night neuralgia, must walk about	< rest, warmth of bed, temperature change; > walking about	The "orphan's remedy"
Mercurius solubilis (Hahnemann's soluble mercury)	Profuse offensive sweat that does not relieve; salivation; flabby tongue with tooth marks; metallic taste	< night, sweating, warmth of bed	The "human thermometer"
Natrum carbonicum (sodium carbonate)	Weak digestion and weak ankles; cannot bear sun, heat or milk; averse to company	< summer heat, sun, mental strain, milk; > movement	Weak ankles; sun headache
Natrum muriaticum (common salt)	Ailments from grief, consolation aggravates; craves salt; thin neck; hammering headache sunrise to sunset; map tongue	< 10 a.m., sun, consolation, seashore; > open air, cold bathing	Consolation aggravates
Natrum sulphuricum (Glauber's salt)	Hydrogenoid constitution, worse damp; effects of head injury; diarrhoea after rising	< damp weather, lying on left side; > dry weather	Head injury; damp asthma

Remedy (source)	Keynote	Worse / better	Exam hook
Nitricum acidum (nitric acid)	Splinter-like pains; fissures where skin meets mucous membrane; urine smells like a horse's; warts bleed easily; vindictive	< evening, night, cold, touch; > riding in a carriage	Splinter pains, strong urine
Petroleum (crude rock oil)	Deep bleeding cracks of skin in winter, healed in summer; motion sickness; diarrhoea only by day	< winter, riding, thunderstorm; > warm dry air	Winter eczema
Phosphorus (phosphorus)	Tall, slender, sympathetic; fears being alone and thunder; craves cold water, vomited once warm in the stomach; bright red bleeding	< lying on left side, twilight; > eating, sleep, cold food	Vomits water once warm
Platinum metallicum (platinum)	Pride, looks down on others; things look smaller than they are; numbness; hypersensitive genitals	< evening, touch; > walking in open air	Objects look small
Silicea (pure flint)	Lacks grit, yielding yet obstinate; chilly; offensive foot sweat; expels foreign bodies; stool slips back	< cold, uncovering, new moon; > warmth, wrapping the head	"Bashful" stool
Spongia tosta (roasted sponge)	Dry, sawing, barking croup cough like a saw through a pine board; goitre; heart symptoms	< before midnight, ascending, cold wind; > warm food and drink	Croup
Stannum metallicum (tin)	Great weakness of the chest, too weak to talk; pains come and go gradually; sweet green sputum	< talking, 10 a.m.; > hard pressure, coughing up	Pains rise and fall slowly
Sulphur (sublimated sulphur)	Untidy philosopher; burning soles and vertex; itching < warmth of bed; craves sweets and fats; hungry at 11 a.m.; red orifices	< bathing, warmth of bed, standing, 11 a.m.; > dry warm weather	King of antipsorics
Sulphuricum acidum (sulphuric acid)	Weakness with inner trembling; always in a hurry; dark bleeding; hot flushes; alcoholism	< cold, open air; > warmth, hot drinks	Weakness and trembling
Zincum metallicum (zinc)	Brain fag; fidgety feet, cannot keep them still; suppressed eruptions	< wine, touch, 5 to 7 p.m.; > discharges, eating	Fidgety feet

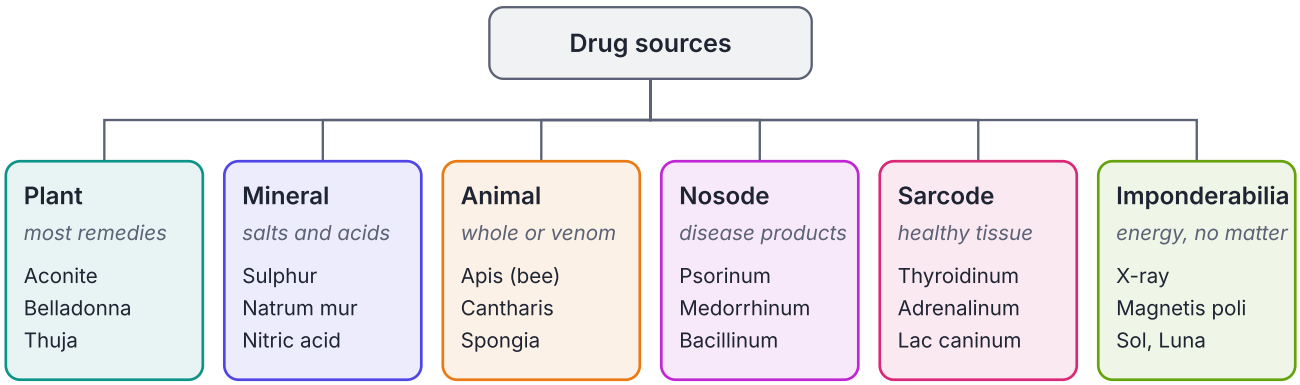
Memory hook: the **Natrum family** splits by weather: **Nat mur** hates the **sun**, **Nat carb** hates the **sun and milk**, **Nat sulph** hates the **damp**. And the **Calc pair** by food: **Calc carb** wants **eggs**, **Calc phos** wants **bacon**.

2.4 Nosodes, sarcodes and special therapeutics

Definitions

- **Nosodes** are prepared from diseased tissue or the products of disease (discharges, cultures, morbid tissue). Hering introduced the idea; Burnett and Swan widened its use.
- **Sarcodes** are prepared from healthy animal tissues and secretions: Thyroidinum, Adrenalinum, Pituitrinum, Insulinum, Lac caninum.
- **Imponderabilia** are energies with no material substance: X-ray, Magnetis poli, Sol, Luna, Electricitas.
- A nosode is often used as an **intercurrent remedy** when the well-chosen remedy stops acting or a miasm blocks progress.

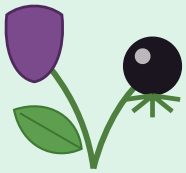
Six sources of homoeopathic drugs: name the source, then the remedy



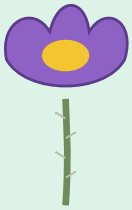
drug sources · 6 kinds with examples

Where the remedies come from

Source, family and one keynote to hang it on



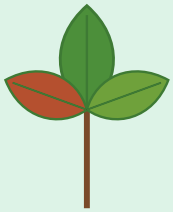
Belladonna
deadly nightshade · Solanaceae
Sudden heat, redness, throbbing; dilated pupils



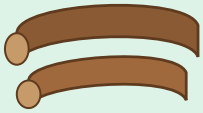
Pulsatilla
wind flower · Ranunculaceae
Mild, weepy, thirstless; better in open air



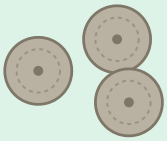
Arnica
leopard's bane · Compositae
Bruised soreness; says he is well when very ill



Rhus tox
poison ivy · Anacardiaceae
Stiff; worse first motion, better continued motion



Cinchona (China)
Peruvian bark · Rubiaceae
Weakness from loss of fluids; first drug proved



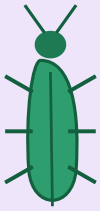
Nux vomica
poison nut (seeds) · Loganiaceae
Irritable, chilly; urging without result



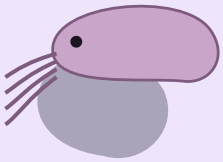
Apis
honey bee
Oedema, stinging pains, thirstless



Lachesis
snake venom
Worse after sleep



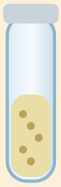
Cantharis
Spanish fly, a beetle
Burning cystitis; burns and scalds



Sepia
cuttlefish ink
Complements Nat mur



Spongia
roasted sponge
Dry, sawing croup cough like a saw through pine



Nosodes
disease products
Psorinum, Medorrhinum, Pyrogenium, Bacillinum

Trap: Apis, Cantharis and Spongia are ANIMAL remedies even when listed with the minerals (purple tiles).

Remedy sources · plants, animals and nosodes, each with one keynote

Source questions often hide a trap: Apis, Cantharis and Spongia are animal remedies even when listed with minerals.

The six named in the syllabus

Remedy (source)	Keynote	Worse / better	Exam hook
Bacillinum (Burnett; tuberculous lung tissue)	Recurrent colds, tubercular tendency; ringworm; catarrh with thick sputum	< night, cold	Burnett's nosode; ringworm

Remedy (source)	Keynote	Worse / better	Exam hook
Medorrhinum (gonorrhoeal discharge)	Sycotic; always in a hurry yet time passes slowly; bites nails; sleeps in the knee-chest position	< sunrise to sunset, heat; > seashore, lying on the abdomen	Better at the seashore
Psorinum (scabies vesicle)	Very chilly, wants warm clothes even in summer; dirty, offensive discharges; hopeless despair; hungry at night	< cold, open air, coffee; > heat, lying with arms spread apart	Wears a fur coat in summer
Pyrogenium (decomposed lean beef)	Septic states; pulse out of proportion to temperature; bed feels too hard; restless	< cold; > heat, motion	Pulse and temperature mismatch
Thuja occidentalis (arbor vitae, a plant)	Anti-sycotic: warts, fig-warts, polypi; fixed idea that the body is brittle or made of glass; sweat on uncovered parts	< cold damp, 3 a.m. and 3 p.m., vaccination; > drawing up a limb	Ill effects of vaccination
Glonoine (nitroglycerine)	Sudden violent bursting, throbbing headache; sunstroke; cannot bear heat on the head	< sun, heat, jar; > cold, open air	Sunstroke

Clinical organ therapeutics

Organ remedies (Rademacher's idea, developed by Burnett as **organopathy**) act on one organ and are often used in low potency or mother tincture alongside the constitutional remedy.

Organ	Organ remedies
Heart	Crataegus (heart tonic), Cactus grandiflorus (constriction as if by an iron band), Digitalis
Liver	Chelidonium (pain under the right scapula), Carduus marianus, Chionanthus
Spleen	Ceanothus americanus
Kidney	Berberis vulgaris, Solidago virgaurea
Bladder	Cantharis, Equisetum
Prostate	Sabal serrulata
Uterus	Caulophyllum, Sabina
Thyroid	Thyroidinum, Iodium, Spongia

Organ	Organ remedies
Bones and teeth	Calcarea phos, Symphytum (fracture)

Memory hook: **N**osode = **N**asty (disease product); **S**arcode = **S**ound (healthy tissue). Then the six in a sentence: "**Medo** loves the sea, **Psor** wears a fur coat, **Pyro**'s pulse lies, **Thuja** hates the needle, **Glonoine** hates the sun, **Bacillinum** keeps catching cold."

Rapid revision

Cue	Remedy or fact
Sudden onset after cold dry wind, fear of death	Aconite
Worse slightest motion, thirst for large quantities	Bryonia
Better continued motion, worse first motion	Rhus tox
Weepy, thirstless, worse warm room	Pulsatilla
Consolation aggravates, craves salt	Natrum mur
Burning better by heat, worse after midnight	Arsenicum album
Collapse, wants to be fanned	Carbo veg
Cold sweat on the forehead, cholera	Veratrum album
Nausea with a clean tongue	Ipecac
Colic better bending double	Colocynth
Punctured wound, part cold yet better cold	Ledum
Gushing painless morning diarrhoea	Podophyllum
Milk vomited in curds by infants	Aethusa
Fear of downward motion	Borax
Vomits water once it warms in the stomach	Phosphorus
Splinter pains, horse-like urine	Nitric acid
Croup like a saw through a board	Spongia

Cue	Remedy or fact
Better at the seashore	Medorrhinum
Pulse out of proportion to temperature	Pyrogenium
Sunstroke, bursting headache	Glonoine
First drug proved by Hahnemann	Cinchona (1790)
King of antipsorics	Sulphur
Inimical pairs	Causticum–Phosphorus, Apis–Rhus tox, Mercurius–Silicea
Nosode / sarcode	Diseased product / healthy tissue

One-glance recall card

Topic 2 recall card: the 15 keynotes asked most

Cover the small lines, name the remedy from the big cue, then check

Cold dry wind
sudden onset, fear of death:
Aconite

Lies still
worse slightest motion,
big thirst: Bryonia

Rusty gate
worse first motion, better
continued motion: Rhus tox

Weepy, thirstless
worse warm room, better
open air: Pulsatilla

Consolation agg.
grief, craves salt,
sun headache: Nat mur

Burning > heat
restless, sips, worse
after midnight: Arsenicum

Clean tongue
nausea not relieved by
vomiting: Ipecac

Bending double
colic better pressure
and bending: Colocynth

Cold yet > cold
punctured wounds,
stings: Ledum

Fan me
collapse, cold breath,
wants air: Carbo veg

Warm water vomit
vomits water once warm
in stomach: Phosphorus

Seashore
better at the sea,
knee-chest sleep: Medorrhinum

Pulse mismatch
pulse out of step with
temperature: Pyrogenium

1790
first drug Hahnemann
proved: Cinchona

Inimical
Causticum-Phosphorus,
Apis-Rhus, Merc-Silicea

Topic 2 recall card · 15 tiles

For the last ten minutes before the exam: cover the small lines of each tile and name the remedy from the big cue.

Practice MCQs

1. A remedy whose complaints come on suddenly after exposure to cold dry wind, with fear of death, is: (a) Belladonna (b) Aconite (c) Bryonia (d) Gelsemium
2. Worse from the slightest motion and better lying on the painful side is typical of: (a) Rhus tox (b) Bryonia (c) Pulsatilla (d) Arnica
3. A thirstless, weepy patient who is worse in a warm room and better in open air needs: (a) Pulsatilla (b) Natrum mur (c) Sepia (d) Ignatia
4. Constant nausea not relieved by vomiting, with a clean tongue, suggests: (a) Nux vomica (b) Antimonium crudum (c) Ipecacuanha (d) Arsenicum album
5. Colic better by bending double and hard pressure is a keynote of: (a) Chamomilla (b) Colocynthis (c) Belladonna (d) Dioscorea
6. The first drug Hahnemann proved on himself was: (a) Aconite (b) Belladonna (c) Cinchona (d) Sulphur
7. Punctured wounds that are cold to touch yet better by cold applications call for: (a) Hypericum (b) Arnica (c) Ledum (d) Calendula
8. An infant who vomits milk in curds soon after nursing and then falls asleep needs: (a) Aethusa (b) Magnesia carb (c) Calcarea carb (d) Chamomilla
9. Fear of downward motion is a keynote of: (a) Gelsemium (b) Borax (c) Argentum nitricum (d) Calcarea carb
10. Water is vomited as soon as it becomes warm in the stomach in: (a) Arsenicum album (b) Bismuth (c) Phosphorus (d) Veratrum album
11. Splinter-like pains with urine smelling like a horse's suggest: (a) Hepar sulph (b) Nitric acid (c) Argentum nitricum (d) Benzoic acid
12. A croupy cough sounding like a saw driven through a pine board needs: (a) Hepar sulph (b) Aconite (c) Spongia tosta (d) Drosera
13. Consolation aggravates, craving for salt and a hammering headache from sunrise to sunset point to: (a) Ignatia (b) Natrum mur (c) Sepia (d) Staphisagria
14. Which pair is inimical? (a) Sulphur and Calcarea carb (b) Causticum and Phosphorus (c) Belladonna and Calcarea carb (d) Nat mur and Sepia
15. A remedy prepared from the products of disease is called a: (a) sarcode (b) nosode (c) imponderabilia (d) tissue salt
16. Medorrhinum patients are characteristically better: (a) in the mountains (b) at the seashore (c) in a warm room (d) after sleep
17. A pulse out of proportion to the temperature in septic fever suggests: (a) Baptisia (b) Pyrogenium (c) Arsenicum album (d) Lachesis

18. Bursting headache from exposure to the sun (sunstroke) is a keynote of: (a) Natrum carb (b) Glonoine (c) Belladonna (d) Gelsemium

19. Collapse with a cold body and a desire to be fanned calls for: (a) Camphor (b) Veratrum album (c) Carbo vegetabilis (d) Arsenicum album

20. Ill effects of vaccination are classically treated with: (a) Thuja (b) Silicea (c) Sulphur (d) Medorrhinum

Answer key

Q	Answer	Why
1	b	Sudden onset, cold dry wind, fear of death
2	b	Bryonia wants to lie still; pressure relieves
3	a	Weepy, thirstless, warm room aggravates
4	c	Ipecac: nausea with a clean tongue
5	b	Colocynth: better bending double
6	c	Cinchona bark, 1790
7	c	Ledum: cold yet better cold
8	a	Aethusa: milk intolerance in infants
9	b	Borax: fear of downward motion
10	c	Phosphorus vomits water once warm
11	b	Nitric acid: splinter pains, strong urine
12	c	Spongia: sawing croup
13	b	Nat mur: grief, salt, sun headache
14	b	Causticum and Phosphorus must not follow each other
15	b	Nosode = product of disease
16	b	Medorrhinum is better at the seashore
17	b	Pyrogenium: pulse–temperature mismatch
18	b	Glonoine: sunstroke

Q	Answer	Why
19	c	Carbo veg: the "corpse reviver"
20	a	Thuja: vaccination and sycosis

Topic 3: Homoeopathic Pharmacy

How to use this guide

Pharmacy is one of the ten Core topics (the Core section carries 25 of the 60 CBRT marks), so expect about 2 to 3 questions. They ask a ratio, a percentage, a licence form, a year or an abbreviation, so this guide keeps the numbers in tables you can learn cold.

Syllabus heading	What it covers	Section
Sources, vehicles and drug collection	Vegetable, mineral, animal, nosodes, sarcodes, imponderabilia; liquid and solid vehicles	3.1
Scales and potentisation	Decimal, centesimal, LM; Hahnemannian and Korsakovian methods; trituration and succussion	3.2
Standards and legislation	Homoeopathic Pharmacopoeia of India; Drugs and Cosmetics Act and Rules	3.3
Prescription writing and dispensing	Parts of a prescription, abbreviations, posology, dosage forms	3.4

Standard books: Mandal and Mandal, *A Textbook of Homoeopathic Pharmacy*; *Homoeopathic Pharmacopoeia of India* (HPI); Hahnemann's *Organon* (§§264 to 271 for preparation); the Drugs and Cosmetics Act 1940 and Rules 1945.

Memory hook: pharmacy is a kitchen: **ingredients** (sources), **pots and water** (vehicles), **the recipe** (scales and potentisation), **the food inspector** (HPI and the Act) and **the menu card** (the prescription).

3.1 Sources, vehicles and drug collection

Sources of drugs

Kingdom	What is used	Examples
Vegetable	Whole plant or a part: root, bark, leaf, flower, seed, resin, juice	Aconite, Belladonna, Cinchona (bark), Nux vomica (seeds)
Mineral	Elements, salts, acids	Sulphur, Natrum mur, Calcarea carb, Nitric acid

Kingdom	What is used	Examples
Animal	Whole animal, venom or secretion	Apis (bee), Cantharis (beetle), Lachesis (snake venom), Sepia (ink)
Nosodes	Diseased tissue or products of disease	Psorinum, Medorrhinum, Tuberculinum
Sarcodes	Healthy animal tissue or secretion	Thyroidinum, Adrenalinum, Pepsinum
Imponderabilia	Energy with no matter	X-ray, Magnetis poli, Sol, Luna

Collecting plants

- Collect from the plant's **natural habitat**, wild where possible, healthy and free from disease.
- Collect in **dry, clear weather**, and use the plant fresh unless the monograph says dried.
- Collect each part when it is most active: **whole plants** usually as flowering begins; **seeds and fruits** when fully ripe.
- Identify the plant correctly by botanical name; the HPI monograph gives the part used and how to prepare it.

Vehicles

A **vehicle** is the inert medium that carries the drug. It must be pure, inert, and keep the drug stable.

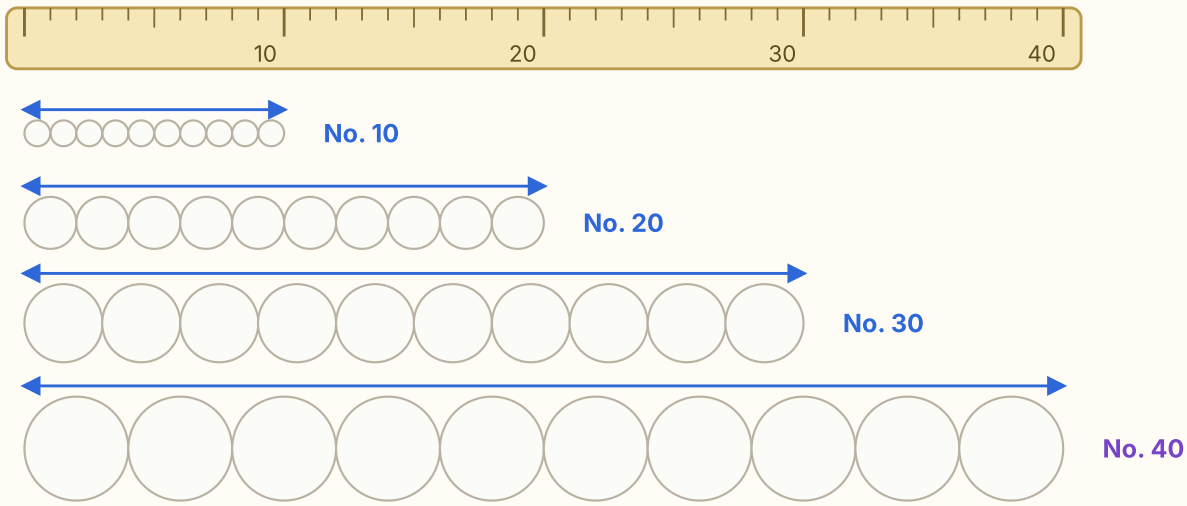
Vehicle	Key facts
Purified water	Used to dissolve water-soluble substances and to make some lower dilutions
Strong alcohol	95% v/v ethanol (limits about 94.7 to 95.2%); the usual base for dilutions
Absolute alcohol	At least 99.4% v/v
Dilute alcohol	Strong alcohol diluted with purified water; used for some mother tinctures
Dispensing alcohol	88% v/v; used to medicate globules
Glycerin	For glycerols and some external preparations
Sugar of milk (lactose)	Solid vehicle for triturations and tablets; about 1 g dissolves in 5 ml of cold water; insoluble in alcohol
Cane sugar globules	Sizes 5 to 80 (commonly 10 to 40); sizes 40 to 80 are called pilules; the size number is the length in mm of 10 globules in a row
Tablets	Lactose, about 1 grain (65 mg) each

Vehicle	Key facts
Cones	Cane sugar and egg; a No. 6 cone absorbs one drop of dispensing alcohol

Globule sizes: read the number

Cane-sugar globules on a millimetre ruler

The size number = the length, in mm, of 10 globules laid in a row



Sizes run 5 to 80 (commonly 10 to 40); 40 to 80 are called pilules.
Globules are cane sugar; they are medicated with 88% dispensing alcohol, then dried.
LM potencies use tiny poppy-seed globules: one drop medicates 500 of them.

Globule sizes · the number is the length in mm of 10 globules in a row

Memory hook: 95, 99.4, 88: "Strong is **95**, absolute **99.4**, the dispenser is **88**." Lactose dissolves in **water**, never in alcohol: so a 3C trituration is dissolved in water first, and alcohol is added after.

Preparing the neutral vehicles

A vehicle is **neutral**: medicinally inert, free from taste and smell, chemically stable, and able to carry the drug energy. Every vehicle is prepared and stored so that it stays pure.

Vehicle	How it is prepared and kept
Distilled water	Made by distillation; used fresh to dissolve triturations (the 3C conversion) and to make dilute alcohols; stored in clean, glass-stoppered bottles
Alcohol	Strong alcohol (95% v/v) is diluted with distilled water to the strength each step needs, for example dispensing alcohol (88%) for medicating globules

Vehicle	How it is prepared and kept
Sugar of milk	Pure, crystalline lactose, free from cane sugar and starch; ground to a fine powder and sieved before trituration; kept dry in closed containers
Globules, pilules, tablets	Globules and pilules are made of cane sugar; tablets are pressed from lactose; all are medicated only after they are made, by soaking in the potency in 88% alcohol and drying

Memory hook: water dissolves, alcohol preserves, lactose grinds, sugar carries. Neutral means it adds nothing of its own.

3.2 Scales and potentisation

Potentisation (dynamisation) is the step-by-step dilution of a drug with **succussion** (vigorous shaking) for liquids or **trituration** (grinding) for solids. Hahnemann describes it in Organon §§269 to 271; §270 gives the LM method.

The three scales

Scale	Ratio per step	Symbol	Introduced by	Note
Decimal	1 : 10	X or D	Constantine Hering	Mainly for low potencies and triturations
Centesimal	1 : 100	C or CH	Hahnemann	The standard scale; 1C = 2X in dilution
Fifty millesimal (LM, Q)	1 : 50,000	LM, 0/1, Q	Hahnemann, 6th edition	100 succussions per step; written 0/1, 0/2 and so on

Ten, hundred, fifty thousand: the three potency scales

Drug to vehicle at each step, who introduced it, and how it is written

Decimal · X

1 : 10

1 part drug + 9 parts vehicle
introduced by Hering
written 6X, 12X

Centesimal · C

1 : 100

1 part drug + 99 parts vehicle
Hahnemann's standard scale
written 30C, 200C, 1M

Fifty millesimal · LM

1 : 50,000

100 succussions per step
Organon 6th edition, §270
written 0/1, 0/2, 0/3

Insoluble drugs: three triturations, then conversion to liquid

Drug + lactose ground → **1C** one hour → **2C** one hour → **3C** one hour → **Dissolve** water first → **4C onwards** in alcohol

potency scales and trituration · 3 scales, 6 steps

Say the big numbers aloud, left to right: ten, hundred, fifty thousand. Then walk the bottom row: three hours of grinding before the drug ever meets alcohol.

The **mother tincture** (Ø) is the starting liquid preparation of a drug. **12C (24X)** is roughly where Avogadro's number is passed, so no molecule of the original substance is likely to remain.

Methods of making liquid potencies

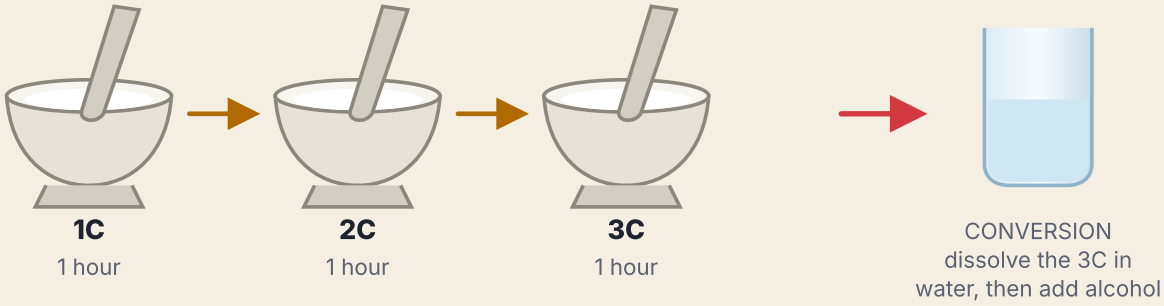
Method	How	Key point
Hahnemannian (multi-vial)	A fresh vial for every potency	The official method; most accurate
Korsakovian (single-vial)	The same vial is emptied and refilled for each step	Faster, less exact; marked K
Continuous fluxion	Fluid flows through a chamber with no separate steps	Used for very high potencies by machine

From drug to potency

Trituration for solids, succussion for liquids

Solids: TRITURATION with sugar of milk

for insoluble drugs: metals, most minerals, many animal products



each hour = 3 rounds of 20 minutes: grind, then scrape

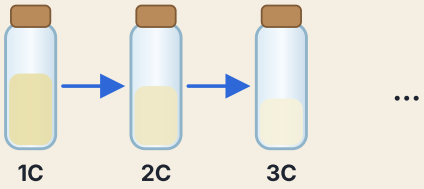
Liquids: DILUTION + SUCCUSSION (vigorous shaking)

centesimal step: 1 part of the previous potency + 99 parts of alcohol, then succussed



HAHNEMANNIAN (multi-vial)

a fresh vial for every potency · official, most accurate



KORSAKOVIAN (single-vial)

the same vial is emptied and refilled each step · faster, marked K



Ratio per step: X 1 : 10 (Hering) · C 1 : 100 (Hahnemann) · LM 1 : 50,000 (6th edition, 100 succussions).

From drug to potency · trituration for solids, succussion for liquids

Trituration

- Used for **insoluble** drugs (metals, most minerals, many animal products): the drug is ground with **sugar of milk**.
- Each potency takes **one hour**: three rounds of 20 minutes, each of grinding and scraping.
- Insoluble drugs are triturated to the 3C (6X); then the 3C is dissolved in water, alcohol is added, and the potency goes on as a liquid (**conversion**).

How LM potency is made (Organon §270)

1. Triturate the drug to the 3C.
2. Dissolve 1 grain of the 3C in 500 drops of a water and alcohol mixture.
3. Take 1 drop of this, add 100 drops of alcohol and give **100 succussions**: this medicates poppy-seed-sized globules, **500 globules per drop**. That is LM 1 (0/1).
4. Dissolve 1 such globule in a drop of water, add 100 drops of alcohol, succuss 100 times: LM 2. And so on.

The ratio is 1/100 × 1/500 = 1 : 50,000 per step.

Memory hook: "**Ten, hundred, fifty thousand**": X is 10, C is 100, LM is 50,000. **Hering** brought the **X**; **Hahnemann** made **C** and **LM**. One **hour** per trituration, **three** triturations, then **convert**.

3.3 Standards and legislation

Homoeopathic Pharmacopoeia of India (HPI)

- A **pharmacopoeia** is the official book of standards for drugs: identity, purity, strength and method of preparation.
- The Government of India set up the **Homoeopathic Pharmacopoeia Committee in 1962**, under **Dr B. K. Sarkar**.
- **HPI Volume I** was published in 1971; further volumes followed. Before it, Indian manufacturers followed the **HPUS** (USA) and the German pharmacopoeia.
- The HPI is listed in the **Second Schedule of the Drugs and Cosmetics Act, 1940**, which makes its standards legally binding.
- Most HPI mother tinctures are made by the **new method**, at a drug strength of about 1/10; Hahnemann's older method sorted drugs into classes by their juice content.
- Drugs are tested at the **Pharmacopoeia Commission for Indian Medicine and Homoeopathy (PCIM&H), Ghaziabad**, which took over the Homoeopathic Pharmacopoeia Laboratory.

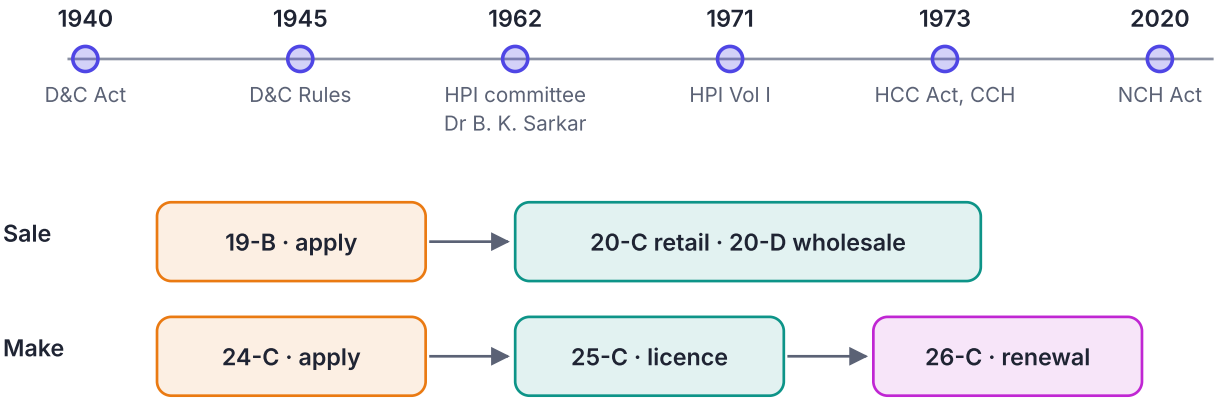
Drugs and Cosmetics Act 1940 and Rules 1945

Provision	What it covers
Rule 2(dd)	Defines homoeopathic medicines (the parenteral route is excluded)
Rule 30AA	Import of new homoeopathic medicines
Rules 67A, 67D, 67E, 67GG	Sale of homoeopathic medicines and the sale licence

Provision	What it covers
Form 19-B	Application for a licence to sell
Form 20-C / 20-D	Retail / wholesale sale licence (valid 5 years)
Form 24-C	Application for a licence to manufacture
Form 25-C	Licence to manufacture
Form 26-C	Certificate of renewal
Schedule M-I	Good manufacturing practice for homoeopathic medicines

From the 1940 Act to the 2020 Commission, and the licence ladder

Milestones in order (not to scale); then which form you apply on and which you receive



The HPI sits in the 2nd Schedule of the 1940 Act; GMP for homoeopathic medicines is Schedule M-I.

pharmacy law timeline and licence forms

Orange boxes are what you apply on, green what you receive; renewal comes last.

Related councils

- **Central Council of Homoeopathy** (CCH) was set up under the **Homoeopathy Central Council Act, 1973**, to regulate education and practice.
- It was replaced by the **National Commission for Homoeopathy** (NCH) under the **NCH Act, 2020**.

Memory hook: "**62** formed, **71** printed": committee 1962, HPI Vol I 1971. Licence forms run like a ladder: **19-B** you ask to sell, **20-C/D** you sell, **24-C** you ask to make, **25-C** you make, **26-C** you renew.

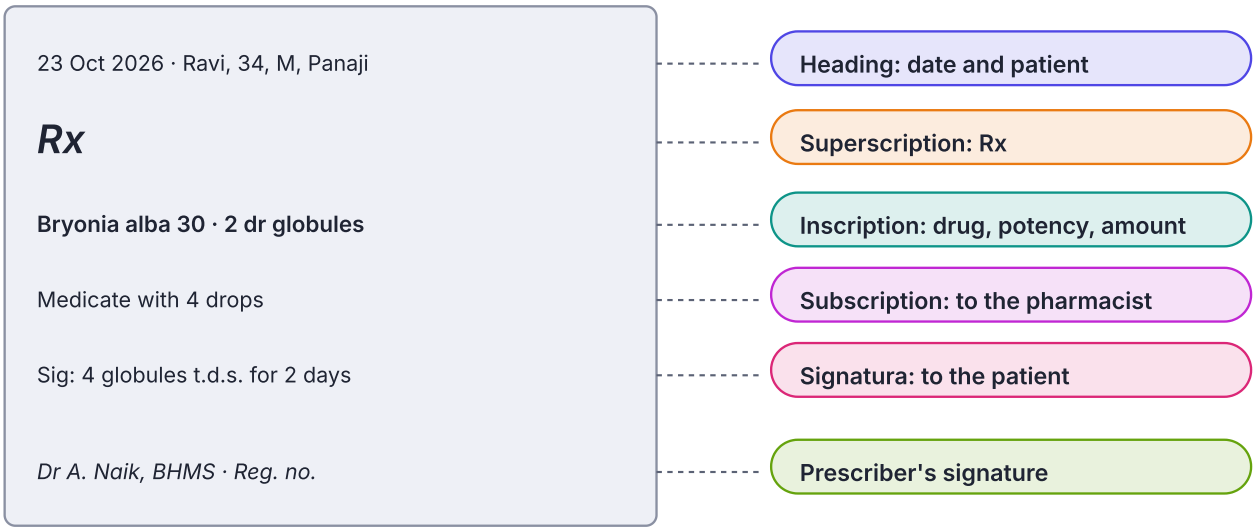
3.4 Prescription writing and dispensing

Parts of a prescription

1. **Heading:** date; the patient's name, age, sex and address; the prescriber's name, qualification and registration number.
2. **Superscription:** the symbol **Rx** ("take thou").
3. **Inscription:** the medicine, its potency and quantity, for example *Bryonia alba 30, 2 dram globules*.
4. **Subscription:** directions to the pharmacist (how to prepare or medicate).
5. **Signatura (signature):** directions to the patient, often marked **Sig.** or **S.**
6. **The prescriber's signature.**

A prescription reads top to bottom like a letter

An example slip on the left; the name of each part on the right



anatomy of a prescription · 6 parts

The two that get mixed up: **sub**scription is for the pharmacist, **sig**natura is for the patient.

Rules

- Write clearly and in full; one remedy at a time, as the Organon teaches (§273).
- Always state the **potency**, the **scale** and the **dose form**.
- Name the medicine by its full Latin name, not a vague abbreviation.
- Give the patient plain directions: when, how many, how long, and what to avoid (strong smells such as camphor or coffee, if the prescriber wishes).

Abbreviations

Abbreviation	Meaning
o.d. / b.d. / t.d.s. / q.i.d.	Once / twice / three times / four times a day
h.s.	At bedtime
s.o.s.	If needed
stat.	At once
a.c. / p.c.	Before food / after food
aq.	Water
gtt.	Drops
glob.	Globules
q.s.	Quantity sufficient
Ø	Mother tincture
S.L.	Saccharum lactis (sugar of milk, placebo)

Dosage forms

- **Internal:** mother tincture, dilutions, triturations, medicated globules (pilules), tablets, cones.
- **External:** ointments, lotions, glycerols, oils, liniments.
- **Medicating globules:** pour enough of the potency, made in **dispensing alcohol (88%)**, to soak all the globules in a clean vial; shake, then let them dry.
- **Storage:** amber glass, well corked, cool and dark, away from strong odours and direct sunlight.

Posology

- Posology is the science of dose: the potency, the quantity and the repetition.
- The **minimum dose** (Organon §§275 to 279) is the smallest dose that cures gently.
- Acute cases may need frequent repetition; chronic cases wait for the dose to act (see Topic 1, 1.3).

Memory hook: a prescription reads like a letter: **date and address**, "**Dear pharmacist**" (Rx), **what to send** (inscription), **how to pack it** (subscription), **how to use it** (signatura), **signed** by you.

Rapid revision

Cue	Fact
Strong alcohol	95% v/v
Absolute alcohol	At least 99.4% v/v
Dispensing alcohol	88% v/v; medicates globules
Sugar of milk	Lactose; soluble in water, insoluble in alcohol
Globule size	Length in mm of 10 globules in a row; 40 to 80 are pilules
Decimal scale	1:10, X; introduced by Hering
Centesimal scale	1:100, C; Hahnemann
LM scale	1:50,000; 100 succussions; Organon §270
Avogadro limit	About 12C (24X)
Trituration	Insoluble drugs with sugar of milk; one hour per potency; to 3C then converted
Hahnemannian method	Separate vial for each potency
Korsakovian method	One vial emptied and refilled
HPI	Committee 1962 (Dr B. K. Sarkar); Vol I 1971; in 2nd Schedule of the D&C Act
Testing laboratory	PCIM&H, Ghaziabad
Sale licence	Form 20-C retail, 20-D wholesale; applied for on 19-B
Manufacture licence	Form 25-C; applied for on 24-C
GMP	Schedule M-I
Superscription	Rx
Signatura	Directions to the patient
S.L.	Sugar of milk placebo

One-glance recall card

Topic 3 recall card: the 15 numbers asked most

Cover the small lines, say the fact from the big cue, then check

95% strong alcohol (v/v); absolute is at least 99.4%	**88%** dispensing alcohol: medicates globules	**Lactose** dissolves in water, not in alcohol
1 : 10 decimal scale, X; introduced by Hering	**1 : 100** centesimal scale, C; Hahnemann's standard	**1 : 50,000** LM scale, Organon §270; 100 successions a step
12C = 24X Avogadro's number passed; no molecule likely left	**3C, 1 hour each** triturate insolubles to 3C, then dissolve in water	**Multi vs single** Hahnemannian: new vial; Korsakovian: same vial
1962 · 1971 HPI committee (B. K. Sarkar); HPI Vol I published	**2nd Schedule** HPI is in the 2nd Schedule of the D&C Act 1940	**19-B → 20-C/D** apply to sell, then retail or wholesale licence
24-C → 25-C apply to make, then the manufacture licence	**Schedule M-I** good manufacturing practice for homoeopathic medicines	**Sub vs Sig** subscription: pharmacist; signatura: patient

Topic 3 recall card · 15 tiles

For the last ten minutes before the exam: cover the small lines of each tile and say the fact from the big cue.

Practice MCQs

1. Strong alcohol as defined in the HPI contains about: (a) 88% v/v (b) 95% v/v (c) 99.4% v/v (d) 62% v/v
2. Dispensing alcohol used to medicate globules is: (a) 95% (b) 70% (c) 88% (d) 60%
3. Absolute alcohol contains not less than: (a) 95% (b) 98% (c) 99.4% (d) 100%
4. The drug-to-vehicle ratio of the decimal scale is: (a) 1:10 (b) 1:100 (c) 1:1000 (d) 1:50,000
5. The decimal scale was introduced by: (a) Hahnemann (b) Hering (c) Korsakoff (d) Kent
6. The 50 millesimal potency is described in the Organon in: (a) §128 (b) §153 (c) §270 (d) §273
7. Each LM potency step receives: (a) 10 successions (b) 50 successions (c) 100 successions (d) 2 successions
8. Insoluble substances are triturated up to: (a) 1C (b) 3C (c) 6C (d) 12C
9. Each trituration potency takes about: (a) 10 minutes (b) 30 minutes (c) one hour (d) three hours

10. The single-vial method of potentisation is the: (a) Hahnemannian (b) Korsakovian (c) fluxion (d) LM
11. Sugar of milk is: (a) soluble in alcohol (b) insoluble in water (c) soluble in water, insoluble in alcohol (d) insoluble in both
12. The Homoeopathic Pharmacopoeia Committee was formed in: (a) 1948 (b) 1962 (c) 1971 (d) 1973
13. HPI Volume I was published in: (a) 1962 (b) 1971 (c) 1973 (d) 1980
14. A retail sale licence for homoeopathic medicines is issued in Form: (a) 19-B (b) 20-C (c) 20-D (d) 25-C
15. A licence to manufacture homoeopathic medicines is issued in Form: (a) 24-C (b) 25-C (c) 26-C (d) 20-D
16. Good manufacturing practice for homoeopathic medicines is laid down in: (a) Schedule M (b) Schedule M-I (c) Schedule H (d) Schedule X
17. The part of a prescription giving directions to the patient is the: (a) superscription (b) inscription (c) subscription (d) signatura
18. In homoeopathic prescriptions, S.L. stands for: (a) sublingual (b) saccharum lactis (c) single liquid (d) solution
19. Globules of size 40 to 80 are called: (a) cones (b) pilules (c) tablets (d) granules
20. The potency beyond which no molecule of the original drug is likely to remain is about: (a) 3C (b) 6C (c) 12C (d) 30C

Answer key

Q	Answer	Why
1	b	95% v/v ethanol
2	c	88% v/v
3	c	At least 99.4% v/v
4	a	X = 1:10
5	b	Constantine Hering
6	c	§270, the LM method
7	c	100 succussions per step
8	b	Triturate to 3C, then convert
9	c	Three rounds of 20 minutes
10	b	Korsakovian: one vial

Q	Answer	Why
11	c	Lactose: water yes, alcohol no
12	b	1962, under Dr B. K. Sarkar
13	b	1971
14	b	20-C retail; 20-D wholesale
15	b	25-C; 24-C is the application
16	b	Schedule M-I
17	d	Signatura: directions to the patient
18	b	Sugar of milk, the placebo
19	b	Pilules
20	c	12C (24X): Avogadro's number passed

Topic 4: Repertory & Repertorisation

How to use this guide

Repertory is one of the ten Core topics (the Core section carries 25 of the 60 CBRT marks), so expect about 2 to 3 questions. The GPSC syllabus splits it into four headings, and this guide follows them in order:

1. **4.1 History and structure:** from Boenninghausen to Kent, then synthetic and computer repertories (RADAR, Synthesis).
2. **4.2 Types:** General, Purposive/Particular, Clinical and Card.
3. **4.3 Methods:** Kentian, Boenninghausen totality and concordance, elimination, rubric conversion.
4. **4.4 Symptom evaluation:** the hierarchy and rubric differentiation.

- Each section ends with a coloured **memory hook**: say it aloud once, and it will carry the facts.
- The pictures are drawn to be recalled as pictures.
- The one-glance recall card near the end is for the last ten minutes before the exam.
- Standard books: S. K. Tiwari, *Essentials of Repertorisation*; Kent's *Repertory* (6th American edition); Boenninghausen's *Therapeutic Pocket Book* (T. F. Allen edition); Boger's *BBCR*.
- The fuller rapid-revision doc you already have, *Homoeopathic Repertory – Rapid Revision*, stays unchanged and goes deeper on rubric locations.

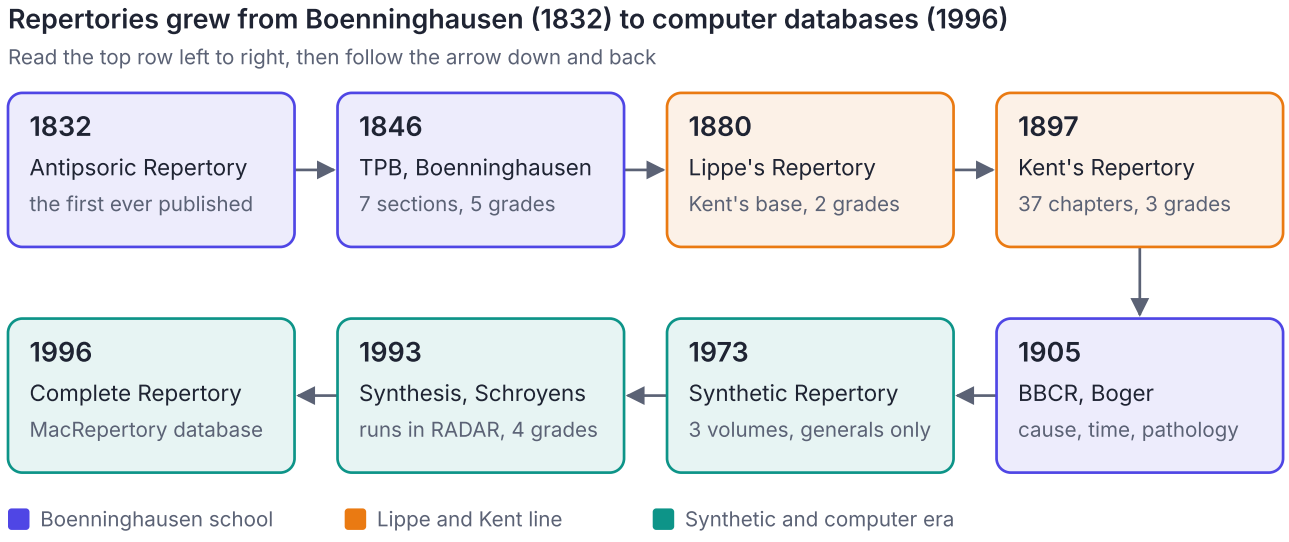
4.1 History and structure of repertories

- A **repertory** (Latin *repertorium*, a storehouse where things can be found) is an index of the symptoms of the Materia Medica.
- It is a means to an end: the final choice is always confirmed in the Materia Medica.
- A **rubric** (Latin *rubrica*, red) is a heading; old books printed headings in red ink.

The timeline

Year	Work	Author	Exam hook
1805	*Fragmenta de Viribus*, Part 2 (index)	Hahnemann	First repertory-like index
1832	Repertory of the Antipsoric Medicines	Boenninghausen	First published repertory
1835	Repertory of the Non-Antipsoric Medicines	Boenninghausen	Companion volume

Year	Work	Author	Exam hook
1846	Therapeutic Pocket Book (TPB)	Boenninghausen	7 sections, 5 grades
1880	Repertory to the More Characteristic Symptoms	C. Lippe	Base for Kent; 2 grades
1890–92	Concordance Repertory (6 vols)	W. D. Gentry	Puritan, key-word index
1891	TPB English edition	T. F. Allen	342 remedies
1896–97	Repertory of Hering's Guiding Symptoms	C. B. Knerr	48 chapters, 4 grades
1897–99	Repertory of the Homoeopathic MM	J. T. Kent	37 chapters, 3 grades
1904	Clinical Repertory	J. H. Clarke	Clinical type
1905	BBCR	C. M. Boger	Causation, time, pathological generals
1915	Synoptic Key	C. M. Boger	Basis for Phatak
1973–74	Synthetic Repertory (3 vols)	Barthel and Klunker	Mind; Generals; Sleep, Dreams, Sex
1993	Synthesis	F. Schroyens	RADAR; 4 grades
1996	Complete Repertory	R. van Zandvoort	MacRepertory



repertory timeline · 8 milestones, 3 schools

Recall the picture as a snake: two Boenninghausen boxes, two Kent-line boxes, Boger turning the corner, then the three modern books.

The big three

- **Boenninghausen's TPB (1846).** It has 7 sections: Mind and intellect; Parts of the body; Sensations and complaints; Sleep and dreams; Fever; Aggravations and ameliorations (modalities); Relationship of remedies (concordance).
- It has 5 grades and about 125 remedies in the original (342 in Allen's edition).
- Its ideas are the **complete symptom** (Location + Sensation + Modality + Concomitant), **grand generalisation** (a modality found in one part belongs to the whole patient) and the **doctrine of analogy** (fill a gap from similar symptoms elsewhere).

BBCR (C. M. Boger, 1905). *Boger Boenninghausen's Characteristics and Repertory* is a Materia Medica plus a repertory on Boenninghausen's plan, with more particular rubrics. Boger added **causation, time and pathological generals** (for example a tendency to suppuration or haemorrhage).

- **Kent's Repertory (1897–99).** Built on Lippe's repertory, Kent's has 37 chapters from **Mind** to **Generalities** in Hahnemann's head-to-foot order.
- It has 3 grades: **bold = 3**, *italics* = 2, roman = 1.
- The plan runs from general to particular (Swedenborg's influence: the inner is above the outer).
- Within a rubric, sub-rubrics run **Side → Time → Conditions (modalities) → Extension → Location → Character (sensation)**.
- Times run morning, forenoon, noon, afternoon, evening, night, midnight.
- Desires, aversions and thirst sit in **Stomach**; food aggravations sit in **Generalities**.
- Editions: 1st 1897; 2nd 1908; 3rd 1924, after Kent's death in 1916; 6th American 1957.

Kent's Repertory: how a page works

Grades by typeface; sub-rubrics in a fixed order

MIND

RUBRIC: **ABC.** *def.*, ghi., jkl., mno.

side:	left · right
time:	morning · forenoon · noon · afternoon · evening · night · midnight
conditions:	the modalities: < and >
extension:	to other parts
location:	where exactly
character:	the sensation

… then the next rubric, alphabetically

— 37 chapters, Mind first, Generalities last —

Reading a page of Kent

GRADES

ABC.	bold = 3 marks
def.	italic = 2 marks
ghi.	roman = 1 mark

SUB-RUBRIC ORDER

1. Side
2. Time
3. Conditions
4. Extension
5. Location
6. Character

Plan: general to particular,
head to foot (Hahnemann).
Desires, aversions, thirst: Stomach.
Food aggravations: Generalities.

Remedy names on the page are placeholders.

Kent's Repertory · grades by typeface, sub-rubrics in a fixed order

Modern and computer repertories

- **Synthetic Repertory** (Barthel and Klunker) covers only general symptoms, compiled from many sources: Vol I Mind, Vol II Generals, Vol III Sleep, Dreams and Sexuality.
- **Synthesis** (Schroyens) is Kent's 6th edition corrected and enlarged, with author references and 4 grades. It runs in **RADAR** (Rapid Aid to Drug Aimed Research; Archibel, Belgium).
- **Complete Repertory** (van Zandvoort) is the database in **MacRepertory** (Kent Homeopathic Associates, USA).
- **Hompath** was developed in India by Dr Jawahar Shah. **ISIS** is by David Witko.
- Computers give speed, many repertories at once and many analysis strategies. But they are only as good as the rubrics chosen: garbage in, garbage out.

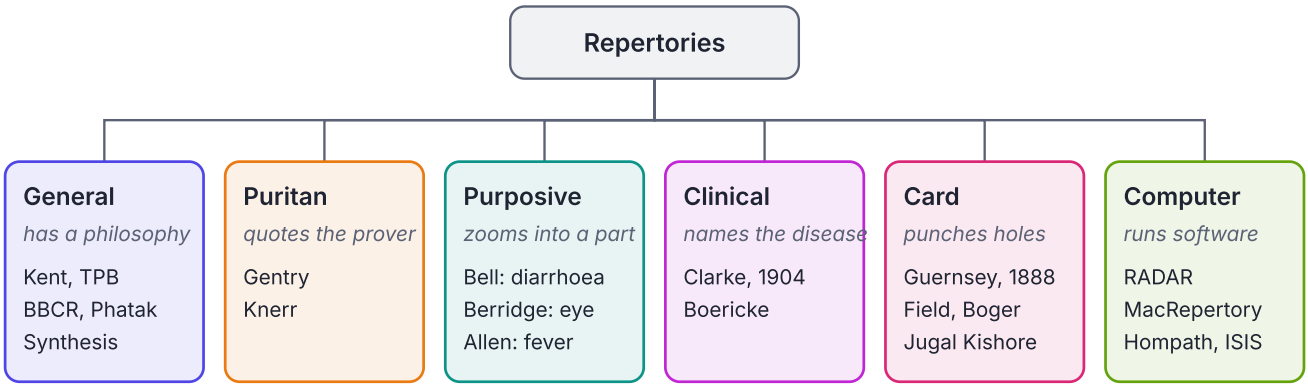
Memory hook: *"Boenninghausen built the pocket, Boger filled it with time, Kent climbed from Mind to Generalities."* Then the numbers in a chant: **TPB 7 and 5, Kent 37 and 3, Knerr 48 and 4, Gentry 6 volumes.**

4.2 Types of repertories

The syllabus groups repertories as **General**, **Purposive/Particular**, **Clinical** and **Card**. MCQs usually name a book and ask its type, or name a type and ask for the odd one out.

Type	What it is	Examples to remember
General (logical / philosophical)	Covers the whole Materia Medica and all body parts, built on a philosophy	TPB, BBCR, Kent, Boger's Synoptic Key, Phatak, Synthesis
Puritan	Keeps the provers' exact words, with no generalising	Gentry's Concordance, Knerr
Purposive / Particular / Regional	One region, one disease or one symptom group	Bell (Diarrhoea); H. C. Allen (Intermittent fever); Berridge (Eye); Minton (Uterine therapeutics); Douglass (Skin)
Clinical	Rubrics are disease names	Clarke's *Clinical Repertory*; Boericke's repertory (with the *Pocket Manual*)
Card (mechanical)	Each card is one rubric, punched at fixed remedy positions	Guernsey (first, 1888); Field; Boger; Jugal Kishore (India)
Computer	Software that runs repertory databases	RADAR, MacRepertory, Hompath, ISIS

Six types of repertory: one verb tells each apart



types of repertory · 6 branches with examples

Read each branch as a verb: if an MCQ names a book, ask what it does (plan, quote, zoom, name, punch or compute) and the type follows.

By arrangement

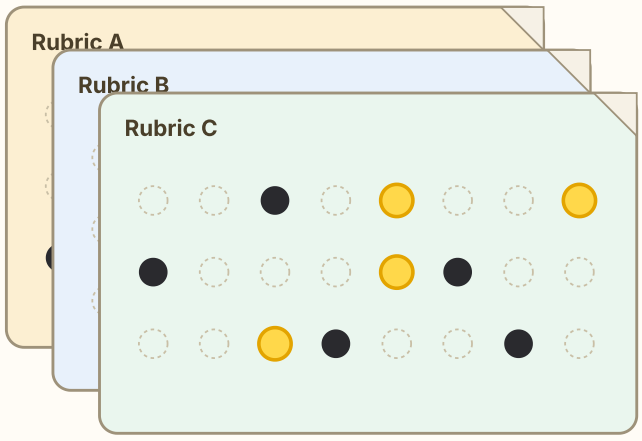
- **Schematic or anatomical** (Hahnemann's head-to-foot order): Kent, Knerr, BBCR.
- **Alphabetical**: Gentry (by key word), Phatak (by rubric), Murphy (by chapter), Clarke.
- The classic two-way split is the **logico-utilitarian** group (symptoms reshaped to fit the book's plan: Kent, TPB, BBCR) against the **Puritan** group (Gentry, Knerr).

How a card repertory works

- Each card carries one rubric, and each remedy has a fixed numbered position on every card.
- A hole is punched where the remedy is present.
- Stack the cards for the chosen rubrics and hold them to the light: holes that pass through every card are the remedies common to all rubrics.
- It is quick and needs no counting.
- Its limits: it ignores grades, and one small rubric can wrongly drop the true remedy.
- **Marcos Jimenez** first brought grading into cards; **Field's** cards are based on Kent.

Card repertory: let the light decide

Punched cards, stacked: the light passes only where every rubric agrees



Hold the stack to the light: it shines only through holes punched in every card.

How a card repertory works

One card = one rubric. Every remedy has a fixed, numbered position on every card; a hole is punched where the remedy is in the rubric.

Light through every card = remedy common to all rubrics

LIMITS

Ignores grades; one small rubric can wrongly drop the true remedy.

NAMES

Guernsey (first, 1888); Field (based on Kent); Boger; Jugal Kishore (India). Marcos Jimenez first graded cards.

Card repertory · holes that line up through every card mark the common remedies

Clinical and regional repertories, in one line each

- **Clinical:** quick in acute and one-sided cases, but prescribing on the disease name ignores individualisation, so it should never be the only basis. G. H. G. **Jahr** is called the father of the clinical repertory.
- **Regional:** useful when one region dominates and generals are scanty; its remedy lists for that field are richer than a general repertory's.

Memory hook: *"Generals Plan, Puritans Quote, Purposives Zoom, Clinics Name, Cards Punch."* One verb per type: the general repertory **plans** (a philosophy), the Puritan **quotes** the prover, the purposive **zooms** into one part, the clinical **names** the disease, and the card **punches** holes.

4.3 Methods of repertorisation and rubric conversion

Repertorisation means finding the remedy, or the small group of remedies, that covers the totality of the case with the help of a repertory. The final choice is made by studying the shortlist in the Materia Medica.

The steps, in order

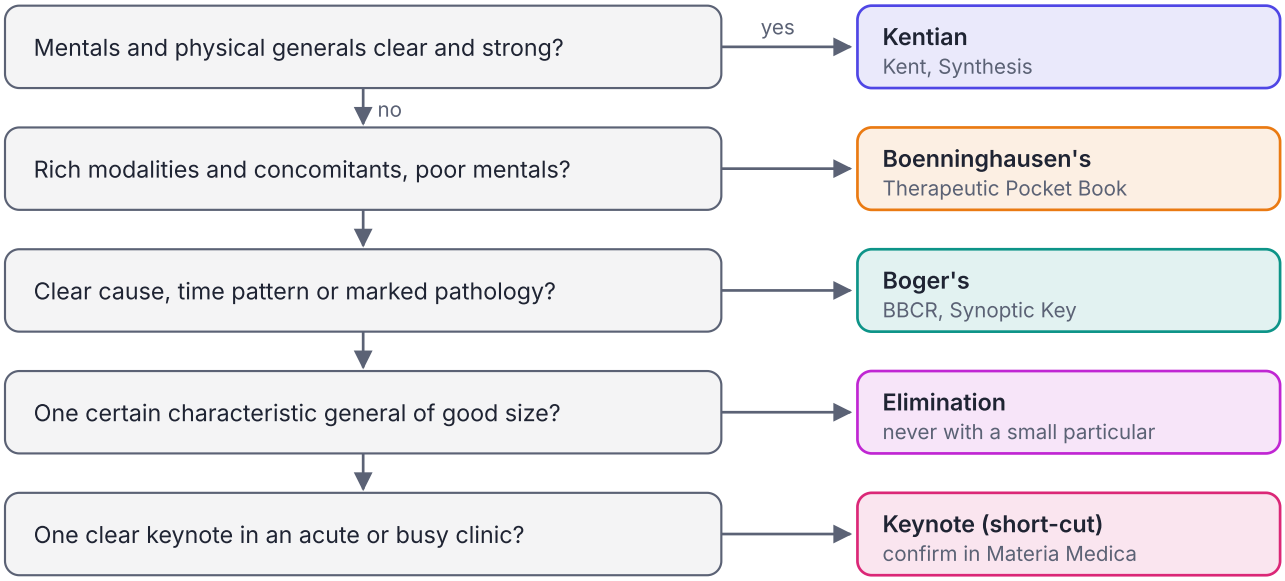
1. Case taking (recording)
2. Analysis of the case (classify the symptoms)
3. Evaluation of symptoms (grade their importance)
4. Erecting the repertorial totality
5. Choosing the repertory and method that suit the case
6. Rubric conversion (patient's words into repertory language)
7. Repertorisation (working out the rubrics)
8. Repertorial result, then differentiation in the Materia Medica, then the prescription

Which method for which case

Method	Core idea	Best when	Repertory
Kentian (totality)	Mental generals, then physical generals, then particulars	Chronic cases with clear mentals and generals	Kent, Synthesis
Boenninghausen's	Complete symptom (L-S-M-C), generalised modalities, concomitants	Rich modalities and concomitants, poor mentals	TPB
Boger's	Causation and time first, pathological generals, then modalities and mentals	Clear cause and time pattern; marked pathology	BBCR, Synoptic Key
Elimination	One strong rubric drops every remedy not in it	One or two very reliable characteristic generals	Any
Keynote (short-cut)	One or two keynotes, confirmed in MM	Clear keynotes; acute or busy practice	Any; clinical
Mechanical (card)	Superimpose punched cards	Quick screening	Card repertories
Computer	Software combines rubrics by chosen strategies	Large cases	RADAR, MacRepertory, Hompath

Pick the method from what the case is rich in

Ask top to bottom; the first yes gives the method. Every result is confirmed in the Materia Medica.



method chooser · 5 questions, 5 methods

In a case-based MCQ, find what the case is rich in, walk down the ladder and stop at the first yes.

Kent's working rules

- Take the strongest mental and physical generals first, then the particulars that are peculiar.
- If no exact rubric exists, go to the larger general rubric: Kent's book is built to be used general first.
- There are two ways to count: **numerical** (how many rubrics a remedy covers) and **qualitative** (the sum of grades 3, 2 and 1). The totals guide the choice; they do not dictate it.

Elimination: the trap MCQs test

- The eliminating rubric should be a **characteristic general** that is certain and of reasonable size.
- Never eliminate with a small particular rubric or a doubtful symptom: the true remedy may be missing only because the rubric is incomplete.
- **Cascading elimination** means each rubric eliminates for the next, all through the exercise.

Rubric conversion

Rubric conversion turns the patient's language into the repertory's language without changing the meaning.

- "I feel worse when people try to comfort me" becomes *Mind – Consolation – aggravates*.
- "I am scared to be alone at home" becomes *Mind – Fear – alone, of being*.
- "My headache is better when I press hard" becomes *Head – Pain – pressure – amel*.

Use synonyms and cross-references, choose the rubric that matches the exact meaning, and when unsure take the broader rubric. Know where things sit: in Kent, thirst and food desires are in Stomach.

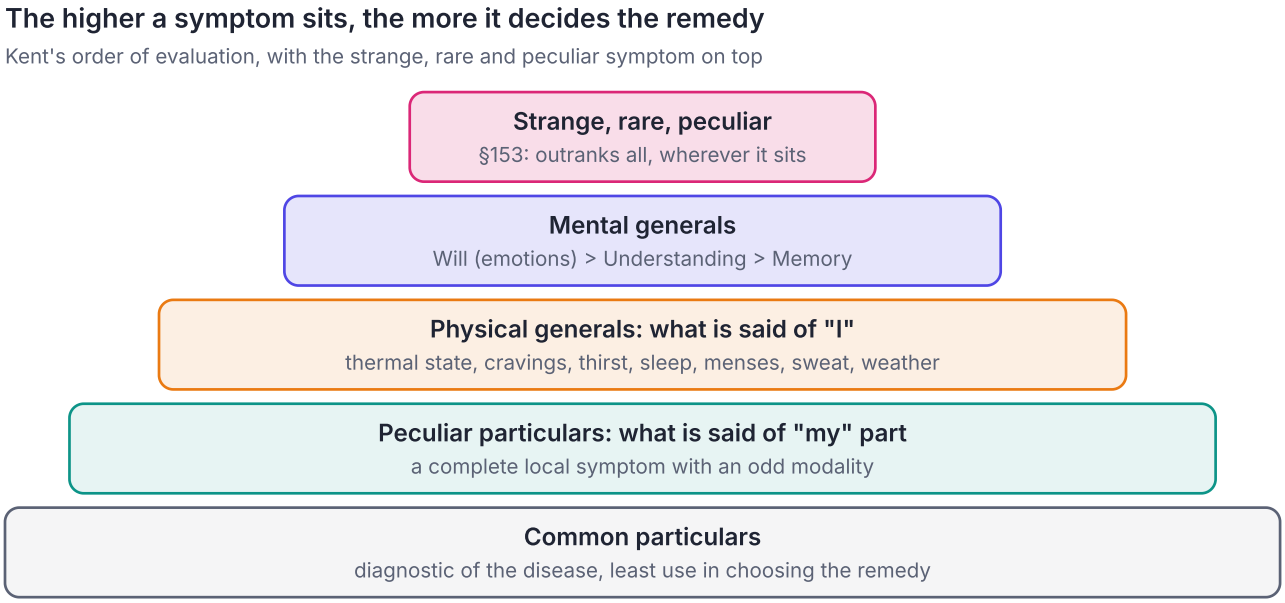
Memory hook: *"Kent climbs down, Boenninghausen spreads out, Boger asks why and when."* Kent goes from the top (mind) down to the parts; Boenninghausen spreads one modality over the whole patient; Boger starts with the **cause** and the **time**.

4.4 Symptom evaluation and rubric differentiation

Evaluation decides which symptoms lead the repertorisation. The most-tested idea: characteristic (strange, rare, peculiar) symptoms rank highest and common diagnostic symptoms rank lowest (Organon §153).

Kent's hierarchy, high to low

1. **Mental generals**, in their own order: **Will** (emotions: loves, hates, fears, desires) above **Understanding** (intellect: delusions, confusion) above **Memory**.
2. **Physical generals**, the things the patient says of "I": thermal state, appetite, cravings and aversions, thirst, sleep and dreams, sexual sphere and menses, perspiration, time and weather modalities.
3. **Particulars**, the things said of a part: peculiar particulars count; common particulars count least.



symptom hierarchy · 5 levels, high to low

Picture a pyramid: the narrow top decides the remedy, the wide base only names the disease.

A strange, rare and peculiar symptom outranks this order wherever it sits.

Symptom classes in one line

Term	Meaning
General	Refers to the whole person ("I am chilly")
Particular	Refers to one part ("my knee is worse in cold")
Common	Found in most patients with the disease; diagnostic, low value
Characteristic	Individualising: striking, singular, peculiar (§153)
Basic and determinative	Garth Boericke's terms: disease-defining versus remedy-deciding
Concomitant	Accompanies the chief complaint with no pathological link (Boenninghausen)
Complete symptom	Location + Sensation + Modality (+ Concomitant)
Keynote	A highly characteristic symptom pointing strongly to one remedy
Eliminating symptom	A strong general used to narrow the remedy list
Pathological general	Pathology affecting the whole system, such as a tendency to suppuration (Boger)
Negative general	An expected general that is absent, such as thirstlessness in fever

Who ranks what first

- **Boenninghausen:** the complete symptom; concomitants are most valuable; modalities can be generalised.
- **Kent:** mentals first, then physical generals, then particulars.
- **Boger:** causation and time first, then pathological generals, then modalities and mentals.

Rubric differentiation (Kent)

- **Consolation aggravates** (Nat-m, Sep, Ign, Sil) against **Consolation ameliorates** (Puls, Phos).
- **Anxiety** is vague uneasiness, **Fear** is of a definite thing, **Anguish** is extreme anxiety.
- **Irritability** (easily annoyed), **Anger** (an outburst) and **Violent** are three different rubrics.
- **Forsaken feeling** is a Mind rubric; *Delusion, deserted* is a false belief.
- A **delusion** is a fixed false idea; **fancies** are imaginations the patient knows are not real.
- *Right side* in a particular chapter refers to that part; *Sides, right* in Generalities refers to the whole body.

Memory hook: *"Will before Wit before Memory; I before My."* Emotions outrank intellect, which outranks memory; what the patient says of **I** (a general) outranks what is said of **my** knee (a particular). And **SRP beats them all**.

Rapid revision

Cue	Fact
First published repertory	Boenninghausen, *Repertory of the Antipsoric Medicines*, 1832
TPB	1846; 7 sections; 5 grades; about 125 remedies (342 in Allen's 1891 edition)
Complete symptom	Boenninghausen: Location, Sensation, Modality, Concomitant
Grand generalisation	TPB: a modality of one part is taken as true of the whole patient
BBCR	C. M. Boger, 1905; causation, time, pathological generals
Kent's Repertory	1897–99; 37 chapters; Mind first, Generalities last; 3 grades
Kent's base	Lippe's repertory (2 grades)
Chapter after Mind	Vertigo
Food desires and thirst (Kent)	Stomach chapter
Knerr	Index to Hering's *Guiding Symptoms*; 48 chapters; 4 grades
Gentry	*Concordance Repertory*, 6 volumes, Puritan, alphabetical key words
Clarke	*Clinical Repertory*, 1904
First card repertory	W. J. Guernsey, 1888
Synthetic Repertory	Barthel and Klunker; 3 volumes (Mind; Generals; Sleep, Dreams, Sex)
Synthesis	Schroyens, 1993; RADAR; 4 grades
Complete Repertory	van Zandvoort; MacRepertory
Best eliminating rubric	A characteristic general of good size, never a small particular
Kent's top mental	Will (emotions), then Understanding, then Memory
Least valuable symptom	Common particular

Cue	Fact
Basic and determinative	Garth Boericke

One-glance recall card

Topic 4 recall card: the 15 facts asked most

Cover the small lines, say the fact from the big cue, then check

1832 first published repertory: Boenninghausen, Antipsoric	**TPB 7 · 5** 1846: 7 sections, 5 grades; ~125 remedies, 342 in Allen	**L + S + M + C** complete symptom: Boenninghausen's idea
Grand generalisation a modality of one part belongs to the whole patient	**BBCR 1905** Boger: causation, time, pathological generals	**Kent 37 · 3** Mind first, Generalities last; bold 3, italic 2, roman 1
Lippe Kent's base repertory, 2 grades (1880)	**Vertigo** the chapter right after Mind in Kent	**Stomach** Kent: desires, aversions and thirst sit here
Knerr 48 · 4 index to Hering's Guiding Symptoms	**Gentry 6 vols** Concordance, Puritan, alphabetical key words	**Guernsey 1888** first card repertory; Jugal Kishore in India
Synthesis Schroyens 1993; RADAR; Complete Rep. = MacRepertory	**Will > Wit > Memory** Kent's order inside the mental generals	**Eliminate wisely** use a big characteristic general, never a small particular

Topic 4 recall card · 15 tiles

For the last ten minutes before the exam: cover the small lines of each tile and say the fact from the big cue.

Practice MCQs

1. The first published repertory in homoeopathy was: (a) Kent's Repertory (b) the Therapeutic Pocket Book (c) the Repertory of the Antipsoric Medicines (d) Knerr's Repertory
2. The number of sections in Boenninghausen's Therapeutic Pocket Book is: (a) 5 (b) 7 (c) 9 (d) 37
3. Kent's Repertory grades remedies in: (a) 2 grades (b) 3 grades (c) 4 grades (d) 5 grades
4. The last chapter of Kent's Repertory is: (a) Skin (b) Sleep (c) Generalities (d) Perspiration
5. Kent's Repertory was developed from the repertory of: (a) Jahr (b) Lippe (c) Gentry (d) Knerr

6. Knerr's Repertory is an index to: (a) Allen's Encyclopedia (b) Hering's Guiding Symptoms (c) Hahnemann's Chronic Diseases (d) Clarke's Dictionary
7. A Puritan repertory is: (a) Kent (b) BBCR (c) Gentry's Concordance (d) Phatak
8. The concept of the complete symptom was introduced by: (a) Kent (b) Boger (c) Boenninghausen (d) Hering
9. Grand generalisation is a feature of: (a) Kent (b) the TPB (c) Knerr (d) Clarke
10. BBCR gives special place to: (a) causation, time and pathological generals (b) only mentals (c) only keynotes (d) disease names
11. In Kent's Repertory, desire for sweets is found in the chapter: (a) Generalities (b) Mind (c) Stomach (d) Mouth
12. The first card repertory was prepared by: (a) Jugal Kishore (b) Boger (c) W. J. Guernsey (d) Field
13. Bell's repertory deals with: (a) intermittent fever (b) diarrhoea (c) the eye (d) the uterus
14. Clarke's repertory is classified as: (a) Puritan (b) Clinical (c) Card (d) General logical
15. Synthesis repertory is linked with the software: (a) Hompath (b) RADAR (c) MacRepertory (d) ISIS
16. The highest rank among mental generals in Kent's evaluation goes to: (a) memory (b) intellect (c) will and emotions (d) delusions
17. The least valuable symptoms in repertorisation are: (a) mental generals (b) common particulars (c) physical generals (d) peculiar symptoms
18. The best eliminating symptom is: (a) a small particular rubric (b) a common symptom (c) a characteristic general of good size (d) a doubtful symptom
19. Converting the patient's language into repertory language is called: (a) repertorisation (b) rubric conversion (c) case analysis (d) elimination
20. The method best for a case rich in modalities and concomitants but poor in mentals is: (a) Kentian (b) Boenninghausen's (c) keynote (d) card

Answer key

Q	Answer	Why
1	c	Boenninghausen, 1832; the TPB came in 1846
2	b	7 sections, from Mind to Relationship of remedies
3	b	Bold 3, italics 2, roman 1
4	c	Mind first, Generalities last
5	b	Lippe's repertory of 1880, with 2 grades

Q	Answer	Why
6	b	48 chapters, 4 grades
7	c	Keeps the provers' exact words
8	c	Location, Sensation, Modality, Concomitant
9	b	A local modality is generalised to the whole patient
10	a	Boger's additions to Boenninghausen's plan
11	c	Kent places desires, aversions and thirst in Stomach
12	c	Guernsey, 1888, "Boenninghausen's Slips"
13	b	A purposive repertory on diarrhoea
14	b	Rubrics are disease names
15	b	Schroyens' Synthesis runs in RADAR
16	c	Will, then Understanding, then Memory
17	b	Common to the disease, so they do not individualise
18	c	A small rubric may wrongly drop the true remedy
19	b	Same meaning, repertory wording
20	b	The TPB generalises modalities and values concomitants

Topic 5: General & Applied Human Biology

How to use this guide

Topic 5 gives about 2 of the 25 core marks, and almost every question is a one-line fact: a normal value, a nerve that supplies a muscle, a hormone and its gland. Learn the tables and the labelled diagrams first; the prose only links them.

Syllabus (Advt 06/2026, Topic 5). The sections below follow it in order:

Syllabus heading	What it covers	Section here
Applied neuro-musculoskeletal anatomy	Clinically relevant osteology, joint classification, brain localization, peripheral nerve pathways	5.1
Cardiovascular, respiratory and renal physiology	Cardiac cycle, ECG, regulation of blood pressure, mechanics of respiration, renal clearance	5.2a–c
Gastrointestinal and endocrine systems	Digestion, absorption, hepatic functions, hypothalamic-pituitary axis, feedback loops	5.3a–b
Body fluids and homeostasis	Fluid-electrolyte balance, acid-base balance, clinical biochemistry (enzymes, lipid profile, blood glucose)	5.4a–b

- **Standard books used.** Guyton and Hall, *Textbook of Medical Physiology*; Ganong, *Review of Medical Physiology*; A. K. Jain, *Textbook of Physiology*; B. D. Chaurasia, *Human Anatomy* (vols 1–4); Vishram Singh, *Textbook of Anatomy*; Vasudevan and Sreekumari, *Textbook of Biochemistry*; Satyanarayana, *Biochemistry*.
- These are the anatomy, physiology and biochemistry books prescribed in the BHMS first-year (NCH/NCISM) course.
- Normal values follow these books; laboratories differ slightly, so learn the range, not the last decimal.

Plan to 1 November (12 days). Two sections a day for days 1–5 (diagrams first, then tables); day 6 the rapid-revision sheet and the practice MCQs; after that, 15 minutes a day on the rapid-revision sheet alongside your mock tests.

5.1 Applied neuro-musculoskeletal anatomy

Examiners ask four kinds of question here: which bone or joint (with its type), which part of the brain does what, which nerve supplies a muscle, and which deformity follows a nerve injury.

Clinically relevant osteology

Bone or landmark	High-yield fact
Clavicle	First bone to ossify (5th–6th week of intrauterine life); ossifies in membrane; commonest bone to fracture, at the junction of its middle and lateral thirds
Scaphoid	Commonest carpal bone fractured (fall on outstretched hand); tenderness in the anatomical snuffbox; risk of avascular necrosis of the proximal fragment
Surgical neck of humerus	Fracture injures the axillary nerve
Shaft of humerus (spiral groove)	Fracture injures the radial nerve: wrist drop
Medial epicondyle of humerus	Ulnar nerve runs behind it (the "funny bone")
Neck of femur	Fracture in the elderly; risk of avascular necrosis of the head (blood supply via retinacular arteries)
Neck of fibula	Common peroneal nerve winds round it: foot drop
Femur	Longest and strongest bone
Stapes	Smallest bone
Pisiform	Sesamoid bone in flexor carpi ulnaris; patella is the largest sesamoid bone
Atlas (C1)	No body, no spine; atlanto-occipital joint allows nodding ("yes")
Axis (C2)	Dens (odontoid process); atlanto-axial joint allows rotation ("no")
Vertebra prominens	C7, first prominent palpable spinous process
Sternal angle (of Louis)	Level of the 2nd costal cartilage and the T4–T5 disc; used to count ribs
Spinal cord	Ends at the lower border of L1 in adults (L3 at birth); lumbar puncture is done at L3–L4

Adults have 206 bones; a newborn about 270–300, many of which fuse.

Joint classification

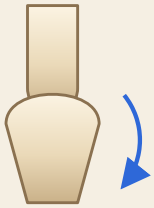
Type	Sub-type	Example
Fibrous	Suture	Skull sutures
Fibrous	Syndesmosis	Inferior tibiofibular joint
Fibrous	Gomphosis	Tooth in its socket
Primary cartilaginous (synchondrosis, hyaline)	Temporary	Epiphyseal plate; 1st chondrosternal joint
Secondary cartilaginous (symphysis, fibrocartilage)	Midline, permanent	Pubic symphysis; intervertebral discs; manubriosternal joint
Synovial	Plane	Intercarpal, acromioclavicular
Synovial	Hinge	Elbow (humero-ulnar), interphalangeal, ankle
Synovial	Pivot	Superior radio-ulnar; atlanto-axial (median)
Synovial	Condylar (bicondylar)	Knee, temporomandibular
Synovial	Ellipsoid	Wrist (radiocarpal), metacarpophalangeal
Synovial	Saddle (sellar)	First carpometacarpal joint of the thumb
Synovial	Ball and socket	Shoulder, hip

Synovial joints by shape

The shape of the bone ends decides the movement



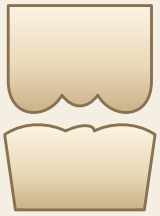
Plane
Intercarpal, acromioclavicular



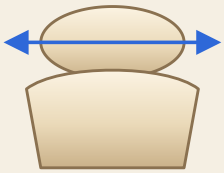
Hinge
Elbow, ankle, interphalangeal



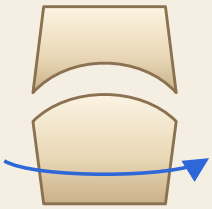
Pivot
Superior radio-ulnar; atlanto-axial



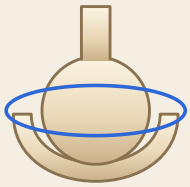
Condylar
Knee, temporomandibular



Ellipsoid
Wrist (radiocarpal), MCP joints



Saddle
1st carpometacarpal (thumb)



Ball and socket
Shoulder, hip

Not synovial

Fibrous: sutures, syndesmosis, gomphosis (tooth in socket)

Primary cartilaginous: epiphyseal plate

Secondary: pubic symphysis, discs

Hook: "Saddle for the thumb, ball for the limb roots." Blue arrows show the movement each shape allows.

Synovial joints by shape · the shape of the bone ends decides the movement

Memory hook: "Saddle for the thumb, ball for the limb roots." The thumb's saddle joint allows opposition; the shoulder and hip are the only ball-and-socket joints in the limbs. The shoulder is the most mobile and most often dislocated joint (anterior dislocation, axillary nerve at risk).

Brain localization

The labelled map below shows which area does what; a lesion causes loss of that function.

Area (Brodmann)	Function	Lesion
Primary motor cortex, precentral gyrus (4)	Voluntary movement of the opposite half of the body	Contralateral upper motor neuron weakness
Premotor area (6)	Planning of movement	Apraxia
Broca's area, inferior frontal gyrus, dominant side (44, 45)	Motor speech	Expressive (non-fluent) aphasia: understands, cannot speak fluently

Area (Brodmann)	Function	Lesion
Prefrontal cortex	Personality, judgement	Disinhibition, personality change
Primary sensory cortex, postcentral gyrus (3, 1, 2)	Touch, pain, position of the opposite side	Contralateral sensory loss
Wernicke's area, superior temporal gyrus, dominant side (22)	Understanding speech	Receptive (fluent) aphasia: fluent but meaningless speech
Primary auditory cortex, Heschl's gyrus (41, 42)	Hearing	Little deafness (bilateral input)
Hippocampus (medial temporal lobe)	New memory	Anterograde amnesia
Primary visual cortex, calcarine sulcus, occipital lobe (17)	Vision	Contralateral homonymous hemianopia with macular sparing
Cerebellum	Coordination, balance, tone	Ipsilateral ataxia, intention tremor, nystagmus, dysdiadochokinesia
Basal ganglia (substantia nigra, corpus striatum)	Initiation and smoothness of movement	Parkinsonism (dopamine loss in the substantia nigra)
Hypothalamus	Temperature, hunger, thirst, pituitary control	Diabetes insipidus, temperature and appetite disorders
Medulla oblongata	Respiratory and cardiovascular centres	Death if destroyed ("vital centres")
Internal capsule (posterior limb)	Corticospinal and sensory fibres	Contralateral hemiplegia; commonest site of stroke

Memory hook: "B for Broca, B for Broken speech; W for Wernicke, W for Word salad." Broca is in front (frontal, motor), Wernicke behind (temporal, sensory).



Brain localization · lobes, key Brodmann areas and lesion signs

Read it as a map: frontal = movement and speech output, parietal = sensation, temporal = hearing, understanding and memory, occipital = vision; the cerebellum is the one part that works on the same side.

Peripheral nerve pathways and injuries

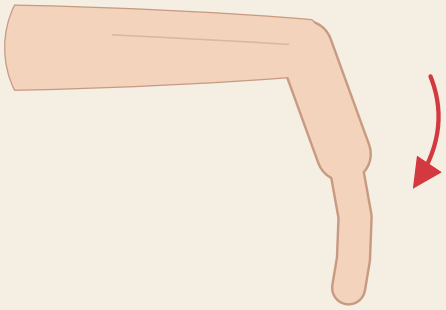
Brachial plexus. Roots C5–T1, then trunks, divisions, cords, branches ("Read That Damn Cadaver Book"). The cords are named for their relation to the axillary artery: lateral, medial and posterior.

Nerve (root value)	Supplies (key muscles)	Injury and sign
Axillary (C5, C6)	Deltoid, teres minor	Surgical-neck fracture or shoulder dislocation; loss of abduction 15–90°; numb "regimental badge" patch
Musculocutaneous (C5–C7)	Biceps, brachialis, coracobrachialis	Weak elbow flexion and supination
Radial (C5–T1)	Extensors of arm and forearm (triceps, wrist and finger extensors)	Spiral-groove fracture or "Saturday night palsy"; **wrist drop**

Nerve (root value)	Supplies (key muscles)	Injury and sign
Median (C5–T1)	Most forearm flexors; thenar muscles; lateral two lumbricals	Carpal tunnel syndrome; **ape-thumb hand**; at the elbow: **hand of benediction** on trying to make a fist
Ulnar (C8, T1)	Most intrinsic hand muscles; flexor carpi ulnaris	Medial-epicondyle injury; **claw hand** (ring and little fingers); Froment's sign
Long thoracic (C5–C7)	Serratus anterior	**Winging of the scapula**
Upper trunk (C5, C6)	—	**Erb's palsy**: "waiter's tip" position (birth injury, fall on shoulder)
Lower trunk (C8, T1)	—	**Klumpke's palsy**: claw hand, may have Horner's syndrome

Nerve injuries you can see in the hand

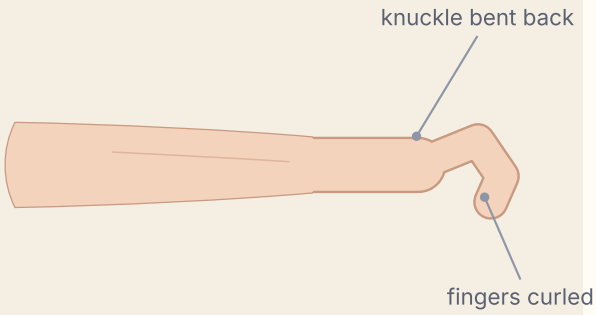
Radial drops the wrist, ulnar claws, median blesses or flattens the thumb



Wrist drop

Radial nerve (C5–T1)

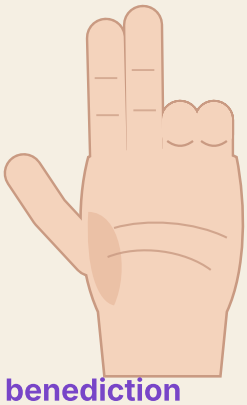
Spiral-groove fracture of the humerus or "Saturday night palsy"
Wrist and finger extensors fail



Claw hand

Ulnar nerve (C8, T1)

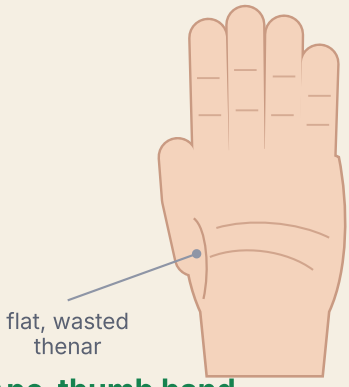
Ring and little fingers: knuckles bent back, finger joints curled
Medial epicondyle injury; Froment's sign



Hand of benediction

Median nerve at the elbow

On trying to make a fist, the index and middle fingers stay straight



Ape-thumb hand

Median nerve at the wrist

Carpal tunnel syndrome: thenar muscles waste, the thumb lies flat in line with the fingers

Nerve injuries in the hand · wrist drop, claw hand, benediction, ape thumb

Lower-limb nerve	Supplies	Injury and sign
Femoral (L2–L4)	Quadriceps	Lost knee extension and knee jerk
Obturator (L2–L4)	Adductors of thigh	Weak adduction
Sciatic (L4–S3)	Hamstrings and all muscles below the knee	Thickest and longest nerve; injured by a wrongly placed gluteal injection (give it in the upper outer quadrant)
Common peroneal (L4–S2)	Dorsiflexors and evertors of the foot	Fibular-neck injury; **foot drop**, high-stepping gait

Lower-limb nerve	Supplies	Injury and sign
Tibial (L4–S3)	Calf muscles, plantar flexors	Loss of plantar flexion; ankle jerk lost
Superior gluteal (L4–S1)	Gluteus medius and minimus	**Trendelenburg sign** (pelvis drops to the opposite side)

Reflex root values. Biceps C5, C6 · Supinator C5, C6 · Triceps C6, C7 · Knee L3, L4 · Ankle S1, S2. **Hook: "1–2 buckle my shoe (ankle S1–S2), 3–4 kick the door (knee L3–L4), 5–6 pick up sticks (biceps C5–C6), 7–8 lay them straight (triceps C7, with C6)."**

- **Cranial nerves to remember.** Longest: vagus (X).
- Thickest: trigeminal (V).
- Longest intracranial course and the only one leaving the back of the brainstem: trochlear (IV).
- A facial-nerve (VII) lesion at or below its nucleus weakens the whole half-face (Bell's palsy); a lesion above it spares the forehead.
- Hook for the eye muscles: **LR6 SO4** (lateral rectus by VI, superior oblique by IV, the rest by III).

5.2a Cardiovascular physiology

Three things carry most marks: the order of events in the cardiac cycle, the normal ECG intervals, and the baroreceptor and RAAS control of blood pressure.

Cardiac cycle (heart rate 75/min, cycle 0.8 s)

Systole lasts 0.27–0.3 s and diastole about 0.5 s, so diastole shortens most when the heart rate rises.

Phase	Valves	What happens	Duration
1. Atrial systole	AV open, semilunar closed	Atria top up the ventricles (about 20–30% of filling); a-wave in JVP; S4 if the ventricle is stiff	0.1 s
2. Isovolumetric contraction	All four closed	Pressure rises, volume constant; begins with closure of AV valves (**S1**); c-wave in JVP	0.05 s
3. Rapid then reduced ejection	Semilunar open	Blood leaves; aortic pressure peaks at about 120 mmHg	0.22 s
4. Isovolumetric relaxation	All four closed	Begins with closure of aortic and pulmonary valves (**S2**); v-wave builds in the atria	0.08 s

Phase	Valves	What happens	Duration
5. Rapid then slow filling (diastasis)	AV open	Passive filling, about 70% of filling; **S3** in rapid filling (normal in children and athletes)	about 0.4 s

Key numbers. End-diastolic volume 120–130 mL; end-systolic volume 50–60 mL; stroke volume about 70 mL; ejection fraction (SV/EDV) 55–70%; cardiac output = SV × HR = about 5 L/min; cardiac index about 3.2 L/min/m².

- **Heart sounds.** S1 "lub" = closure of mitral and tricuspid valves, start of systole.
- S2 "dub" = closure of aortic and pulmonary valves, end of systole; it splits on inspiration (physiological splitting).
- S3 = rapid ventricular filling.
- S4 = atrial contraction into a stiff ventricle (always abnormal).

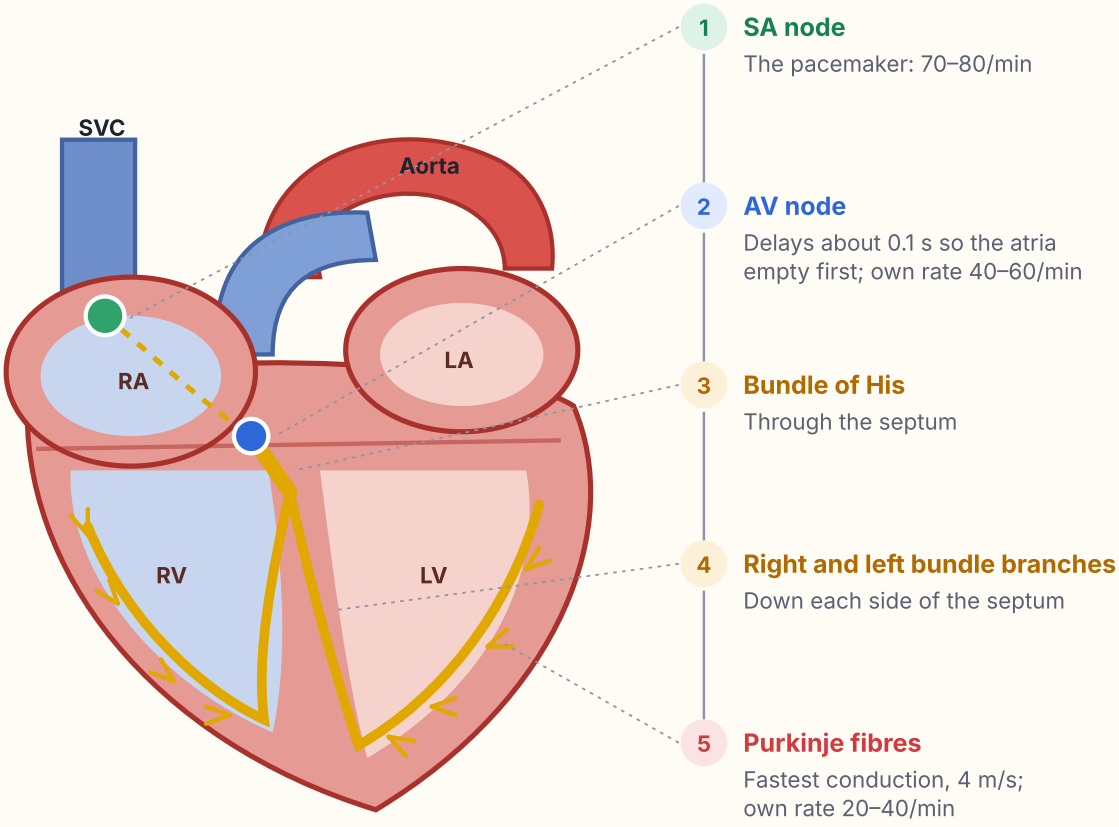
Laws. Frank–Starling: the greater the stretch (EDV), the stronger the contraction. Laplace: wall tension = pressure × radius / (2 × wall thickness), so a dilated heart works harder.

Conducting system and ECG

SA node (pacemaker, 70–80/min) → atria → AV node (delay of about 0.1 s, so atria empty first; intrinsic rate 40–60/min) → bundle of His → right and left bundle branches → Purkinje fibres (fastest conduction, 4 m/s; intrinsic rate 20–40/min).

How the heartbeat spreads

The conducting system, from pacemaker to ventricular wall



Front view, chambers opened. The impulse runs top to bottom: SA → AV → His → branches → Purkinje.

Conducting system · SA node → AV node → His → bundle branches → Purkinje

ECG item	Represents	Normal
P wave	Atrial depolarisation	Up to 0.1 s (2.5 small squares)
PR interval	Atrial depolarisation + AV-node delay	0.12–0.20 s (3–5 small squares)
QRS complex	Ventricular depolarisation (atrial repolarisation is hidden in it)	0.08–0.10 s, under 0.12 s
ST segment	Ventricles fully depolarised (plateau)	Isoelectric
T wave	Ventricular repolarisation	Upright in most leads
QT interval	Whole of ventricular depolarisation and repolarisation	About 0.40 s; corrected QTc under 0.44 s
U wave	Repolarisation of papillary muscles or Purkinje fibres	Prominent in hypokalaemia

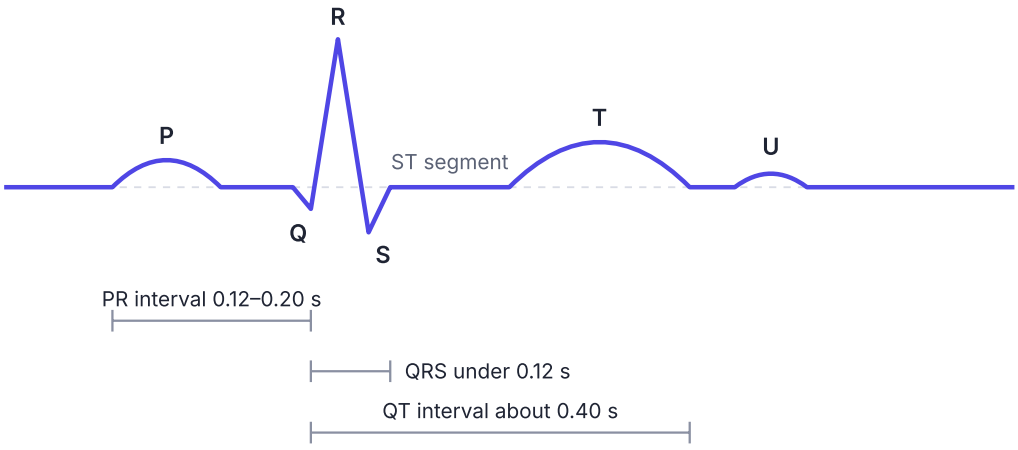
Paper. At 25 mm/s one small square = 0.04 s and one large square = 0.2 s; 1 mV = 10 mm. Heart rate = 300 ÷ number of large squares between two R waves (or 1500 ÷ small squares). Standard 12 leads: 3 bipolar limb leads (I, II, III; Einthoven's triangle), 3 augmented unipolar limb leads (aVR, aVL, aVF), 6 chest leads (V1–V6).

Change	Seen in
Tall peaked T, wide QRS	Hyperkalaemia
Flat T, prominent U wave, ST depression	Hypokalaemia
ST elevation in the leads facing the infarct	Acute myocardial infarction
Prolonged PR interval	First-degree heart block
Short QT / long QT	Hypercalcaemia / hypocalcaemia

Memory hook: "P for Pump-up of atria, QRS for the big ventricles, T for Tidy-up (repolarisation)."

One heartbeat on the ECG: P, QRS and T in order

Lead II, normal sinus beat (schematic) with the normal intervals



P wave = atria depolarise (contract)
QRS complex = ventricles depolarise; atrial repolarisation is hidden in it
T wave = ventricles repolarise · U wave = prominent in low potassium

At 25 mm/s: 1 small square = 0.04 s, 1 large square = 0.2 s; heart rate = 300 ÷ large squares between two R waves

ECG · one normal beat with its waves and intervals

The PR interval includes the AV-node delay; the QT interval covers the whole of ventricular activity, depolarisation plus repolarisation.

Regulation of blood pressure

Mean arterial pressure = diastolic pressure + one third of pulse pressure (about 93 mmHg at 120/80). Blood pressure = cardiac output × total peripheral resistance; resistance is set mainly by the arterioles (Poiseuille: flow varies with the fourth power of the radius).

Control	Speed	How it works
Baroreceptor reflex	Seconds	Stretch receptors in the carotid sinus (via glossopharyngeal, IX) and aortic arch (via vagus, X) → vasomotor centre in the medulla. Rise in BP → more vagal tone, less sympathetic tone → slower heart, vasodilatation. Works best between 60 and 180 mmHg.
Chemoreceptor reflex	Seconds	Carotid and aortic bodies respond to low O_2, high CO_2, low pH; raise BP and breathing when BP falls below about 80 mmHg
CNS ischaemic response	Seconds	Powerful sympathetic discharge when the brain is ischaemic (Cushing reflex: high BP, slow pulse with raised intracranial pressure)
Renin–angiotensin–aldosterone system	Minutes to hours	Low renal perfusion → renin from juxtaglomerular cells → angiotensinogen to angiotensin I → ACE (mostly in lungs) → angiotensin II: vasoconstriction, aldosterone (salt and water retention), ADH, thirst
ADH (vasopressin)	Minutes	From the posterior pituitary on high osmolality or low volume; water retention and vasoconstriction
Atrial natriuretic peptide	Hours	Released when the atria stretch; sodium and water loss, vasodilatation
Renal–body fluid mechanism (pressure natriuresis)	Days (long-term)	The kidneys excrete salt and water until BP returns to its set point; the dominant long-term control

Memory hook: "Baroreceptors are the Buffer, the Kidneys are the Keeper." Baroreceptors smooth out moment-to-moment swings; long-term BP is set by the kidneys.

5.2b Respiratory physiology

The exam asks the muscles of breathing, the lung volumes, surfactant and compliance, and how O_2 and CO_2 are carried.

Mechanics of breathing

- **Inspiration is active.** The diaphragm (phrenic nerve, C3–C5: "C3, 4, 5 keeps the diaphragm alive") does about 75% of quiet breathing; external intercostals lift the ribs. Forced inspiration adds sternocleidomastoid, scalenes and pectorals.
- **Quiet expiration is passive**, from elastic recoil of the lungs and chest wall. Forced expiration uses the internal intercostals and abdominal muscles.
- **Pressures.** Intrapleural pressure is about −2.5 to −3 mmHg at the end of expiration and about −6 mmHg at the end of inspiration; it keeps the lungs expanded. Air enters because alveolar pressure falls below atmospheric (about −1 mmHg).
- **Compliance** = change in volume ÷ change in pressure (about 200 mL/cmH_2O for lungs alone). Falls in fibrosis and pulmonary oedema; rises in emphysema.
- **Surfactant** (dipalmitoyl phosphatidylcholine), secreted by **type II alveolar cells** from about 24 weeks of gestation, lowers surface tension, keeps small alveoli open (Laplace: P = 2T/r) and prevents oedema. Lack of it in preterm babies = respiratory distress syndrome (hyaline membrane disease).
- **Airway resistance** is highest in the medium-sized bronchi. Sympathetic (β_2) stimulation dilates bronchi; parasympathetic (vagus) constricts them.

Lung volumes and capacities (adult male)

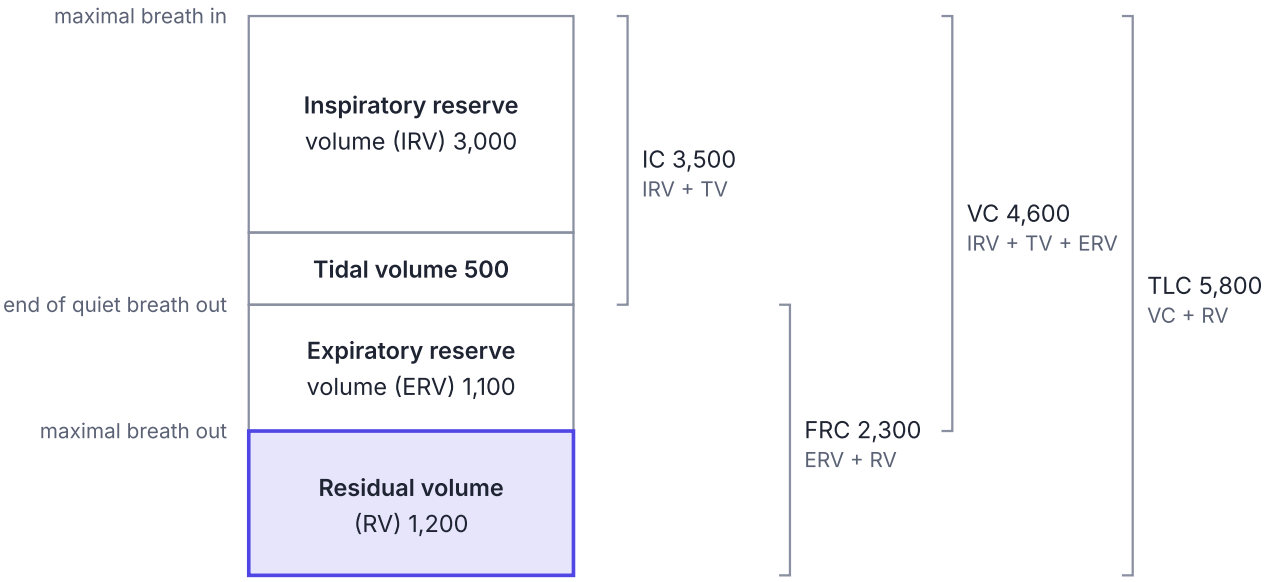
Volume or capacity	Definition	Value
Tidal volume (TV)	Air in or out per quiet breath	500 mL
Inspiratory reserve volume (IRV)	Extra air inhaled after a normal inspiration	3,000 mL
Expiratory reserve volume (ERV)	Extra air exhaled after a normal expiration	1,100 mL
Residual volume (RV)	Air left after maximal expiration; **cannot be measured by spirometry**	1,200 mL
Inspiratory capacity	TV + IRV	3,500 mL
Functional residual capacity (FRC)	ERV + RV; volume at rest	2,300 mL
Vital capacity (VC)	IRV + TV + ERV	4,600 mL

Volume or capacity	Definition	Value
Total lung capacity (TLC)	VC + RV	5,800 mL

Hook: a "capacity" is always two or more volumes added together. RV, FRC and TLC need helium dilution, nitrogen washout or body plethysmography. FEV_1/FVC is normally above 0.75–0.80; it falls in obstructive disease (asthma, COPD) and stays normal or high in restrictive disease (fibrosis).

Capacities are sums of volumes; residual volume always stays in the lungs

Adult male, values in mL (not to scale)



A capacity is two or more volumes added together. Residual volume (highlighted), and so FRC and TLC, cannot be measured with a spirometer: they need helium dilution, nitrogen washout or body plethysmography.

Lung volumes and capacities · 4 volumes, 4 capacities

The four volumes stack from top to bottom; each bracket on the right is a capacity, the sum of the volumes it spans.

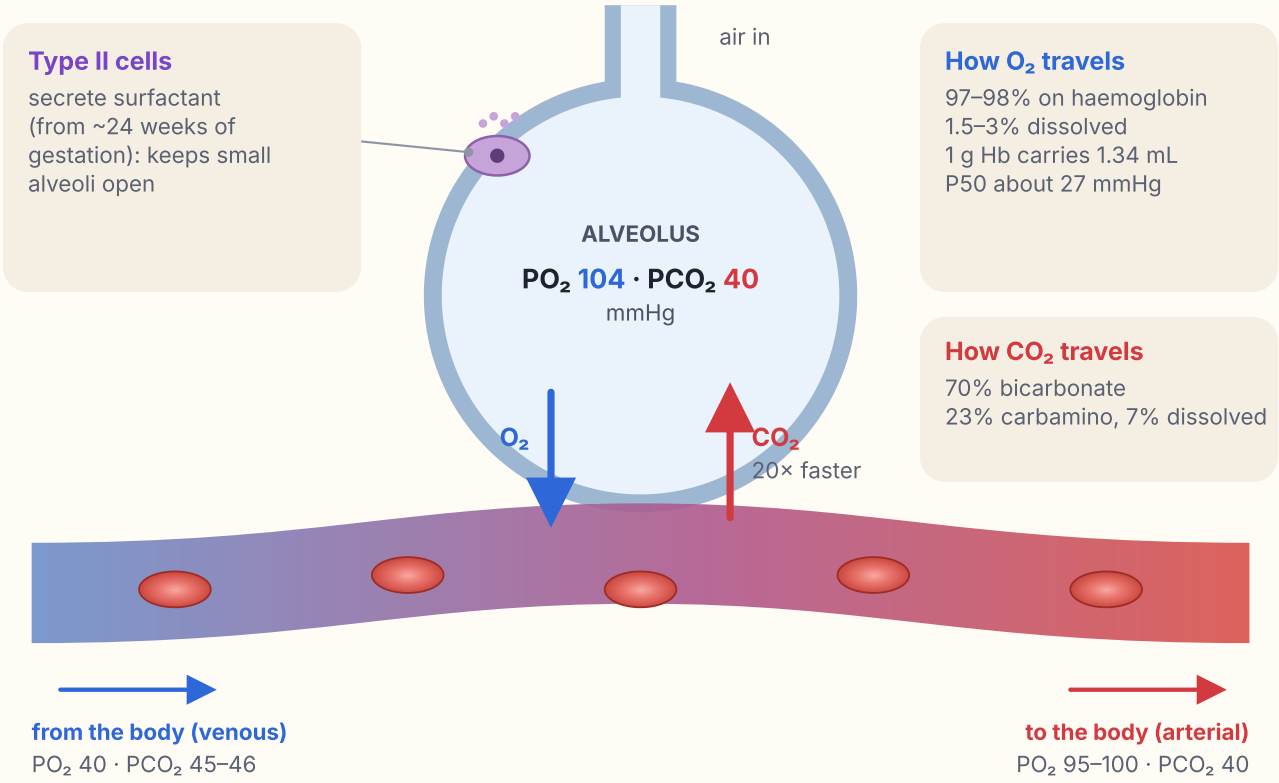
Dead space. Anatomical dead space about 150 mL (about 1 mL per pound of body weight). Alveolar ventilation = (TV − dead space) × rate = (500 − 150) × 12 ≈ 4.2 L/min.

Gas exchange and transport

Item	Value or fact
Alveolar PO_2 / PCO_2	104 / 40 mmHg
Arterial PO_2 / PCO_2	95–100 / 40 mmHg
Mixed venous PO_2 / PCO_2	40 / 45–46 mmHg
O_2 carried	97–98% bound to haemoglobin, 1.5–3% dissolved; 1 g Hb carries 1.34 mL O_2
CO_2 carried	About 70% as bicarbonate (carbonic anhydrase in red cells, chloride shift), 23% as carbamino compounds, 7% dissolved
P50	PO_2 at which Hb is 50% saturated, about 27 mmHg
Diffusion	CO_2 diffuses about 20 times faster than O_2

Gas exchange in the lung

Partial pressures across one alveolus and its capillary (mmHg)



The thin alveolar wall and capillary wall together form the respiratory membrane.

Gas exchange · partial pressures across one alveolus and its capillary

- **Oxygen–haemoglobin dissociation curve** (sigmoid).
- **Shift to the right** (Hb gives up O_2 more easily): raised CO_2, H^+ (low pH: the **Bohr effect**), temperature and 2,3-BPG, as in exercising muscle.

- **Shift to the left**: the opposite, plus fetal haemoglobin and carbon monoxide.
- **Hook: "CADET, face Right"** — CO_2, Acid, 2,3-DPG, Exercise, Temperature shift the curve right.
- The **Haldane effect**: deoxygenated blood carries more CO_2.

Control of breathing

- The dorsal respiratory group in the medulla sets the inspiratory rhythm; the pneumotaxic centre in the pons limits inspiration.
- **Central chemoreceptors** (medulla) respond to H^+ in the CSF, that is, to arterial CO_2: the main moment-to-moment drive.
- **Peripheral chemoreceptors** (carotid bodies via IX, aortic bodies via X) respond mainly to low arterial PO_2 (below about 60 mmHg).
- The **Hering–Breuer reflex** (stretch receptors, vagus) stops over-inflation.

5.2c Renal physiology

Most questions ask the GFR, which substance measures what, and which part of the nephron does what.

Basic numbers

Item	Value
Nephrons per kidney	About 1 million (cortical 85%, juxtamedullary 15%)
Renal blood flow	About 1,200–1,250 mL/min, about 20–25% of cardiac output
Renal plasma flow	About 625–650 mL/min
Glomerular filtration rate (GFR)	About 125 mL/min, or 180 L/day
Filtration fraction (GFR ÷ RPF)	About 0.2 (20%)
Urine output	1–1.5 L/day; 99% of filtrate is reabsorbed
Net filtration pressure	Glomerular capillary pressure 60 − Bowman's capsule pressure 18 − plasma oncotic pressure 32 = about 10 mmHg

Renal clearance

Clearance is the volume of plasma completely cleared of a substance per minute.

$$C = (U \times V) \div P$$

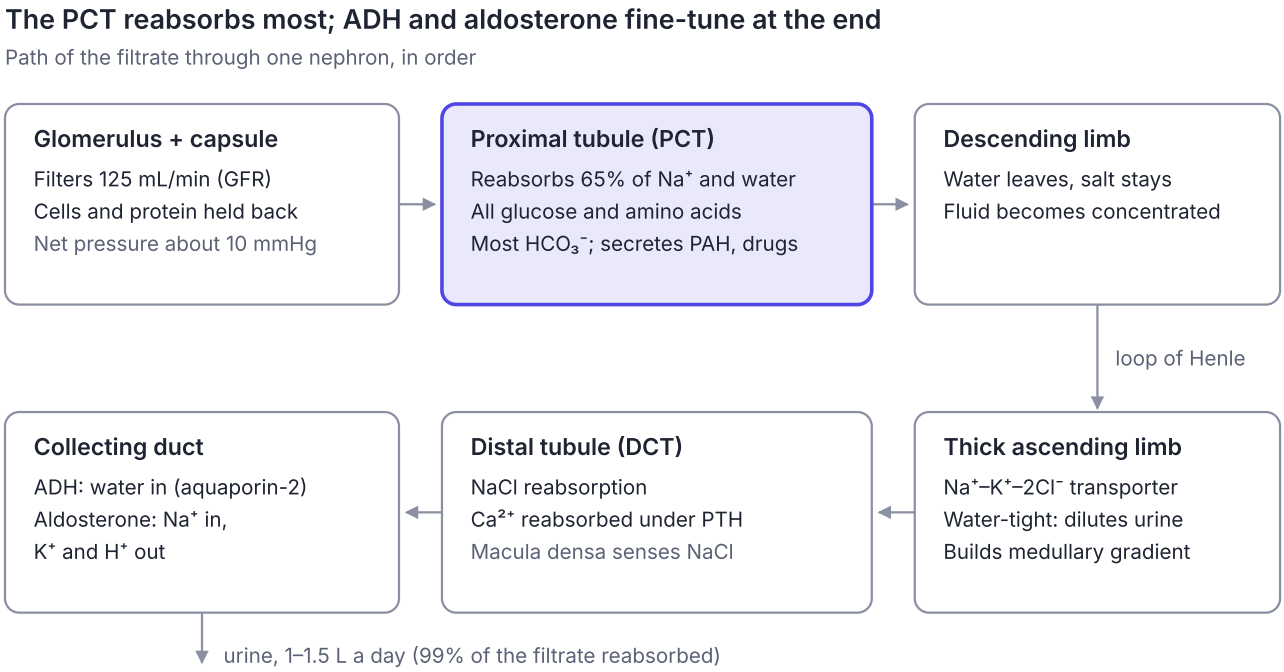
U = urine concentration, V = urine flow (mL/min), P = plasma concentration.

Substance	Handling	Clearance tells you
Inulin	Filtered only; not reabsorbed or secreted	**GFR** (gold standard, 125 mL/min)
Creatinine	Filtered + slightly secreted	GFR in clinical practice (slightly over-estimates)
Para-aminohippuric acid (PAH)	Filtered + almost fully secreted	**Effective renal plasma flow** (about 625 mL/min)
Glucose	Filtered + fully reabsorbed (up to the transport maximum)	Clearance 0; appears in urine above the renal threshold of about 180 mg/dL
Urea	Filtered + partly reabsorbed	About 75 mL/min (less than GFR)

Hook: clearance less than inulin = reabsorbed; more than inulin = secreted.

Nephron, segment by segment

Segment	Main job
Glomerulus and Bowman's capsule	Filtration; proteins and cells held back by the filtration barrier
Proximal convoluted tubule	Reabsorbs about 65% of Na^+ and water, **all glucose and amino acids**, most bicarbonate; secretes drugs and PAH
Descending limb of loop of Henle	Permeable to water, not salt: fluid becomes concentrated
Thick ascending limb	Na^+–K^+–$2Cl^-$ co-transport; impermeable to water ("diluting segment"); builds the medullary gradient (countercurrent multiplier)
Distal convoluted tubule	Na^+–Cl^- reabsorption; calcium reabsorption under PTH
Collecting duct	**Aldosterone**: Na^+ reabsorption, K^+ and H^+ secretion. **ADH**: inserts aquaporin-2, water reabsorption, concentrated urine
Juxtaglomerular apparatus	JG cells (afferent arteriole) secrete renin; the macula densa senses NaCl in the distal tubule



Nephron · what each segment reabsorbs or secretes

Follow the arrows: filtration at the glomerulus, bulk reabsorption in the PCT (highlighted), concentration and dilution in the loop, and hormone-controlled fine-tuning in the distal tubule and collecting duct.

Autoregulation. GFR and renal blood flow stay nearly constant between mean arterial pressures of about 80 and 180 mmHg, through the myogenic response and tubuloglomerular feedback.

Hormones the kidney makes. Renin, erythropoietin (from peritubular interstitial cells; falls in chronic kidney disease, causing anaemia), and calcitriol (1,25-dihydroxy vitamin D, made by 1α-hydroxylase in the proximal tubule).

Micturition. A spinal reflex at S2–S4, controlled by the pons. Parasympathetic (pelvic splanchnic nerves) contracts the detrusor; sympathetic relaxes it; the external sphincter is voluntary (pudendal nerve). The urge is first felt at about 150 mL and becomes marked at 400 mL.

5.3a Gastrointestinal physiology and hepatic functions

Learn one table for digestion (enzyme, source, substrate), one for the GI hormones, the sites of absorption, and the liver's jobs with bilirubin metabolism.

Digestion

Juice (daily volume, pH)	Enzyme	Acts on	Product
Saliva (1–1.5 L, pH 6.8–7)	Salivary amylase (ptyalin)	Starch	Maltose, dextrins
Saliva	Lingual lipase	Fats	Fatty acids (important in infants)
Gastric juice (2–3 L, pH 1–2)	Pepsin (from pepsinogen, activated by HCl)	Proteins	Peptides
Gastric juice	Gastric lipase; rennin (infants)	Fats; milk casein	—
Pancreatic juice (1.5 L, pH 7.8–8.3)	Trypsin, chymotrypsin (trypsinogen activated by enterokinase)	Proteins	Peptides
Pancreatic juice	Pancreatic amylase	Starch	Maltose
Pancreatic juice	Pancreatic lipase + colipase	Triglycerides	Monoglycerides, fatty acids
Bile (0.5–1 L, pH 7.8)	No enzyme; bile salts	Fat	Emulsification and micelles
Intestinal brush border	Maltase, sucrase, lactase, peptidases	Disaccharides, peptides	Monosaccharides, amino acids

Gastric cells. Parietal (oxyntic) cells: HCl and intrinsic factor (needed for vitamin B12; lack = pernicious anaemia). Chief (peptic) cells: pepsinogen. G cells (antrum): gastrin. Mucous neck cells: mucus.

Phases of gastric secretion. Cephalic (sight, smell; vagus; about 30%), gastric (food in the stomach; gastrin; about 60%), intestinal (about 10%).

Gastrointestinal hormones

Hormone	Made by	Stimulus	Main actions
Gastrin	G cells, antrum	Protein, stomach distension, vagus	Acid secretion, gastric motility, mucosal growth
Secretin (first hormone discovered, 1902)	S cells, duodenum	Acid in the duodenum	**Bicarbonate-rich** pancreatic juice; inhibits gastric acid
Cholecystokinin (CCK)	I cells, duodenum and jejunum	Fat and protein	Gall-bladder contraction, **enzyme-rich** pancreatic juice; slows gastric emptying; satiety

Hormone	Made by	Stimulus	Main actions
GIP (glucose-dependent insulinotropic peptide)	K cells	Glucose, fat	Insulin release; inhibits gastric acid
Motilin	M cells	Fasting	Migrating motor complex
Somatostatin	D cells	Acid	Inhibits most GI hormones
Ghrelin	Stomach	Fasting	Hunger

Hook: "Secretin = Soda (bicarbonate); CCK = Contract the gall bladder + Cook (enzymes)."

Absorption

Site	What is absorbed
Stomach	Alcohol, aspirin, little water
Duodenum	**Iron** and **calcium** (active), folate (upper jejunum)
Jejunum	Most carbohydrates, proteins, fats, water-soluble vitamins
Terminal ileum	**Vitamin B12** (with intrinsic factor) and **bile salts** (enterohepatic circulation)
Colon	Water, sodium; vitamin K and biotin made by gut bacteria

Glucose and galactose enter by **SGLT-1** (sodium co-transport, the basis of oral rehydration solution); fructose by facilitated diffusion (GLUT-5). Fats enter as micelles, are rebuilt into chylomicrons and enter the lacteals (lymph), not blood.

Motility. Swallowing: oral (voluntary), pharyngeal and oesophageal (reflex) phases. Peristalsis is run by the myenteric (Auerbach's) plexus; secretion by the submucosal (Meissner's) plexus. Gastrocolic reflex: food in the stomach triggers mass movement in the colon. Defecation reflex: sacral S2–S4.

Hepatic functions

Function	Details
Carbohydrate	Glycogenesis, glycogenolysis, gluconeogenesis: keeps fasting blood glucose steady
Protein	Makes albumin (keeps oncotic pressure), clotting factors (I, II, V, VII, IX, X), urea from ammonia (urea cycle)

Function	Details
Fat	Makes cholesterol, lipoproteins (VLDL), bile salts; beta-oxidation; ketone bodies
Storage	Glycogen, vitamins A, D, B12, iron (ferritin), copper
Detoxification	Drugs, alcohol, ammonia, hormones; conjugation of bilirubin
Bile	Bile salts, bilirubin, cholesterol; about 600 mL/day
Immune	Kupffer cells (macrophages) remove bacteria from portal blood
Fetal blood	The liver makes red cells in fetal life

Vitamin K-dependent factors: "1972" = II, VII, IX, X (plus proteins C and S). A raised prothrombin time is an early sign of poor liver synthesis; low albumin a late one (half-life about 20 days).

Bilirubin metabolism. Old red cells → haem → biliverdin → **unconjugated bilirubin** (fat-soluble, carried on albumin, not in urine) → liver conjugates it with glucuronic acid (UDP-glucuronyl transferase) → **conjugated bilirubin** (water-soluble) → bile → gut bacteria make **urobilinogen** → stercobilin (colours stool brown) and some reabsorbed and excreted in urine as urobilin. Normal serum bilirubin 0.3–1.2 mg/dL; jaundice shows above about 2–2.5 mg/dL.

Jaundice	Raised bilirubin	Urine	Stool
Pre-hepatic (haemolytic)	Unconjugated	No bilirubin; urobilinogen raised	Dark
Hepatic (hepatitis)	Both	Bilirubin present	Normal or pale
Post-hepatic (obstructive)	Conjugated	Bilirubin present (dark); urobilinogen absent	**Clay-coloured**, with itching

5.3b Endocrine: hypothalamic-pituitary axis and feedback loops

The hypothalamus sends releasing hormones down the portal veins to the anterior pituitary, the anterior pituitary drives the target glands, and the target-gland hormone switches both off. That loop, negative feedback, answers most questions.

Hypothalamus → anterior pituitary → target

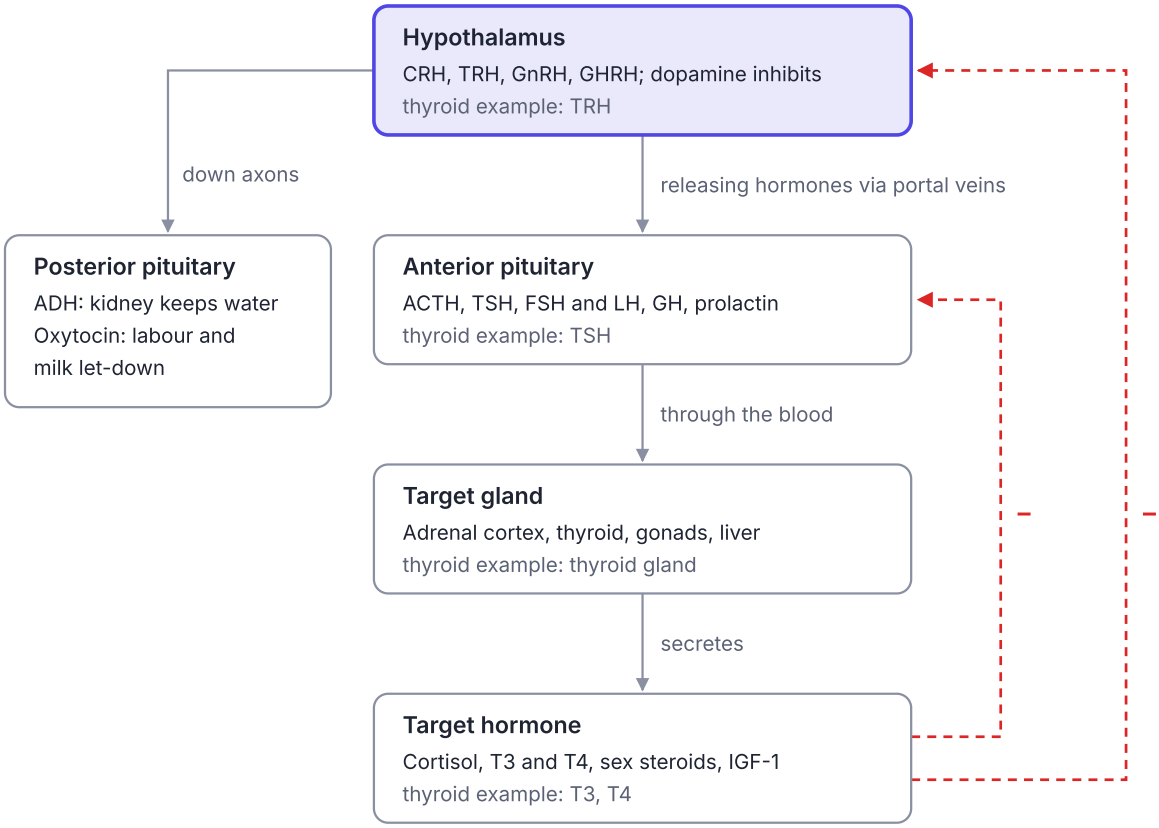
Hypothalamic hormone	Anterior pituitary hormone	Target and effect	Feedback
TRH	TSH	Thyroid: T3, T4 (metabolic rate)	T3/T4 inhibit TRH and TSH
CRH	ACTH	Adrenal cortex: cortisol	Cortisol inhibits CRH and ACTH
GnRH (pulsatile)	FSH, LH	Gonads: oestrogen, progesterone, testosterone; follicles, sperm	Sex steroids, inhibin
GHRH (stimulates); somatostatin (inhibits)	Growth hormone	Liver: IGF-1; growth, protein synthesis, raises glucose	IGF-1, GH
Dopamine (inhibits); TRH (stimulates)	Prolactin	Breast: milk production	**Prolactin is the only pituitary hormone mainly under inhibitory control**

- **Posterior pituitary (neurohypophysis).** Stores hormones made in the hypothalamus (supraoptic and paraventricular nuclei) and sent down axons.
- **ADH (vasopressin)**: water reabsorption in the collecting duct; lack = diabetes insipidus (large volumes of dilute urine).
- **Oxytocin**: milk ejection (let-down) and uterine contraction in labour.

Feedback types. Negative feedback (most hormones): the product switches off its own drive. **Positive feedback** (rare): the LH surge before ovulation (rising oestrogen), oxytocin in labour (Ferguson reflex), clotting cascade.

The target hormone switches off its own drive (negative feedback)

Hypothalamic-pituitary axis, with the thyroid as the worked example



Red dashed arrows: the target hormone inhibits both the pituitary and the hypothalamus (negative feedback). Exceptions with positive feedback: the LH surge before ovulation and oxytocin in labour.

Hypothalamic-pituitary axis · 4 levels and 2 feedback loops

In hypothyroidism from a failing thyroid, T4 falls, feedback is lost, and TSH rises; that is why a high TSH is the first sign of primary hypothyroidism.

Other glands at a glance

Gland	Hormone	Key action	Too little / too much
Thyroid follicular cells	T4 (mostly), T3 (more active)	Basal metabolic rate, growth, brain development	Cretinism, myxoedema / Graves' disease
Thyroid parafollicular (C) cells	Calcitonin	Lowers blood calcium	—
Parathyroid (chief cells)	Parathormone (PTH)	Raises blood calcium (bone resorption, kidney reabsorption, activates vitamin D)	Tetany (after thyroid surgery) / bone loss, kidney stones

Gland	Hormone	Key action	Too little / too much
Adrenal cortex: zona glomerulosa	Aldosterone (mineralocorticoid)	Na^+ retention, K^+ loss	Addison's disease / Conn's syndrome
Zona fasciculata	Cortisol (glucocorticoid)	Raises glucose, anti-inflammatory, stress	Addison's / Cushing's syndrome
Zona reticularis	Androgens (DHEA)	Pubic and axillary hair	—
Adrenal medulla	Adrenaline (80%), noradrenaline	Fight or flight	— / phaeochromocytoma
Pancreas: β cells (about 70%)	Insulin	Lowers glucose	Diabetes mellitus / hypoglycaemia
Pancreas: α cells	Glucagon	Raises glucose	—
Pancreas: δ cells	Somatostatin	Inhibits insulin and glucagon	—
Pineal	Melatonin	Sleep–wake rhythm	—

Adrenal cortex hook: "Salt, Sugar, Sex — the deeper you go, the sweeter it gets" (glomerulosa, fasciculata, reticularis: "GFR"). The cortex comes from mesoderm, the medulla from neural crest.

Thyroid facts. Iodide is trapped by the sodium–iodide symporter, oxidised by thyroid peroxidase and attached to thyroglobulin. T4 is the main product; T3 is 3–5 times more potent and is mostly made from T4 in the tissues. Normal TSH is about 0.4–4.5 mIU/L.

Calcium. Normal serum calcium 9–11 mg/dL (about half ionised). PTH and vitamin D raise it; calcitonin lowers it.

Memory hook: the adrenal cortex from outside in is **salt, sugar, sex**: zona **G**lomerulosa makes aldosterone (salt), **F**asciculata cortisol (sugar), **R**eticularis androgens (sex), "**GFR**". The deeper you go, the sweeter it gets.

5.4a Body fluids, electrolytes and acid-base balance

The numbers to learn are the 60–40–20 rule, the main ion inside and outside the cell, and normal arterial blood gas values; acid-base questions are solved by the four-step flowchart below.

Fluid compartments (70 kg adult male)

Compartment	Share of body weight	Volume	Measured with
Total body water	60%	42 L	Heavy water (D_2O), antipyrine
Intracellular fluid (ICF)	40% (two-thirds of TBW)	28 L	TBW − ECF
Extracellular fluid (ECF)	20% (one-third of TBW)	14 L	Inulin, mannitol, sucrose
— Interstitial fluid	15%	about 10.5 L	ECF − plasma
— Plasma	5%	about 3.5 L	Evans blue (T-1824), radio-iodinated albumin
Blood volume	about 7–8%	5 L	Plasma volume ÷ (1 − haematocrit)

Hook: "60–40–20" (total, inside, outside). Women and the elderly have less water (more fat); infants more (about 75%).

Ion	ECF (mEq/L)	ICF (mEq/L)
Na^+	142 (main ECF cation)	10–14
K^+	4	140 (main ICF cation)
Cl^-	103 (main ECF anion)	4
HCO_3^-	24–28	10
Phosphate, proteins	low	high (main ICF anions)

The Na^+–K^+ ATPase (3 Na^+ out, 2 K^+ in per ATP) keeps this difference. Normal plasma osmolality is 285–295 mOsm/kg; estimated as 2 × Na^+ + glucose/18 + urea/2.8 (mg/dL). 0.9% NaCl is isotonic with plasma.

Starling forces decide movement between plasma and interstitium: capillary hydrostatic pressure pushes fluid out; plasma oncotic pressure (albumin, about 25–28 mmHg) pulls it in. Oedema follows high venous pressure (heart failure), low albumin (nephrotic syndrome, liver disease), leaky capillaries (inflammation) or blocked lymphatics (filariasis).

Water balance. Daily intake about 2.5 L (drink 1.5 L, food 0.8 L, metabolic water 0.3 L) equals output (urine 1.5 L, skin and lungs about 0.9 L as insensible loss, faeces 0.1 L). Thirst and ADH respond to a 1–2% rise in

osmolality.

Acid-base balance

Arterial blood	Normal
pH	7.35–7.45 (7.40)
$PaCO_2$	35–45 mmHg (40)
HCO_3^-	22–26 mEq/L (24)
Anion gap = $Na^+ - (Cl^- + HCO_3^-)$	8–16 mEq/L (about 12)

Henderson–Hasselbalch.

$$pH = 6.1 + \log\left([HCO_3^-] \div (0.03 \times PaCO_2) \right)$$

The ratio of bicarbonate to dissolved CO_2 is 20 : 1 at pH 7.4.

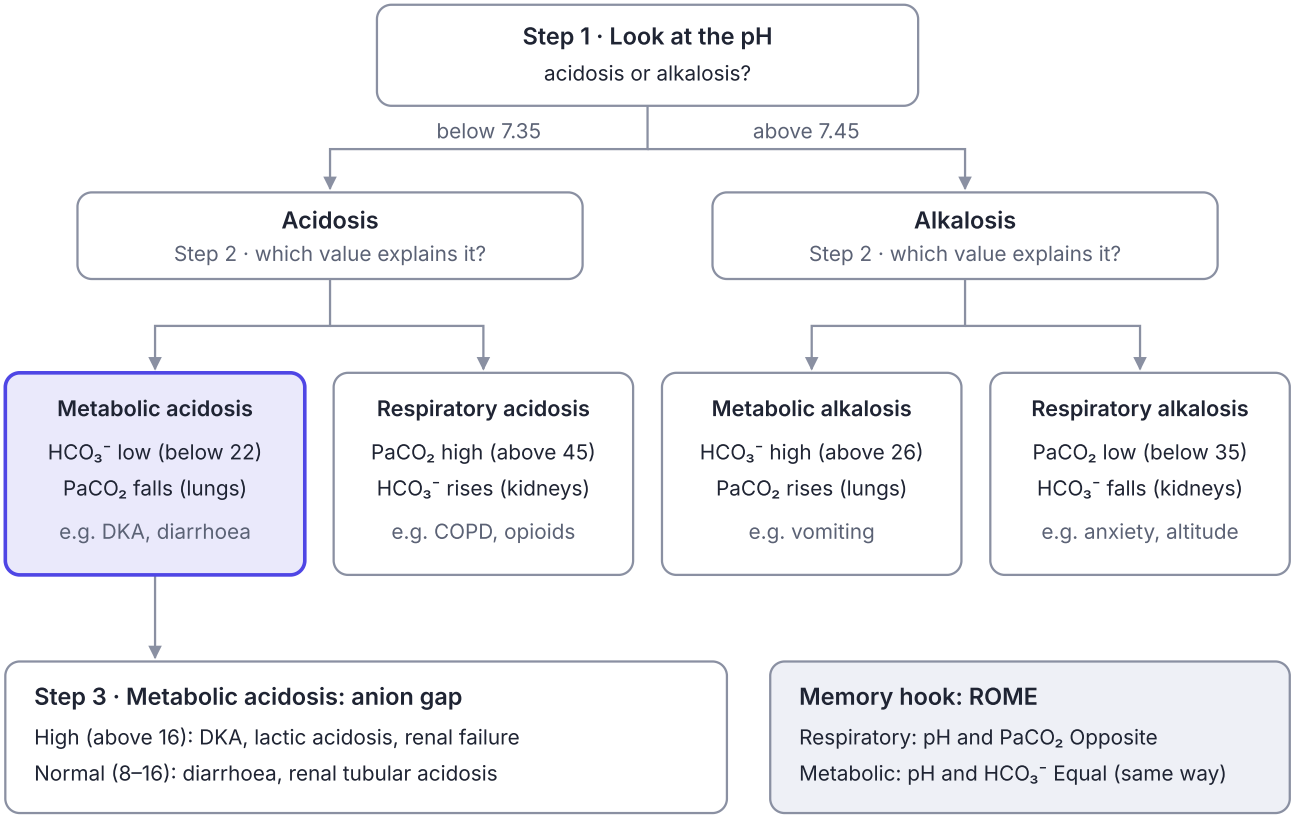
Buffers. Bicarbonate is the main buffer of the ECF; haemoglobin of the blood (red cells); phosphate and proteins inside cells, phosphate also in urine. Lines of defence: buffers (seconds), lungs (minutes, by changing CO_2), kidneys (days, by excreting H^+ as ammonium and titratable acid and making new HCO_3^-).

Disorder	Primary change	Compensation	Common causes
Metabolic acidosis	HCO_3^- falls	Hyperventilation, $PaCO_2$ falls (Kussmaul breathing)	Diabetic ketoacidosis, lactic acidosis, renal failure (high anion gap); diarrhoea (normal anion gap)
Metabolic alkalosis	HCO_3^- rises	Hypoventilation, $PaCO_2$ rises	Vomiting, nasogastric suction, diuretics, hypokalaemia
Respiratory acidosis	$PaCO_2$ rises	Kidneys keep HCO_3^-	COPD, opioid overdose, chest-wall weakness
Respiratory alkalosis	$PaCO_2$ falls	Kidneys excrete HCO_3^-	Anxiety hyperventilation, high altitude, early salicylate poisoning

Hook: "ROME" — Respiratory Opposite, Metabolic Equal. In respiratory disorders pH and $PaCO_2$ move in opposite directions; in metabolic disorders pH and HCO_3^- move in the same direction. High-anion-gap metabolic acidosis causes: **"MUDPILES"** (methanol, uraemia, diabetic ketoacidosis, paraldehyde, iron or isoniazid, lactic acidosis, ethylene glycol, salicylates).

Three steps to any blood gas: pH, cause, anion gap

Normal: pH 7.35–7.45 · $PaCO_2$ 35–45 mmHg · HCO_3^- 22–26 mEq/L



Acid-base flowchart · 3 steps to the primary disorder

Worked example: pH 7.25, $PaCO_2$ 30, HCO_3^- 14. The pH is low (acidosis); HCO_3^- is low, which explains it (metabolic); $PaCO_2$ is low because the lungs are compensating.

Key electrolytes

Ion	Normal serum	Low	High
Sodium	135–145 mEq/L	Confusion, seizures (SIADH, diuretics)	Thirst, confusion (water loss, diabetes insipidus)
Potassium	3.5–5.0 mEq/L	Muscle weakness, flat T and U waves	Peaked T waves, cardiac arrest

Ion	Normal serum	Low	High
Calcium	9–11 mg/dL	Tetany: Chvostek's and Trousseau's signs	Stones, bones, groans, moans
Chloride	96–106 mEq/L	Metabolic alkalosis (vomiting)	Normal anion gap acidosis
Magnesium	1.5–2.5 mg/dL	Tetany, arrhythmia	Drowsiness, loss of reflexes

5.4b Clinical biochemistry: enzymes, lipid profile, blood glucose

For each test, know the normal range and the organ a raised value points to.

Enzymes

- **Basics.** Enzymes are protein catalysts that lower activation energy; they do not change the equilibrium.
- **Km** (Michaelis constant) = substrate concentration at half the maximum velocity; a low Km means high affinity.
- Competitive inhibitors raise Km, Vmax unchanged; non-competitive inhibitors lower Vmax, Km unchanged.
- Isoenzymes catalyse the same reaction but differ in structure (for example, LDH and CK).

Six classes (IUBMB): "Over The HILL" — Oxidoreductases, Transferases, Hydrolases, Isomerases, Lyases, Ligases.

Enzyme	Normal (approx.)	Raised in
ALT (SGPT)	7–56 U/L	Liver cell injury; **more specific for the liver**
AST (SGOT)	10–40 U/L	Liver, heart, muscle; AST/ALT above 2 suggests alcoholic liver disease
Alkaline phosphatase (ALP)	40–130 U/L	Obstructive jaundice, bone disease (rickets, Paget's), normal growth and pregnancy
Gamma-glutamyl transferase (GGT)	under 50 U/L	Alcohol, cholestasis (confirms a liver source of raised ALP)
CK-MB	under 5% of total CK	Myocardial infarction: rises 4–6 h, peaks 24 h, normal in 2–3 days

Enzyme	Normal (approx.)	Raised in
Cardiac troponin I or T	below assay cut-off	Myocardial infarction: most specific; stays up 7–10 days
LDH (LDH-1 > LDH-2 "flipped")	140–280 U/L	Late marker of MI; haemolysis
Amylase and lipase	Amylase 30–110 U/L	Acute pancreatitis (lipase more specific, stays raised longer)
Acid phosphatase (prostatic)	—	Prostate cancer (now replaced by PSA)

Hook for liver tests: "ALT = Liver (L for Liver); ALP = Pipes (obstruction, bile ducts and bone)."

Lipid profile (fasting 9–12 hours)

Test	Desirable	High
Total cholesterol	under 200 mg/dL	240 mg/dL or more
LDL cholesterol ("bad")	under 100 mg/dL	160 mg/dL or more
HDL cholesterol ("good")	40 mg/dL or more (men), 50 or more (women); 60 or more protective	—
Triglycerides	under 150 mg/dL	200 mg/dL or more
VLDL	about TG ÷ 5	—

Friedewald formula: LDL = total cholesterol − HDL − (triglycerides ÷ 5), in mg/dL, valid when triglycerides are below 400.

Lipoprotein	Carries	From	Note
Chylomicrons	Dietary triglycerides	Intestine	Largest, least dense; make fasting serum milky
VLDL	Liver-made triglycerides	Liver	Becomes IDL, then LDL

Lipoprotein	Carries	From	Note
LDL	Cholesterol to tissues	From VLDL	Atherogenic; taken up by LDL receptors (apo B-100)
HDL	Cholesterol back to the liver (reverse transport)	Liver, intestine	Smallest, densest; protective (apo A-I)

Hook: "HDL = Healthy, LDL = Lethal." HMG-CoA reductase is the rate-limiting enzyme of cholesterol synthesis.

Blood glucose regulation

Test	Normal	Prediabetes	Diabetes (ADA, WHO)
Fasting plasma glucose	70–99 mg/dL	100–125 mg/dL	126 mg/dL or more
2-hour glucose after 75 g oral glucose	under 140 mg/dL	140–199 mg/dL	200 mg/dL or more
HbA1c	under 5.7%	5.7–6.4%	6.5% or more
Random plasma glucose with symptoms	—	—	200 mg/dL or more

Hypoglycaemia: below 70 mg/dL (symptoms usually below about 55). The renal threshold for glucose is about 180 mg/dL. HbA1c reflects the average glucose of the past 2–3 months (red-cell lifespan 120 days).

Hormone	Effect on glucose	How
Insulin (the only hormone that **lowers** glucose)	Lowers	Glucose uptake into muscle and fat (GLUT-4), glycogen synthesis, fat storage; stops gluconeogenesis
Glucagon	Raises	Glycogenolysis and gluconeogenesis in the liver
Adrenaline	Raises	Glycogenolysis in liver and muscle (fast stress response)
Cortisol	Raises	Gluconeogenesis
Growth hormone	Raises	Opposes insulin action

- **Fed and fasting states.** After a meal: insulin high, glycogen stored.
- Overnight fast: liver glycogenolysis keeps glucose steady.

- Longer fasting (about 24 hours and beyond): gluconeogenesis from amino acids, lactate and glycerol; after several days, ketone bodies feed the brain.
- Liver glycogen lasts about 12–18 hours.
- Glucose enters the brain and red cells without insulin (GLUT-1, GLUT-3).

Rapid revision: normal values and one-line facts

Read this sheet every day until the exam; most Topic 5 questions are one of these lines.

System	Value or fact
Blood	Hb 13–17 g/dL (men), 12–15 (women); RBC 4.5–5.5 million/µL; WBC 4,000–11,000/µL; platelets 1.5–4 lakh/µL; RBC lifespan 120 days; platelet 7–10 days
Blood	Universal donor O negative; universal recipient AB positive; ESR (Westergren) men 0–15, women 0–20 mm/h
Heart	HR 60–100/min; BP 120/80 mmHg; cardiac output 5 L/min; SV 70 mL; EF 55–70%
Heart	PR 0.12–0.20 s; QRS under 0.12 s; QT about 0.40 s; SA node = pacemaker; AV node = delay
Lungs	TV 500 mL; VC 4.6 L; TLC 5.8 L; RV 1.2 L (not by spirometry); dead space 150 mL; RR 12–16/min
Lungs	Surfactant from type II pneumocytes; O_2–Hb curve shifts right with CO_2, acid, 2,3-BPG, exercise, heat
Kidney	GFR 125 mL/min (inulin); RPF 625 mL/min (PAH); RBF 1.2 L/min; filtration fraction 20%; renal glucose threshold 180 mg/dL
Kidney	Glucose and amino acids fully reabsorbed in the PCT; ADH acts on the collecting duct; renin from JG cells
GI	Gastric pH 1–2; HCl and intrinsic factor from parietal cells; B12 and bile salts absorbed in the terminal ileum; iron in the duodenum
GI	Secretin = bicarbonate juice; CCK = gall-bladder contraction and enzymes; gastrin = acid
Liver	Bilirubin 0.3–1.2 mg/dL; albumin 3.5–5 g/dL; vitamin K factors II, VII, IX, X; Kupffer cells = macrophages
Endocrine	Prolactin under dopamine inhibition; ADH and oxytocin made in the hypothalamus, stored in the posterior pituitary
Endocrine	Adrenal cortex: glomerulosa aldosterone, fasciculata cortisol, reticularis androgens; insulin from β cells
Fluids	TBW 60% of body weight; ICF 40%; ECF 20%; plasma 5% (Evans blue); main ICF cation K^+, main ECF cation Na^+

System	Value or fact
Acid-base	pH 7.35–7.45; $PaCO_2$ 35–45 mmHg; HCO_3^- 22–26 mEq/L; anion gap 8–16; HCO_3^- : dissolved CO_2 = 20 : 1
Electrolytes	Na^+ 135–145; K^+ 3.5–5.0 mEq/L; Ca^{2+} 9–11 mg/dL; serum osmolality 285–295 mOsm/kg
Biochemistry	Fasting glucose 70–99 mg/dL; diabetes fasting 126 or more, HbA1c 6.5% or more; total cholesterol under 200; LDL under 100; HDL 40 or more; TG under 150
Biochemistry	ALT liver-specific; ALP raised in obstruction and bone disease; troponin most specific for MI
Anatomy	Spinal cord ends at L1 (adult); lumbar puncture at L3–L4; sternal angle = 2nd rib, T4–T5
Anatomy	Wrist drop = radial; claw hand = ulnar; ape thumb = median; foot drop = common peroneal; winged scapula = long thoracic
Anatomy	Broca = motor speech (frontal); Wernicke = understanding (temporal); vision = occipital (area 17); cerebellum = same-side ataxia

One-glance recall card

Topic 5 recall card: the 15 facts asked most

Cover the small lines, say the fact from the big cue, then check

Broca 44, 45
motor speech, frontal;
Wernicke 22, temporal

LR6 SO4
lateral rectus VI,
superior oblique IV

C5 to T1
roots of the
brachial plexus

PR 0.12-0.20 s
QRS under 0.12 s;
1 small square = 0.04 s

CADET, face right
CO2, acid, DPG, exercise,
temperature shift O2 curve

TV 500 mL
tidal volume; VC is
IRV + TV + ERV

GFR 125
mL/min; inulin clearance
measures it

PAH clearance
measures renal
plasma flow

Parietal cells
acid and intrinsic
factor (B12)

Glucose: SGLT-1
sodium-linked uptake
in the small gut

Salt, sugar, sex
adrenal cortex: glomerulosa,
fasciculata, reticularis

60 · 40 · 20
body water %, ICF %, ECF %
of body weight

20 : 1
bicarbonate to CO2
keeps pH at 7.4

MUDPILES
causes of high anion
gap acidosis

LDL = TC-HDL-TG/5
Friedewald formula,
mg/dL

Topic 5 recall card · 15 tiles

For the last ten minutes before the exam: cover the small lines of each tile and say the fact from the big cue.

Practice MCQs

Thirty questions in the CBRT style, grouped by syllabus heading; the answer key with a one-line reason follows. Option E in the real paper is always "None of the above".

Neuro-musculoskeletal anatomy

1. The first carpometacarpal joint of the thumb is a: (a) hinge joint (b) saddle joint (c) pivot joint (d) plane joint
2. Expressive (non-fluent) aphasia follows a lesion of: (a) Wernicke's area (b) Broca's area (c) angular gyrus (d) occipital lobe
3. Wrist drop after a fracture of the shaft of the humerus is due to injury of the: (a) median nerve (b) ulnar nerve (c) radial nerve (d) axillary nerve
4. The pubic symphysis is an example of a: (a) primary cartilaginous joint (b) secondary cartilaginous joint (c) syndesmosis (d) synovial joint
5. Trendelenburg's sign is positive in paralysis of the: (a) gluteus maximus (b) gluteus medius and minimus (c) quadriceps (d) hamstrings
6. The knee jerk tests spinal segments: (a) S1, S2 (b) L3, L4 (c) C5, C6 (d) C7, C8

Cardiovascular, respiratory and renal

7. The first heart sound is caused by closure of the: (a) aortic and pulmonary valves (b) mitral and tricuspid valves (c) mitral and aortic valves (d) pulmonary valve only
8. The PR interval on the ECG mainly represents: (a) ventricular repolarisation (b) atrial depolarisation and the delay in the AV node (c) ventricular depolarisation (d) atrial repolarisation
9. The main long-term regulator of arterial blood pressure is the: (a) baroreceptor reflex (b) chemoreceptor reflex (c) renal–body fluid mechanism (d) CNS ischaemic response
10. The lung volume that cannot be measured with a simple spirometer is the: (a) tidal volume (b) vital capacity (c) residual volume (d) inspiratory reserve volume
11. Pulmonary surfactant is secreted by: (a) type I pneumocytes (b) type II pneumocytes (c) alveolar macrophages (d) Clara cells
12. A shift of the oxygen–haemoglobin dissociation curve to the right is caused by: (a) fall in temperature (b) fetal haemoglobin (c) rise in PCO_2 (d) carbon monoxide

13. The clearance of which substance measures the glomerular filtration rate most accurately? (a) PAH (b) urea (c) inulin (d) glucose
14. Glucose is normally reabsorbed almost completely in the: (a) proximal convoluted tubule (b) loop of Henle (c) distal convoluted tubule (d) collecting duct

Gastrointestinal, hepatic and endocrine

15. Intrinsic factor is secreted by the: (a) chief cells (b) parietal cells (c) G cells (d) mucous neck cells
16. Vitamin B12 is absorbed mainly in the: (a) duodenum (b) jejunum (c) terminal ileum (d) colon
17. Cholecystokinin causes: (a) bicarbonate-rich pancreatic juice (b) contraction of the gall bladder (c) gastric acid secretion (d) relaxation of the gall bladder
18. Clay-coloured stools with dark urine suggest: (a) haemolytic jaundice (b) obstructive jaundice (c) Gilbert syndrome (d) physiological jaundice
19. The pituitary hormone mainly under inhibitory control of the hypothalamus is: (a) TSH (b) ACTH (c) prolactin (d) growth hormone
20. ADH acts mainly on the: (a) proximal tubule (b) loop of Henle (c) collecting duct (d) glomerulus
21. Aldosterone is secreted by the: (a) zona glomerulosa (b) zona fasciculata (c) zona reticularis (d) adrenal medulla
22. An example of positive feedback is: (a) cortisol suppressing ACTH (b) the LH surge before ovulation (c) T4 suppressing TSH (d) insulin lowering glucose

Body fluids, acid-base and biochemistry

23. Intracellular fluid forms about what share of body weight? (a) 20% (b) 40% (c) 60% (d) 5%
24. The main cation of intracellular fluid is: (a) sodium (b) potassium (c) calcium (d) magnesium
25. Plasma volume is measured with: (a) inulin (b) heavy water (c) Evans blue (d) antipyrine
26. pH 7.25, $PaCO_2$ 30 mmHg and HCO_3^- 14 mEq/L indicate: (a) respiratory acidosis (b) metabolic acidosis with respiratory compensation (c) metabolic alkalosis (d) respiratory alkalosis
27. Persistent vomiting typically produces: (a) metabolic acidosis (b) metabolic alkalosis (c) respiratory acidosis (d) respiratory alkalosis
28. The enzyme most specific for liver-cell injury is: (a) AST (b) ALT (c) ALP (d) LDH
29. Diabetes mellitus is diagnosed when HbA1c is: (a) 5.7% or more (b) 6.0% or more (c) 6.5% or more (d) 7.5% or more
30. The lipoprotein that carries cholesterol from the tissues back to the liver is: (a) chylomicron (b) VLDL (c) LDL (d) HDL

Answer key

Q	Answer	Why
1	b	Saddle joint allows opposition of the thumb
2	b	Broca's area (frontal) = motor speech
3	c	Radial nerve runs in the spiral groove
4	b	Symphysis = secondary cartilaginous, midline
5	b	Superior gluteal nerve palsy; pelvis drops to the opposite side
6	b	Knee jerk L3, L4
7	b	S1 = AV-valve closure at the start of systole
8	b	Includes the AV-node delay; 0.12–0.20 s
9	c	The kidneys set long-term pressure (pressure natriuresis)
10	c	RV stays in the lungs; needs helium dilution or plethysmography
11	b	Type II alveolar cells
12	c	CADET: CO_2, acid, 2,3-BPG, exercise, temperature
13	c	Inulin is filtered, not reabsorbed or secreted
14	a	PCT reabsorbs all glucose up to the transport maximum
15	b	Parietal cells make HCl and intrinsic factor
16	c	B12–intrinsic factor complex in the terminal ileum
17	b	CCK contracts the gall bladder; secretin gives bicarbonate
18	b	Bile cannot reach the gut; conjugated bilirubin spills into urine
19	c	Dopamine inhibits prolactin
20	c	Aquaporin-2 in the collecting duct
21	a	Glomerulosa = salt (mineralocorticoid)
22	b	Rising oestrogen triggers more LH

Q	Answer	Why
23	b	60–40–20 rule
24	b	K^+ about 140 mEq/L inside the cell
25	c	Evans blue binds albumin and stays in plasma
26	b	Low pH, low HCO_3^- (same direction = metabolic); low $PaCO_2$ = compensation
27	b	Loss of gastric HCl
28	b	ALT is mainly in the liver
29	c	6.5% or more (ADA/WHO)
30	d	HDL = reverse cholesterol transport

Topic 6: Pathology, Microbiology & Diagnostics

How to use this in 2 days

Topic 6 is one of ten core topics, so expect roughly 2–4 of the 25 core marks from it. GPSC asks one-line recall: a stain, an organism, a vector, a normal value, a test of choice. Learn the tables, not the prose.

Session	Read	Then do
Day 1 morning	GPSC questions already asked; cell injury; inflammation; haemodynamics	Mark every fact you got wrong
Day 1 afternoon	Neoplasia; immunity and amyloid; general microbiology	First 25 practice MCQs
Day 1 evening	Bacteria; viruses	Rapid revision, sections 1–6
Day 2 morning	Parasites and fungi; haematology	—
Day 2 afternoon	Clinical pathology (LFT, RFT, urine, CSF); imaging and serology	Last 25 practice MCQs
Day 2 evening	Rapid revision page twice; your marked errors	Sleep

MCQ habits that pay here:

- **'Most common' and 'investigation of choice'** are the favourite stems. Each table below has a column for them.
- **Stains and media** are pure recall: learn the pairs (Ziehl–Neelsen → AFB, Giemsa → malaria, India ink → Cryptococcus).
- **Normal values**: learn one number per test, not ranges from every book.
- **Goa context**: leptospirosis, scrub typhus, dengue, malaria and KFD-type monsoon fevers come up because they are local.

What GPSC has already asked

- GPSC's Medical Officer papers of 28 Sep 2024 and 28 Sep 2025 asked about 17 of their 60 core questions from this topic's ground.
- Almost all of them test a single linked fact: a sign and its cancer, a test of choice, a radiology sign, an organism.
- The answers below are GPSC's keys, checked.

Paper	Question (shortened)	Answer	The fact to carry away
MO 2025	Rat-tail appearance on barium swallow	Carcinoma oesophagus	Rat tail / shouldering = carcinoma; bird's beak = achalasia
MO 2025	Best diagnostic modality for TB spine	CT-guided biopsy	Tissue (histology + culture/CBNAAT) confirms; MRI is best imaging
MO 2025	Whitish tonsillar membrane that bleeds on scraping	Diphtheria	Corynebacterium diphtheriae; Albert stain, Loeffler's serum slope, tellurite medium
MO 2025	Suspected COVID-19 with loss of smell: next step	Nasopharyngeal swab for RT-PCR	NP swab RT-PCR is the reference test
MO 2025	Investigation of choice, acute subarachnoid haemorrhage	Non-contrast CT	CT first; lumbar puncture (xanthochromia) if CT negative
MO 2025	Screening method for CIN — the exception	Colposcopic biopsy	Pap smear, HPV-DNA and VIA screen; biopsy diagnoses
MO 2025	Intraocular metallic foreign body: first investigation	Thin-cut CT orbit	MRI is contraindicated for metallic foreign bodies
MO 2024	Paget disease of the nipple is	Malignant	Underlying ductal carcinoma; large clear Paget cells in epidermis
MO 2024	Lesion with higher risk of invasive breast cancer	Atypical ductal hyperplasia	Atypical hyperplasia ≈ 4–5× risk; sclerosing adenosis, papillomatosis lower
MO 2024	Troisier's sign	Carcinoma stomach	Left supraclavicular node (Virchow's node)
MO 2024	Leukonychia is seen in	Hypoalbuminaemia	Koilonychia = iron deficiency
MO 2024	Most common cause of infantile diarrhoea worldwide	Rotavirus	Rotavirus vaccine is in India's UIP
MO 2024	Commonest heart defect in congenital rubella	PDA	Rubella triad: PDA, cataract, deafness
MO 2024	Bagel sign on ultrasound	Ectopic pregnancy	Tubal ring sign; 'snowstorm' = molar pregnancy
MO 2024	Graves' disease: thyroid profile	Low TSH	High T3/T4, suppressed TSH; TRAb positive

Paper	Question (shortened)	Answer	The fact to carry away
MO 2024	Cannot cross the placenta	Albumin and IgM	IgG is the only immunoglobulin that crosses
MO 2024	Not a structure on the right heart border on X-ray	Right ventricle	Right border = SVC, right atrium, IVC

Related items from GPSC's other medical papers: 'visible morphological change at cell level' = **atypia** (Public Health Dentist, 2024); perineural invasion = **adenoid cystic carcinoma** (same paper); the commonest primary metastasising to jaws = **breast** (same paper); ocular surface squamous neoplasia → test for **HIV** (Junior Ophthalmologist, 2023).

⚠ **One key to distrust:** the MO 2024 key gives Little's area as 'posterosuperior'. Every ENT text puts it in the **anteroinferior** septum. If it is asked again, mark anteroinferior.

Cell injury, adaptation and death

The single most-tested line: **irreversible injury = membrane damage + nuclear changes** (pyknosis → karyorrhexis → karyolysis). Swelling and fatty change are reversible.

Adaptations

Change	Meaning	MCQ example
Hypertrophy	Bigger cells	Heart in hypertension, uterus in pregnancy (also hyperplasia), skeletal muscle
Hyperplasia	More cells	Endometrium, BPH, breast at puberty; pathological hyperplasia can precede cancer
Atrophy	Smaller cells	Brain in old age, limb in plaster; ubiquitin–proteasome + autophagy
Metaplasia	One adult cell type replaced by another	Smoker's bronchus (columnar → squamous); Barrett oesophagus (squamous → intestinal columnar)
Dysplasia	Disordered growth: pleomorphism, high N:C ratio, loss of polarity	Cervix CIN; reversible until it crosses the basement membrane
Anaplasia	Lack of differentiation	Hallmark of malignancy

Necrosis types

Type	Where	Remember
Coagulative	Infarcts of heart, kidney, spleen (all solid organs except brain)	Ghost outlines of cells
Liquefactive	Brain infarct, abscess	Enzymes digest tissue
Caseous	Tuberculosis	Cheesy, amorphous, inside granulomas
Fat	Acute pancreatitis, breast trauma	Calcium soaps = saponification
Fibrinoid	Malignant hypertension, vasculitis	Bright pink vessel wall
Gangrenous	Limb	Dry = arterial; wet = venous + infection; gas = Clostridium perfringens

Apoptosis vs necrosis: apoptosis is programmed, single cells, **no inflammation**, cell shrinkage, chromatin condensation, apoptotic bodies, DNA ladder on electrophoresis, caspases. Intrinsic (mitochondrial, cytochrome c, BCL-2 family) and extrinsic (Fas–FasL, TNF) pathways. Example: Councilman bodies in viral hepatitis.

Free radicals and reperfusion: injury worsens when blood returns (reperfusion injury). Superoxide, H2O2, hydroxyl radical (most reactive); defences = SOD, catalase, glutathione peroxidase, vitamins E, C, A.

Intracellular accumulations and pigments

- Fatty change (steatosis): liver in alcohol, diabetes, obesity, kwashiorkor.
- **Lipofuscin** = 'wear-and-tear' brown pigment of ageing (heart, liver).
- **Haemosiderin** = iron; Prussian blue (Perls) stain; haemochromatosis.
- Melanin; bilirubin; anthracotic carbon in lungs.
- **Hyaline**: Russell bodies (plasma cells), Mallory–Denk bodies (alcoholic liver).

Calcification

Dystrophic	Metastatic
Normal serum calcium	High serum calcium

Dystrophic	Metastatic
In dead/damaged tissue: TB caseation, atheroma, damaged valves, psammoma bodies	In normal tissue: lungs, kidneys, gastric mucosa, blood vessels
Psammoma bodies: papillary thyroid Ca, serous ovarian Ca, meningioma, mesothelioma	Causes: hyperparathyroidism, vitamin D excess, bone destruction, renal failure

Stain for calcium: **von Kossa** or Alizarin red.

Memory hook: apoptosis is a **quiet suicide** (one cell, shrinks, no inflammation); necrosis is a **messy murder** (groups of cells, swelling, inflammation). The dying nucleus goes **P-R-L**: **P**yknosis shrinks it, karyo**R**rhexis breaks it, karyo**L**ysis dissolves it.

Inflammation and repair

Acute inflammation is neutrophil-led and lasts hours to days; chronic inflammation is led by macrophages, lymphocytes and plasma cells, with tissue destruction and fibrosis going on together.

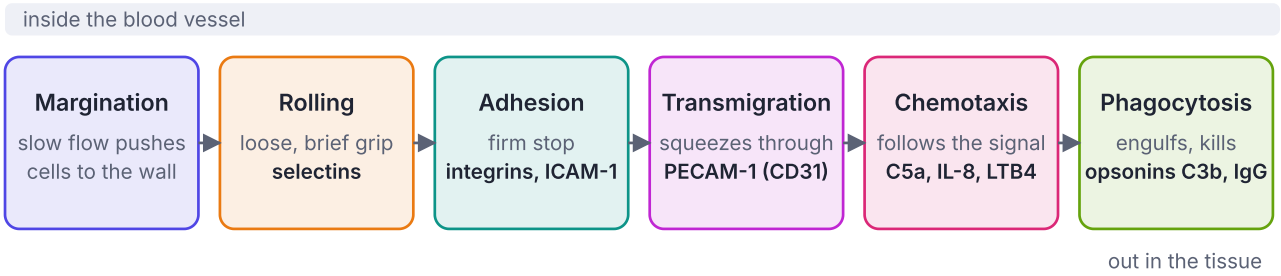
Cardinal signs: rubor, tumor, calor, dolor (Celsus); functio laesa added by Virchow.

Sequence of acute inflammation: transient vasoconstriction → vasodilatation → increased permeability (exudate) → stasis → leucocyte margination → rolling (**selectins**) → firm adhesion (**integrins** to ICAM-1/VCAM-1) → transmigration (**PECAM-1/CD31**, mainly in venules) → chemotaxis → phagocytosis → killing.

- Chemotactic agents: **C5a, LTB4, IL-8, bacterial products (fMLP)**.
- Opsonins: **IgG Fc and C3b** (also collectins).
- Most efficient microbicidal system: **H2O2–MPO–halide** (myeloperoxidase) system.
- Chronic granulomatous disease = defect in NADPH oxidase (no respiratory burst); NBT test negative.
- Chédiak–Higashi = defective phagolysosome fusion, giant granules.
- LAD type 1 = integrin (CD18) defect, delayed umbilical cord separation.

A white cell leaves the vessel in six steps

"My Royal Army Takes Charge", then eats; the molecule that drives each step is underneath



leucocyte exit cascade · 6 steps

Selectins make cells roll, integrins make them stop: MCQs swap these two.

Mediators

Mediator	Key action
Histamine (mast cells)	First mediator: vasodilatation and increased permeability (immediate transient response)
Serotonin	Like histamine; from platelets
Prostaglandins PGI2, PGE2	Vasodilatation; PGE2 = fever and pain
Thromboxane A2	Vasoconstriction, platelet aggregation
Leukotrienes C4, D4, E4	Bronchospasm, permeability (slow-reacting substance of anaphylaxis)
Bradykinin	Pain, permeability
C3a, C5a	Anaphylatoxins (histamine release); C5a also chemotactic
C5b–9	Membrane attack complex
IL-1, TNF, IL-6	Fever, acute-phase reactants, leucocytosis
NO	Vasodilatation; kills microbes

Aspirin and NSAIDs block **cyclo-oxygenase**; steroids block **phospholipase A2**.

Outcomes and patterns: resolution, abscess (pus), fibrosis, or chronic inflammation. Serous (blister, effusion), fibrinous (pericarditis — 'bread and butter'), suppurative (abscess, appendicitis), ulcer.

Granulomatous inflammation: a granuloma is a collection of **epithelioid cells** (activated macrophages) with or without Langhans giant cells, rimmed by lymphocytes.

Caseating	Non-caseating
Tuberculosis (classic); some fungi	Sarcoidosis, Crohn's disease, leprosy (tuberculoid), foreign body, berylliosis

Giant cells: **Langhans** = nuclei at the periphery in a horseshoe (TB); **foreign body** = scattered nuclei; **Touton** = xanthoma; **Reed–Sternberg** = Hodgkin lymphoma; **Warthin–Finkeldey** = measles; **Aschoff bodies / Anitschkow cells** = rheumatic carditis.

Systemic effects: fever (IL-1, TNF, PGE2 on the hypothalamus), acute-phase proteins (CRP, fibrinogen, serum amyloid A), raised ESR, leucocytosis. Neutrophilia = bacteria; lymphocytosis = viruses, TB, pertussis; eosinophilia = allergy, parasites; leucopenia = typhoid, dengue, some viral infections.

Repair

- Labile cells (skin, mucosa, bone marrow) always divide; stable cells (liver, kidney) divide when needed; **permanent cells (neurons, cardiac muscle)** cannot — they heal by scar.
- **Granulation tissue** = new capillaries (angiogenesis, VEGF) + fibroblasts + inflammatory cells; it is the hallmark of healing.
- Myofibroblasts cause **wound contraction**.
- First intention (clean, sutured): epithelium covers in 24–48 h; collagen type III early, replaced by type I; strength ~70–80% of normal by 3 months.
- Second intention: larger gap, more granulation tissue, contraction.
- Delayed healing: infection (commonest), poor blood supply, diabetes, steroids, vitamin C deficiency (hydroxylation of proline/lysine), zinc deficiency, foreign body.
- **Keloid** extends beyond the wound margin; **hypertrophic scar** stays within it.

Memory hook: the four Latin signs **rubor, tumor, calor, dolor** come from Celsus; Virchow added the fifth, **functio laesa**. White cells leave the vessel as "**M**y **R**oyal **A**rmy **T**akes **C**harge": **M**argination, **R**olling (selectins), **A**dhesion (integrins), **T**ransmigration (PECAM-1), **C**hemotaxis.

Haemodynamics: oedema, thrombosis, embolism, infarction, shock

The three classics here are Virchow's triad, the commonest source of pulmonary embolism (deep veins of the legs) and the organs that get red versus pale infarcts.

Oedema

Transudate	Exudate
Protein < 3 g/dL, specific gravity < 1.012	Protein > 3 g/dL, specific gravity > 1.020
Few cells	Many cells (inflammatory)
Raised hydrostatic pressure (CHF), low oncotic pressure (nephrotic syndrome, cirrhosis)	Inflammation, increased permeability

Light's criteria for an exudative pleural effusion: fluid/serum protein > 0.5, fluid/serum LDH > 0.6, fluid LDH > two-thirds of the upper normal serum value.

Thrombosis

- **Virchow's triad**: endothelial injury (most important, alone can cause thrombosis), abnormal flow (stasis or turbulence), hypercoagulability.
- **Lines of Zahn** = alternating pale platelet–fibrin and dark red-cell layers; they prove a thrombus formed in flowing blood (ante-mortem). Post-mortem clots are rubbery, 'chicken-fat' over 'currant jelly', and not attached.
- Commonest inherited hypercoagulable state: **Factor V Leiden** (resistance to activated protein C). Acquired: antiphospholipid syndrome, cancer (Trousseau migratory thrombophlebitis), pregnancy, OCPs, immobilisation, surgery.
- Fate: propagation, embolisation, dissolution, organisation and recanalisation.

Embolism

Type	Key facts
Pulmonary thromboembolism	Commonest embolism; > 95% from deep leg veins (popliteal and above); saddle embolus at the bifurcation can kill instantly
Systemic	Mostly from heart (mural thrombus after MI, AF, valve vegetations); lodge in lower limbs > brain
Fat	Long-bone fractures; 1–3 days later: breathlessness, confusion, petechial rash; fat globules stained by Oil Red O / Sudan
Air	> 100 mL needed; decompression sickness (nitrogen) = caisson disease, 'bends', 'chokes'

Type	Key facts
Amniotic fluid	Labour; squames and lanugo hair in pulmonary vessels; DIC

Infarction

White (pale, anaemic)	Red (haemorrhagic)
Arterial occlusion in solid organs with end-arteries: heart, kidney, spleen	Venous occlusion (testicular/ovarian torsion), loose tissue, dual blood supply: **lung, liver**, small intestine; reperfusion

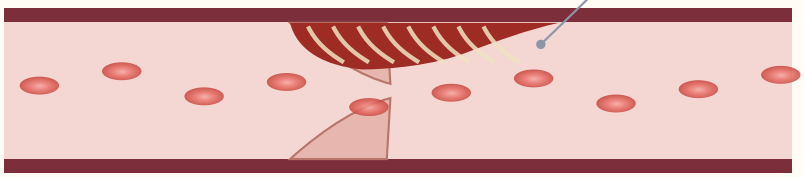
Infarcts are wedge-shaped with the apex at the occluded vessel. Brain infarcts liquefy; others undergo coagulative necrosis. Neurons die after 3–4 minutes of ischaemia, myocardium after 20–30 minutes.

Thrombosis, embolism and infarction

Why clots form, where they travel, what they kill

VIRCHOW'S TRIAD

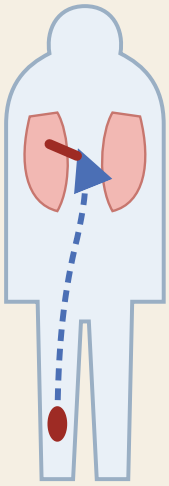
I — Endothelial injury
Most important: alone can cause thrombosis



S — Stasis or turbulence
Abnormal flow: blood pooling or swirling

H — Hypercoagulability
Factor V Leiden (inherited); pregnancy, OCPs, cancer, immobilisation, surgery

Embolus: a clot that travels

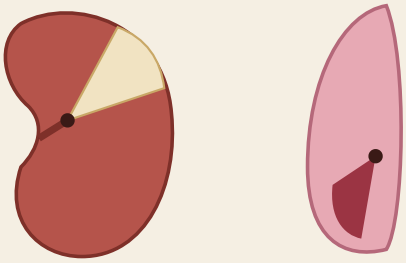


Pulmonary embolism: commonest embolism; > 95% from deep leg veins (popliteal and above)

Saddle embolus at the bifurcation can kill instantly

Others: fat (long-bone fractures), air, amniotic fluid

Infarct: wedge-shaped, apex at the block



White (pale) infarct
arterial block in solid organs with end-arteries: heart, kidney, spleen

Red infarct
venous block, loose tissue or dual supply: lung, liver, small intestine

Thrombosis to infarct · Virchow's triad, the travelling clot, pale vs red infarcts

Myocardial infarction timeline (histology): up to 4 h nothing visible; 4–24 h coagulative necrosis, wavy fibres; 1–3 days neutrophils; 3–7 days macrophages (rupture risk highest); 7–10 days granulation tissue; 2 months scar.

Shock

Type	Mechanism	Example
Hypovolaemic	Low blood volume	Haemorrhage, burns, diarrhoea
Cardiogenic	Pump failure	MI, arrhythmia, tamponade
Septic (distributive)	Vasodilatation from bacterial products (endotoxin = LPS of Gram-negatives)	Gram-negative sepsis
Neurogenic / anaphylactic	Loss of vascular tone	Spinal injury; IgE-mediated

Stages: non-progressive (compensated) → progressive (lactic acidosis) → irreversible. Morphology: 'shock lung' (diffuse alveolar damage / ARDS), acute tubular necrosis, adrenal lipid depletion; Waterhouse–Friderichsen syndrome (adrenal haemorrhage) in meningococcal sepsis.

DIC: widespread microthrombi + bleeding; low platelets, prolonged PT and aPTT, low fibrinogen, **raised D-dimer / FDP**, schistocytes on smear. Causes: sepsis, obstetric (abruption, amniotic fluid embolism, dead fetus), malignancy (acute promyelocytic leukaemia), snakebite (viper).

Memory hook: Virchow's triad is **I-S-H**: endothelial **I**njury, **S**tasis, **H**ypercoagulability. A thrombus forms in a living vessel, a clot can form anywhere, and an embolus is anything that travels and lodges.

Neoplasia

The single feature that defines malignancy is **invasion/metastasis**; anaplasia is its microscopic hallmark. Carcinomas spread by lymphatics, sarcomas by blood.

Naming traps

Sounds benign, is malignant	Is benign despite the name
Lymphoma, melanoma, seminoma, mesothelioma, hepatoma	Hamartoma (disorganised normal tissue), choristoma (normal tissue in wrong place, e.g. pancreatic rest)

Teratoma: all three germ layers; mature cystic teratoma (dermoid) of ovary is benign.

Benign vs malignant: differentiation, rate of growth, local invasion (capsule), metastasis. **Carcinoma in situ** = full-thickness dysplasia, basement membrane intact. Basal cell carcinoma and gliomas rarely metastasise.

Spread

- Lymphatic: carcinomas; first node = sentinel node; **Virchow's (Troisier's) node** = left supraclavicular, gastric carcinoma.
- Haematogenous: sarcomas, renal cell carcinoma (renal vein), hepatocellular carcinoma (hepatic vein), follicular thyroid carcinoma, choriocarcinoma. Liver and lungs are the commonest sites.
- Transcoelomic: ovary; **Krukenberg tumour** = signet-ring metastasis to ovaries from stomach; Sister Mary Joseph nodule = umbilical metastasis.
- Bone metastases: osteolytic mostly; **osteoblastic in prostate** (and breast sometimes).

Carcinogens

Agent	Cancer
Tobacco (polycyclic hydrocarbons, nitrosamines)	Lung, oral, larynx, oesophagus, bladder, pancreas
Aflatoxin B1 (Aspergillus flavus)	Hepatocellular carcinoma (p53 mutation)
Asbestos	Mesothelioma (and lung, with smoking)
Vinyl chloride	Angiosarcoma of liver
Aromatic amines, aniline dyes, Schistosoma haematobium	Bladder (Schistosoma → squamous)
Benzene	Acute myeloid leukaemia
Ionising radiation	Leukaemia, papillary thyroid carcinoma
UV light (UVB)	Skin: BCC, SCC, melanoma; xeroderma pigmentosum

Oncogenic microbes

Microbe	Cancer
HPV 16, 18	Cervix, anal, oropharyngeal carcinoma (E6 → p53; E7 → RB)

Microbe	Cancer
EBV	Burkitt lymphoma, nasopharyngeal carcinoma, Hodgkin lymphoma
HBV, HCV	Hepatocellular carcinoma
HTLV-1	Adult T-cell leukaemia/lymphoma
HHV-8	Kaposi sarcoma
Helicobacter pylori	Gastric adenocarcinoma, MALT lymphoma

Genes: oncogenes need one hit (RAS — commonest mutated oncogene, MYC, ERBB2/HER2, BCR–ABL in CML from t(9;22) Philadelphia chromosome). Tumour suppressors need two hits (Knudson): **RB** (retinoblastoma, osteosarcoma), **p53** (guardian of the genome; commonest mutated gene in human cancer; Li–Fraumeni), APC (colon), BRCA1/2 (breast, ovary), WT1 (Wilms), NF1.

Tumour markers

Marker	Cancer
AFP	Hepatocellular carcinoma, yolk sac tumour
β-hCG	Choriocarcinoma, molar pregnancy, testicular tumours
CEA	Colorectal carcinoma (follow-up)
CA-125	Ovarian (epithelial) carcinoma
CA 19-9	Pancreatic carcinoma
CA 15-3	Breast carcinoma
PSA	Prostate carcinoma
Calcitonin	Medullary thyroid carcinoma
Thyroglobulin	Follow-up of papillary/follicular thyroid cancer
LDH, β2-microglobulin	Lymphoma, myeloma

Immunohistochemistry: **cytokeratin** = carcinoma; **vimentin** = sarcoma; **desmin** = muscle; **S-100 / HMB-45** = melanoma; **LCA (CD45)** = lymphoma; **GFAP** = glial tumours; **chromogranin, synaptophysin** = neuroendocrine.

Grading and staging: grade = degree of differentiation (microscopic); **stage = size + spread (TNM)** and is the better predictor of prognosis.

Paraneoplastic syndromes: SIADH and Cushing's (small-cell lung carcinoma); hypercalcaemia via PTHrP (squamous-cell lung carcinoma); polycythaemia (renal cell carcinoma, hepatoma); Lambert–Eaton (small-cell); acanthosis nigricans (gastric cancer); Trousseau sign (pancreas).

Commonest cancers in Indian men: lip and oral cavity, lung. **In women**: breast, then cervix. Pap smear screens for cervical cancer; VIA is the low-cost screening tool used in India.

Memory hook: **grade** is how ugly the cells look; **stage** is how far the tumour has travelled, and stage predicts the outcome better. For stains: **C**ytokeratin for **C**arcinoma, **V**imentin for sarcoma ("**V**ery soft tissue"), **D**esmin for muscle, **S**-100 for melanoma.

Immunity, hypersensitivity and amyloidosis

The hypersensitivity table answers most questions here; learn one example per type.

Immunoglobulins

Class	Remember
IgG	Most abundant in serum (~75%); only one to cross the placenta; secondary response; 4 subclasses
IgA	Secretions (milk, saliva, tears, gut); dimer with secretory component; colostrum
IgM	Largest (pentamer); first in primary response; best complement activator; IgM in serum = recent infection; does not cross placenta (fetal IgM = congenital infection)
IgE	Lowest level; binds mast cells; allergy and helminths; heat-labile
IgD	B-cell surface receptor

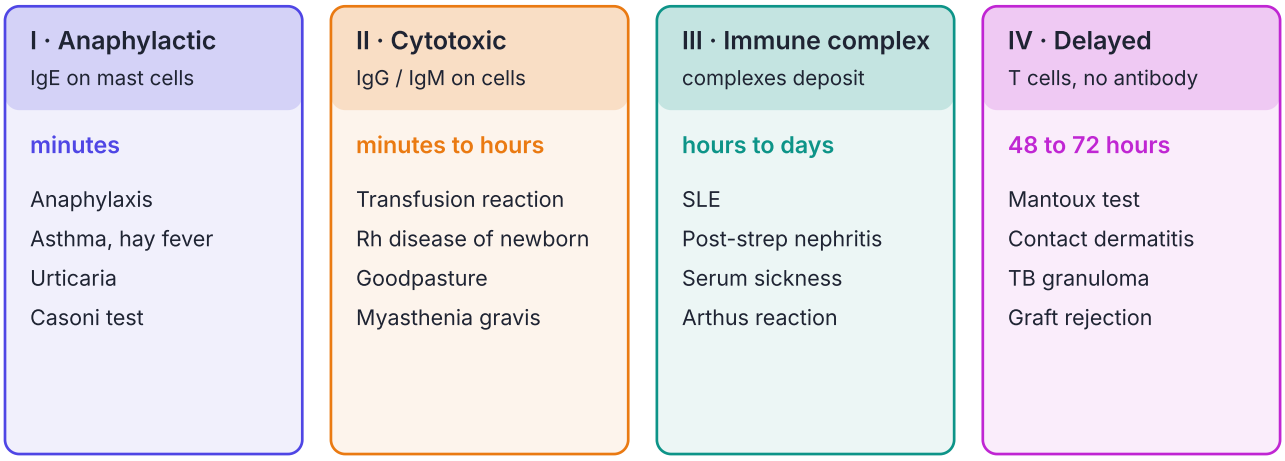
T cells mature in the **thymus**; B cells in the bone marrow. CD4 = helper (MHC II); CD8 = cytotoxic (MHC I). HIV binds CD4 with CCR5/CXCR4 co-receptors. NK cells kill without prior sensitisation (CD16, CD56).

Hypersensitivity (Gell and Coombs)

Type	Mechanism	Examples
I — immediate (anaphylactic)	IgE on mast cells → histamine	Anaphylaxis, asthma, hay fever, urticaria; Casoni test (old)
II — cytotoxic	IgG/IgM against cell or tissue antigens	Transfusion reaction, haemolytic disease of newborn, Goodpasture, Graves' (stimulating), myasthenia gravis, autoimmune haemolytic anaemia, rheumatic fever
III — immune complex	Antigen–antibody complexes deposit, activate complement	SLE, post-streptococcal glomerulonephritis, serum sickness, Arthus reaction, polyarteritis nodosa
IV — delayed (cell-mediated)	Sensitised T cells, 48–72 h	Tuberculin (Mantoux) test, contact dermatitis, granulomas, transplant rejection, lepromin test

Hypersensitivity spells ACID: four types, four speeds

What drives each type, how fast it appears, and the examples asked most



Only type IV involves no antibody; the Mantoux test is read at 48 to 72 hours for that reason.

hypersensitivity types I to IV · ACID

Read the speed row first: minutes, hours, days, then two to three days for the T-cell type.

Vaccines (types): live attenuated = BCG, OPV, MMR, rotavirus, yellow fever, chickenpox; killed/inactivated = IPV, hepatitis A, rabies, pertussis (whole cell); toxoid = tetanus, diphtheria; subunit/recombinant = hepatitis B, HPV. Live vaccines are avoided in pregnancy and immunosuppression.

Autoantibodies worth memorising

Antibody	Disease
Anti-dsDNA, anti-Smith	SLE (most specific)

Antibody	Disease
ANA	SLE screening (most sensitive)
Anti-histone	Drug-induced lupus
Anti-CCP, rheumatoid factor	Rheumatoid arthritis (anti-CCP more specific)
Anti-centromere	Limited systemic sclerosis (CREST)
Anti-Scl-70 (topoisomerase)	Diffuse systemic sclerosis
Anti-Ro, anti-La	Sjögren syndrome; congenital heart block
c-ANCA (PR3) / p-ANCA (MPO)	Granulomatosis with polyangiitis / microscopic polyangiitis
Anti-TPO	Hashimoto thyroiditis
TSH-receptor antibody (TRAb)	Graves' disease
Anti-tTG IgA	Coeliac disease
Anti-AChR	Myasthenia gravis

Amyloidosis: extracellular deposition of misfolded fibrillar protein in a **β-pleated sheet**.

- Stains: **Congo red** — pink-red, and **apple-green birefringence under polarised light** (the diagnostic test). Also crystal violet/methyl violet (metachromasia), thioflavin T.
- Types: **AL** (light chains: multiple myeloma, primary); **AA** (serum amyloid A: chronic inflammation — TB, rheumatoid arthritis, bronchiectasis, osteomyelitis, leprosy); Aβ2-microglobulin (long-term haemodialysis); Aβ (Alzheimer's); ATTR (senile cardiac, familial); calcitonin (medullary thyroid carcinoma).
- Biopsy sites: **abdominal fat pad aspiration** or rectal biopsy (least invasive); kidney is the commonest organ affected (nephrotic syndrome).
- Spleen: 'sago spleen' (follicular deposits) and 'lardaceous spleen' (red pulp/sinusoidal).

Transplant rejection: hyperacute (minutes; preformed antibodies), acute (weeks–months; T cells), chronic (months–years; vascular fibrosis). Graft-versus-host disease follows bone-marrow transplant (skin, liver, gut).

Memory hook: hypersensitivity is **ACID**: Type I **A**naphylactic (IgE), II **C**ytotoxic (antibody on cells), III **I**mmune complex, IV **D**elayed (T cells). For antibodies: Ig**M** is **M**ade first, Ig**G** alone **G**oes across the placenta, Ig**A** guards the secretions, Ig**E** causes allergy.

General microbiology: sterilisation, stains, media, tests

Three pairings carry most marks here: the sterilising method for each material, the stain for each organism and the medium for each organism.

Sterilisation and disinfection

Method	Conditions	Used for
Autoclave (moist heat)	**121 °C, 15 psi (1.05 kg/cm²), 15–20 min**	Surgical instruments, dressings, culture media; best method. Test organism: **Geobacillus stearothermophilus** spores
Hot-air oven (dry heat)	**160 °C for 2 h** (or 180 °C, 30 min)	Glassware, oils, powders, sharp instruments. Test: Bacillus atrophaeus (subtilis) spores
Pasteurisation	63 °C 30 min (holder) or 72 °C 15–20 s (flash)	Milk; kills Brucella, Mycobacterium bovis, Coxiella (most resistant of milk pathogens)
Tyndallisation	100 °C, 20 min on 3 successive days	Sugar and egg media
Inspissation	80–85 °C, 30 min, 3 days	Löwenstein–Jensen and Löffler serum slope
Incineration	—	Anatomical waste, soiled dressings
Ethylene oxide	Gas	Heat-sensitive items: plastics, heart-lung machines, catheters
Glutaraldehyde 2% (Cidex)	Liquid	Endoscopes, cystoscopes
Ionising (gamma) radiation	Cold sterilisation	Disposable syringes, catheters, sutures
Filtration (0.22 µm membrane)	—	Sera, antibiotic and sugar solutions

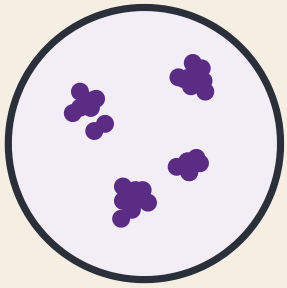
Prions resist routine sterilisation (need autoclaving at 134 °C for longer, plus NaOH). Bacterial **spores** are the most resistant living form; endospores are found in Bacillus and Clostridium.

Stains

Stain	Detects
Gram	Gram-positive (purple, thick peptidoglycan) vs Gram-negative (pink, LPS outer membrane). Steps: crystal violet → Gram's iodine (mordant) → decolourise (acetone/alcohol) → safranin
Ziehl–Neelsen (acid-fast)	M. tuberculosis (20% H2SO4), M. leprae (5% H2SO4); Nocardia (1%); Cryptosporidium, Isospora oocysts (modified)
Albert / Neisser	Metachromatic (volutin) granules of C. diphtheriae — 'Chinese letter' arrangement
India ink (negative stain)	Capsule of Cryptococcus neoformans in CSF
Giemsa / Leishman / JSB	Malaria parasites, blood films; Giemsa also Chlamydia inclusions, Donovan bodies
KOH mount (10%)	Fungal hyphae in skin, hair, nails
Lactophenol cotton blue	Fungal morphology in culture
Silver (Levaditi, Fontana)	Spirochaetes: Treponema, Leptospira
Dark-ground microscopy	Live Treponema pallidum, Leptospira
Auramine–rhodamine (fluorescent)	AFB screening — faster than ZN

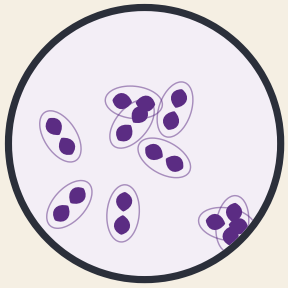
Down the microscope

Shape, arrangement and stain identify the organism



Gram-positive cocci in clusters

Staphylococcus aureus



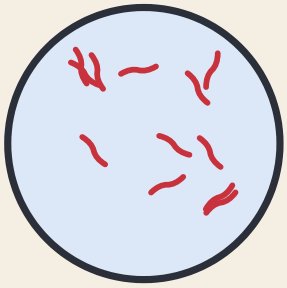
Lancet-shaped diplococci

Streptococcus pneumoniae



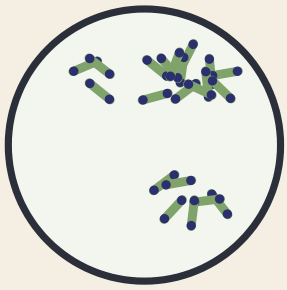
Gram-negative rods

E. coli
lactose fermenter



Acid-fast bacilli (Ziehl–Neelsen)

M. tuberculosis
red rods on blue



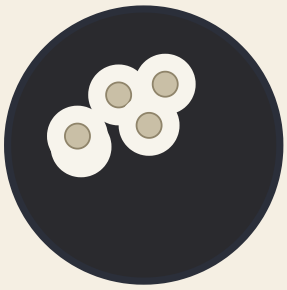
Albert stain: "Chinese letters"

C. diphtheriae,
volutin granules



Spirochaetes (dark ground)

Treponema pallidum,
Leptospira



India ink: clear capsule

Cryptococcus neoformans in CSF

Gram stain: 4 steps

- Crystal violet
- Gram's iodine
- Decolourise
- Safranin

Positive stays purple (thick wall); negative turns pink (LPS)

Down the microscope · shape, arrangement and stain name the organism

Culture media

Medium	Organism / type
Löwenstein–Jensen (LJ)	M. tuberculosis (6–8 weeks); malachite green inhibits contaminants
Löffler serum slope; potassium tellurite	C. diphtheriae
TCBS, alkaline peptone water (enrichment)	Vibrio cholerae (yellow colonies)
MacConkey	Gram-negative enterics; lactose fermenters pink (E. coli, Klebsiella)

Medium	Organism / type
Blood agar	Haemolysis: β (complete, Strep. pyogenes), α (green, Strep. pneumoniae, viridans), γ (none)
Chocolate agar	Haemophilus (needs X and V factors), Neisseria
Thayer–Martin	Neisseria gonorrhoeae
Wilson–Blair, DCA, XLD, SS agar	Salmonella / Shigella
Bile broth / blood culture	Salmonella Typhi (first week)
EMJH, Korthof's	Leptospira
Sabouraud dextrose agar	Fungi
Robertson's cooked meat	Anaerobes (Clostridium)
Bordet–Gengou	Bordetella pertussis ('bisected pearl' colonies)
NNN	Leishmania

Endotoxin = lipid A of the LPS in Gram-negative walls (fever, septic shock); Gram-positives lack it. **Exotoxins** = proteins, mostly Gram-positive, toxoids possible (tetanus, diphtheria, botulinum, cholera). Most potent toxin: **botulinum**.

Serological and lab tests

Test	Principle / use
ELISA	Screening (HIV, hepatitis, dengue NS1) — high sensitivity
Western blot	Confirmation of HIV (older algorithm)
PCR / RT-PCR	DNA / RNA detection; RT-PCR for COVID-19, HIV viral load
CBNAAT (GeneXpert) / TrueNat	TB + rifampicin resistance in ~2 h
Widal	Tube agglutination, typhoid (O and H antibodies); rising titre matters
VDRL / RPR	Non-treponemal flocculation test for syphilis; false positives (SLE, pregnancy, leprosy)
TPHA / FTA-ABS	Treponemal, confirmatory for syphilis

Test	Principle / use
ASO titre	Recent streptococcal infection (rheumatic fever)
Weil–Felix	Heterophile agglutination with Proteus strains for rickettsiae (scrub typhus = OX-K)
Paul–Bunnell / Monospot	Infectious mononucleosis (EBV)
Coombs (direct / indirect)	Antibody on RBCs / in serum: haemolytic anaemia, Rh disease
Mantoux (tuberculin, 5 TU PPD)	Type IV; read at 48–72 h; ≥ 10 mm positive
Schick test (historic)	Immunity to diphtheria
Lepromin	Prognosis, not diagnosis, of leprosy (Mitsuda reaction at 3–4 weeks)

Sensitivity vs specificity: sensitivity = true positives found among the diseased (screening tests need high sensitivity); specificity = true negatives among the healthy (confirmatory tests need high specificity).

Memory hook: the autoclave rule is **121, 15, 15**: 121 °C at 15 lb per square inch for 15 minutes kills spores. Prions shrug off even that, which is why they are the hardest of all to sterilise.

Bacteria

Tuberculosis and typhoid are asked most; leptospirosis and scrub typhus are the Goa monsoon specials.

Tuberculosis (Mycobacterium tuberculosis)

- Acid-fast, slow growing (generation time ~18 h), obligate aerobe; cord factor (virulence); grows on LJ in 6–8 weeks, faster in liquid MGIT culture.
- **Primary TB**: Ghon focus (subpleural, lower part of upper lobe / upper part of lower lobe) + hilar node = **Ghon complex**; healed and calcified = Ranke complex.
- **Secondary (post-primary)**: apex of the lung (high oxygen) — **Simon focus** = apical seeding; **Assmann focus** = infraclavicular; cavitation; Rasmussen aneurysm causes haemoptysis.
- Miliary TB: millet-seed lesions; spine = **Pott's disease** (thoracolumbar, cold abscess, gibbus); TB of the bone: best diagnosis = biopsy (MO 2025).

- Diagnosis under NTEP: sputum smear for AFB (two samples) and **NAAT (CBNAAT/TrueNat) as initial test**; culture is the gold standard. Mantoux and IGRA show infection, not disease. ADA raised in TB pleural fluid.
- MDR-TB = resistant to isoniazid + rifampicin; XDR adds fluoroquinolone + one Group A drug (bedaquiline/linezolid).

Leprosy (M. leprae): cannot be grown in culture (grown in mouse footpad, nine-banded armadillo); prefers cool parts and nerves (Schwann cells); slit-skin smear, ZN with 5% H2SO4; 'globi' in lepromatous; lepromin test negative in lepromatous, positive in tuberculoid.

Enteric fever (Salmonella Typhi, Paratyphi A and B)

- Faeco-oral; incubation 10–14 days; step-ladder fever, relative bradycardia, rose spots, splenomegaly, **leucopenia**.
- Pathology: Peyer's patches of terminal ileum; ulcers are **longitudinal (oval, along the long axis of gut)** — perforation in week 3. (TB ulcers are transverse/girdle.)
- Investigation by week (**BASU**): week 1 **B**lood culture (also bone-marrow culture — most sensitive overall), week 2 **A**gglutination (Widal), week 3 **S**tool, week 4 **U**rine. Carrier site: gall bladder.

Other key bacteria

Organism	Disease and key fact
Vibrio cholerae O1 (El Tor) and O139	Rice-water stools; toxin raises cAMP; darting motility; TCBS; treatment = ORS
Corynebacterium diphtheriae	Grey pseudomembrane that bleeds on removal; bull neck; toxin inhibits EF-2; Albert stain; Elek's gel precipitation test for toxin
Streptococcus pyogenes (group A)	Pharyngitis → rheumatic fever (ASO), post-streptococcal glomerulonephritis; bacitracin-sensitive
Streptococcus pneumoniae	Lobar pneumonia; lancet-shaped diplococci; optochin-sensitive; bile soluble; Quellung
Staphylococcus aureus	Coagulase-positive; boils, osteomyelitis, toxic shock, food poisoning (preformed toxin, 1–6 h); MRSA
Clostridium tetani	Drumstick spores; tetanospasmin blocks inhibitory transmitters (GABA, glycine); risus sardonicus

Organism	Disease and key fact
Clostridium perfringens	Gas gangrene; Nagler reaction (alpha toxin = lecithinase)
Clostridium botulinum	Blocks acetylcholine release; descending flaccid paralysis; canned food
Neisseria meningitidis	Meningitis with petechial rash; Waterhouse–Friderichsen
Treponema pallidum	Syphilis: painless chancre (primary), condylomata lata (secondary), gumma (tertiary); dark-ground microscopy; VDRL then TPHA
Helicobacter pylori	Peptic ulcer, gastric cancer; urease breath test; Warthin–Starry silver stain
Bordetella pertussis	Whooping cough; lymphocytosis; cough-plate culture

Leptospirosis (Leptospira interrogans) — common in Goa after monsoon paddy work.

- Spread through **rat urine** in water/soil entering through abraded skin or mucosa.
- Biphasic illness; **conjunctival suffusion**, calf tenderness; severe form = **Weil's disease** (jaundice, renal failure, bleeding); pulmonary haemorrhage.
- Diagnosis: first week blood/CSF culture (EMJH) or PCR; after day 7 **IgM ELISA**; MAT (microscopic agglutination test) = reference; dark-ground microscopy. Treatment doxycycline/penicillin; doxycycline chemoprophylaxis for high-risk exposure.

Scrub typhus (Orientia tsutsugamushi) — transmitted by the **larval trombiculid mite (chigger)**; painless **eschar** at the bite site; fever, lymphadenopathy, pneumonitis. Diagnosis IgM ELISA (Weil–Felix OX-K is old). Drug: **doxycycline** (azithromycin in pregnancy and children).

Other rickettsiae: epidemic typhus (louse, R. prowazekii), endemic typhus (flea), Indian tick typhus (tick, R. conorii), Q fever (Coxiella, inhalation, no rash).

Memory hook: typhoid tests follow the weeks as **BASU**: **B**lood culture in week 1, **A**ntibody (Widal) in week 2, **S**tool culture in week 3, **U**rine in week 4. TB is **red rods on blue** with Ziehl–Neelsen.

Viruses

Hepatitis serology and dengue lab tests are the high-yield core; know which marker means what.

Hepatitis viruses

Virus	Genome / family	Spread	Remember
HAV	RNA, picornavirus	Faeco-oral	No carrier state; anti-HAV IgM = acute
HBV	**DNA** (partly double-stranded), hepadnavirus	Blood, sexual, vertical	Carrier state, chronic hepatitis, cirrhosis, HCC; vaccine available
HCV	RNA, flavivirus	Blood (transfusion, IV drug use)	Most chronicity (~80%); no vaccine; direct-acting antivirals cure
HDV	Defective RNA	Needs HBsAg	Co-infection or superinfection with HBV
HEV	RNA	Faeco-oral, water-borne epidemics	**High mortality in pregnant women (~20%)**

Hepatitis B serology

Marker	Meaning
HBsAg	First marker; infection (acute or chronic); > 6 months = chronic carrier
HBeAg	Active replication, high infectivity
Anti-HBc IgM	Acute (recent) infection; the only marker in the 'window period'
Anti-HBc IgG	Past or chronic infection
Anti-HBs	Immunity (after recovery or vaccination); vaccinated persons have anti-HBs only
HBV DNA	Replication; guides treatment

Dengue (flavivirus, 4 serotypes) — vector **Aedes aegypti** (day-biter, breeds in clean stored water).

- Day 1–5: **NS1 antigen** or RT-PCR. After day 5: **IgM ELISA**. Secondary infection with another serotype risks severe dengue (antibody-dependent enhancement).
- Labs: **thrombocytopenia**, leucopenia, **rising haematocrit (≥ 20% = plasma leakage)**; positive tourniquet test (≥ 10–20 petechiae per square inch).
- Warning signs: abdominal pain, persistent vomiting, mucosal bleed, lethargy, liver > 2 cm, rising haematocrit with falling platelets.
- Chikungunya (alphavirus, also Aedes): severe, long-lasting arthralgia.

- Japanese encephalitis: Culex; pigs = amplifier hosts, ardeid birds = reservoir; JE vaccine in UIP for endemic districts.
- **Kyasanur Forest Disease** (flavivirus, tick Haemaphysalis spinigera; monkeys die first): reported in Goa's Sattari taluka and Karnataka's Western Ghats.

- **HIV**: retrovirus (RNA → DNA via **reverse transcriptase**); gp120 binds CD4, co-receptors CCR5/CXCR4.
- Window period ~3–12 weeks (shorter with 4th-generation antigen–antibody tests).
- Indian testing strategy: three different rapid/ELISA tests for diagnosis (no Western blot needed).
- AIDS = CD4 < 200/µL or an AIDS-defining illness.
- Monitoring: CD4 count and **viral load**.
- Infant diagnosis < 18 months = DNA PCR (maternal antibodies confuse ELISA).

Other viruses to recognise

Virus	Key fact
Rabies (rhabdovirus, bullet-shaped)	**Negri bodies** in hippocampus and Purkinje cells; corneal impression smear / skin biopsy for antigen
Measles (paramyxovirus)	Koplik spots; Warthin–Finkeldey giant cells; SSPE late
Mumps	Parotitis, orchitis; aseptic meningitis
Rubella	Congenital rubella: PDA, cataract, deafness (MO 2024)
Varicella-zoster	Tzanck smear: multinucleated giant cells (also HSV); zoster = reactivation in dorsal root ganglion
EBV	Infectious mononucleosis, atypical lymphocytes (Downey cells), Monospot
Rotavirus	Commonest cause of infantile diarrhoea (MO 2024); wheel-shaped
Poliovirus	Faeco-oral; anterior horn cells; OPV + IPV
Influenza	Antigenic shift (pandemics, influenza A) vs drift (epidemics)
SARS-CoV-2	NP swab RT-PCR (MO 2025); ACE-2 receptor
HPV	Koilocytes in Pap smear; types 16, 18 cancer; 6, 11 warts

Inclusion bodies: **Negri** (rabies), **Guarnieri** (vaccinia/smallpox), **Cowdry type A** (herpes), **owl's eye** (CMV), **Molluscum bodies** (molluscum contagiosum).

Memory hook: "**The vowels hit the bowels**": hepatitis **A** and **E** spread by the faecal–oral route; **B, C and D** by blood and sex. Hepatitis D needs B to exist. Negri bodies mean **rabies**.

Parasites and fungi

Malaria carries the most marks: the species differences, the smear, and which stage is infective to whom.

Malaria (Plasmodium)

- Vector: female **Anopheles** (in Goa, urban malaria vector *An. stephensi* breeds in construction-site water). Man = intermediate host; mosquito = definitive host.
- Infective to man: **sporozoite** (mosquito bite). Infective to mosquito: **gametocyte**. Hypnozoites (dormant liver stage) in **vivax and ovale** cause relapse → primaquine.
- **Gold standard: peripheral smear** — thick smear to detect, thin smear to identify species; Giemsa/JSB/Leishman. RDT: HRP-2 = falciparum; pLDH = pan-species.

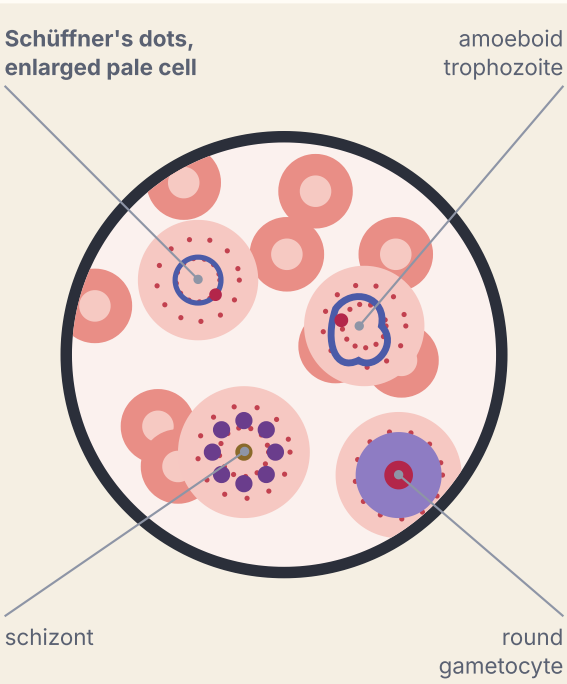
Feature	P. vivax	P. falciparum
RBC size	Enlarged, pale	Normal
Dots	Schüffner's dots	Maurer's clefts
Ring forms	Single, in one RBC	Multiple rings per RBC, accolé forms, double chromatin
Gametocyte	Round	**Crescent / banana shaped** (diagnostic)
Schizonts in smear	Seen	Usually not (sequestered in capillaries)
Fever cycle	Tertian (48 h)	Malignant tertian; cerebral malaria, blackwater fever

P. malariae = quartan (72 h), band forms, nephrotic syndrome. P. knowlesi = 24 h (monkey malaria, SE Asia).

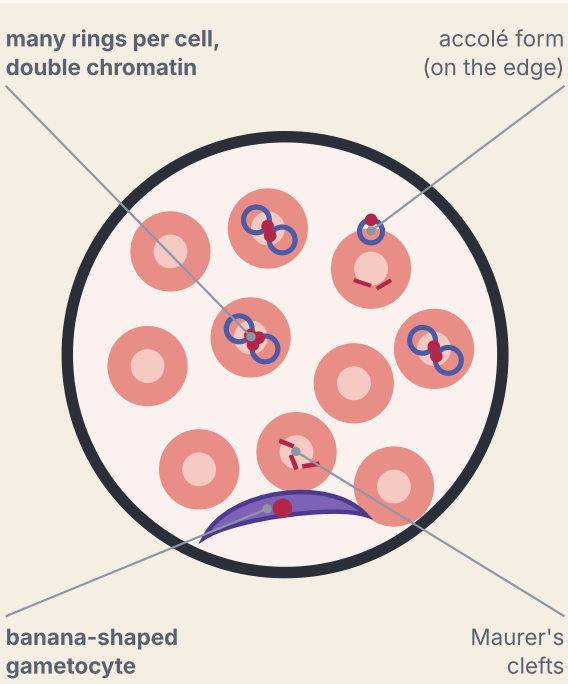
Malaria smear: vivax vs falciparum

Thin film, Giemsa stain

Plasmodium vivax



Plasmodium falciparum



	Plasmodium vivax	Plasmodium falciparum
RBC size	Enlarged, pale	Normal
Dots	Schüffner's dots	Maurer's clefts
Rings	Single, one per RBC	Many per RBC, accolé, double chromatin
Gametocyte	Round	Crescent / banana (diagnostic)
Schizonts in smear	Seen	Usually not (sequestered)
Fever	Tertian, 48 h	Malignant tertian; cerebral malaria

Relapse: vivax and ovale hide in the liver as hypnozoites → primaquine.
Infective to man: sporozoite (mosquito bite). Infective to mosquito: gametocyte.
Thick smear detects, thin smear identifies the species (Giemsa, JSB, Leishman).

Malaria smear · vivax vs falciparum on a thin film

Protozoa

Parasite	Disease and lab fact
Entamoeba histolytica	Amoebic dysentery (flask-shaped ulcers), **amoebic liver abscess ('anchovy sauce' pus)**, right lobe; trophozoite with ingested RBCs; cyst with 4 nuclei

Parasite	Disease and lab fact
Giardia lamblia	Malabsorption, steatorrhoea; 'falling-leaf' motility, pear-shaped trophozoite
Leishmania donovani	Kala-azar (sandfly *Phlebotomus*); LD bodies (amastigotes) in splenic/bone-marrow aspirate; rK39 test; NNN medium
Toxoplasma gondii	Cat = definitive host; congenital (chorioretinitis, hydrocephalus, intracranial calcification)
Trichomonas vaginalis	Frothy greenish discharge, 'strawberry cervix'; jerky motility on wet mount
Cryptosporidium	Diarrhoea in AIDS; modified ZN-positive oocysts

Helminths

Worm	Key fact
Ascaris lumbricoides	Commonest worm; Löffler syndrome (lung migration, eosinophilia); intestinal obstruction
Ancylostoma duodenale / Necator	Skin penetration; **iron-deficiency anaemia**
Enterobius vermicularis	Perianal itch at night; **NIH swab / cellophane tape** test
Wuchereria bancrofti	Lymphatic filariasis (Culex); **nocturnal periodicity** — night blood smear 10 pm–2 am; DEC; ICT card test
Taenia solium	Pork tapeworm; **neurocysticercosis** — commonest cause of acquired epilepsy in India
Echinococcus granulosus	Hydatid cyst (dog = definitive host, man = accidental intermediate); liver > lung; Casoni test (old)
Strongyloides	Autoinfection, hyperinfection in steroid users
Schistosoma haematobium	Haematuria; bladder squamous carcinoma

Eosinophilia + worm: think of tissue-migrating helminths (Ascaris, hookworm, filaria, Strongyloides).

Fungi

Fungus	Key fact
Dermatophytes (Trichophyton, Microsporum, Epidermophyton)	Tinea (ringworm); KOH mount; Wood's lamp fluorescence for Microsporum
Candida albicans	Oral thrush (white patches that wipe off), vaginitis; **germ-tube test** positive; pseudohyphae
Cryptococcus neoformans	Meningitis in AIDS; **India ink** capsule; pigeon droppings; latex antigen test
Aspergillus	Fungus ball in old TB cavity; ABPA in asthmatics; **acute-angle (45°) septate branching**
Mucor / Rhizopus	Rhinocerebral mucormycosis in diabetic ketoacidosis (post-COVID surge); **broad aseptate, right-angle branching**
Pneumocystis jirovecii	Pneumonia in AIDS (CD4 < 200); silver stain; co-trimoxazole
Histoplasma	Intracellular yeasts in macrophages
Sporothrix	Rose gardener's disease; lymphocutaneous nodules
Malassezia furfur	Pityriasis versicolor; 'spaghetti and meatballs' on KOH
Rhinosporidium seeberi	Nasal polyps (strawberry-like), common in India; pond bathing

Memory hook: **V**ivax and **O**vale come back ("**VO**lunteer to return"): their liver hypnozoites cause relapses. **Falciparum** is the killer, with banana-shaped gametocytes and many rings in one cell.

Haematology: CBC, anaemia, ESR, coagulation, blood groups

Learn one normal value per test and the anaemia table; most haematology MCQs are built on these.

Normal values (adult)

Test	Normal
Haemoglobin	Men 13–17 g/dL; women 12–15 g/dL; WHO anaemia cut-off in pregnancy < 11 g/dL
RBC count	4.5–5.5 million/µL
Total leucocyte count	4,000–11,000/µL

Test	Normal
Differential	Neutrophils 40–75%, lymphocytes 20–40%, monocytes 2–10%, eosinophils 1–6%, basophils < 1%
Platelets	1.5–4.5 lakh/µL
MCV	80–100 fL
MCH / MCHC	27–32 pg / 32–36 g/dL
Haematocrit (PCV)	Men 40–50%, women 36–46%
Reticulocytes	0.5–2.5%
ESR (Westergren)	Men 0–15, women 0–20 mm in 1st hour
Bleeding time	2–7 min (Ivy)
Clotting time	4–9 min (Lee–White)
PT / INR	11–16 s / ~1.0
aPTT	25–40 s

ESR: raised in TB, rheumatoid arthritis, multiple myeloma (very high, with rouleaux), pregnancy, anaemia, malignancy, temporal arteritis. **Low in polycythaemia, sickle cell disease, CHF, hypofibrinogenaemia**. Wintrobe and Westergren methods; anticoagulant for ESR = 3.8% trisodium citrate (1:4).

Anticoagulant tubes: **EDTA (lavender)** = CBC; **sodium citrate (blue)** = PT/aPTT (1:9); **fluoride (grey)** = glucose (inhibits glycolysis); heparin (green) = some biochemistry and blood gases; plain/red = serum.

Anaemias

Type	Indices and smear	Key tests
Iron deficiency (commonest worldwide and in India)	Microcytic hypochromic; pencil cells; **koilonychia**	Low ferritin (best single test), high TIBC, low serum iron; Mentzer index > 13
Thalassaemia trait	Microcytic, target cells, high RBC count	**HbA2 > 3.5%** on electrophoresis/HPLC; Mentzer < 13; normal ferritin

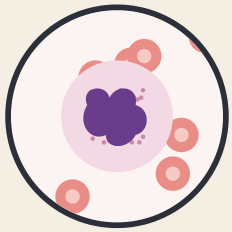
Type	Indices and smear	Key tests
Anaemia of chronic disease	Normo- or microcytic	High ferritin, low TIBC (hepcidin)
Sideroblastic	Microcytic	Ring sideroblasts (Prussian blue on marrow)
Megaloblastic (B12/folate)	**Macrocytic, hypersegmented neutrophils (> 5 lobes)**, macro-ovalocytes	Low B12 / folate; B12 deficiency = subacute combined degeneration; vegans (MO 2025)
Haemolytic	Normocytic; high reticulocytes, unconjugated bilirubin, LDH; low haptoglobin	Coombs test (autoimmune); spherocytes in hereditary spherocytosis (osmotic fragility raised)
Sickle cell (HbS: glutamate → valine at β6)	Sickle cells; Howell–Jolly bodies after autosplenectomy	Sickling test (sodium metabisulphite); HPLC; tribal belts of central and western India
G6PD deficiency	Bite cells, **Heinz bodies** (supravital stain) after oxidant drugs (primaquine, dapsone) or fava beans	G6PD assay
Aplastic	Pancytopenia	Hypocellular bone-marrow **biopsy** (trephine)

Red-cell inclusions: Howell–Jolly (DNA remnants, asplenia); basophilic stippling (lead poisoning, thalassaemia); Pappenheimer (iron); Heinz (denatured Hb); Cabot rings.

Reading a blood film

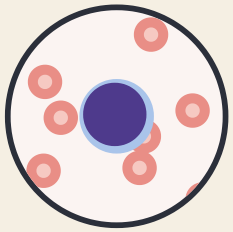
Normal white cells, and what anaemia does to red cells

WHITE CELLS · normal differential count



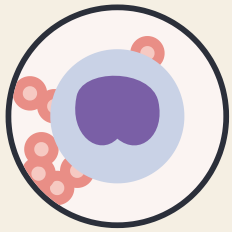
Neutrophil
40–75%

3–5 lobes; leads acute inflammation. Neutrophilia: bacteria



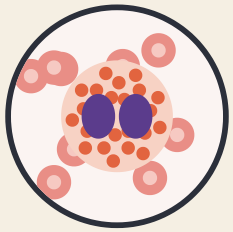
Lymphocyte
20–40%

Round nucleus fills the cell. Raised in viruses, TB, pertussis



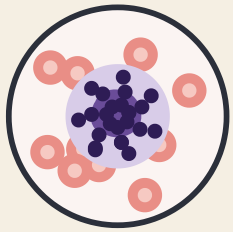
Monocyte
2–10%

Largest white cell; kidney-shaped nucleus



Eosinophil
1–6%

Two lobes, red granules. Eosinophilia: allergy, parasites



Basophil
under 1%

Dark granules hide the nucleus; rarest white cell

RED CELLS · what anaemia does to their shape



Normal

Biconcave disc, pale centre



Microcytic, hypochromic

Iron deficiency: small, pale, pencil cells



Target cells

Thalassaemia trait: Mentzer index < 13



Macro-ovalocytes

B12 / folate; with neutrophils > 5 lobes



Spherocytes

Hereditary spherocytosis: no pale centre



Sickle cells

HbS: glutamate → valine at β6



Bite cells

G6PD deficiency after oxidant drugs, fava beans

Inclusions

Howell–Jolly: asplenia
Basophilic stippling: lead, thalassaemia
Heinz bodies: G6PD
Ring sideroblasts: sideroblastic (marrow)

Blood film · normal white cells and the red-cell changes of anaemia

White cells and leukaemia

Disease	Key finding
AML	Myeloblasts with **Auer rods**; MPO positive; M3 (APL) t(15;17), DIC, treat with ATRA
ALL	Commonest childhood cancer/leukaemia; lymphoblasts; PAS positive; CD10 (CALLA)

Disease	Key finding
CML	**Philadelphia chromosome t(9;22), BCR–ABL**; very high TLC with all stages of maturation, basophilia; low LAP score; imatinib
CLL	Elderly; **smudge (basket) cells**; CD5 + CD23 B cells
Hodgkin lymphoma	**Reed–Sternberg cells** (owl's eye, CD15 + CD30); nodular sclerosis commonest
Multiple myeloma	Bone pain, hypercalcaemia, renal failure, anaemia (CRAB); **M-spike** on electrophoresis, **Bence Jones proteins** in urine, punched-out lytic skull lesions, rouleaux

Leukaemoid reaction vs CML: leukaemoid has **high LAP score**, toxic granules, no Philadelphia chromosome.

Bleeding and clotting

Test	Prolonged in
Bleeding time	Platelet disorders: thrombocytopenia (ITP, dengue), von Willebrand disease, aspirin
PT / INR (extrinsic: factor VII)	Warfarin therapy (monitored by INR), liver disease, vitamin K deficiency, DIC
aPTT (intrinsic: VIII, IX, XI, XII)	Haemophilia A (VIII) and B (IX), heparin (monitored by aPTT), von Willebrand, DIC

Haemophilia: X-linked recessive; normal bleeding time and PT, prolonged aPTT; haemarthrosis. **ITP**: low platelets, normal PT/aPTT, megakaryocytes increased in marrow. Vitamin K-dependent factors: **II, VII, IX, X** (and proteins C, S).

Blood groups and transfusion

- ABO antibodies are naturally occurring IgM; Rh antibodies are IgG (cross the placenta).
- **Universal donor = O negative** (red cells); **universal recipient = AB positive**. Universal plasma donor = AB.
- Most dangerous acute transfusion reaction = ABO incompatibility (acute intravascular haemolysis, type II). Commonest = febrile non-haemolytic reaction.
- Storage: whole blood/PRBC 2–6 °C (CPDA-1, 35 days; SAGM, 42 days); **platelets 20–24 °C with agitation, 5 days**; FFP ≤ −30 °C (1 year). Cryoprecipitate = fibrinogen, factor VIII, vWF, XIII.

- Rh-negative mother: **anti-D** at 28 weeks and within 72 hours of delivery of an Rh-positive baby; indirect Coombs test to monitor.

Memory hook: tube colours: **L**avender for the b**L**ood count (EDTA), **B**lue for **B**leeding tests (citrate), **G**rey for **G**lucose (fluoride). Small red cells are **TICS**: **T**halassaemia, **I**ron deficiency, **C**hronic disease, **S**ideroblastic.

Clinical pathology: LFT, RFT, urine, CSF, glucose, lipids, enzymes

Questions here are pattern-recognition: which enzyme rises, which jaundice, which CSF picture.

Liver function tests

Test	Normal	Raised in
Total bilirubin	0.3–1.2 mg/dL (clinical jaundice > 2–3)	Jaundice
ALT (SGPT)	< 40 U/L	**Hepatocellular damage (most liver-specific)**; viral hepatitis ALT > AST
AST (SGOT)	< 40 U/L	Liver, heart, muscle; **AST:ALT > 2 = alcoholic liver disease**
Alkaline phosphatase	40–130 U/L	**Cholestasis / obstruction**, bone disease (Paget's, rickets, metastases), pregnancy, growing children
GGT	—	Alcohol, cholestasis (confirms hepatic origin of ALP)
Albumin	3.5–5 g/dL	Low in chronic liver disease, nephrotic syndrome, malnutrition (synthetic function)
PT/INR	—	Prolonged = liver synthetic failure (best acute marker)

Jaundice patterns

Type	Bilirubin	Urine	Stool
Pre-hepatic (haemolytic)	Unconjugated ↑	Urobilinogen ↑, no bilirubin (acholuric)	Dark
Hepatic	Both ↑	Bilirubin +, urobilinogen variable	Normal/pale

Type	Bilirubin	Urine	Stool
Post-hepatic (obstructive)	Conjugated ↑, ALP ↑ ↑	**Bilirubin +, urobilinogen absent**; dark urine	**Clay-coloured**; itching

Three jaundices, told apart by urine, stool and enzymes

Read across a row to compare; read down a column to recall one type

	Pre-hepatic (haemolytic)	Hepatic (liver cell)	Post-hepatic (obstructive)
Bilirubin raised	unconjugated	both kinds	conjugated
Urine bilirubin	absent (acholuric)	present	present: dark urine
Urobilinogen	increased	normal or raised	absent
Stool	normal or dark	normal to pale	pale, clay-coloured
Key test	reticulocytes, LDH up	ALT, AST very high	ALP, GGT very high

jaundice types · pre-hepatic, hepatic, obstructive

The right-hand column is the one asked most: dark urine with pale stools means the bile cannot reach the gut.

Gilbert syndrome = mild unconjugated rise with fasting/stress; Crigler–Najjar = UGT deficiency; Dubin–Johnson = conjugated, black liver.

Renal function tests

- Blood urea 15–40 mg/dL (BUN 7–20); serum creatinine 0.6–1.2 mg/dL. Creatinine is better than urea; **eGFR** (from creatinine) staging CKD.
- Urea:creatinine ratio > 20 = pre-renal (dehydration, GI bleed).
- GFR ~125 mL/min; **inulin clearance = gold standard GFR**; creatinine clearance in practice; PAH = renal plasma flow.
- Uric acid: men 3.5–7.2 mg/dL; raised in gout, tumour lysis, leukaemia, thiazides.
- Electrolytes: Na 135–145, K 3.5–5.0 mmol/L. **Hyperkalaemia ECG: tall peaked T waves**; hypokalaemia: U waves.

Urine examination

Finding	Meaning
Normal output 1–2 L/day; oliguria < 400 mL; anuria < 100 mL; polyuria > 3 L	—
Specific gravity 1.003–1.030; fixed 1.010 (isosthenuria)	Chronic renal failure
Proteinuria > 3.5 g/day	**Nephrotic syndrome** (with oedema, hypoalbuminaemia, hyperlipidaemia, lipiduria — oval fat bodies, Maltese cross)
Microalbuminuria 30–300 mg/day	Earliest sign of diabetic nephropathy
RBC casts, dysmorphic RBCs	Glomerulonephritis (nephritic syndrome)
WBC casts, pus cells	Pyelonephritis
Granular / muddy-brown casts	Acute tubular necrosis
Hyaline casts	Normal / concentrated urine
Bence Jones protein (heat test: precipitates at 40–60 °C, redissolves at 100 °C)	Multiple myeloma
Ketones (Rothera's test)	DKA, starvation
Glucose (Benedict's test)	Glycosuria above renal threshold (~180 mg/dL)
Significant bacteriuria ≥ 10^5 CFU/mL (midstream urine)	UTI (E. coli commonest)
Haemoglobinuria vs haematuria	Dipstick positive without RBCs = haemoglobin/myoglobin

Crystals: calcium oxalate (envelope), uric acid (rhomboid), triple phosphate (coffin-lid, Proteus infection), cystine (hexagonal).

CSF

Condition	Cells	Protein	Glucose	Clue
Normal	0–5 lymphocytes	15–45 mg/dL	40–70 mg/dL (≈ 2/3 blood)	Clear, opening pressure 5–18 cmH2O
Bacterial (pyogenic)	Neutrophils ↑↑	↑↑	↓↓	Turbid; Gram stain

Condition	Cells	Protein	Glucose	Clue
Tuberculous	Lymphocytes ↑	↑ ↑	↓	**Cobweb clot** on standing; ADA ↑
Viral	Lymphocytes ↑	Normal/slightly ↑	**Normal**	Clear
Fungal (Cryptococcus)	Lymphocytes	↑	↓	India ink
Subarachnoid haemorrhage	RBCs	↑	Normal	**Xanthochromia**
Guillain–Barré	Normal	↑ ↑	Normal	**Albuminocytological dissociation**

Glucose and diabetes

- Diabetes: fasting plasma glucose ≥ 126 mg/dL, 2-h OGTT ≥ 200, random ≥ 200 with symptoms, or **HbA1c ≥ 6.5%**. Prediabetes: FPG 100–125 (IFG); 2-h 140–199 (IGT); HbA1c 5.7–6.4%.
- HbA1c reflects the average glucose of the previous ~**2–3 months** (RBC life 120 days). Fructosamine = 2–3 weeks.
- Type 1 = ketoacidosis-prone (MO 2025 trap: type 2 is *not* more ketosis-prone); type 2 deaths are mostly cardiovascular.
- Gestational diabetes in India: single-step 75 g OGTT (DIPSI), 2-h ≥ 140 mg/dL.

Lipids: total cholesterol < 200, **LDL < 100** (optimal), HDL > 40 (men) / > 50 (women), triglycerides < 150 mg/dL. Friedewald: LDL = TC − HDL − TG/5.

Enzymes and markers

Marker	Use
Troponin I/T	Myocardial infarction — most sensitive and specific; rises 3–4 h, stays 7–10 days
CK-MB	MI; returns to normal by 48–72 h (good for re-infarction)
LDH (LDH1 > LDH2 'flipped')	Late MI marker; haemolysis, lymphoma
Amylase, **lipase** (more specific)	Acute pancreatitis (> 3× normal)

Marker	Use
Acid phosphatase / PSA	Prostate
ALP	Bone, cholestasis
ADA	TB effusions
CRP, procalcitonin	Inflammation; procalcitonin = bacterial sepsis
BNP / NT-proBNP	Heart failure
D-dimer	DVT / PE exclusion; DIC
TSH	Best screening test for thyroid function; low TSH + high T4 = hyperthyroidism (Graves', MO 2024)

Body fluids: pleural fluid — exudate by Light's criteria (see haemodynamics); ADA > 40 U/L suggests TB pleurisy; semen analysis (WHO 2021 lower limits): volume 1.4 mL, count 16 million/mL, progressive motility 30%, normal forms 4%.

Memory hook: obstructive jaundice is **dark urine, pale stools, itching and a very high ALP**. In CSF, **bacteria eat the sugar** (low glucose, neutrophils); **viruses leave it alone** (normal glucose, lymphocytes); **TB** eats it slowly (low glucose, lymphocytes, cobweb clot).

Imaging and serology: X-ray, ultrasound, markers

GPSC asks named radiological signs and 'investigation of choice' — both appeared in the 2024 and 2025 Medical Officer papers.

Chest X-ray

Sign / finding	Meaning
Right heart border = SVC, right atrium (IVC low)	Right ventricle is NOT on the right border (MO 2024); left border = aortic knuckle, pulmonary trunk, left auricle, left ventricle
Cardiothoracic ratio > 0.5 (PA view)	Cardiomegaly
Air under the diaphragm (erect film)	Perforated hollow viscus (peptic ulcer)

Sign / finding	Meaning
Blunting of costophrenic angle; meniscus sign	Pleural effusion (≥ 200–300 mL on PA view; lateral decubitus detects less)
Air bronchogram, lobar opacity	Consolidation (pneumonia)
Apical cavity, fibrosis	Post-primary TB
Millet-seed shadows	Miliary TB
Kerley B lines, bat-wing opacity, upper-lobe diversion	Pulmonary oedema / left heart failure
Hyperinflation, flat diaphragm	COPD/emphysema
Visceral pleural line, no lung markings	Pneumothorax
Westermark sign, Hampton's hump	Pulmonary embolism (CT pulmonary angiography = investigation of choice)
Egg-shell calcification of hilar nodes	Silicosis (also sarcoidosis)
Boot-shaped heart	Tetralogy of Fallot
Egg-on-string	Transposition of great arteries
Snowman / figure-of-8	TAPVC

Other X-ray and barium signs

Sign	Condition
Bird's beak	Achalasia cardia
Rat tail / shouldering	Carcinoma oesophagus (MO 2025)
Apple-core lesion	Colorectal carcinoma
String sign of Kantor	Crohn's disease
Lead-pipe colon	Ulcerative colitis
Coffee-bean sign	Sigmoid volvulus

Sign	Condition
Double-bubble	Duodenal atresia
Step-ladder fluid levels	Small-bowel obstruction
Cupping, fraying, splaying of metaphysis	Rickets (sub-periosteal haemorrhage = scurvy, MO 2024)
Hair-on-end skull	Thalassaemia, sickle cell
Pepper-pot skull	Hyperparathyroidism
Punched-out lytic skull lesions	Multiple myeloma
Sun-ray appearance, Codman's triangle	Osteosarcoma
Onion-peel periosteal reaction	Ewing sarcoma
Soap-bubble appearance	Giant-cell tumour (osteoclastoma)
Bamboo spine	Ankylosing spondylitis

Ultrasound and other imaging

Finding	Meaning
Ultrasound = investigation of choice	Gallstones (acoustic shadow), obstetrics, renal calculi screening, liver abscess
Snowstorm appearance	Hydatidiform mole
Bagel / tubal ring sign	Ectopic pregnancy (MO 2024)
String of pearls ovaries	PCOS
Target / pseudokidney sign	Intussusception
Water-lily sign	Hydatid cyst (collapsed membrane)
Investigation of choice: acute subarachnoid haemorrhage	**Non-contrast CT** (MO 2025)
Intraocular metallic foreign body	CT orbit; MRI contraindicated (MO 2025)
Spine TB / cord compression imaging	MRI (diagnosis confirmed by biopsy)
Pulmonary embolism	CT pulmonary angiography

Finding	Meaning
Bronchiectasis	HRCT
Renal stones	Non-contrast CT (KUB)
Ventricular function	Echocardiography

Serological markers (rapid recall)

Disease	Marker / test
Typhoid	Blood culture (week 1), Widal (week 2), Typhidot (IgM)
Dengue	NS1 (day 1–5), IgM ELISA (after day 5)
Malaria	Smear (gold standard); RDT: HRP-2 (falciparum), pLDH
Leptospirosis	IgM ELISA; MAT (reference)
Scrub typhus	IgM ELISA; Weil–Felix OX-K
HIV	ELISA/rapid tests × 3 (India); DNA PCR in infants
Hepatitis B	HBsAg (screen); anti-HBc IgM (acute); anti-HBs (immunity)
Syphilis	VDRL/RPR (screen, follow-up); TPHA (confirm)
Rheumatic fever	ASO titre; Jones criteria
Rheumatoid arthritis	Anti-CCP, RF
SLE	ANA (screen), anti-dsDNA (specific)
Infectious mononucleosis	Paul–Bunnell / Monospot
Kala-azar	rK39 rapid test
Filariasis	Night smear; ICT card for antigen
COVID-19	RT-PCR on nasopharyngeal swab

Memory hook: X-ray pictures have names: **millet seeds** = miliary TB, **bat wings** = pulmonary oedema, **Kerley B lines** = interstitial oedema, **steeple sign** = croup, **snowstorm** on ultrasound = molar pregnancy.

Rapid revision: 60 one-liners

Read this page the night before and again on the morning of the exam.

'Most common' and 'gold standard'

1. Commonest type of necrosis: coagulative. Brain: liquefactive.
2. Commonest embolism: pulmonary thromboembolism from deep leg veins.
3. Commonest inherited thrombophilia: factor V Leiden.
4. Commonest mutated gene in cancer: p53; commonest oncogene: RAS.
5. Commonest site of metastasis: liver (then lungs); lymph nodes for carcinomas.
6. Commonest anaemia: iron deficiency; best test serum ferritin.
7. Commonest childhood malignancy: ALL.
8. Commonest cause of infantile diarrhoea: rotavirus.
9. Commonest cause of acquired epilepsy in India: neurocysticercosis.
10. Commonest UTI organism: E. coli.
11. Commonest site of amoebic liver abscess: right lobe.
12. Commonest worm: Ascaris.
13. Gold standard for malaria: peripheral smear. For TB: culture. For GFR: inulin clearance. For amyloid: Congo red with apple-green birefringence.
14. Investigation of choice: gallstones = ultrasound; SAH = non-contrast CT; PE = CT pulmonary angiography; bronchiectasis = HRCT.

Pairs (stain, medium, test)

15. ZN stain → AFB; Albert → diphtheria granules; India ink → Cryptococcus; Giemsa → malaria.
16. Congo red → amyloid; Perls (Prussian blue) → iron; von Kossa → calcium; PAS → glycogen, fungi; Sudan/Oil Red O → fat.
17. LJ → TB; TCBS → cholera; Löffler/tellurite → diphtheria; Sabouraud → fungi; EMJH → Leptospira; NNN → Leishmania.
18. Autoclave 121 °C, 15 psi, 15 min; hot-air oven 160 °C, 2 h.
19. Spore indicators: autoclave = Geobacillus stearothermophilus; hot-air oven = Bacillus atrophaeus.
20. Endoscopes: 2% glutaraldehyde. Plastics: ethylene oxide. Disposable syringes: gamma rays. Sera: filtration.
21. EDTA → CBC; citrate → PT/aPTT and ESR; fluoride → glucose.

Bodies, cells and signs

22. Lines of Zahn → ante-mortem thrombus.
23. Langhans giant cells → TB; Reed–Sternberg → Hodgkin; Aschoff bodies → rheumatic fever.
24. Negri bodies → rabies; owl's-eye inclusions → CMV; Councilman bodies → viral hepatitis (apoptosis); Mallory bodies → alcoholic hepatitis.
25. Auer rods → AML; smudge cells → CLL; Philadelphia chromosome → CML.
26. Psammoma bodies → papillary thyroid, serous ovarian, meningioma.
27. Koilocytes → HPV; Tzanck giant cells → herpes/varicella.
28. Crescent gametocytes → falciparum; Schüffner's dots → vivax.
29. Hypersegmented neutrophils → megaloblastic anaemia; target cells → thalassaemia, liver disease; Heinz bodies → G6PD deficiency; Howell–Jolly → asplenia.
30. Troisier's / Virchow's node → gastric carcinoma; Krukenberg → stomach to ovary.
31. Conjunctival suffusion + calf pain → leptospirosis; eschar → scrub typhus; rice-water stool → cholera.
32. Anchovy-sauce pus → amoebic liver abscess; cobweb CSF → TB meningitis; xanthochromia → SAH.

Numbers

33. Fasting glucose ≥ 126 mg/dL; HbA1c ≥ 6.5% = diabetes; HbA1c shows 2–3 months.
34. Haemoglobin: anaemia < 13 (men), < 12 (women), < 11 (pregnancy).
35. TLC 4,000–11,000/µL; platelets 1.5–4.5 lakh/µL.
36. Serum creatinine 0.6–1.2 mg/dL; urea 15–40 mg/dL.
37. Nephrotic proteinuria > 3.5 g/day; microalbuminuria 30–300 mg/day.
38. Significant bacteriuria ≥ 10^5 CFU/mL.
39. HbA2 > 3.5% → β-thalassaemia trait.
40. Mantoux ≥ 10 mm positive, read at 48–72 h.
41. Transudate protein < 3 g/dL; exudate > 3 g/dL.
42. Platelets stored at 20–24 °C (5 days); PRBC 2–6 °C.
43. AIDS: CD4 < 200/µL.
44. Rising haematocrit ≥ 20% in dengue = plasma leakage.

Classic traps

45. Lymphoma, melanoma, seminoma, mesothelioma are malignant despite '-oma'.
46. Typhoid ulcers are longitudinal; TB ulcers are transverse.

47. ESR is LOW in polycythaemia and sickle cell disease.
48. Hepatitis B is a DNA virus; A, C, D, E are RNA.
49. HEV kills pregnant women; HCV goes chronic most often.
50. IgM is the first antibody; IgG alone crosses the placenta; IgA is in secretions.
51. Type 1 diabetes, not type 2, is ketosis-prone.
52. Stage, not grade, predicts prognosis.
53. Mantoux and lepromin test cell-mediated (type IV) immunity; lepromin gives prognosis, not diagnosis.
54. Bleeding time is normal in haemophilia; aPTT is prolonged.
55. PT/INR monitors warfarin; aPTT monitors heparin.
56. Red infarcts in lung and liver (dual supply); white infarcts in heart, kidney, spleen.
57. Prostate bone metastases are osteoblastic.
58. Paget disease of the nipple is malignant.
59. Little's area is anteroinferior on the nasal septum (GPSC's 2024 key is wrong).
60. Mosquito is the definitive host of malaria; man is the intermediate host.

One-glance recall card

Topic 6 recall card: the 15 facts asked most

Cover the small lines, say the fact from the big cue, then check

P-R-L nucleus dies: pyknosis, karyorrhexis, karyolysis	**Roll vs stop** selectins roll the cell; integrins stop it	**ISH** Virchow: injury, stasis, hypercoagulability
Grade vs stage stage (spread) predicts outcome better than grade	**ACID** I IgE, II cytotoxic, III complex, IV T cell	**IgG** the only antibody that crosses the placenta
121 · 15 · 15 autoclave: °C, lb/sq in, minutes; kills spores	**BASU** typhoid: blood, antibody, stool, urine by week	**Vowels to bowels** hepatitis A and E spread faecal-oral; B, C, D by blood
Negri bodies rabies; Guarnieri = vaccinia, owl's eye = CMV	**Vivax, Ovale** hypnozoites cause relapse; falciparum kills	**TICS** microcytic: thalassaemia, iron, chronic, sideroblastic
Lavender tube EDTA for blood count; blue citrate; grey fluoride	**Dark urine, pale stool** obstructive jaundice, ALP very high	**CSF sugar low** bacterial or TB; normal in viral

Topic 6 recall card · 15 tiles

For the last ten minutes before the exam: cover the small lines of each tile and say the fact from the big cue.

Practice MCQs (50)

Time yourself: 50 questions in 35 minutes. Cover the key below until you finish.

1. The earliest change of irreversible cell injury seen on light microscopy is: (A) nuclear pyknosis (B) cell swelling (C) fatty change (D) loss of microvilli
2. Caseous necrosis is typical of: (A) myocardial infarction (B) acute pancreatitis (C) brain infarct (D) tuberculosis
3. Fat necrosis is characteristic of: (A) TB (B) gangrene (C) acute pancreatitis (D) malignant hypertension
4. Apoptosis differs from necrosis because it: (A) causes inflammation (B) affects sheets of cells (C) always follows ischaemia (D) shows cell shrinkage without inflammation

5. Squamous change in the bronchus of a smoker is: (A) dysplasia (B) hyperplasia (C) metaplasia (D) anaplasia
6. Dystrophic calcification occurs with: (A) normal serum calcium in damaged tissue (B) high serum calcium (C) vitamin D excess (D) hyperparathyroidism
7. Psammoma bodies are seen in: (A) follicular carcinoma thyroid (B) medullary carcinoma thyroid (C) anaplastic carcinoma thyroid (D) papillary carcinoma thyroid
8. The chief chemotactic factor among these is: (A) C5a (B) C3b (C) histamine (D) bradykinin
9. The cell typical of a granuloma is the: (A) epithelioid cell (B) neutrophil (C) eosinophil (D) mast cell
10. Langhans giant cells have nuclei arranged: (A) scattered (B) at the periphery in a horseshoe (C) in the centre (D) around a lipid core
11. The hallmark of healing is: (A) fibrin (B) pus (C) granulation tissue (D) necrosis
12. Lines of Zahn indicate: (A) a post-mortem clot (B) an infarct (C) an ante-mortem thrombus (D) a fat embolus
13. The commonest source of pulmonary emboli is: (A) right atrium (B) pelvic veins only (C) deep veins of the legs (D) arm veins
14. A red (haemorrhagic) infarct is typical of the: (A) lung (B) heart (C) kidney (D) spleen
15. Fat embolism most often follows: (A) fracture of long bones (B) caisson disease (C) labour (D) IV injection
16. Neutrophils dominate a myocardial infarct at: (A) 0–4 hours (B) 7–10 days (C) 1–3 days (D) 2 months
17. Which of these is malignant? (A) hamartoma (B) choristoma (C) lipoma (D) seminoma
18. The best predictor of prognosis in cancer is: (A) stage (B) grade (C) mitotic count (D) tumour marker level
19. Aflatoxin is linked to carcinoma of the: (A) stomach (B) liver (C) bladder (D) lung
20. A Krukenberg tumour is a metastasis to the ovary usually from the: (A) breast (B) colon (C) uterus (D) stomach
21. CA-125 is a marker for carcinoma of the: (A) pancreas (B) colon (C) prostate (D) ovary
22. Amyloid shows apple-green birefringence with: (A) PAS (B) Perls stain (C) Congo red under polarised light (D) von Kossa
23. The tuberculin (Mantoux) reaction is hypersensitivity type: (A) I (B) IV (C) II (D) III
24. The immunoglobulin that crosses the placenta is: (A) IgA (B) IgM (C) IgE (D) IgG
25. Standard autoclave conditions are: (A) 100 °C for 30 min (B) 160 °C for 1 h (C) 60 °C for 30 min (D) 121 °C, 15 psi, 15 min
26. Endoscopes are best disinfected with: (A) autoclaving (B) hot-air oven (C) 2% glutaraldehyde (D) boiling
27. India ink preparation is used to detect: (A) Candida (B) Cryptococcus (C) Aspergillus (D) Mucor
28. Metachromatic granules are characteristic of: (A) Mycobacterium (B) Corynebacterium diphtheriae (C) Clostridium (D) Streptococcus
29. TCBS medium is used to culture: (A) Vibrio cholerae (B) Salmonella (C) Shigella (D) E. coli

30. In the first week of typhoid the most useful test is: (A) Widal (B) blood culture (C) urine culture (D) stool culture
31. Typhoid ulcers in the ileum are: (A) longitudinal (B) transverse (C) circular girdle (D) flask-shaped
32. Leptospirosis is mainly acquired through: (A) mosquito bite (B) droplets (C) blood transfusion (D) water or soil contaminated with rat urine
33. An eschar with fever after field work suggests: (A) scrub typhus (B) leptospirosis (C) dengue (D) malaria
34. The window period of acute hepatitis B is detected by: (A) HBsAg (B) anti-HBs (C) anti-HBc IgM (D) HBeAg
35. The hepatitis with the highest mortality in pregnancy is: (A) hepatitis A (B) hepatitis B (C) hepatitis C (D) hepatitis E
36. Dengue on day 2 of fever is best confirmed by: (A) IgG ELISA (B) NS1 antigen (C) Widal (D) IgM ELISA
37. Negri bodies are diagnostic of: (A) measles (B) CMV (C) rabies (D) herpes
38. Banana-shaped gametocytes are seen in: (A) P. vivax (B) P. falciparum (C) P. malariae (D) P. ovale
39. Relapse in vivax malaria is due to: (A) hypnozoites (B) gametocytes (C) merozoites (D) sporozoites in blood
40. Anchovy-sauce pus is typical of: (A) pyogenic liver abscess (B) amoebic liver abscess (C) hydatid cyst (D) hepatoma
41. A night blood smear is needed for: (A) malaria (B) filariasis (C) kala-azar (D) typhoid
42. The commonest anaemia in India is: (A) megaloblastic (B) sickle cell (C) aplastic (D) iron deficiency
43. Hypersegmented neutrophils suggest: (A) iron deficiency (B) thalassaemia (C) vitamin B12 or folate deficiency (D) lead poisoning
44. HbA2 above 3.5% indicates: (A) iron deficiency (B) sickle cell anaemia (C) β-thalassaemia trait (D) aplastic anaemia
45. ESR is LOW in: (A) polycythaemia (B) TB (C) multiple myeloma (D) pregnancy
46. In haemophilia A, the test that is prolonged is: (A) bleeding time (B) PT (C) thrombin time (D) aPTT
47. The Philadelphia chromosome t(9;22) is the hallmark of: (A) AML (B) CML (C) CLL (D) ALL
48. Absent urinary bilirubin with raised urobilinogen suggests: (A) obstructive jaundice (B) haemolytic jaundice (C) hepatitis (D) Dubin–Johnson syndrome
49. CSF with lymphocytes, high protein, low sugar and a cobweb clot suggests: (A) pyogenic meningitis (B) tuberculous meningitis (C) viral meningitis (D) subarachnoid haemorrhage
50. The most specific marker of myocardial infarction is: (A) troponin (B) CK total (C) LDH (D) AST

Answer key

Q	Ans	Why
1	A	Nuclear changes mark irreversibility

Q	Ans	Why
2	D	Cheesy necrosis in granulomas
3	C	Lipase causes saponification
4	D	No inflammation, single cells
5	C	Columnar to squamous
6	A	Dead tissue, normal calcium
7	D	Papillary thyroid carcinoma
8	A	C5a (also LTB4, IL-8)
9	A	Epithelioid cell = activated macrophage
10	B	Horseshoe at the periphery
11	C	New capillaries + fibroblasts
12	C	Formed in flowing blood
13	C	Over 95% from deep leg veins
14	A	Dual blood supply (lung, liver)
15	A	Long-bone fractures
16	C	Days 1 to 3
17	D	Seminoma is malignant despite '-oma'
18	A	TNM stage
19	B	Hepatocellular carcinoma
20	D	Signet-ring gastric carcinoma
21	D	Epithelial ovarian cancer
22	C	Congo red, polarised light
23	B	Delayed, T-cell mediated
24	D	IgG only
25	D	121 °C, 15 psi, 15–20 min

Q	Ans	Why
26	C	Cidex
27	B	Capsule in CSF
28	B	Volutin granules (Albert stain)
29	A	Yellow colonies
30	B	Bacteraemia in week 1
31	A	Along Peyer's patches
32	D	Rat urine in water or soil
33	A	Chigger (larval mite) bite
34	C	Only marker in the window
35	D	About 20% mortality in pregnancy
36	B	NS1 positive in days 1–5
37	C	Hippocampus and Purkinje cells
38	B	Crescent gametocytes
39	A	Dormant liver stage
40	B	Entamoeba histolytica
41	B	Nocturnal periodicity
42	D	Low ferritin
43	C	Megaloblastic anaemia
44	C	HPLC / electrophoresis
45	A	Also sickle cell disease, CHF
46	D	Factor VIII is intrinsic pathway
47	B	BCR–ABL fusion
48	B	Unconjugated bilirubin is not filtered
49	B	Tuberculous meningitis

Q	Ans	Why
50	A	Troponin I or T

Score guide: 40+ is exam-ready; below 30, reread the tables for the questions you missed.

Sources

GPSC question papers with answer keys (read in full; questions quoted in shortened form):

- Medical Officer CBRT, 28 Sep 2025
- Medical Officer CBRT, 28 Sep 2024
- Public Health Dentist CBRT, 18 Feb 2024
- Junior Ophthalmologist (ESI) CBRT, 5 Nov 2023
- GPSC question paper archive

Standard textbooks the notes follow (facts written from them, not copied; check a doubtful point against your own edition):

- Robbins and Cotran, *Pathologic Basis of Disease* — cell injury, inflammation, haemodynamics, neoplasia, amyloid.
- Harsh Mohan, *Textbook of Pathology* — Indian emphasis, haematology, clinical pathology.
- Ananthanarayan and Paniker, *Textbook of Microbiology* — sterilisation, stains, media, bacteria, viruses.
- C. K. Jayaram Paniker, *Textbook of Medical Parasitology* — malaria, protozoa, helminths.
- Godkar, *Textbook of Medical Laboratory Technology* — normal values, urine, CSF, body fluids.

Indian programme rules (NTEP TB testing, dengue and malaria testing, HIV testing strategy, UIP vaccines) are as of mid-2026; confirm any figure that the paper words differently.

Topic 7: Practice of Medicine & Homoeopathic Therapeutics

How to use this in 2 days

Topic 7 is the largest clinical topic in the core: expect about 3 of the 25 core marks, plus medicine-flavoured questions hidden under pathology and community medicine. GPSC asks the diagnostic number, the drug of choice, the classic sign and one keynote remedy per disease.

Session	Read	Then do
Day 1 morning	GPSC questions already asked; hypertension and IHD; asthma and COPD	Write the five cut-off numbers from memory
Day 1 afternoon	Stroke and epilepsy; GI (peptic ulcer, GERD, IBS); liver and gallbladder	First 25 practice MCQs
Day 1 evening	Diabetes; thyroid, gout, nutrition	Rapid revision 1–30
Day 2 morning	Skin (eczema, psoriasis, urticaria); fevers and seasonal outbreaks	—
Day 2 afternoon	Homoeopathic therapeutics table, twice	Last 25 practice MCQs
Day 2 evening	Rapid revision page; your wrong answers	Sleep

How GPSC frames medicine:

- **Numbers**: BP 140/90, fasting glucose 126, HbA1c 6.5%, FEV1/FVC 0.7, thrombolysis 4.5 h.
- **'Investigation of choice' and 'drug of choice'**: CT for acute stroke/SAH, spirometry for COPD, PTU in first-trimester hyperthyroidism, salbutamol for acute asthma.
- **'All EXCEPT' and 'which is FALSE'**: the 2025 Medical Officer paper opened with one on diabetes.
- **Homoeopathic therapeutics**: one disease, one leading remedy, one keynote. The therapeutics table near the end is built for this.

What GPSC has already asked

The Medical Officer papers of 28 Sep 2024 and 28 Sep 2025 asked 17 medicine questions between them. All are one-fact questions: a target, a drug of choice, a mechanism, a pattern. The answers are GPSC's keys, checked against standard texts.

Paper	Question (shortened)	GPSC answer	The fact to carry away
MO 2025	Diabetes: which statement is false?	Type 2 more prone to ketoacidosis than type 1	Type 1 is ketosis-prone; type 2 deaths are mostly cardiovascular; FBS ≥ 126 diagnoses
MO 2025	Vitamin most likely deficient in strict vegans	Vitamin B12	B12 comes only from animal foods
MO 2025	Antithyroid drug safer in the 1st trimester	Propylthiouracil	Switch to carbimazole/methimazole after the 1st trimester (PTU liver toxicity)
MO 2025	Non-modifiable risk factor for hypertension	Gender	Also age, family history, ethnicity
MO 2025	Investigation of choice in acute SAH	Non-contrast CT	LP for xanthochromia if CT negative
MO 2025	RHD with AF, sudden left hemiparesis, infarct on CT	Embolic stroke	AF is the commonest source of cardio-embolism
MO 2025	Loss of smell and taste with cough (2020)	Nasopharyngeal swab RT-PCR	Reference test for SARS-CoV-2
MO 2025	Contraindication to parenteral nutrition	Active infection	GPSC's key; uncontrolled sepsis is a relative contraindication
MO 2025	Kwashiorkor differs from marasmus by	Oedema	Protein deficiency with oedema; marasmus = severe wasting
MO 2024	BP goal in a diabetic with hypertension	120–130 / 70–79 mmHg	Aim < 130/80 in diabetes
MO 2024	True about furosemide	Indicated in acute pulmonary oedema	Loop diuretic (thick ascending limb); potent; oral and IV
MO 2024	Key mechanism in GERD	Transient LES relaxations	Not hiatus hernia, not LES hypertension
MO 2024	Commonest pattern in IBS	Constipation alternating with diarrhoea	Rome IV: pain related to defecation
MO 2024	Graves' disease thyroid profile	Low TSH	High T3/T4, TSH-receptor antibodies
MO 2024	Statin effective at any time of day	Atorvastatin	Long half-life (rosuvastatin too); simvastatin and lovastatin at night

Paper	Question (shortened)	GPSC answer	The fact to carry away
MO 2024	Leukonychia	Hypoalbuminaemia	Koilonychia = iron deficiency; clubbing = lung/heart disease
MO 2024	Mood stabiliser with anti-suicidal action	Lithium	Narrow therapeutic range; check levels, thyroid and renal function

Pattern: GPSC writes long clinical stems but asks one textbook fact. Read the last line of the stem first.

Hypertension and ischaemic heart disease

Hypertension is diagnosed at **≥ 140/90 mmHg** on repeated clinic readings (Indian and ISH/WHO guidelines); ACC/AHA 2017 uses ≥ 130/80. Troponin is the marker of choice for myocardial infarction.

Hypertension

Point	Fact
Grades (ISH 2020)	Grade 1: 140–159 / 90–99; Grade 2: ≥ 160 / ≥ 100
Commonest type	Essential (primary), ~90–95%
Secondary causes	Renal (commonest secondary: renal parenchymal disease), renal artery stenosis, Cushing's, Conn's (low K+), phaeochromocytoma (episodic headache, sweating, palpitations), coarctation (radio-femoral delay), OCPs, NSAIDs
Non-modifiable risks	Age, gender, family history, ethnicity (MO 2025)
Modifiable risks	Salt > 5 g/day, obesity, alcohol, inactivity, stress, smoking
Targets	General < 140/90 (< 130/80 if tolerated); diabetes or CKD < 130/80 (MO 2024)
Lifestyle	DASH diet, salt < 5 g/day (WHO), weight loss, exercise 150 min/week
First-line drugs	ACE inhibitor/ARB, calcium-channel blocker (amlodipine), thiazide-like diuretic
In pregnancy	Labetalol, nifedipine, methyldopa; **ACE inhibitors/ARBs are contraindicated**
Hypertensive emergency	BP ≥ 180/120 with organ damage: IV labetalol, nitroprusside, nicardipine; lower by no more than 25% in the first hour
Target-organ damage	LVH, retinopathy (Keith–Wagener grades; papilloedema = grade IV), nephrosclerosis, stroke

Drug side-effects asked often: ACE inhibitors → **dry cough** (bradykinin), angio-oedema, hyperkalaemia; amlodipine → ankle oedema; thiazides → hypokalaemia, hyperuricaemia, hyperglycaemia; beta-blockers → avoid in asthma.

Ischaemic heart disease

Condition	Key features	Diagnosis / treatment
Stable angina	Exertional retrosternal pain < 10 min, relieved by rest or sublingual nitrate	ECG stress test; coronary angiography = gold standard
Unstable angina / NSTEMI	Pain at rest or crescendo; NSTEMI has raised troponin	Antiplatelets, anticoagulant, early angiography
STEMI	Pain > 20 min, sweating; ST elevation in contiguous leads or new LBBB	**Primary PCI** within 90–120 min, or thrombolysis (streptokinase, tenecteplase) within 12 h
Prinzmetal (variant) angina	Rest pain from coronary spasm, transient ST elevation	Calcium-channel blockers, nitrates

- **Biomarkers**: troponin I/T most sensitive and specific (rises 3–4 h, stays 7–10 days); CK-MB rises 4–6 h, normal by 48–72 h (useful for re-infarction).
- **Infarct territory on ECG**: II, III, aVF = inferior (right coronary); V1–V4 = anterior (LAD); I, aVL, V5–V6 = lateral (circumflex).
- **MONA-B immediate care**: aspirin 300 mg chewed (all patients), clopidogrel/ticagrelor, nitrates (avoid in right ventricular infarct or hypotension), oxygen only if saturation < 90%, morphine for pain; plus statin and beta-blocker.
- **Complications**: arrhythmia (VF — commonest cause of early death), heart failure, cardiogenic shock, papillary-muscle rupture (mitral regurgitation), septal rupture, free-wall rupture (days 3–7), Dressler syndrome (weeks later, pericarditis).
- **Secondary prevention**: aspirin, statin (high-intensity: atorvastatin 40–80 mg; any time of day — MO 2024), beta-blocker, ACE inhibitor, stop smoking.
- **Heart failure basics**: breathlessness, orthopnoea, PND, raised JVP, basal crepitations, pedal oedema; BNP/NT-proBNP raised; echo for ejection fraction; furosemide for congestion and acute pulmonary oedema (MO 2024).

Memory hook: **A**CE inhibitors give an **A**nnoying cough (bradykinin). After an MI, **troponin is the tortoise**: it rises in a few hours and stays for about a week or two; **CK-MB is the hare**: gone in 2 to 3 days, so it is the marker for a fresh re-infarction.

Bronchial asthma and COPD

The one-line difference: asthma is **variable and reversible** airflow obstruction (inflammation, often allergic, young); COPD is **persistent, not fully reversible** obstruction (smoking or biomass smoke, older).

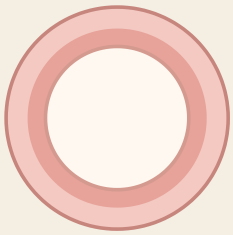
Feature	Bronchial asthma	COPD
Onset	Often childhood, atopic	> 40 years, smoker
Symptoms	Episodic wheeze, chest tightness, night/early-morning cough	Chronic productive cough, progressive breathlessness
Triggers	Allergens (house dust mite, pollen), cold air, exercise, aspirin/NSAIDs, beta-blockers, infections	Smoking (commonest cause), biomass fuel (common in Indian women), occupational dust
Spirometry	Reversibility: FEV1 rises ≥ 12% and ≥ 200 mL after bronchodilator; PEF variability > 10%	**Post-bronchodilator FEV1/FVC < 0.70** (GOLD)
Cells	Eosinophils, mast cells, Th2	Neutrophils, macrophages, CD8
Sputum/pathology	Curschmann spirals, Charcot–Leyden crystals, Creola bodies	Goblet-cell hyperplasia; Reid index > 0.4 (chronic bronchitis)
Imaging	Usually normal	Hyperinflation, flat diaphragm, bullae (emphysema)

Definitions

- **Chronic bronchitis**: productive cough for ≥ 3 months a year for 2 consecutive years ('blue bloater').
- **Emphysema**: permanent enlargement of airspaces distal to terminal bronchioles with wall destruction ('pink puffer'). Centriacinar = smokers (upper lobes); panacinar = **alpha-1 antitrypsin deficiency** (lower lobes, young).
- **Status asthmaticus / acute severe asthma**: PEF 33–50% predicted, can't complete sentences, pulse ≥ 110, RR ≥ 25. **Life-threatening**: PEF < 33%, SpO2 < 92%, silent chest, cyanosis, normal or rising PaCO2.

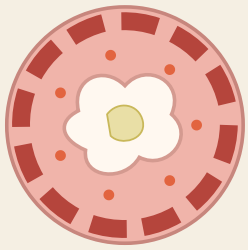
Asthma and COPD in the airway

What narrows the airway, and what destroys the alveoli



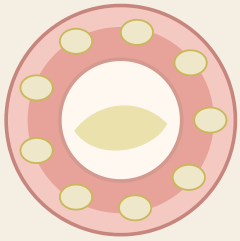
Normal airway

Open lumen, thin wall



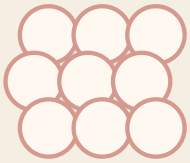
Asthma

Smooth muscle squeezes, wall swells, mucus plug; eosinophils, mast cells



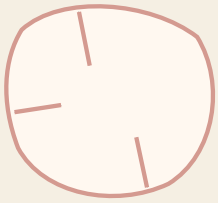
Chronic bronchitis

Enlarged mucus glands, goblet cells (Reid index > 0.4); "blue bloater"



Normal alveoli

Many small sacs, large surface for gas exchange



Emphysema

Walls destroyed, airspaces enlarged; "pink puffer"

Centriacinar: smokers, upper lobes
Panacinar: α-1 antitrypsin, lower lobes

Asthma: variable and REVERSIBLE (FEV1 rises ≥ 12% and ≥ 200 mL after a bronchodilator).

COPD: persistent, not fully reversible (post-bronchodilator FEV1/FVC < 0.70); smoking is the commonest cause.

Asthma and COPD · what narrows the airway, and what destroys the alveoli

Treatment

Situation	Treatment
Acute asthma attack	Inhaled **short-acting beta-2 agonist (salbutamol)** by nebuliser/spacer + ipratropium; oxygen to SpO2 93–95%; systemic steroid (prednisolone/hydrocortisone); IV magnesium sulphate if severe
Long-term asthma (GINA)	Inhaled corticosteroid is the controller; ICS–formoterol as reliever and controller; add LABA, then LAMA or montelukast; anti-IgE (omalizumab) in severe allergic asthma
COPD maintenance	Long-acting bronchodilators: LAMA (tiotropium) and/or LABA; ICS only if frequent exacerbations with eosinophilia

Situation	Treatment
COPD exacerbation	Bronchodilators, steroids, antibiotics if purulent sputum; **controlled oxygen (target SpO2 88–92%)** to avoid CO2 retention; NIV for hypercapnic acidosis
Proven to improve COPD survival	**Smoking cessation** and **long-term oxygen therapy** (≥ 15 h/day if PaO2 ≤ 55 mmHg)
Vaccines	Influenza yearly; pneumococcal

Complications of COPD: cor pulmonale (right heart failure, raised JVP, oedema), secondary polycythaemia, type 2 respiratory failure (raised CO2: bounding pulse, flapping tremor, headache), pneumothorax from bullae.

- Related respiratory facts often clubbed in: **pulmonary TB** — cough > 2 weeks, fever, night sweats, weight loss, haemoptysis; NAAT (CBNAAT/TrueNat) is the initial test under NTEP.
- **Community-acquired pneumonia** — commonest organism Streptococcus pneumoniae; CURB-65 for severity.
- **Bronchiectasis** — copious foul sputum; HRCT is investigation of choice; 'signet ring' sign.

Memory hook: **A**sthma comes and goes **A**gain (variable, reversible); **COPD Clings** (persistent). An FEV1/FVC ratio **below 0.7** means obstruction; a rise after a bronchodilator means asthma.

Stroke and epilepsy

Non-contrast CT is the first investigation in every acute stroke: it separates haemorrhage from infarction before any clot-buster is given.

Stroke

Point	Fact
Definition	Sudden focal neurological deficit of vascular origin lasting > 24 h or causing death; TIA resolves without infarction (most within 1 h)
Types	Ischaemic ~85% (thrombotic, embolic, lacunar); haemorrhagic ~15% (intracerebral, subarachnoid)
Commonest risk factor	**Hypertension**; then diabetes, smoking, AF, dyslipidaemia
Commonest cardiac source of emboli	**Atrial fibrillation** (rheumatic mitral stenosis in India) — MO 2025 case

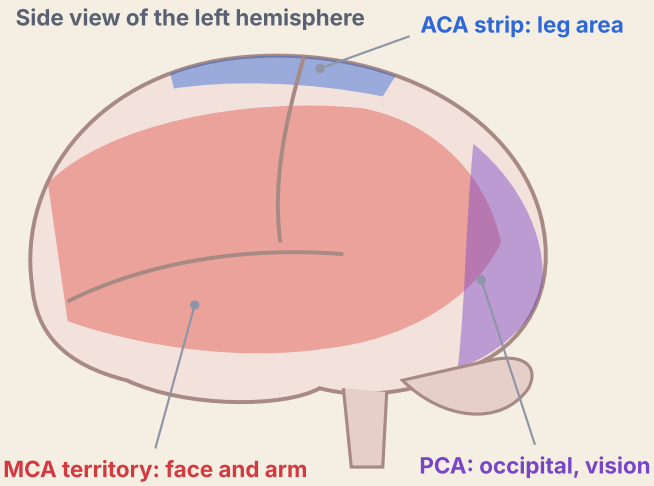
Point	Fact
Commonest site	Middle cerebral artery: contralateral hemiplegia (face and arm > leg), aphasia if dominant side
ACA stroke	Leg > arm weakness
Internal capsule	Dense hemiplegia of face, arm and leg (lenticulostriate arteries; hypertensive haemorrhage)
Brainstem / PCA	Crossed signs; vertigo, diplopia; homonymous hemianopia with macular sparing (PCA)
Investigation of choice	**Non-contrast CT** (haemorrhage white/hyperdense); MRI-DWI best for early infarct
Thrombolysis	IV alteplase/tenecteplase within 4.5 h of onset if CT excludes bleed; mechanical thrombectomy up to 6–24 h for large-vessel occlusion
After the acute phase	Aspirin, statin, BP control; anticoagulate for AF (warfarin/DOAC)
Scores	FAST for recognition (Face, Arm, Speech, Time); NIHSS for severity; GCS for consciousness (3–15)

Subarachnoid haemorrhage: 'worst headache of my life' (thunderclap), neck stiffness, photophobia; commonest cause **ruptured berry aneurysm** (circle of Willis, anterior communicating artery). CT first (MO 2025); if negative, lumbar puncture after 12 h for **xanthochromia**. Nimodipine prevents vasospasm.

Bell's palsy (LMN facial palsy): whole half of face including forehead; steroids within 72 h. In UMN (stroke) the forehead is spared.

Stroke territories and facial palsy

Which artery, which weakness; does the forehead move?



Middle cerebral artery
COMMONEST site. Opposite-side weakness, face and arm > leg; aphasia if dominant side

Anterior cerebral artery
Leg > arm weakness

Posterior cerebral artery
Homonymous hemianopia, macular sparing

Internal capsule
Dense weakness of face, arm AND leg (lenticulostriate; hypertensive bleed)

Stroke (UMN facial palsy)



Lower face droops on one side
FOREHEAD SPARED: wrinkles on both sides

Bell's palsy (LMN)



WHOLE half of the face, forehead too: no wrinkles, eye will not close.
Steroids within 72 h

Non-contrast CT first in every acute stroke: it rules out bleeding before a clot-buster.

Stroke territories and facial palsy · MCA face and arm, ACA leg; is the forehead spared?

Epilepsy

Seizure type	Features	Drug of choice
Generalised tonic-clonic	Tonic then clonic phase, tongue bite, incontinence, post-ictal confusion	Valproate (avoid in women of child-bearing age → levetiracetam/lamotrigine)
Absence (petit mal)	Children; brief staring spells; **3 Hz spike-and-wave** EEG; hyperventilation provokes	**Ethosuximide** or valproate

Seizure type	Features	Drug of choice
Focal (partial)	Aura, automatisms (temporal lobe: déjà vu, lip-smacking)	Carbamazepine or lamotrigine
Myoclonic (juvenile myoclonic epilepsy)	Morning jerks, teens	Valproate (carbamazepine worsens it)
Infantile spasms (West syndrome)	Flexor spasms, hypsarrhythmia on EEG	ACTH, vigabatrin
Febrile seizure	6 months–5 years, with fever, usually simple and benign	Treat fever; no long-term drugs

Status epilepticus: seizure > 5 minutes or repeated seizures without recovery. Treatment: airway, glucose, then **IV lorazepam** (or midazolam/diazepam), then phenytoin/fosphenytoin, valproate or levetiracetam; if refractory, anaesthesia.

In India the **commonest cause of acquired seizures is neurocysticercosis** (Taenia solium; ring-enhancing lesion on CT). Investigations: EEG, MRI brain. Pregnancy: valproate is the most teratogenic (neural-tube defects); give folic acid. Phenytoin side-effects: gum hypertrophy, hirsutism, ataxia, nystagmus.

Memory hook: plain CT first: it rules out bleeding before any clot-busting drug. "Worst headache of my life" = subarachnoid haemorrhage. In facial palsy, an **upper** motor neurone lesion **spares the forehead**; **Bell's palsy** (lower motor neurone) takes the whole half-face.

Peptic ulcer, GERD and irritable bowel syndrome

H. pylori and NSAIDs cause most peptic ulcers; transient LES relaxation causes GERD; IBS is a functional disorder diagnosed by the Rome IV criteria with no structural disease.

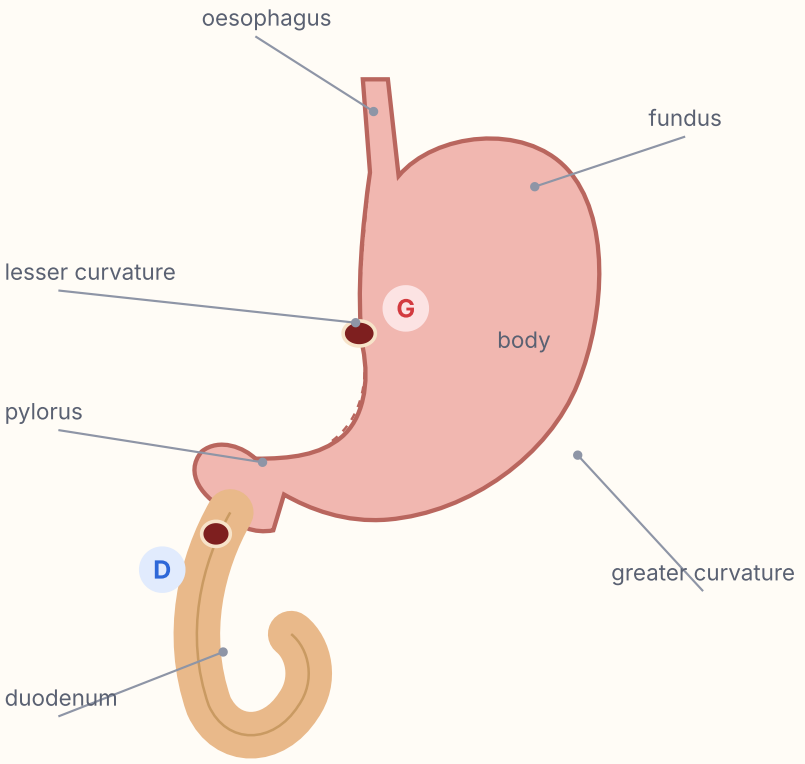
Peptic ulcer disease

Point	Duodenal ulcer	Gastric ulcer
Frequency	Commoner (about 4:1)	Less common
Pain	Epigastric, **relieved by food**, hunger pain, wakes at night	**Worse after food**, weight loss
Cause	H. pylori in ~90%; high acid	H. pylori ~70%, NSAIDs

Point	Duodenal ulcer	Gastric ulcer
Cancer risk	Almost never malignant	Biopsy all gastric ulcers (can be malignant)
Site	First part of duodenum, anterior wall commonest	Lesser curvature

Peptic ulcer: duodenal or gastric?

Site, cause and how food changes the pain



D Duodenal ulcer

Commoner (about 4 : 1)
1st part of duodenum, anterior wall commonest
H. pylori in ~90%; high acid
Almost never malignant

G Gastric ulcer

Lesser curvature
H. pylori ~70%, NSAIDs
Biopsy every gastric ulcer: it can be malignant

PAIN AND FOOD

Duodenal: relieved by food, hunger pain, wakes at night

Gastric: worse after food, weight loss

"Gastric Gets worse with food; Duodenal is Dulled by food."

Peptic ulcer · duodenal vs gastric: site, cause and the effect of food

- H. pylori tests: **urea breath test** and stool antigen (non-invasive; best for confirming eradication), rapid urease (CLO) test on biopsy, histology.
- Eradication: PPI + clarithromycin + amoxicillin (or metronidazole) for 14 days; bismuth quadruple therapy if resistance.
- Complications: **bleeding (commonest)**, perforation (anterior duodenal ulcer; air under diaphragm), gastric outlet obstruction (projectile non-bilious vomiting, succussion splash, hypokalaemic hypochloraemic metabolic alkalosis), penetration (posterior ulcer into pancreas).
- Zollinger–Ellison syndrome: gastrinoma, multiple/refractory ulcers, diarrhoea; raised fasting gastrin.

GERD

- Mechanism: **transient lower oesophageal sphincter relaxations** (MO 2024); hiatus hernia, obesity, pregnancy, smoking, alcohol, fatty food contribute.
- Symptoms: heartburn, acid regurgitation worse after meals and lying down; atypical: chronic cough, hoarseness, asthma, dental erosion.
- Diagnosis: clinical trial of PPI; endoscopy for alarm features (dysphagia, weight loss, bleeding, anaemia, age > 60); **24-hour pH monitoring = gold standard**.
- Treatment: lifestyle (raise head end, avoid late meals, lose weight), **PPIs** (most effective), H2 blockers; surgery = Nissen fundoplication.
- Complications: oesophagitis, stricture, **Barrett's oesophagus** (intestinal metaplasia) → adenocarcinoma.
- Oesophagus contrasts: achalasia = **bird's beak**, dysphagia to solids and liquids from the start; carcinoma = **rat tail** (MO 2025), progressive dysphagia solids → liquids.

Irritable bowel syndrome

- **Rome IV**: recurrent abdominal pain on average ≥ 1 day/week for the last 3 months, with ≥ 2 of: related to defecation, change in stool frequency, change in stool form; onset ≥ 6 months before diagnosis.
- Types: IBS-C, IBS-D, IBS-M (mixed); the commonest pattern keyed by GPSC is **constipation alternating with diarrhoea** (MO 2024).
- Features that are NOT IBS (red flags): age > 50 at onset, rectal bleeding, weight loss, nocturnal diarrhoea, anaemia, family history of colon cancer.
- Management: reassurance, diet (low-FODMAP), fibre for constipation, antispasmodics (mebeverine, dicyclomine), loperamide for diarrhoea, low-dose tricyclics; stress is a major trigger.

Inflammatory bowel disease (for contrast)

Ulcerative colitis	Crohn's disease
Rectum upwards, continuous	Mouth to anus, skip lesions; terminal ileum commonest
Mucosal; bloody diarrhoea	Transmural; fistulae, strictures, non-caseating granulomas
Lead-pipe colon; pseudopolyps	String sign of Kantor; cobblestone mucosa
Smoking protects; p-ANCA	Smoking worsens; ASCA

Ulcerative colitis	Crohn's disease
Higher colon-cancer risk; toxic megacolon	Perianal disease, malabsorption (B12)

Memory hook: **G**astric ulcer pain **G**ets worse with food; **D**uodenal ulcer pain is **D**ulled by food and wakes the patient at night. Most ulcers trace back to **H. pylori** or **NSAIDs**.

Hepatitis, cirrhosis and cholecystitis

Viral hepatitis transmission and serology, the complications of cirrhosis, and Murphy's sign with ultrasound carry most marks here.

Viral hepatitis

Virus	Route	Chronicity	Clinical point
A	Faeco-oral	None	Children, mild; anti-HAV IgM = acute; vaccine
B (DNA)	Blood, sexual, perinatal	5–10% adults; ~90% of infected newborns	HBsAg = infection; anti-HBc IgM = acute; anti-HBs = immunity; vaccine at birth (UIP)
C	Blood (transfusion, IV drugs, dialysis)	~80%	Cirrhosis, HCC; cured by direct-acting antivirals (sofosbuvir); no vaccine
D	Needs HBV	—	Superinfection worsens HBV
E	Faeco-oral, water-borne epidemics	Rare	**~20% mortality in pregnancy** (fulminant failure)

Acute hepatitis: prodrome (anorexia, nausea, distaste for smoking), then jaundice; **ALT > AST**, often > 1000 U/L. Alcoholic hepatitis: **AST:ALT > 2**, raised GGT. Fulminant hepatic failure: encephalopathy within 8 weeks, prolonged INR (best prognostic marker). Paracetamol overdose antidote: **N-acetylcysteine**.

Cirrhosis

- Causes: alcohol, chronic HBV/HCV, **NAFLD/MASLD** (rising), autoimmune hepatitis, haemochromatosis, Wilson's disease (Kayser–Fleischer rings, low caeruloplasmin), primary biliary cholangitis (anti-mitochondrial antibody).
- Signs: palmar erythema, spider naevi (SVC area), gynaecomastia, testicular atrophy, Dupuytren's contracture (alcohol), clubbing, leukonychia (low albumin; MO 2024), jaundice, ascites, caput

medusae, splenomegaly.

- Severity: **Child–Pugh** (bilirubin, albumin, INR, ascites, encephalopathy); MELD for transplant listing.

Complication	Key facts
Portal hypertension, oesophageal varices	Haematemesis; endoscopic band ligation, IV octreotide/terlipressin; non-selective beta-blocker (propranolol) prevents bleeds
Ascites	SAAG ≥ 1.1 g/dL = portal hypertension; salt restriction, **spironolactone** + furosemide; large-volume paracentesis with albumin
Spontaneous bacterial peritonitis	Ascitic neutrophils > 250/mm^3; cefotaxime
Hepatic encephalopathy	**Asterixis (flapping tremor)**, confusion, fetor hepaticus; precipitated by GI bleed, infection, constipation, sedatives, high protein; **lactulose**, rifaximin
Hepatorenal syndrome	Functional renal failure; terlipressin + albumin
Hepatocellular carcinoma	Raised **AFP**; ultrasound + AFP surveillance every 6 months

Gallbladder: cholelithiasis and cholecystitis

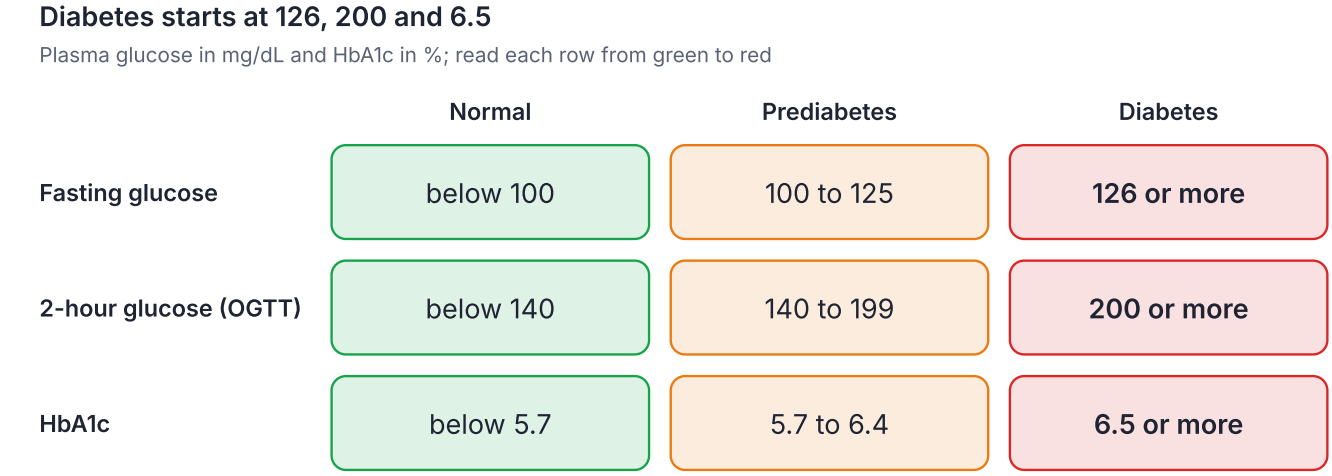
- Gallstone risk: '**fat, female, fertile, forty, fair**'; cholesterol stones commonest; pigment stones in haemolysis.
- **Biliary colic**: right upper quadrant/epigastric pain after fatty meals, radiating to right scapula.
- **Acute cholecystitis**: persistent pain, fever, **Murphy's sign** (inspiration arrested on palpating the right subcostal area); leucocytosis.
- **Investigation of choice: ultrasound** (stones with acoustic shadow, wall thickening, pericholecystic fluid). HIDA scan if doubt.
- Treatment: analgesia, antibiotics, **laparoscopic cholecystectomy** (early, within 72 h ideally).
- Complications: empyema, gangrene, perforation, **Mirizzi syndrome**, gallstone ileus.
- **Charcot's triad** (ascending cholangitis): fever with rigors, jaundice, RUQ pain; add hypotension and confusion = **Reynolds' pentad**.
- **Courvoisier's law**: a palpable, non-tender gallbladder with jaundice is unlikely to be due to stones — think carcinoma head of pancreas.
- Acute pancreatitis: epigastric pain radiating to the back, relieved by leaning forward; **lipase** (more specific) / amylase > 3× normal; commonest causes gallstones and alcohol; Cullen's (periumbilical) and Grey Turner's (flank) signs.

Memory hook: viral hepatitis lifts **ALT above AST**; alcohol flips it, **AST:ALT above 2** ("**S**cotch makes A**S**T soar"). Gallstones love the **5 Fs**: fat, female, fertile, forty, fair.

Diabetes mellitus

Diagnosis rests on four cut-offs: fasting plasma glucose ≥ 126 mg/dL, 2-h OGTT ≥ 200, HbA1c ≥ 6.5%, or random ≥ 200 with symptoms. Type 1 is ketosis-prone; type 2 patients die mostly of cardiovascular disease.

Test	Normal	Prediabetes	Diabetes
Fasting plasma glucose (mg/dL)	< 100	100–125 (IFG)	≥ 126
2-h OGTT, 75 g (mg/dL)	< 140	140–199 (IGT)	≥ 200
HbA1c	< 5.7%	5.7–6.4%	≥ 6.5%
Random glucose	—	—	≥ 200 with classic symptoms



diabetes cut-offs · normal, prediabetes, diabetes

The red column is the one examiners ask: 126, 200, 6.5.

An abnormal result must be repeated on another day unless symptoms are clear-cut.

Type 1 vs type 2

Type 1	Type 2
Autoimmune β-cell destruction (GAD, ICA antibodies); HLA-DR3/DR4	Insulin resistance + relative deficiency; obesity, family history (stronger genetic link)

Type 1	Type 2
Young, thin, abrupt onset	Usually > 30–40 years, obese, insidious (Indians at younger age and lower BMI)
Prone to diabetic ketoacidosis	Prone to hyperosmolar hyperglycaemic state
Low C-peptide; insulin essential	Lifestyle + metformin first line

Other types: gestational diabetes (India: DIPSI single-step 75 g OGTT, 2-h ≥ 140 mg/dL); secondary diabetes from **acromegaly, Cushing's syndrome**, phaeochromocytoma, glucagonoma, chronic pancreatitis, steroids (all endocrine causes asked in MO 2025); MODY.

Acute complications

Condition	Features	Treatment
Diabetic ketoacidosis	Glucose > 250, pH < 7.3, HCO3 < 18, ketones; Kussmaul breathing, fruity breath, abdominal pain, dehydration	**IV normal saline first**, then insulin infusion 0.1 U/kg/h; potassium replacement (check K before insulin)
Hyperosmolar hyperglycaemic state	Older type 2; glucose > 600, osmolality > 320, little ketosis, severe dehydration, confusion	Fluids, cautious insulin
Hypoglycaemia	Glucose < 70 mg/dL; sweating, tremor, confusion; Whipple's triad	15–20 g oral glucose; IV 25% dextrose or glucagon if unconscious

Chronic complications: microvascular — retinopathy (commonest cause of blindness in working adults; screen yearly), nephropathy (earliest sign **microalbuminuria 30–300 mg/day**; Kimmelstiel–Wilson nodules), neuropathy (glove-and-stocking, diabetic foot); macrovascular — coronary disease (commonest cause of death), stroke, peripheral vascular disease.

Treatment and targets

- Targets: HbA1c < 7% for most adults; BP < 130/80 (MO 2024); LDL < 100 (< 70 with heart disease); statin for most diabetics over 40.
- **Metformin** = first-line oral drug; does not cause hypoglycaemia; side-effects GI upset, B12 deficiency, **lactic acidosis** (avoid if eGFR < 30).
- Sulphonylureas (glimepiride, gliclazide): hypoglycaemia, weight gain. DPP-4 inhibitors (sitagliptin): weight-neutral. **SGLT2 inhibitors** (empagliflozin, dapagliflozin): heart-failure and kidney protection;

genital infections, euglycaemic DKA. GLP-1 agonists (liraglutide, semaglutide): weight loss. Pioglitazone: fluid retention, avoid in heart failure.

- Insulins: rapid (lispro, aspart), short (regular), intermediate (NPH), long-acting (glargine, degludec). Insulin is the drug of choice in pregnancy.
- Somogyi effect = morning hyperglycaemia after night hypoglycaemia; dawn phenomenon = early-morning rise from growth hormone and cortisol.

Memory hook: diabetes cut-offs read **126, 200, 6.5**: fasting glucose 126 mg/dL or more, 2-hour (or random with symptoms) 200 mg/dL or more, HbA1c 6.5% or more.

Thyroid disorders, gout and nutritional deficiencies

TSH is the best single screening test for thyroid function; serum uric acid alone does not diagnose gout (crystals do); and India still carries both deficiency diseases and obesity.

Thyroid

Condition	Labs	Key features	Treatment
Primary hypothyroidism	**High TSH, low free T4**	Weight gain, cold intolerance, constipation, dry skin, bradycardia, menorrhagia, delayed relaxation of ankle jerk; commonest cause worldwide = iodine deficiency, in iodine-sufficient areas = **Hashimoto's** (anti-TPO)	Levothyroxine on an empty stomach; recheck TSH after 6–8 weeks
Subclinical hypothyroidism	High TSH, normal T4	Often none	Treat if TSH > 10, pregnancy, symptoms
Hyperthyroidism (Graves')	**Low TSH**, high T3/T4 (MO 2024); TSH-receptor antibodies	Weight loss with good appetite, heat intolerance, tremor, palpitations/AF, **exophthalmos, pretibial myxoedema** (Graves'-specific), diffuse goitre with bruit	Carbimazole/methimazole; **PTU in the first trimester** (MO 2025); beta-blocker for symptoms; radioiodine; surgery
Toxic multinodular goitre	Low TSH	Elderly, nodules; no eye signs	Radioiodine or surgery

Condition	Labs	Key features	Treatment
Subacute (de Quervain) thyroiditis	Low TSH early, low radioiodine uptake	Painful tender goitre after viral infection, raised ESR	NSAIDs, steroids
Thyroid storm	—	Fever, tachycardia, delirium after surgery/infection	PTU, iodine (after PTU), propranolol, steroids
Congenital hypothyroidism	Neonatal TSH screening	Prolonged jaundice, large tongue, umbilical hernia, delayed milestones — **preventable cause of intellectual disability** (MO 2025)	Early levothyroxine

Thyroid cancers: **papillary commonest** (psammoma bodies, Orphan Annie nuclei, lymphatic spread, radiation); follicular (blood spread); medullary (calcitonin, MEN 2); anaplastic (elderly, worst). Fine-needle aspiration is the first test for a nodule.

Gout

- Cause: hyperuricaemia (uric acid > 7 mg/dL men) from under-excretion (commonest) or over-production; triggers — alcohol (beer), purine-rich meat and seafood, fructose drinks, thiazides, low-dose aspirin, dehydration, surgery.
- Acute attack: sudden, very painful, red hot swollen **first metatarsophalangeal joint (podagra)**, often at night.
- Diagnosis: **needle-shaped, negatively birefringent monosodium urate crystals** in joint fluid (gold standard). Pseudogout = rhomboid, positively birefringent calcium pyrophosphate crystals (knee).
- Chronic: tophi (ear helix, elbows), 'punched-out' erosions with overhanging edges on X-ray, urate kidney stones.
- Treatment: acute — NSAIDs, colchicine, steroids (do not start allopurinol during an attack); long-term — **allopurinol** (xanthine oxidase inhibitor; target uric acid < 6 mg/dL), febuxostat; lifestyle.

Nutritional deficiencies

Deficiency	Disease and signs
Protein–energy	**Kwashiorkor: oedema** (MO 2025), flaky-paint dermatosis, flag sign of hair, fatty liver; **marasmus**: severe wasting, 'old man' face, no oedema; SAM = MUAC < 11.5 cm

Deficiency	Disease and signs
Vitamin A	Night blindness (earliest), Bitot's spots, xerophthalmia, keratomalacia; 1 lakh IU at 9 months, then 2 lakh IU every 6 months to 5 years
Vitamin B1 (thiamine)	Beriberi (wet = heart failure; dry = neuropathy); Wernicke's encephalopathy in alcoholics (give thiamine before glucose)
Vitamin B2	Angular stomatitis, cheilosis, magenta tongue
Niacin (B3)	**Pellagra: dermatitis, diarrhoea, dementia** (death); maize/jowar diets; Casal's necklace
Vitamin B12	Megaloblastic anaemia, **subacute combined degeneration**; strict vegans (MO 2025), pernicious anaemia, metformin
Folate	Megaloblastic anaemia; neural tube defects — 400 µg periconception
Vitamin C	Scurvy: bleeding gums, perifollicular haemorrhage, corkscrew hair, subperiosteal haemorrhage
Vitamin D	Rickets (children: widened wrists, rachitic rosary, bow legs), osteomalacia (adults)
Vitamin K	Bleeding; haemorrhagic disease of newborn — 1 mg IM at birth
Iron	Commonest deficiency: microcytic anaemia, koilonychia, pica; Anemia Mukt Bharat
Iodine	Goitre, cretinism; universal salt iodisation
Zinc	Acrodermatitis enteropathica, poor wound healing, taste loss; ORS + zinc for diarrhoea

Limiting amino acids: cereals (wheat) lack **lysine** (and threonine — MO 2025), pulses lack methionine; mixing them gives complementary protein.

Memory hook: TSH first, always. **P**apillary carcinoma is the **P**opular one (commonest, best outlook, psammoma bodies); **M**edullary **M**akes calcitonin. Gout crystals are **needles** (negatively birefringent) and love the **big toe**.

Eczema, psoriasis and urticaria

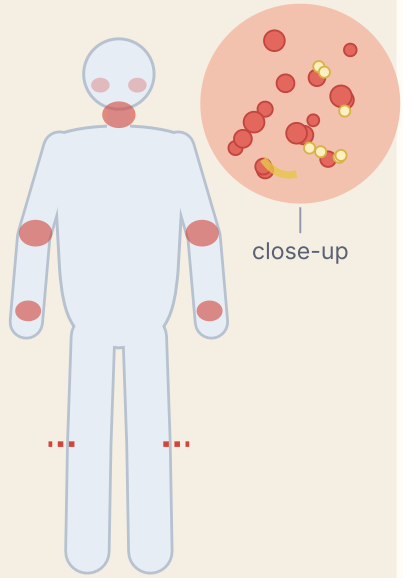
Psoriasis is silvery scales on extensors with Auspitz and Koebner signs; eczema is itchy, flexural and oozing; urticaria is wheals that vanish within 24 hours.

Feature	Eczema (dermatitis)	Psoriasis	Urticaria
Lesion	Itchy papulovesicles, oozing (acute) → lichenified (chronic)	Well-defined erythematous plaques with **silvery-white scales**	Itchy, raised **wheals** with flare; each lasts < 24 h
Sites	Atopic: face/cheeks in infants, **flexures** (elbow, knee folds) in children/adults	**Extensors** (elbows, knees), scalp, lumbosacral area, nails	Anywhere; angio-oedema of lips, eyelids
Signs	Dennie–Morgan fold; white dermographism	**Auspitz sign** (pinpoint bleeding on scraping), **Koebner phenomenon** (lesions at injury sites), candle-grease sign	Dermographism (scratch → wheal)
Nails	Rarely	**Pitting**, onycholysis, oil-drop sign, subungual hyperkeratosis	—
Mechanism	Skin-barrier defect (filaggrin) + type 2 inflammation; contact = type IV	T-cell (Th17, IL-23) driven; rapid epidermal turnover (3–4 days)	Mast-cell histamine release (type I or non-immune)
Associations	Asthma, allergic rhinitis ('atopic march'); raised IgE	**Psoriatic arthritis** (DIP joints, 'pencil-in-cup'), metabolic syndrome	Food, drugs (NSAIDs, penicillin), infections, insect stings; chronic if > 6 weeks
Triggers	Irritants, soaps, wool, heat, sweat	Stress, streptococcal throat (guttate), beta-blockers, lithium, antimalarials, **steroid withdrawal** (pustular)	Food, drugs, cold, pressure, heat, exercise
Treatment	Emollients (mainstay), topical steroids, tacrolimus; antihistamines for itch	Topical steroids, vitamin D analogues (calcipotriol), coal tar, dithranol; phototherapy; methotrexate, biologics	Non-sedating H1 antihistamines (cetirizine); adrenaline for anaphylaxis

Eczema, psoriasis, urticaria

Where it appears, and what the lesion looks like

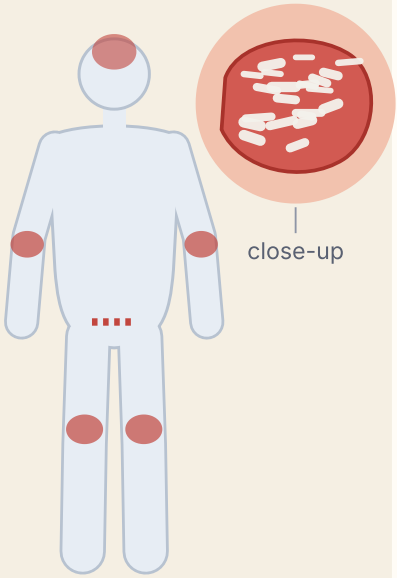
Eczema (atopic)



LESION
Itchy papulovesicles, oozing; chronic: thick, lichenified

SITES
FLEXURES: elbow and knee folds; cheeks in infants

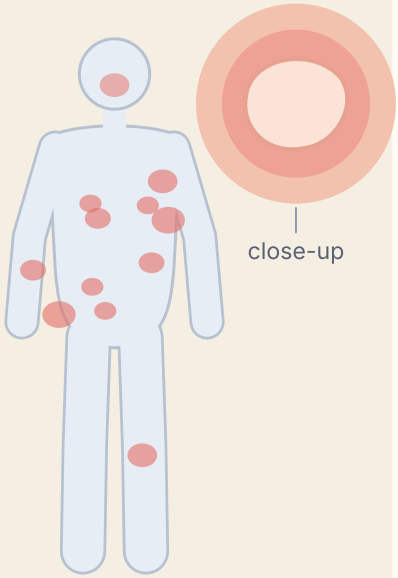
Psoriasis



LESION
Well-defined red plaques, silvery-white scales

SITES
EXTENSORS: elbows, knees; scalp, lower back, nails

Urticaria



LESION
Itchy raised wheals with a red flare; each fades < 24 h

SITES
Anywhere; angio-oedema of lips and eyelids

Psoriasis is silvery scales on extensors with Auspitz and Koebner signs; eczema is itchy, flexural and oozing; urticaria is wheals that vanish within 24 hours.

Eczema, psoriasis, urticaria · where each appears and what the lesion looks like

Eczema types: atopic; contact (irritant vs allergic — nickel, the commonest contact allergen; patch test); seborrhoeic (scalp, nasolabial folds, Malassezia; 'cradle cap' in infants); nummular (coin-shaped); stasis (venous, lower legs); dyshidrotic (pompholyx — palms/soles vesicles).

Psoriasis types: plaque (commonest), guttate (drop-like, after strep throat in children), pustular (von Zumbusch — emergency), erythrodermic, inverse (flexural), nail and scalp. Histology: Munro microabscesses, parakeratosis, elongated rete ridges.

Urticaria and anaphylaxis: acute < 6 weeks, chronic > 6 weeks (often idiopathic/autoimmune). **Anaphylaxis** = urticaria with airway or circulatory compromise: **IM adrenaline 0.5 mg (1:1000) into the anterolateral thigh** is the first drug; then fluids, antihistamine, steroid.

- Other skin MCQs: **scabies** — burrows in finger webs, wrists, genitals; night itch; family members affected; **permethrin 5%**.
- **Tinea** — annular, scaly, active edge; KOH mount.
- **Vitiligo** — depigmented patches, Wood's lamp; Koebner positive.
- **Lichen planus** — '6 Ps' (purple, polygonal, pruritic, planar papules), Wickham's striae.
- **Pemphigus vulgaris** — flaccid intraepidermal blisters, Nikolsky positive; **bullous pemphigoid** — tense subepidermal blisters, elderly.
- **Leprosy** — hypopigmented anaesthetic patches, thickened nerves (ulnar, common peroneal).

Memory hook: pemphig**us** blisters are flaccid and burst on **us** (Nikolsky sign positive, high in the skin); pemphig**oid** blisters are tense and sit deep. Lichen planus is the **6 Ps**: purple, pruritic, polygonal, planar papules and plaques.

Exanthematous fevers and seasonal outbreaks

The order in which rashes appear (chickenpox day 1, scarlet fever day 2, smallpox day 3, measles day 4, typhus day 5) and the monsoon fevers of Goa are the most-asked items.

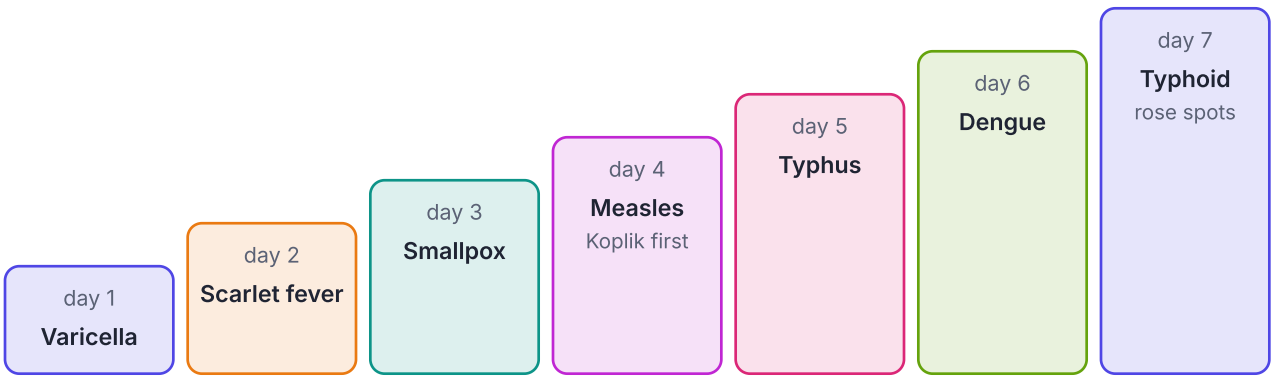
Exanthematous fevers

Disease	Agent	Rash and clues	Complication / note
Measles	Paramyxovirus	Prodrome of the 3 Cs (cough, coryza, conjunctivitis); **Koplik spots** on buccal mucosa; maculopapular rash on day 4, behind ears → face → trunk, leaves brown staining	Pneumonia (commonest cause of death), diarrhoea, SSPE years later; vitamin A; MR vaccine at 9–12 and 16–24 months
Rubella (German measles)	Togavirus	Mild; rash day 1; **postauricular/suboccipital lymph nodes**	Congenital rubella: **PDA**, cataract, deafness (MO 2024)
Chickenpox	Varicella-zoster	**Pleomorphic rash** (macule, papule, vesicle, crust at once), centripetal (trunk), 'dewdrop on a rose petal', spares palms/soles	Secondary infection, pneumonia (adults); zoster later; acyclovir
Smallpox (eradicated 1980)	Variola	Centrifugal, all lesions at the same stage, palms and soles involved	Last case India 1975

Disease	Agent	Rash and clues	Complication / note
Scarlet fever	Group A Streptococcus	Sandpaper rash, **strawberry tongue**, circumoral pallor, Pastia's lines, desquamation	Rheumatic fever, glomerulonephritis
Roseola infantum	HHV-6	High fever 3–4 days, rash appears **as fever falls**	Febrile seizures
Erythema infectiosum (fifth disease)	Parvovirus B19	'**Slapped cheek**' then lacy rash	Aplastic crisis in sickle cell; hydrops fetalis
Hand, foot and mouth disease	Coxsackie A16, enterovirus 71	Oral ulcers + vesicles on palms and soles	Common in Indian toddlers, monsoon
Dengue	Flavivirus	Fever, headache, retro-orbital pain, myalgia ('breakbone'), flushing; convalescent rash 'isles of white in a sea of red'	See below

The rash staircase: which fever shows its rash on which day

"Very Sick Persons Must Take Double Tablets", one step per day of fever



rash staircase · day 1 to day 7

Climb the stairs one day at a time: the later the rash, the higher the step.

Seasonal outbreaks in Goa (monsoon and post-monsoon)

Disease	Spread	Clinical clue	Key test	Drug / control
Dengue	**Aedes aegypti** (day-biter, clean stored water)	Fever, myalgia, rash, low platelets; warning signs (abdominal pain, vomiting, bleeding, lethargy, rising haematocrit)	**NS1 (day 1–5)**, IgM ELISA (after day 5)	Fluids, paracetamol; **avoid NSAIDs/aspirin**; source reduction
Chikungunya	Aedes	Fever with severe, lasting polyarthralgia	IgM ELISA / RT-PCR	Symptomatic
Malaria	Female Anopheles (An. stephensi in urban Goa)	Periodic fever with chills	Smear, RDT	Vivax: chloroquine + primaquine 14 days; falciparum: ACT (artesunate + sulfadoxine–pyrimethamine; artemether–lumefantrine in NE states); severe: IV artesunate
Leptospirosis	Rat urine in water/soil (paddy fields, floods)	Conjunctival suffusion, calf pain, jaundice + renal failure (Weil's)	IgM ELISA, MAT	Doxycycline/penicillin; doxycycline prophylaxis 200 mg weekly
Scrub typhus	Larval mite (chigger)	**Eschar**, fever, lymphadenopathy	IgM ELISA	Doxycycline
Typhoid	Faeco-oral	Step-ladder fever, relative bradycardia, coated tongue	Blood culture (week 1)	Ceftriaxone, azithromycin
Cholera / acute gastroenteritis	Water	Rice-water stools	Stool culture	**ORS** + zinc (children); doxycycline
Hepatitis A and E	Water	Jaundice outbreaks	IgM antibodies	Supportive; safe water
Influenza (H1N1)	Droplets	Fever, cough, sore throat	RT-PCR	Oseltamivir for high-risk (category B/C)
Kyasanur Forest Disease	Tick (Haemaphysalis); monkey deaths	Fever, bleeding, later neurological phase; Sattari taluka	RT-PCR	KFD vaccine in endemic areas

Disease	Spread	Clinical clue	Key test	Drug / control
Japanese encephalitis	Culex; pigs amplify	Encephalitis in children	IgM ELISA (CSF)	JE vaccine

Dengue classification (WHO 2009): dengue without warning signs, dengue with warning signs, severe dengue (shock from plasma leakage, severe bleeding, organ impairment). Rising haematocrit ≥ 20% with falling platelets = plasma leakage. Platelet transfusion only for bleeding or very low counts.

Genus epidemicus: in an outbreak, the homoeopath studies many cases to find the remedy covering most of them (§100–102), then uses it for treatment and prophylaxis — for example Eupatorium perfoliatum in dengue-like fevers, Arsenicum album was promoted by AYUSH as prophylaxis during COVID-19.

Memory hook: the day the rash appears: "**V**ery **S**ick **P**ersons **M**ust **T**ake **D**ouble **T**ablets": **V**aricella day 1, **S**carlet fever day 2, small**P**ox day 3, **M**easles day 4, **T**yphus day 5, **D**engue day 6, **T**yphoid day 7. Koplik spots come before the measles rash.

Homoeopathic therapeutics: leading remedies by disease

GPSC asks a disease plus a keynote and expects one remedy. Learn the keynote column; the remedy follows from it. Prescribe always on the totality — these are exam pointers, not a substitute for individualisation.

Disease	Remedy	Keynote that identifies it
Hypertension	Glonoinum	Bursting, throbbing head; worse sun and heat; surging of blood
Hypertension	Aurum metallicum	High BP with depression, suicidal thoughts, arteriosclerosis
Hypertension (mother tincture)	Rauwolfia serpentina	Organ remedy for high BP
Angina pectoris	Cactus grandiflorus	Heart constricted as if by an iron band
Angina pectoris	Latrodectus mactans	Violent precordial pain to left arm and fingers, numbness, anxiety

Disease	Remedy	Keynote that identifies it
Heart failure / weak heart	Digitalis	Very slow, irregular, weak pulse; fears heart will stop if he moves
Heart tonic (mother tincture)	Crataegus oxyacantha	Weak myocardium, irregular pulse
Bronchial asthma	Arsenicum album	Worse after midnight (1–3 a.m.), restless, anxious, must sit up, better warmth
Bronchial asthma	Ipecacuanha	Wheeze with constant nausea, rattling but nothing comes up
Asthma (hydrogenoid)	Natrum sulphuricum	Worse damp weather and at 4–5 a.m.; loose morning stool
COPD / chronic bronchitis	Antimonium tartaricum	Coarse rattling, little expectoration, drowsy, cyanosed (elderly)
COPD / air hunger	Carbo vegetabilis	Wants to be fanned, cold breath, collapse
Stroke / paralysis	Causticum	Gradual paralysis, right-sided, hoarseness, sympathetic
Stroke (haemorrhage)	Arnica montana	Bruised feeling, apoplexy, bed feels too hard
Stroke (apoplectic stupor)	Opium	Stertorous breathing, red face, painless
Epilepsy	Cuprum metallicum	Violent cramps, aura from knees; begins in fingers and toes
Epilepsy	Cicuta virosa	Opisthotonos, after head injury
Epilepsy	Bufo rana	Fits during sleep or with sexual excitement; foolish
Peptic ulcer / acidity	Nux vomica	Sedentary, stimulants, irritable; ineffectual urging
Peptic ulcer	Argentum nitricum	Craves sweets which disagree; belching; ulcer pain radiates
Peptic ulcer (bleeding)	Kali bichromicum	Round punched-out ulcers; pain in small spots
GERD / heartburn	Robinia pseudacacia	Intense sour acidity, worse at night
IBS	Argentum nitricum	Diarrhoea from anticipation, sweets

Disease	Remedy	Keynote that identifies it
IBS	Lycopodium	Bloating 4–8 p.m., flatulence, anxiety about performance
IBS / diarrhoea	Aloe socotrina	Morning urging, insecurity of rectum
Hepatitis / jaundice	Chelidonium majus	Pain under the right scapula; liver remedy
Hepatitis	Phosphorus	Craving for cold water, bleeding, fatty liver
Cirrhosis	Lycopodium	Liver remedy with ascites, right-sided
Gallstone colic	Berberis vulgaris / Calcarea carb	Radiating pains (Berberis); stone diathesis
Cholecystitis	Bryonia alba	Worse slightest motion, better pressure, thirst
Diabetes (mother tincture)	Syzygium jambolanum	Organ remedy; lowers glucose
Diabetes	Uranium nitricum	Great emaciation, debility, polyuria
Diabetes	Phosphoric acid	Diabetes after grief, milky urine at night, apathy
Hypothyroidism	Calcarea carbonica	Fat, flabby, chilly, sweaty head, slow
Hyperthyroidism	Iodium	Ravenous hunger yet emaciation, hot, restless
Hyperthyroidism / exophthalmos	Lycopus virginicus	Palpitation with exophthalmic goitre
Gout	Colchicum autumnale	Great toe, nausea from the smell of food, worse touch
Gout	Ledum palustre	Ascending gout, better cold applications
Gout (chronic, uric acid)	Urtica urens	Uric acid diathesis, burning itch
Atopic eczema	Graphites	Honey-like sticky discharge, cracks in folds
Eczema / itch	Sulphur	Burning itch worse warmth and washing
Eczema with crusts	Mezereum	Thick crusts with pus beneath
Psoriasis	Arsenicum album	Dry, scaly, burning, better warmth
Psoriasis	Petroleum	Deep cracks, worse winter

Disease	Remedy	Keynote that identifies it
Urticaria	Apis mellifica	Stinging, burning wheals, better cold
Urticaria	Urtica urens	Burning, stinging wheals after shellfish
Urticaria	Rhus toxicodendron	Urticaria from getting wet, restless
Measles	Pulsatilla	Bland catarrh, thirstless, weepy; also Euphrasia for eye signs
Measles (early)	Aconite / Belladonna	Sudden high fever
Chickenpox	Rhus toxicodendron / Antimonium crudum	Itchy vesicles; restless (Rhus)
Mumps	Pilocarpinus / Mercurius	Parotitis with salivation
Dengue / bone-breaking fever	Eupatorium perfoliatum	Bone-breaking pains
Influenza	Gelsemium	Drowsy, dull, trembling, thirstless
Influenza / cold	Arsenicum album	Watery burning coryza, restless, anxious
Malaria	China officinalis	Periodicity, weakness from loss of fluids
Malaria	Natrum muriaticum	Chill at 10 a.m., bursting headache
Typhoid	Baptisia tinctoria	Besotted face, body feels scattered
Leptospirosis / jaundice with myalgia	Bryonia, Phosphorus	By totality

The chief epidemic-prophylaxis examples GPSC can ask: **Belladonna** for scarlet fever (Hahnemann, 1801), **Eupatorium** for dengue-like fevers, **Arsenicum album 30** promoted by the AYUSH ministry for COVID-19, and **Influenzinum** for flu.

Memory hook: the **genus epidemicus** pairs to learn: **B**elladonna for scarlet fever ("**B**right red"), **E**upatorium for dengue ("break-bon**E**"), **A**rsenicum album for the AYUSH COVID-19 advisory, **I**nfluenzinum for flu.

Rapid revision: 60 one-liners

Read this the night before and on the morning of the exam.

Numbers

1. Hypertension ≥ 140/90; target in diabetes < 130/80.
2. Hypertensive emergency ≥ 180/120 with organ damage; lower by ≤ 25% in the first hour.
3. Salt < 5 g/day (WHO).
4. Diabetes: FPG ≥ 126, 2-h OGTT ≥ 200, HbA1c ≥ 6.5%, random ≥ 200 with symptoms.
5. Prediabetes: FPG 100–125, 2-h 140–199, HbA1c 5.7–6.4%.
6. HbA1c target < 7%; it reflects 2–3 months.
7. Hypoglycaemia < 70 mg/dL.
8. COPD: post-bronchodilator FEV1/FVC < 0.70.
9. Asthma reversibility: FEV1 ↑ ≥ 12% and ≥ 200 mL.
10. COPD oxygen target 88–92%; LTOT if PaO2 ≤ 55 mmHg, ≥ 15 h/day.
11. Stroke thrombolysis within 4.5 h; status epilepticus > 5 min.
12. GCS 3–15; ≤ 8 = intubate.
13. Uric acid target < 6 mg/dL on allopurinol.
14. SAM: MUAC < 11.5 cm.
15. Microalbuminuria 30–300 mg/day.

Drugs of choice

16. Acute asthma: salbutamol (inhaled SABA).
17. Asthma controller: inhaled corticosteroid.
18. Anaphylaxis: IM adrenaline 0.5 mg (1:1000).
19. Type 2 diabetes first line: metformin; pregnancy: insulin.
20. DKA: IV normal saline first, then insulin; watch potassium.
21. Hyperthyroidism in first trimester: propylthiouracil.
22. Hypothyroidism: levothyroxine on an empty stomach.
23. Absence seizures: ethosuximide; status epilepticus: IV lorazepam.
24. Hepatic encephalopathy: lactulose (+ rifaximin).
25. Ascites: spironolactone.
26. Variceal bleed: octreotide/terlipressin + band ligation; prevention: propranolol.
27. Acute gout: NSAIDs/colchicine; long-term: allopurinol.
28. H. pylori: PPI + clarithromycin + amoxicillin.
29. Scabies: permethrin 5%.
30. Scrub typhus and leptospirosis: doxycycline.

31. Paracetamol poisoning: N-acetylcysteine.
32. Hypertension in pregnancy: labetalol, nifedipine, methyldopa (no ACE inhibitors).

Investigations of choice

33. Acute stroke / SAH: non-contrast CT.
34. Gallstones: ultrasound.
35. GERD gold standard: 24-h pH study.
36. Bronchiectasis: HRCT.
37. Gout: urate crystals in joint fluid.
38. Thyroid screening: TSH; thyroid nodule: FNAC.
39. MI: troponin.
40. Pulmonary TB (NTEP): NAAT (CBNAAT/TrueNat).
41. Dengue day 1–5: NS1.
42. Heart failure marker: BNP.

Signs and associations

43. Koplik spots = measles; strawberry tongue = scarlet fever; slapped cheek = parvovirus B19.
44. Rash on day 1 chickenpox, day 4 measles; rash as fever falls = roseola.
45. Auspitz sign, Koebner phenomenon, nail pitting = psoriasis.
46. Murphy's sign = acute cholecystitis; Charcot's triad = cholangitis.
47. Asterixis = hepatic encephalopathy (also CO_2 retention, uraemia).
48. Kussmaul breathing + fruity breath = DKA.
49. Exophthalmos + pretibial myxoedema = Graves'.
50. Podagra = gout; needle-shaped negatively birefringent crystals.
51. Pellagra = dermatitis, diarrhoea, dementia.
52. Dry cough with ACE inhibitors; ankle oedema with amlodipine.

⚠ Traps

53. Type 1, not type 2, diabetes is ketosis-prone.
54. Duodenal ulcer pain is relieved by food; gastric ulcer pain worsens.
55. Bell's palsy involves the forehead; stroke (UMN) spares it.
56. Valproate is the most teratogenic antiepileptic; carbamazepine worsens myoclonic epilepsy.
57. Avoid aspirin/NSAIDs in dengue.

58. Do not start allopurinol during an acute gout attack.

59. Rhus tox: better from continued motion; Bryonia: worse from any motion.

60. Arsenicum asthma is after midnight; Natrum sulph asthma is 4–5 a.m. in damp weather.

One-glance recall card

Topic 7 recall card: the 15 facts asked most

Cover the small lines, say the fact from the big cue, then check

140/90 hypertension on repeated readings	**ACE-i cough** dry cough from bradykinin; switch to an ARB	**Troponin** rises in hours, stays about 1 to 2 weeks
Reversible asthma comes and goes; COPD persists	**FEV1/FVC < 0.7** spirometry shows airflow obstruction	**Plain CT first** rules out bleeding before thrombolysis
Forehead spared upper motor neurone; Bell's palsy takes it all	**Food helps?** duodenal ulcer eases with food; gastric worsens	**AST:ALT > 2** alcoholic liver disease; viral: ALT higher
126 · 200 · 6.5 fasting, 2-hour, HbA1c cut-offs	**TSH first** best single thyroid test; papillary commonest	**Needle crystals** gout: negatively birefringent, big toe
6 Ps lichen planus: purple, pruritic, polygonal...	**VSPMTDT** rash days 1-7: varicella, scarlet ... typhoid	**Genus epidemicus** Bell scarlet, Eup dengue, Ars COVID, Influenzinum

Topic 7 recall card · 15 tiles

For the last ten minutes before the exam: cover the small lines of each tile and say the fact from the big cue.

Practice MCQs (50)

Time yourself: 50 questions in 35 minutes. Cover the key until you finish.

1. Hypertension is diagnosed at a repeated clinic blood pressure of at least: (A) 120/80 mmHg (B) 130/85 mmHg (C) 140/90 mmHg (D) 160/100 mmHg
2. The BP goal for a diabetic with hypertension is: (A) below 130/80 mmHg (B) below 150/90 mmHg (C) below 160/100 mmHg (D) below 110/70 mmHg

3. A dry, persistent cough is a typical side-effect of: (A) amlodipine (B) ACE inhibitors (C) thiazides (D) beta-blockers
4. Which antihypertensive is contraindicated in pregnancy? (A) labetalol (B) nifedipine (C) methyldopa (D) enalapril
5. The most specific biomarker of myocardial infarction is: (A) CK total (B) LDH (C) troponin (D) AST
6. ST elevation in leads II, III and aVF indicates infarction of the: (A) anterior wall (B) lateral wall (C) septum only (D) inferior wall
7. The commonest cause of death in the first hours after MI is: (A) cardiogenic shock (B) free-wall rupture (C) Dressler syndrome (D) ventricular fibrillation
8. Furosemide is indicated in: (A) hypokalaemia (B) gout (C) acute pulmonary oedema (D) hypovolaemic shock
9. COPD is confirmed by a post-bronchodilator FEV1/FVC ratio below: (A) 0.70 (B) 0.90 (C) 0.85 (D) 0.50
10. The drug for an acute asthma attack is inhaled: (A) budesonide alone (B) montelukast (C) theophylline (D) salbutamol
11. Panacinar emphysema in a young non-smoker suggests: (A) cystic fibrosis (B) asthma (C) alpha-1 antitrypsin deficiency (D) silicosis
12. Interventions proven to improve survival in COPD are smoking cessation and: (A) inhaled steroids (B) long-term oxygen therapy (C) antibiotics (D) theophylline
13. In a COPD exacerbation, the oxygen saturation target is: (A) 88–92% (B) 98–100% (C) 80–84% (D) 94–98%
14. The commonest risk factor for stroke is: (A) diabetes (B) hypertension (C) smoking (D) alcohol
15. The first investigation in suspected acute stroke is: (A) MRI angiography (B) EEG (C) lumbar puncture (D) non-contrast CT brain
16. IV thrombolysis for ischaemic stroke is given within: (A) 12 hours (B) 24 hours (C) 4.5 hours (D) 48 hours
17. Thunderclap headache with neck stiffness most suggests: (A) migraine (B) tension headache (C) sinusitis (D) subarachnoid haemorrhage
18. A cardiac source of embolic stroke in rheumatic heart disease is usually: (A) atrial fibrillation (B) ventricular septal defect (C) aortic stenosis (D) pericarditis
19. The drug of choice for absence seizures is: (A) phenytoin (B) carbamazepine (C) phenobarbitone (D) ethosuximide
20. The first drug for status epilepticus is IV: (A) phenytoin (B) lorazepam (C) valproate (D) levetiracetam
21. The commonest cause of acquired epilepsy in India is: (A) tuberculoma (B) neurocysticercosis (C) head injury (D) brain tumour
22. Bell's palsy differs from stroke facial weakness because it: (A) involves the forehead (B) spares the forehead (C) is always bilateral (D) has no taste loss
23. Epigastric pain relieved by food and waking the patient at night suggests: (A) duodenal ulcer (B) gastric ulcer (C) GERD (D) cholecystitis

24. The best non-invasive test to confirm H. pylori eradication is: (A) serology (B) barium meal (C) urea breath test (D) CT
25. The main mechanism of GERD is: (A) LES hypertension (B) transient lower oesophageal sphincter relaxation (C) hiatus hernia alone (D) delayed gastric emptying
26. The gold standard test for GERD is: (A) 24-hour pH monitoring (B) barium swallow (C) endoscopy (D) manometry
27. A red-flag feature that argues against IBS is: (A) rectal bleeding (B) pain relieved by defecation (C) bloating (D) stress-related symptoms
28. Hepatitis with the highest mortality in pregnancy is: (A) hepatitis A (B) hepatitis B (C) hepatitis C (D) hepatitis E
29. Asterixis in a cirrhotic patient indicates: (A) hepatic encephalopathy (B) portal vein thrombosis (C) SBP (D) hepatorenal syndrome
30. The drug of first choice for cirrhotic ascites is: (A) furosemide alone (B) mannitol (C) spironolactone (D) acetazolamide
31. Murphy's sign is positive in: (A) acute cholecystitis (B) acute appendicitis (C) pancreatitis (D) peptic ulcer
32. The investigation of choice for gallstones is: (A) plain X-ray (B) ultrasonography (C) CT (D) HIDA scan
33. Diabetes is diagnosed by an HbA1c of at least: (A) 5.7% (B) 7.5% (C) 6.5% (D) 8.0%
34. The diabetic emergency most typical of type 1 diabetes is: (A) hyperosmolar state (B) diabetic ketoacidosis (C) lactic acidosis (D) hypothyroid coma
35. The first step in treating DKA is: (A) IV insulin bolus alone (B) IV normal saline (C) IV bicarbonate (D) oral glucose
36. The first-line oral drug for type 2 diabetes is: (A) glimepiride (B) pioglitazone (C) acarbose (D) metformin
37. The earliest sign of diabetic nephropathy is: (A) raised creatinine (B) haematuria (C) microalbuminuria (D) oedema
38. In primary hypothyroidism the thyroid profile shows: (A) low TSH, high T4 (B) low TSH, low T4 (C) high TSH, low T4 (D) normal TSH, high T4
39. The antithyroid drug preferred in the first trimester is: (A) carbimazole (B) methimazole (C) propylthiouracil (D) radioactive iodine
40. Exophthalmos and pretibial myxoedema are specific to: (A) toxic multinodular goitre (B) thyroid adenoma (C) Hashimoto's thyroiditis (D) Graves' disease
41. Needle-shaped negatively birefringent crystals in joint fluid are diagnostic of: (A) pseudogout (B) rheumatoid arthritis (C) gout (D) septic arthritis
42. Allopurinol acts by inhibiting: (A) uricase (B) xanthine oxidase (C) cyclo-oxygenase (D) tubular secretion
43. Dermatitis, diarrhoea and dementia indicate deficiency of: (A) thiamine (B) riboflavin (C) vitamin B12 (D) niacin

14. Silvery scales with pinpoint bleeding on scraping (Auspitz sign) indicate: (A) psoriasis (B) eczema (C) lichen planus (D) tinea
15. The first drug in anaphylaxis is: (A) IV hydrocortisone (B) IM adrenaline (C) oral cetirizine (D) IV fluids alone
16. Koplik spots are seen in: (A) measles (B) rubella (C) chickenpox (D) scarlet fever
17. A pleomorphic vesicular rash, centripetal and sparing palms and soles, is: (A) smallpox (B) chickenpox (C) measles (D) hand, foot and mouth disease
18. Drug to be avoided in dengue fever: (A) paracetamol (B) aspirin (C) ORS (D) IV fluids
19. For dengue with bone-breaking pains, the leading homoeopathic remedy is: (A) Bryonia (B) Rhus tox (C) Arnica (D) Eupatorium perfoliatum
20. Asthma worse after midnight with anxiety, restlessness and relief from warmth calls for: (A) Arsenicum album (B) Natrum sulph (C) Ipecac (D) Antimonium tart

Answer key

Q	Ans	Why
1	C	Indian and ISH guidelines
2	A	GPSC MO 2024 keyed 120–130/70–79
3	B	Bradykinin accumulation
4	D	ACE inhibitors and ARBs are teratogenic
5	C	Rises in 3–4 h, stays 7–10 days
6	D	Usually the right coronary artery
7	D	Arrhythmia
8	C	Loop diuretic (GPSC MO 2024)
9	A	GOLD criterion
10	D	Short-acting beta-2 agonist
11	C	Lower lobes
12	B	LTOT ≥ 15 h/day
13	A	Avoid CO_2 retention

Q	Ans	Why
14	B	Most important modifiable risk
15	D	Excludes haemorrhage before thrombolysis
16	C	Alteplase or tenecteplase
17	D	Ruptured berry aneurysm
18	A	GPSC MO 2025 case
19	D	Or valproate; 3 Hz spike-wave EEG
20	B	Benzodiazepine first
21	B	Taenia solium
22	A	Lower motor neurone lesion
23	A	Hunger pain
24	C	Or stool antigen
25	B	GPSC MO 2024
26	A	Endoscopy for alarm features
27	A	Also weight loss, nocturnal diarrhoea, age > 50
28	D	Fulminant failure, about 20%
29	A	Flapping tremor
30	C	Aldosterone antagonist
31	A	Inspiration arrested on RUQ palpation
32	B	Most stones are not radio-opaque
33	C	5.7–6.4% = prediabetes
34	B	GPSC MO 2025 trap
35	B	Then insulin infusion and potassium
36	D	No hypoglycaemia
37	C	30–300 mg/day

Q	Ans	Why
38	C	Hashimoto's commonest where iodine is sufficient
39	C	GPSC MO 2025
40	D	TSH-receptor antibodies
41	C	Monosodium urate
42	B	Start after the acute attack settles
43	D	Pellagra
44	A	Extensor plaques
45	B	0.5 mg of 1:1000 in the thigh
46	A	Buccal mucosa, before the rash
47	B	Lesions at different stages
48	B	Bleeding risk
49	D	Bone-breaking pains
50	A	1–3 a.m. aggravation

Score guide: 40+ is exam-ready; below 30, reread the tables for the questions you missed.

Sources

GPSC question papers with answer keys (questions quoted in shortened form):

- Medical Officer CBRT, 28 Sep 2025
- Medical Officer CBRT, 28 Sep 2024
- GPSC question paper archive

Standard texts the notes follow (facts written from them, not copied; check a doubtful figure against your own edition):

- *Davidson's Principles and Practice of Medicine* — cardiology, respiratory, GI, endocrine, infections.
- *API Textbook of Medicine* — Indian context: tropical fevers, neurocysticercosis, snake bite.
- *Harrison's Principles of Internal Medicine* — definitions and cut-offs.

- Guidelines: ISH 2020 hypertension, GINA and GOLD (asthma, COPD), ADA standards of care (diabetes), WHO 2009 dengue classification, Indian national programme treatment guidelines (malaria, TB).
- Homoeopathic therapeutics: Boericke's *Pocket Manual of Homoeopathic Materia Medica with Repertory*, Kent's *Lectures on Materia Medica*, Clarke's *Dictionary of Practical Materia Medica*.

Guideline numbers are as of mid-2026; if a question words a cut-off differently, follow the Indian guideline it names.

Topic 8: Obstetrics, Gynaecology & Paediatrics

How to use this guide

Topic 8 gives about 2 of the 25 core marks. Questions are one-line facts: a number (weeks, days, mL, mg/dL), a definition, a milestone age, a vaccine schedule, or the homoeopathic remedy for a clear picture. Learn the tables and diagrams first.

Syllabus heading (Advt 06/2026)	What it covers	Section
Antenatal care and labour	Normal pregnancy, high-risk conditions (preeclampsia, gestational diabetes), stages of labour	8.1
Gynaecological pathologies	Dysmenorrhoea, abnormal uterine bleeding, PCOD, pelvic inflammatory disease, common pelvic neoplasms	8.2
Neonatology and child development	Developmental milestones, neonatal jaundice, infant feeding, malnutrition, rickets	8.3
Paediatric therapeutics and emergencies	Dentition disorders, respiratory infections, ADHD, enuresis, national child immunisation	8.4

- **Standard books.** D. C. Dutta, *Textbook of Obstetrics*; D. C. Dutta, *Textbook of Gynecology*; *Shaw's Textbook of Gynaecology*; O. P. Ghai, *Essential Pediatrics*; K. Park, *Textbook of Preventive and Social Medicine* (immunisation).
- Homoeopathic therapeutics: Boericke's *Materia Medica with Repertory*; Allen's *Keynotes*; Lilienthal, *Homoeopathic Therapeutics*; Dewey, *Practical Homoeopathic Therapeutics*; Guernsey, *Application of the Principles and Practice of Homoeopathy to Obstetrics*.
- Vaccine schedules follow the Universal Immunization Programme (UIP), Ministry of Health and Family Welfare.

Plan to 1 November. One section a day for four days, then the rapid-revision sheet and the MCQs; after that, ten minutes a day on the rapid-revision sheet.

8.1 Antenatal care and labour

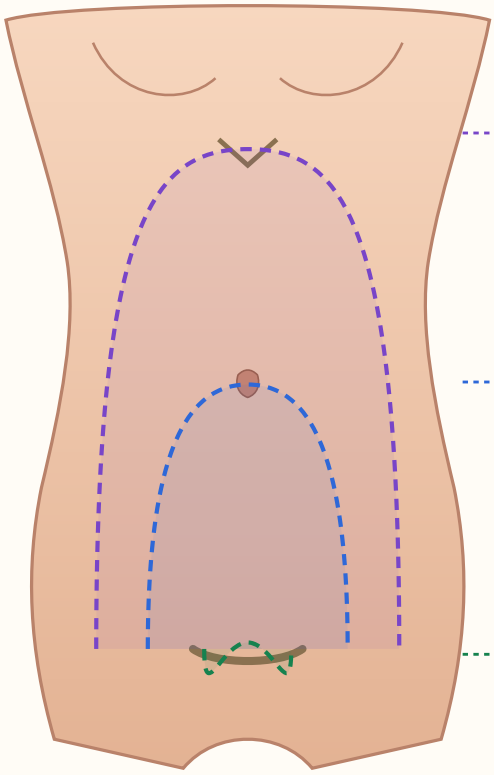
Most questions here are numbers: pregnancy length, number of visits, blood-pressure and glucose cut-offs, and the start and end of each stage of labour.

Normal pregnancy

Item	Fact
Duration	280 days (40 weeks) from the first day of the last menstrual period (LMP); 266 days from ovulation
Naegele's rule	Expected date of delivery = LMP + 9 months + 7 days (or LMP − 3 months + 7 days + 1 year)
Term	37–42 completed weeks; preterm before 37; post-term after 42
Trimesters	First up to 12 weeks, second 13–28, third 29–40
Earliest sign	Missed period; urine β-hCG positive about 10–14 days after conception (peaks at 8–10 weeks)
Quickening	First fetal movements felt at 18–20 weeks (primigravida), 16–18 weeks (multigravida)
Fetal heart	Heard on Doppler at 10–12 weeks; seen on ultrasound at 6 weeks; normal rate 110–160/min
Fundal height	Pubic symphysis at 12 weeks, umbilicus at 24 weeks, xiphisternum at 36 weeks
Weight gain	About 11–12 kg over the pregnancy
Signs	Hegar's sign (softening of the lower uterine segment, 6–10 weeks); Chadwick's sign (bluish vagina and cervix); Goodell's sign (soft cervix)

Fundal height: weeks you can feel

Where the top of the uterus lies as pregnancy advances



36 weeks
fundus at the xiphisternum

24 weeks
fundus at the umbilicus

12 weeks
fundus at the pubic symphysis

Also asked
Quickening: 18–20 weeks (primi), 16–18 weeks (multi)
Fetal heart: 110–160/min
Trimesters: to 12, 13–28, 29–40

Fundal height · symphysis at 12 weeks, umbilicus at 24, xiphisternum at 36

- **Antenatal care in India.** At least **4 antenatal visits** (first in the first trimester, ideally before 12 weeks); the WHO now advises 8 contacts.
- Each visit: weight, blood pressure, urine for protein and sugar, haemoglobin, abdominal examination.
- **Iron and folic acid**: 1 tablet daily (60 mg elemental iron + 500 µg folic acid) for at least 180 days from the second trimester, continued 180 days after delivery; **folic acid 400 µg from before conception** prevents neural tube defects.
- **Calcium** 1 g daily from 14 weeks.
- **Tetanus–diphtheria (Td)** vaccine: 2 doses 4 weeks apart in the first pregnancy (Td-1 early, Td-2 four weeks later), or 1 booster if vaccinated in the previous 3 years.

Hook: "4 visits, 180 days, 2 doses" — four antenatal visits, 180 days of iron, two Td doses.

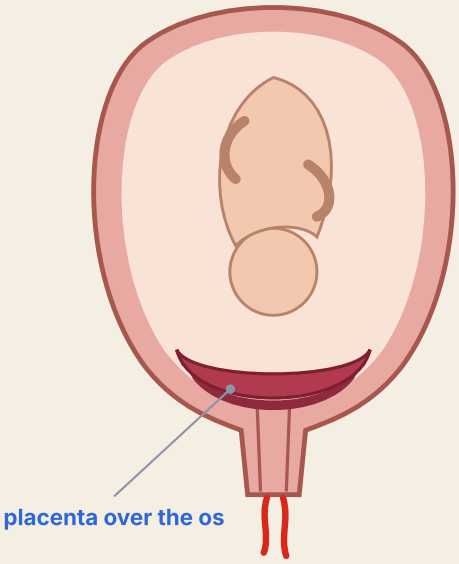
High-risk conditions

Condition	Definition	Key facts
Gestational hypertension	BP 140/90 mmHg or more after 20 weeks, no proteinuria	Returns to normal within 12 weeks after delivery
Preeclampsia	Hypertension after 20 weeks **with proteinuria** (300 mg or more in 24 h) or other organ involvement	Severe: BP 160/110 or more, headache, visual disturbance, epigastric pain, low platelets; HELLP syndrome (haemolysis, elevated liver enzymes, low platelets)
Eclampsia	Preeclampsia + **convulsions**	An obstetric emergency; delivery is the definitive cure
Gestational diabetes (GDM)	Glucose intolerance first recognised in pregnancy	India (DIPSI, single step): 75 g oral glucose, **2-hour value 140 mg/dL or more** = GDM, regardless of fasting state; screen at the first visit and again at 24–28 weeks
Anaemia in pregnancy	Hb below 11 g/dL (WHO)	Severe below 7 g/dL; commonest cause is iron deficiency
Antepartum haemorrhage	Bleeding after 28 weeks (India)	**Placenta praevia**: painless, causeless, recurrent bright red bleeding. **Abruptio placentae**: painful bleeding, tense tender uterus
Ectopic pregnancy	Implantation outside the uterine cavity	Commonest site: **ampulla of the fallopian tube**; triad: amenorrhoea, abdominal pain, vaginal bleeding
Hyperemesis gravidarum	Severe vomiting with dehydration, weight loss, ketosis	Commoner in multiple pregnancy and molar pregnancy
Rh isoimmunisation	Rh-negative mother, Rh-positive fetus	Haemolytic disease of the newborn; anti-D given to Rh-negative unsensitised mothers

Painless or painful bleeding?

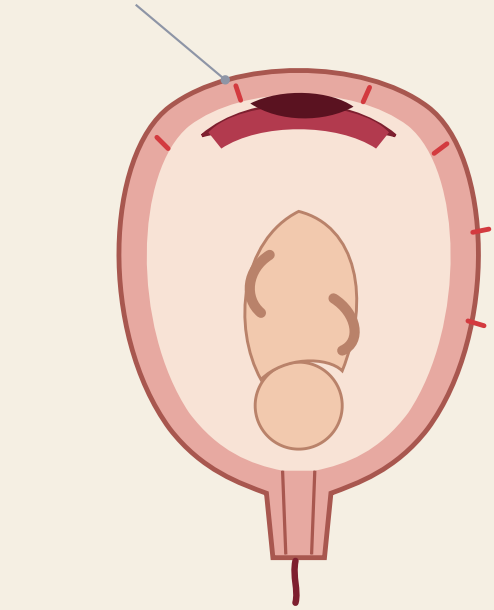
Placenta praevia vs abruptio placentae

Placenta praevia



PLACENTA	Low, over or near the cervix
BLEEDING	PAINLESS, causeless, recurrent, bright red
UTERUS	Soft, not tender

Abruptio placentae



PLACENTA	Normal site, separates early; clot builds up behind it
BLEEDING	PAINFUL
UTERUS	Tense and tender

Antepartum haemorrhage = bleeding after 28 weeks of pregnancy (Indian definition).

Antepartum haemorrhage · placenta praevia is painless, abruptio is painful

Complications of GDM for the baby. Macrosomia (birth weight over 4 kg), shoulder dystocia, neonatal hypoglycaemia, respiratory distress.

Stages of labour

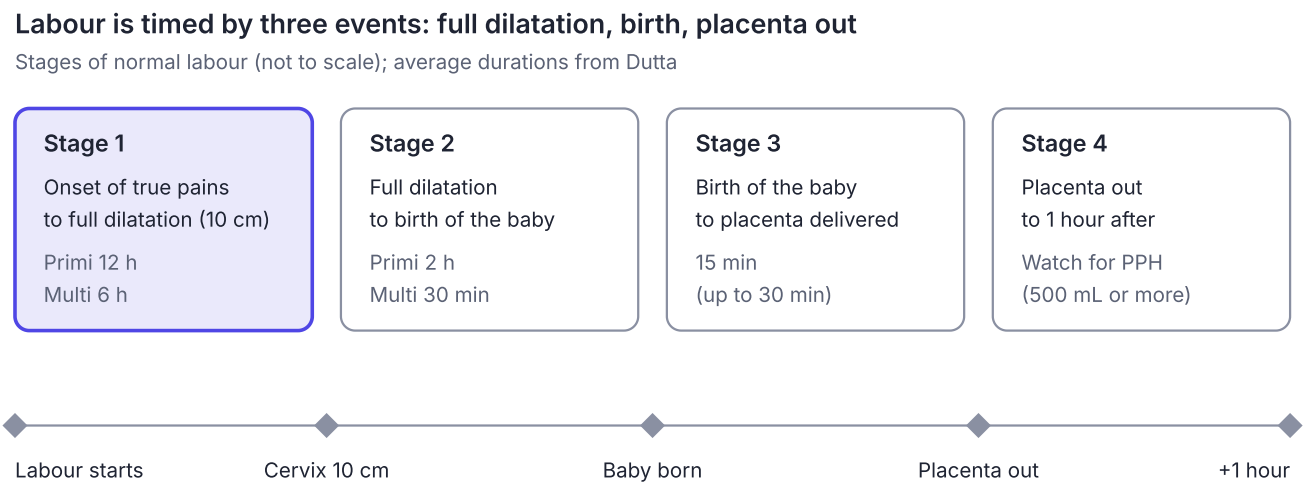
Stage	From	To	Average duration
First	Onset of true labour pains	**Full dilatation of the cervix (10 cm)**	Primigravida 12 h; multipara 6 h
Second	Full dilatation	**Delivery of the baby**	Primigravida 2 h (1 h before epidural era); multipara 30 min
Third	Delivery of the baby	**Delivery of the placenta and membranes**	15 min (up to 30 min)

Stage	From	To	Average duration
Fourth	Delivery of the placenta	1 hour after (observation)	1 h

The first stage has a **latent phase** (up to 4 cm, slow) and an **active phase** (from 4–5 cm, about 1 cm per hour in a primigravida). **True labour pains** are regular, increase in frequency and strength, cause cervical dilatation and are not relieved by sedation; false pains are irregular and do not dilate the cervix.

Cardinal movements of the fetus (vertex): Engagement, Descent, Flexion, Internal rotation, Extension, Restitution, External rotation, Expulsion. **Hook: "Every Decent Fetus Is Exceptionally Restless, Externally Expelled."**

- **Postpartum haemorrhage (PPH).** Blood loss of 500 mL or more after vaginal delivery (1,000 mL after caesarean).
- Commonest cause: **uterine atony**.
- Causes: the "4 Ts" — Tone, Trauma, Tissue (retained placenta), Thrombin (clotting).
- **Puerperium** lasts 6 weeks; lochia: rubra (days 1–4), serosa (5–9), alba (10–15).



Stages of labour · the event that ends each stage and its duration

Each stage ends with one event on the line below it: remember the events and you remember the stages. Stage 1 (highlighted) is the longest.

Homoeopathic therapeutics in pregnancy and labour

Clinical picture	Remedy
Morning sickness: constant nausea not relieved by vomiting, clean tongue	Ipecacuanha

Clinical picture	Remedy
Nausea from the smell or thought of food, with indifference to the family and bearing-down	Sepia
Nausea and vomiting better from eating, worse in the morning, with irritability	Nux vomica
False labour pains, spasmodic, flying about; rigid cervix in a nervous woman	Caulophyllum, Gelsemium
Labour pains with weeping, changeable mood, wants fresh air; malposition	Pulsatilla
Labour with intolerable pain, angry, "cannot bear it"	Chamomilla
After-pains and bruised soreness after delivery	Arnica
Bright red, profuse bleeding with nausea	Ipecacuanha

In eclampsia, antepartum haemorrhage and PPH, the first duty is emergency obstetric care and referral; remedies support, never replace, it.

8.2 Gynaecological pathologies

Know each condition's definition, its classic triad or criteria, and one or two remedies with a clear picture.

Menstrual cycle in one line

- Cycle 28 ± 7 days; flow 2–7 days; blood loss about 35 mL (over 80 mL = heavy).
- Follicular phase (FSH, oestrogen) → **ovulation on about day 14, 14 days before the next period, triggered by the LH surge** → luteal phase (**progesterone** from the corpus luteum keeps the endometrium secretory).
- Menarche about 12–13 years; menopause about 45–51 years (12 months without a period).

Dysmenorrhoea

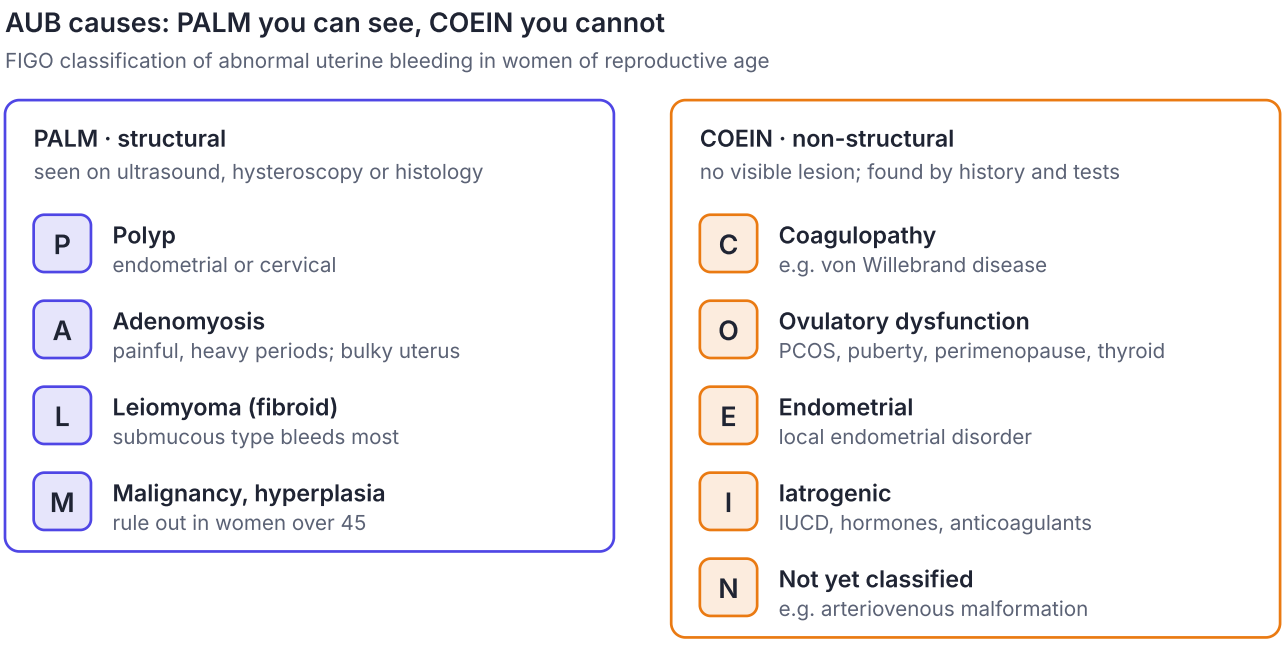
	Primary (spasmodic)	Secondary (congestive)
Cause	No pelvic disease; excess prostaglandins	Pelvic pathology: endometriosis, adenomyosis, PID, fibroid, IUCD
Onset	Within 6–12 months of menarche (once cycles become ovulatory)	Later in life, often after 30

	Primary (spasmodic)	Secondary (congestive)
Pain	Cramping, starts with the flow, lasts 12–72 h; lower abdomen and thighs	Dull, starts days before the flow, may continue after it
Course	Improves after childbirth and with age	Worsens with the disease

Abnormal uterine bleeding (AUB)

AUB is any bleeding abnormal in frequency, regularity, duration or volume. FIGO classifies its causes as **PALM-COEIN** (diagram below): the PALM causes are structural and seen on imaging or histology; the COEIN causes are not.

Old term	Meaning
Menorrhagia (heavy menstrual bleeding)	Regular cycles, heavy or prolonged flow (over 80 mL or over 7 days)
Metrorrhagia	Irregular bleeding between periods
Polymenorrhoea	Cycles shorter than 21 days
Oligomenorrhoea	Cycles longer than 35 days
Amenorrhoea	Primary: no menarche by 15 years (with secondary sex characters) or by 13 (without them). Secondary: no period for 3 cycles or 6 months; **commonest cause is pregnancy**



PALM-COEIN · 4 structural and 5 non-structural causes of AUB

The commonest causes of AUB at puberty and perimenopause are ovulatory (O); in a woman over 45, malignancy (M) must be excluded first.

Polycystic ovarian disease (PCOD / PCOS)

- **Rotterdam criteria (2003): any 2 of 3.** (1) Oligo- or anovulation; (2) clinical or biochemical hyperandrogenism (hirsutism, acne, raised testosterone); (3) polycystic ovaries on ultrasound (12 or more follicles of 2–9 mm, or ovarian volume over 10 mL).
- Features: irregular periods, obesity, acanthosis nigricans (insulin resistance), infertility. **LH : FSH ratio often over 2 : 1.**
- Long-term risks: type 2 diabetes, endometrial hyperplasia and cancer, cardiovascular disease.
- First-line management is lifestyle: a 5–10% weight loss often restores ovulation.

Pelvic inflammatory disease (PID)

Ascending infection of the upper genital tract (endometritis, salpingitis, tubo-ovarian abscess), mostly **Chlamydia trachomatis and Neisseria gonorrhoeae**. Signs: lower abdominal pain, fever, purulent discharge, **cervical motion tenderness**, adnexal tenderness. Complications: **infertility, ectopic pregnancy**, chronic pelvic pain, Fitz-Hugh–Curtis syndrome (perihepatitis).

Common pelvic neoplasms

Tumour	Key facts
Leiomyoma (fibroid)	Commonest benign tumour of the uterus (and of women); oestrogen-dependent, shrinks after menopause. Types: intramural (commonest), subserous, submucous (bleeds most). Heavy menstrual bleeding, pressure symptoms, infertility. Red degeneration in pregnancy; sarcomatous change under 0.5%
Adenomyosis	Endometrium within the myometrium; uniformly enlarged, tender uterus; painful, heavy periods in multiparous women over 35
Endometriosis	Endometrium outside the uterus, commonest on the ovary ("chocolate cyst"); dysmenorrhoea, dyspareunia, infertility
Ovarian tumours	Commonest benign: serous cystadenoma; commonest in young women: dermoid (mature cystic teratoma); **CA-125** marker for epithelial ovarian cancer
Carcinoma cervix	Commonest genital cancer in Indian women; caused by **HPV 16 and 18**; screening by Pap smear and VIA (visual inspection with acetic acid); HPV vaccine prevents it

Tumour	Key facts
Endometrial carcinoma	Postmenopausal bleeding; risk: obesity, nulliparity, diabetes, unopposed oestrogen

Hook: "Fibroid = Fat uterus that Floods; Endometriosis = Elsewhere."

Homoeopathic therapeutics

Clinical picture	Remedy
Cramping dysmenorrhoea, better bending double and from hard pressure	Colocynthis, Magnesia phosphorica (better from warmth)
Bearing-down sensation as if everything would come out, must cross the legs; indifference	Sepia
Scanty, late, changeable periods in a mild, weepy, thirstless girl who wants consolation	Pulsatilla
Profuse, bright red flow with gushes, worse from motion	Sabina
Heavy, dark, clotted menses; pain from back to pubis	Sabina, Trillium
Sudden profuse bright bleeding with nausea	Ipecacuanha
Fibroids with heavy bleeding in a chilly, fat woman who sweats on the head	Calcarea carbonica
Right-sided ovarian pain, hot, throbbing	Belladonna (acute), Lycopodium (chronic)
Left-sided ovarian pain, worse after sleep, cannot bear tight clothing	Lachesis

8.3 Neonatology and child development

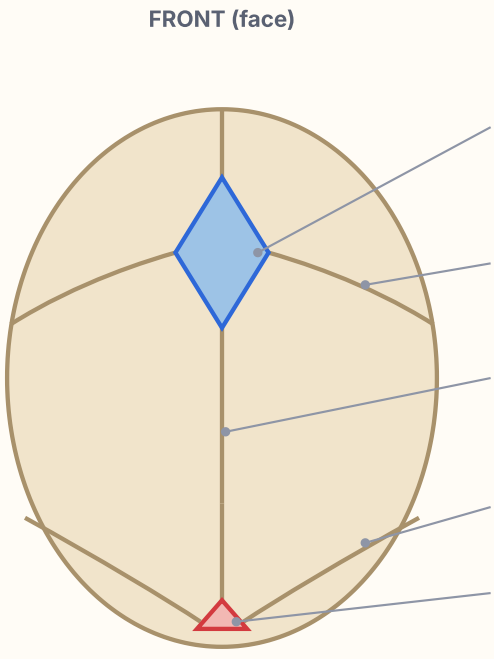
Milestone ages and growth numbers are the most-asked facts in paediatrics; neonatal jaundice is asked as "physiological or pathological".

The newborn

Item	Normal
Birth weight	2.5–4 kg (India average about 2.8 kg); **low birth weight under 2.5 kg**
Length / head circumference	50 cm / 34–35 cm
Heart rate / respiratory rate	120–160 / 40–60 per minute
APGAR score	At **1 and 5 minutes**: Appearance (colour), Pulse, Grimace (reflex), Activity (tone), Respiration; each 0–2; 7–10 normal
Weight changes	Loses up to 10% in the first week, regains birth weight by **10–14 days**; doubles by **5 months**, triples by **1 year**, quadruples by **2 years**
Fontanelles	**Posterior closes by 2–3 months; anterior by 12–18 months**
First milk tooth	Lower central incisor at about **6 months**; all 20 milk teeth by 2–2½ years; first permanent tooth (first molar) at 6 years
Primitive reflexes	Moro, rooting, sucking, grasp: present at birth, disappear by 3–6 months

Fontanelles of the newborn skull

Seen from above



Anterior fontanelle
diamond-shaped; closes by 12–18 months

Coronal suture

Sagittal suture
between the parietal bones

Lambdoid suture

Posterior fontanelle
triangular; closes by 2–3 months

Late-closing fontanelles: rickets; in homoeopathy, the Calcarea phosphorica child

Fontanelles · posterior closes by 2–3 months, anterior by 12–18 months

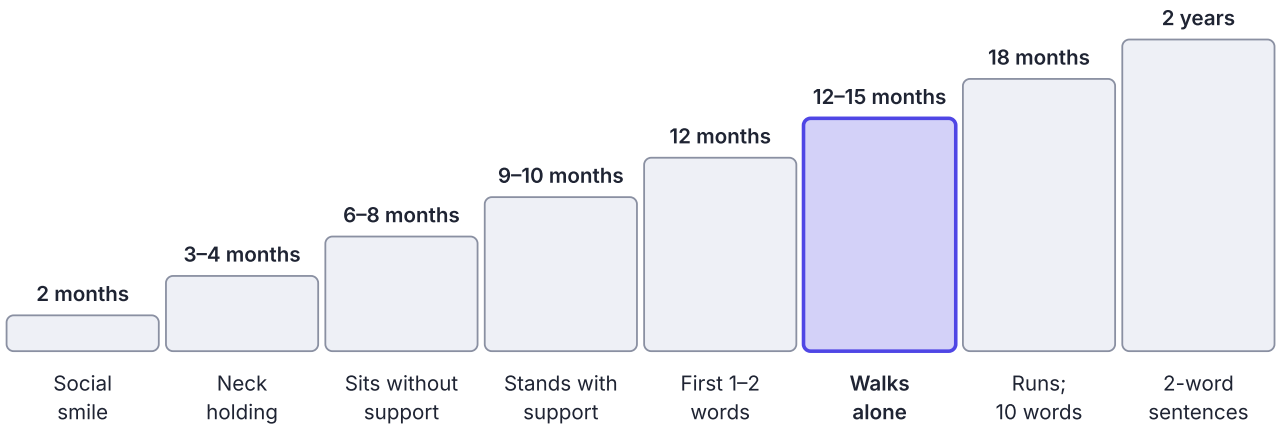
Kangaroo mother care: skin-to-skin contact and exclusive breastfeeding for low-birth-weight and preterm babies.

Developmental milestones

The ladder below shows the key milestone ages. **Red flags**: no social smile by 3 months, no head control by 4 months, not sitting by 9 months, not walking by 18 months, no words by 18 months.

Milestones build on each other: smile, neck, sit, stand, talk, walk

Usual ages from Ghai (steps not to scale); walking alone is highlighted as the most-asked milestone



Developmental milestones · 8 key steps from 2 months to 2 years

Climb the staircase left to right: each skill needs the one before it, so the order is as important as the age.

Age	Gross motor	Fine motor	Language	Social
2 months	Lifts head in prone	Follows objects	Coos	**Social smile**
3–4 months	**Neck holding**	Hands open, reaches	Laughs aloud	Recognises mother
6 months	Sits with support; rolls over	Transfers objects hand to hand	Monosyllables (ba, da)	Stranger anxiety begins
8–9 months	**Sits without support**; crawls	Pincer grasp begins (9 months)	Bisyllables (mama, dada)	Waves bye-bye
12 months	**Walks with support / stands alone**	Mature pincer grasp	**1–2 meaningful words**	Plays simple games
15 months	**Walks alone**	Tower of 2 cubes; scribbles	Jargon	Points to wants

Age	Gross motor	Fine motor	Language	Social
18 months	Runs, climbs stairs with help	Tower of 3–4 cubes	10 words	Copies parents
2 years	Up and down stairs, 2 feet per step	Tower of 6–7 cubes	**2-word sentences**	Parallel play
3 years	Rides tricycle; climbs stairs one foot per step	Copies a circle; tower of 9	Sentences, knows age and sex	Shares toys
4–5 years	Hops, skips	Copies a square (4), triangle (5)	Tells a story	Cooperative play

Hook: "Smile 2, Neck 3, Sit 6–8, Stand 9–10, Walk 12–15, Words 12, Phrases 24."

Neonatal jaundice

	Physiological	Pathological
Appears	**After 24 hours** (day 2–3)	**Within 24 hours** of birth (always pathological)
Peak	Day 3–5 (term)	Any time; rises fast (over 5 mg/dL per day)
Lasts	Disappears by **14 days** (term), 21 days (preterm)	Beyond 2 weeks (term)
Bilirubin	Unconjugated, under about 15 mg/dL	Conjugated over 2 mg/dL is always pathological
Causes	Immature liver conjugation, short RBC life	Rh or ABO incompatibility, sepsis, G6PD deficiency, biliary atresia, hypothyroidism

Very high unconjugated bilirubin can cross into the brain: **kernicterus** (bilirubin encephalopathy). Treated with phototherapy (blue light) and, if severe, exchange transfusion. Breast-milk jaundice appears after the first week and is harmless.

Infant feeding

- **Breastfeeding within 1 hour of birth**; colostrum (first 3–4 days) is rich in protein, IgA and vitamin A.
- **Exclusive breastfeeding for 6 months**, then complementary feeding from **6 months**, continuing breastfeeding to **2 years or beyond**.

- Breast milk has more lactose and less protein than cow's milk; whey : casein about 60 : 40; protective factors: secretory IgA, lactoferrin, lysozyme, bifidus factor.
- Energy of breast milk about 67 kcal/100 mL; a baby needs about 150 mL/kg/day.

Malnutrition and rickets

Condition	Key facts
Severe acute malnutrition (6–59 months)	**MUAC under 11.5 cm**, or weight-for-height under −3 SD, or bilateral pitting oedema
Kwashiorkor	Protein deficiency with adequate calories: **oedema**, flaky-paint dermatosis, sparse pale hair (flag sign), fatty liver, apathy; weight 60–80% of expected
Marasmus	Deficiency of total calories: **severe wasting, no oedema**, "old man" face, alert and hungry; weight under 60%
Rickets	Vitamin D deficiency before the growth plates fuse: bowed legs, **rachitic rosary** (costochondral beading), Harrison's sulcus, craniotabes, delayed fontanelle closure, widened wrists; raised **alkaline phosphatase**, low phosphate
Osteomalacia	Vitamin D deficiency after the growth plates fuse (adults)
Vitamin A deficiency	Night blindness first, then Bitot's spots, keratomalacia; dose under the national programme: 9 doses from 9 months to 5 years

Hook: "Kwashiorkor = Kwater (oedema); Marasmus = Muscle wasting."

Homoeopathic therapeutics. Marasmus with emaciation from the neck downward, ravenous appetite yet thin: **Natrum muriaticum**; emaciation beginning in the legs: **Abrotanum**. Rickets with large head, open fontanelles, sour sweat on the head and slow teething in a fat, flabby child: **Calcarea carbonica**; in a thin child craving smoked meat: **Calcarea phosphorica**; with offensive foot sweat and weak ankles: **Silicea**.

8.4 Paediatric therapeutics and emergencies

The vaccine schedule is the single most-asked item in this section; the rest is definitions plus a remedy picture.

Dentition disorders

Teething begins at about 6 months with drooling, irritability and gum rubbing. Fever above 38.5 °C or diarrhoea should not be blamed on teething without looking for another cause.

Picture	Remedy
Painful teething, one cheek red and hot, the other pale; angry, wants to be carried; green, hot, offensive stools	**Chamomilla**
Slow, delayed dentition; late-closing fontanelles; thin child craving smoked meat	**Calcarea phosphorica**
Delayed teething in a fat, flabby child with sour sweat on the head	**Calcarea carbonica**
Teeth decay as soon as they appear; black teeth	**Kreosotum**
Teething with convulsions, hot head, cold limbs, dilated pupils	Belladonna

Respiratory infections

WHO/IMNCI fast-breathing cut-offs (count for a full minute): **60 or more** under 2 months, **50 or more** from 2 to 12 months, **40 or more** from 1 to 5 years. Chest indrawing or any general danger sign (unable to drink, vomits everything, convulsions, lethargy) means severe illness: refer urgently.

Picture	Remedy
Sudden high fever after exposure to dry cold wind, anxious, restless, thirsty (first stage)	**Aconitum napellus**
High fever with flushed face, throbbing, dilated pupils, no thirst	**Belladonna**
Coarse rattling of mucus in the chest, but cannot cough it up; drowsy	**Antimonium tartaricum**
Croup: dry, barking, "sawing" cough, better from warm food and drink	**Spongia tosta**
Croup with loose rattling and choking, worse from cold air, very sensitive	**Hepar sulphuris**
Whooping cough: spasmodic, barking, worse after midnight, with nosebleed	**Drosera**
Bland coryza and cough, thick yellow-green discharge, wants open air	Pulsatilla

Behavioural problems

ADHD (attention deficit hyperactivity disorder). Inattention, hyperactivity and impulsivity before the age of 12, in two or more settings (home and school), for at least 6 months; commoner in boys.

Remedies for restless, destructive, hurried children: **Tarentula hispanica** (restless, destructive, music ameliorates), **Stramonium** (fears, violence, night terrors), **Hyoscyamus** (jealous, silly, exposing), **Tuberculinum** (wants change, destructive).

- **Enuresis.** Involuntary wetting **after the age of 5 years** (developmental bladder control).
- Primary (never dry) is commonest and often familial; secondary (dry for 6 months, then wetting) suggests stress, infection or diabetes.
- Remedies: **Equisetum** (enuresis with dreams of urinating), **Causticum** (in the first sleep, also on coughing), **Kreosotum** (deep sleep, hard to wake), **Sepia** (in the first sleep, girls).

Other. Breath-holding spells (6 months to 5 years, harmless); night terrors; **encopresis** (soiling after age 4); pica (eating non-food items, look for iron deficiency).

National immunisation (Universal Immunization Programme)

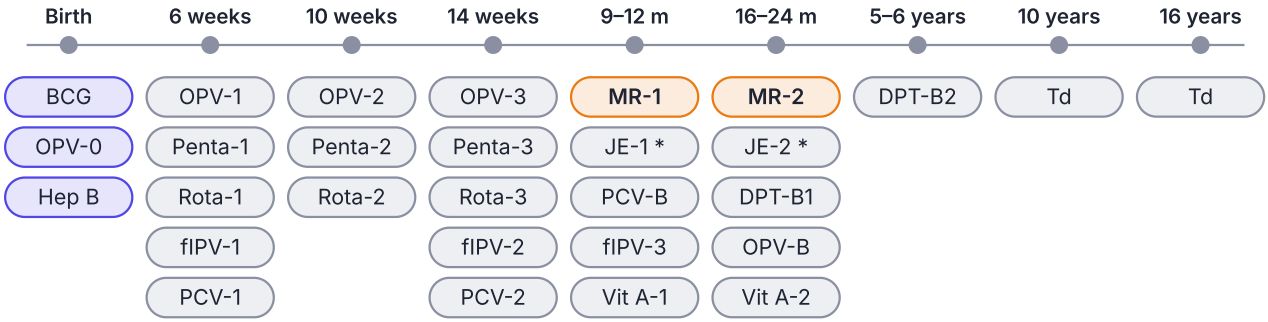
Age	Vaccines
Birth	**BCG**, **OPV-0**, **Hepatitis B birth dose** (within 24 h)
6 weeks	OPV-1, **Pentavalent-1** (DPT + Hep B + Hib), Rotavirus-1, **fIPV-1**, PCV-1
10 weeks	OPV-2, Pentavalent-2, Rotavirus-2
14 weeks	OPV-3, Pentavalent-3, Rotavirus-3, fIPV-2, PCV-2
9–12 months	**MR-1** (measles–rubella), JE-1 (endemic districts), PCV booster, fIPV-3, **Vitamin A (1st dose)**
16–24 months	**MR-2**, JE-2, **DPT booster-1**, OPV booster, Vitamin A (2nd dose; then every 6 months to 5 years)
5–6 years	**DPT booster-2**
10 years and 16 years	**Td**
Pregnant women	Td-1, Td-2 (or booster)

- **Facts asked.** BCG is given intradermally on the left upper arm, up to 1 year of age.
- OPV is the only oral vaccine besides rotavirus.
- Measles vaccine is live and given at 9 months because maternal antibodies interfere earlier.
- Mission Indradhanush targets children and pregnant women missed by routine immunisation; Pulse Polio gives OPV to all children under 5.
- India was declared polio-free in 2014.

- **Hook: "Birth = BOH (BCG, OPV, Hep B); 6-10-14 weeks = POR (Penta, OPV, Rota); 9 months = MR."**

Most vaccines are given in the first year: birth, 6, 10, 14 weeks, 9 months

Universal Immunization Programme, India (not to scale)



Penta = DPT + Hepatitis B + Hib. fIPV = fractional injectable polio vaccine. PCV-B = pneumococcal booster. * JE: endemic areas. Vitamin A: 1st dose at 9 months, then every 6 months to 5 years (9 doses). Pregnant women: Td-1 and Td-2 four weeks apart. Blue = birth doses; orange = measles–rubella (MR), the most-asked schedule item.

National immunisation schedule · 9 visits from birth to 16 years

Read the chart column by column: the three primary doses at 6, 10 and 14 weeks are identical except that PCV and fIPV skip 10 weeks.

Rapid revision: numbers and one-line facts

Area	Fact
Pregnancy	280 days from LMP; EDD = LMP + 9 months + 7 days; term 37–42 weeks
Pregnancy	Quickening 18–20 weeks (primi); fundus at umbilicus 24 weeks; FHR 110–160/min
Antenatal	4 visits minimum; IFA 180 days; Td 2 doses; folic acid 400 µg before conception
High risk	Preeclampsia = BP 140/90 after 20 weeks + proteinuria; eclampsia = + convulsions
High risk	GDM (DIPSI) = 2-hour glucose 140 mg/dL or more after 75 g; anaemia in pregnancy Hb under 11
High risk	Painless bleeding = placenta praevia; painful = abruptio; ectopic commonest in the ampulla
Labour	1st stage ends at full dilatation (10 cm); 2nd at delivery of the baby; 3rd at delivery of the placenta
Labour	PPH = 500 mL or more (vaginal); commonest cause atony; puerperium 6 weeks
Gynaecology	Ovulation 14 days before the next period (LH surge); progesterone keeps the luteal phase
Gynaecology	Primary dysmenorrhoea: no pelvic disease, starts with the flow; secondary: endometriosis, PID, fibroid
Gynaecology	AUB = PALM-COEIN; PCOS = Rotterdam 2 of 3; PID causes infertility and ectopic pregnancy

Area	Fact
Gynaecology	Commonest uterine tumour = fibroid; cervical cancer = HPV 16, 18; CA-125 = ovarian cancer
Newborn	LBW under 2.5 kg; APGAR at 1 and 5 min; regains birth weight by 10–14 days
Growth	Weight doubles 5 months, triples 1 year; anterior fontanelle closes 12–18 months
Milestones	Smile 2 m; neck 3–4 m; sit 6–8 m; stand 9–10 m; walk 12–15 m; 2-word phrases 2 years
Jaundice	Within 24 hours = pathological; physiological after 24 h, gone by 14 days
Feeding	Breastfeed within 1 hour; exclusive 6 months; continue to 2 years
Malnutrition	SAM = MUAC under 11.5 cm; kwashiorkor = oedema; marasmus = wasting
Rickets	Rosary, bowed legs, wide wrists; ALP raised
Paediatrics	Enuresis after 5 years; ADHD onset before 12 years, 6 months, two settings
Breathing	Fast breathing: 60 (under 2 m), 50 (2–12 m), 40 (1–5 y) per minute
Vaccines	Birth BCG, OPV-0, Hep B; 6-10-14 weeks Penta, OPV, Rota (+fIPV, PCV); 9 m MR-1; 16–24 m MR-2, DPT-B1; 5 y DPT-B2; 10 and 16 y Td
Remedies	Teething = Chamomilla; slow dentition = Calc phos; croup = Spongia; rattling = Ant tart; enuresis in first sleep = Causticum
Remedies	Dysmenorrhoea bending double = Colocynth; bearing down = Sepia; bright red gushing = Sabina

One-glance recall card

Topic 8 recall card: the 15 facts asked most

Cover the small lines, say the fact from the big cue, then check

4 ANC visits at least four antenatal check-ups	**Folic acid** before and early in pregnancy: no neural tube defect	**500 mL** PPH after vaginal birth; commonest cause: atony
6 weeks length of the puerperium	**10 cm** full dilatation ends the first stage of labour	**PALM-COEIN** 4 structural and 5 non-structural AUB causes
2 of 3 Rotterdam criteria for PCOS	**Chlamydia** with gonorrhoea, the usual cause of PID	**APGAR** 5 signs at 1 and 5 min; 7 to 10 is normal
First 24 hours jaundice then is never physiological	**6 months** exclusive breastfeeding	**Oedema** kwashiorkor has it; marasmus is wasting
60 · 50 · 40 fast breathing: under 2 mo, 2-12 mo, 1-5 yr	**2 months** social smile; walks alone 12-15 months	**BCG at birth** intradermal, left upper arm

Topic 8 recall card · 15 tiles

For the last ten minutes before the exam: cover the small lines of each tile and say the fact from the big cue.

Practice MCQs

Twenty questions grouped by syllabus heading; the answer key with a one-line reason follows.

Antenatal care and labour

1. If the LMP was 10 January, the expected date of delivery is about: (a) 3 October (b) 17 October (c) 10 October (d) 24 October
2. Preeclampsia is hypertension after 20 weeks with: (a) glycosuria (b) proteinuria or organ involvement (c) polyhydramnios (d) anaemia
3. In India, gestational diabetes is diagnosed (DIPSI) when the 2-hour glucose after 75 g is: (a) 120 mg/dL or more (b) 140 mg/dL or more (c) 180 mg/dL or more (d) 200 mg/dL or more

4. The second stage of labour ends with: (a) full dilatation of the cervix (b) delivery of the baby (c) delivery of the placenta (d) onset of pains
5. Painless, recurrent, bright red bleeding in the third trimester suggests: (a) abruptio placentae (b) placenta praevia (c) ruptured uterus (d) cervical polyp

Gynaecology

6. Primary dysmenorrhoea is defined as painful menstruation: (a) due to endometriosis (b) without pelvic pathology (c) only after childbirth (d) due to fibroid
7. In the PALM-COEIN classification, "L" stands for: (a) luteal phase defect (b) leiomyoma (c) lymphoma (d) lichen sclerosus
8. Polycystic ovarian syndrome is diagnosed by the Rotterdam criteria when present are: (a) all 3 criteria (b) any 2 of 3 (c) any 1 of 3 (d) raised LH only
9. The most important long-term complication of pelvic inflammatory disease is: (a) menorrhagia (b) infertility (c) fibroid (d) PCOS
10. Bearing-down sensation with indifference to family and better from vigorous exercise suggests: (a) Pulsatilla (b) Sepia (c) Sabina (d) Belladonna

Neonatology and development

11. A child usually sits without support at: (a) 3 months (b) 6–8 months (c) 12 months (d) 15 months
12. Jaundice appearing on the first day of life is: (a) physiological (b) pathological (c) breast-milk jaundice (d) normal in term babies
13. The anterior fontanelle closes by: (a) 2–3 months (b) 6 months (c) 12–18 months (d) 3 years
14. Oedema with flaky-paint skin and a fatty liver in a malnourished child indicates: (a) marasmus (b) kwashiorkor (c) rickets (d) scurvy
15. Exclusive breastfeeding is recommended for the first: (a) 2 months (b) 4 months (c) 6 months (d) 12 months

Paediatric therapeutics and immunisation

16. Which vaccines are given at birth under the UIP? (a) BCG, OPV-0, Hepatitis B (b) DPT, OPV, Hib (c) MR, Vitamin A (d) Rotavirus, PCV
17. The first dose of measles–rubella vaccine is given at: (a) birth (b) 6 weeks (c) 9–12 months (d) 5 years
18. Enuresis is diagnosed only after the age of: (a) 2 years (b) 3 years (c) 5 years (d) 10 years

19. Painful dentition in an angry child who wants to be carried, with one cheek red: (a) Calcarea phos (b) Chamomilla (c) Belladonna (d) Kreosotum

20. Croup with dry, barking, sawing cough better from warm drinks: (a) Spongia tosta (b) Antimonium tart (c) Drosera (d) Pulsatilla

Answer key

Q	Answer	Why
1	b	10 Jan + 9 months + 7 days = 17 October
2	b	Proteinuria (300 mg/24 h) or end-organ involvement
3	b	DIPSI single-step 75 g test, 2-hour value 140 mg/dL
4	b	2nd stage = full dilatation to delivery of the baby
5	b	Praevia is painless; abruptio is painful
6	b	No pelvic pathology; prostaglandin-mediated
7	b	PALM = polyp, adenomyosis, leiomyoma, malignancy
8	b	Any 2 of anovulation, hyperandrogenism, polycystic ovaries
9	b	Tubal damage: infertility and ectopic pregnancy
10	b	Sepia keynotes
11	b	Sits without support 6–8 months
12	b	Jaundice within 24 hours is always pathological
13	c	Anterior 12–18 months; posterior 2–3 months
14	b	Oedema = kwashiorkor
15	c	6 months, then complementary feeding
16	a	BCG, OPV-0, Hep B birth dose
17	c	MR-1 at 9–12 months, MR-2 at 16–24 months
18	c	Bladder control is expected by 5 years
19	b	Chamomilla keynote

Q	Answer	Why
20	a	Spongia: dry sawing croup better warm drinks

Topic 9: Surgery, Allied Specialties & First-Aid

How to use this guide

Topic 9 gives about 2 of the 25 core marks. Questions ask a definition (sinus or fistula), a classic sign (Colles' fracture, halos round lights), a first-aid step, or the homoeopathic remedy for a type of injury.

Syllabus heading (Advt 06/2026)	What it covers	Section
General surgical conditions	Wounds, abscesses, shock, ulcers, burns, perioperative care	9.1
Traumatology and orthopaedics	Fractures, dislocations, sprains, osteomyelitis, osteoarthritis, cervical and lumbar spondylosis	9.2
ENT and ophthalmology	Sinusitis, otitis media, tonsillitis, conjunctivitis, cataract, glaucoma	9.3
Surgical therapeutics and first aid	Homoeopathic remedies for mechanical injuries, pre- and post-operative care, snakebite and poisoning	9.4

- **Standard books.** S. Das, *A Concise Textbook of Surgery* and *A Manual on Clinical Surgery*; *Bailey & Love's Short Practice of Surgery*; Maheshwari, *Essential Orthopaedics*; P. L. Dhingra, *Diseases of Ear, Nose and Throat*; A. K. Khurana, *Comprehensive Ophthalmology*; St John Ambulance / Indian Red Cross first-aid manual; national snakebite protocol (MoHFW).
- Homoeopathic surgical therapeutics: Boericke's *Materia Medica*; Allen's *Keynotes*; E. B. Nash, *Leaders in Homoeopathic Therapeutics*; Shepherd and Lilienthal for injuries.
- Drug treatments from conventional pharmacology are left out; first aid and surgical principles are kept because the syllabus asks for them.

9.1 General surgical conditions

Definitions carry most marks here: the types of wound healing, sinus versus fistula, the kinds of shock and ulcer, and burn depth with the rule of nines.

Wounds and healing

Term	Meaning
Healing by first intention (primary)	Clean incised wound, edges brought together; minimal scar (surgical incision)

Term	Meaning
Healing by second intention (secondary)	Wound left open or infected; fills with granulation tissue from the base; wide scar, contraction
Third intention (delayed primary closure)	Contaminated wound cleaned and closed after 3–5 days
Phases of healing	Haemostasis → inflammation (days 1–4) → proliferation, granulation (day 4 to 3 weeks) → remodelling (up to a year or more)
Types of wound	Incised (clean edges, bleeds freely), lacerated (torn, irregular), abrasion (graze), contusion (bruise, skin intact), punctured (deep, risk of tetanus)
Keloid	Scar growing **beyond** the wound margins, recurs after excision; hypertrophic scar stays **within** the margins
Wound dehiscence	Bursting open of a surgical wound (burst abdomen), often days 5–8

Factors delaying healing: infection (commonest), poor blood supply, diabetes, malnutrition (protein, vitamin C, zinc), steroids, foreign body, old age.

Abscess, cellulitis, sinus and fistula

Term	Definition
Abscess	Localised collection of pus in a cavity lined by granulation tissue (pyogenic membrane); commonest organism *Staphylococcus aureus*; "where there is pus, let it out" (incision and drainage)
Cellulitis	Diffuse spreading infection of skin and subcutaneous tissue; commonest organism *Streptococcus pyogenes*
Sinus	A blind track lined by granulation tissue, opening onto an epithelial surface (e.g. pilonidal sinus)
Fistula	An abnormal communication between **two epithelial surfaces** (e.g. fistula-in-ano)
Carbuncle	Infective necrosis of several adjacent hair follicles; back of the neck in diabetics
Furuncle (boil)	Infection of a single hair follicle

Hook: "A Sinus is a Single opening; a Fistula is a Further connection."

Shock

Shock is inadequate tissue perfusion. The commonest type after injury is **hypovolaemic**.

Type	Cause	Signs
Hypovolaemic	Blood or fluid loss (haemorrhage, burns, vomiting, diarrhoea)	Cold clammy skin, rapid thready pulse, low BP, low urine output
Cardiogenic	Pump failure (myocardial infarction)	Cold skin, raised JVP
Septic (distributive)	Infection releasing toxins	Early warm flushed skin and bounding pulse, later cold
Anaphylactic (distributive)	Allergy	Urticaria, stridor, wheeze, low BP
Neurogenic (distributive)	Spinal cord injury	Low BP with **slow** pulse, warm skin
Obstructive	Tension pneumothorax, cardiac tamponade	Raised JVP, low BP

- **Blood loss and signs (ATLS classes)**: up to 15% (750 mL) little change; 15–30% pulse over 100; 30–40% BP falls; over 40% life-threatening.
- Urine output under 0.5 mL/kg/h signals poor perfusion.
- **Hook for shock first aid: lie flat, legs raised, keep warm, nothing by mouth, shift to hospital.**

Ulcers

An ulcer is a break in the continuity of an epithelial surface.

Ulcer	Edge	Typical site
Healing ulcer	Sloping, bluish-pink	—
Tuberculous	**Undermined**	Neck (lymph nodes)
Rodent ulcer (basal cell carcinoma)	**Rolled (raised, beaded)**	Upper face, inner canthus
Squamous cell carcinoma (epithelioma)	**Everted**	Lower lip, old scar (Marjolin's ulcer)
Syphilitic (gumma)	**Punched out**	Leg, palate
Varicose (venous)	Sloping	**Above the medial malleolus (gaiter area)**

Ulcer	Edge	Typical site
Trophic (neuropathic)	Punched out, painless	Heel, ball of the foot (diabetes, leprosy)

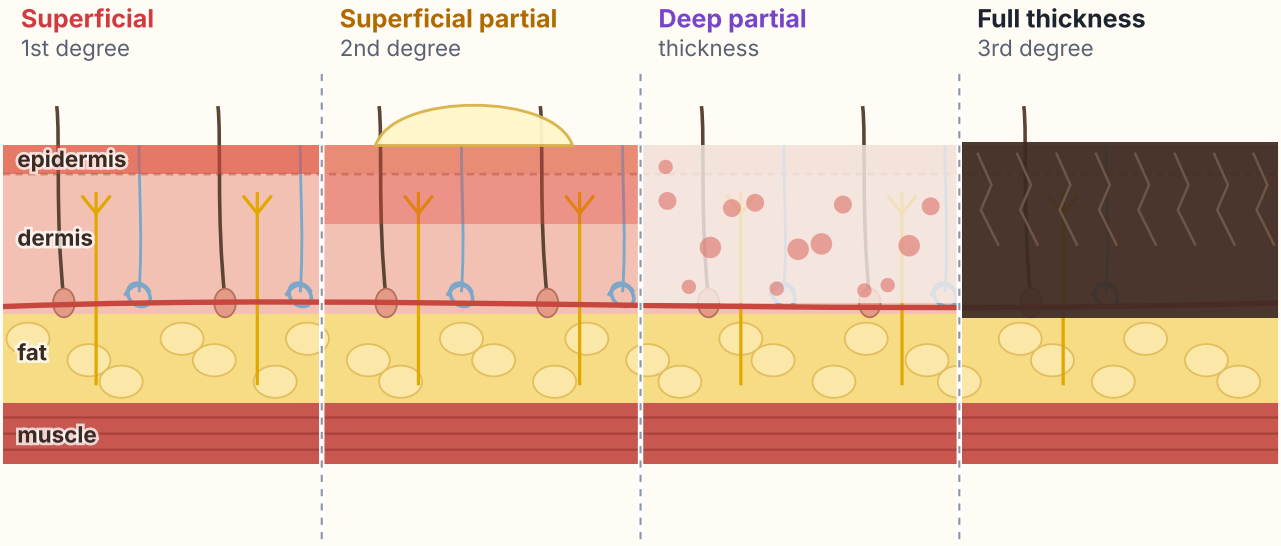
Hook: "TB Undermines, Rodent Rolls, Squamous Spills (everted), Syphilis Punches."

Burns

Depth	Layer	Look	Pain	Heals
Superficial (first degree)	Epidermis	Red, no blisters (sunburn)	Painful	5–7 days, no scar
Superficial partial thickness (second degree)	Upper dermis	**Blisters**, moist, pink	Very painful	2–3 weeks
Deep partial thickness	Deep dermis	Pale, mottled	Less painful	Over 3 weeks, scars
Full thickness (third degree)	Whole skin	White or charred, leathery (eschar)	**Painless** (nerves destroyed)	Needs grafting

Burns: how deep is the damage?

Depth decides the look, the pain and the healing



Superficial (1st degree)	Superficial partial (2nd degree)	Deep partial thickness	Full thickness (3rd degree)
REACHES: Epidermis only	REACHES: Upper dermis	REACHES: Deep dermis	REACHES: Whole skin
LOOKS: Red, no blisters	LOOKS: Blisters, moist, pink	LOOKS: Pale, mottled	LOOKS: White/charred, leathery
PAIN: Painful	PAIN: Very painful	PAIN: Less painful	PAIN: Painless: nerves gone
HEALS: 5–7 days, no scar	HEALS: 2–3 weeks	HEALS: Over 3 weeks, scars	HEALS: Needs grafting

First aid: cool under running tap water for 20 minutes; no ice, butter or oil. Blisters = superficial partial thickness.

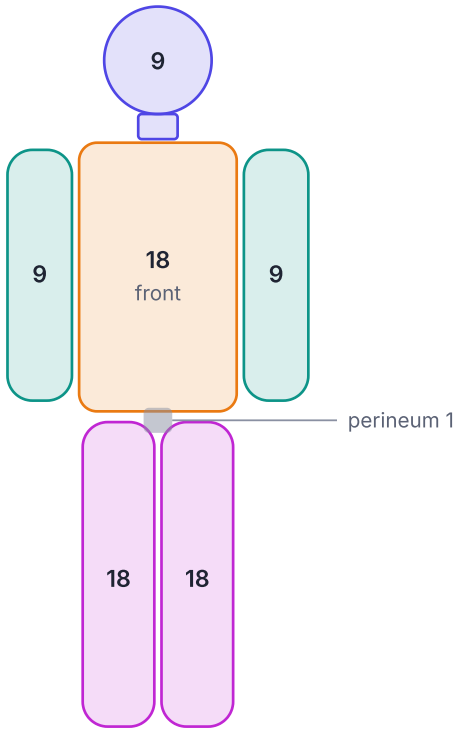
Burn depth · how far down the damage goes decides the look, the pain and the healing

- **Rule of nines (adult)**: head and neck 9%, each upper limb 9%, front of trunk 18%, back of trunk 18%, each lower limb 18%, perineum 1%.
- The patient's own palm (with fingers) is about 1% of body surface.
- In children the head is larger (about 18%) and the legs smaller (Lund and Browder chart).
- Admit adults with over 15% burns (children over 10%), and any burn of the face, hands, feet, perineum or airway.

First aid for burns: stop the burning, **cool with running tap water for 20 minutes** (no ice, butter or oil), remove rings and tight clothing, cover with cling film or a clean cloth, keep the patient warm. Face and neck burns are dangerous because of **airway swelling**.

Rule of nines: each body region is 9% or a multiple of 9%

Adult body, front view; percentages of total body surface area burnt



Region	% of body
Head and neck	9
Each upper limb (×2)	9 + 9
Front of trunk	18
Back of trunk	18
Each lower limb (×2)	18 + 18
Perineum	1
Total	**100**

Patient's palm with fingers = about 1%

Children: head is larger (about 18%) and each leg smaller (about 14%): use the Lund and Browder chart.

Admit adults with burns over 15% (children over 10%).

Rule of nines · adult body surface by region

Add the regions that are burnt: a burn of the whole front of the trunk and one arm is 18 + 9 = 27%.

Perioperative care

- **Before surgery**: informed written consent, fasting (solids 6 hours, clear fluids 2 hours), assessment of fitness (ASA grade I–VI), blood sugar and blood pressure control, marking the site.
- **After surgery**: watch airway, breathing, circulation, pain, urine output; early mobilisation prevents deep-vein thrombosis and chest infection.
- **Post-operative fever**: day 1–2 lung collapse (atelectasis); day 3–5 urinary tract infection or wound infection; day 5–7 deep-vein thrombosis; remember the "5 Ws": wind, water, wound, walking, wonder drugs.

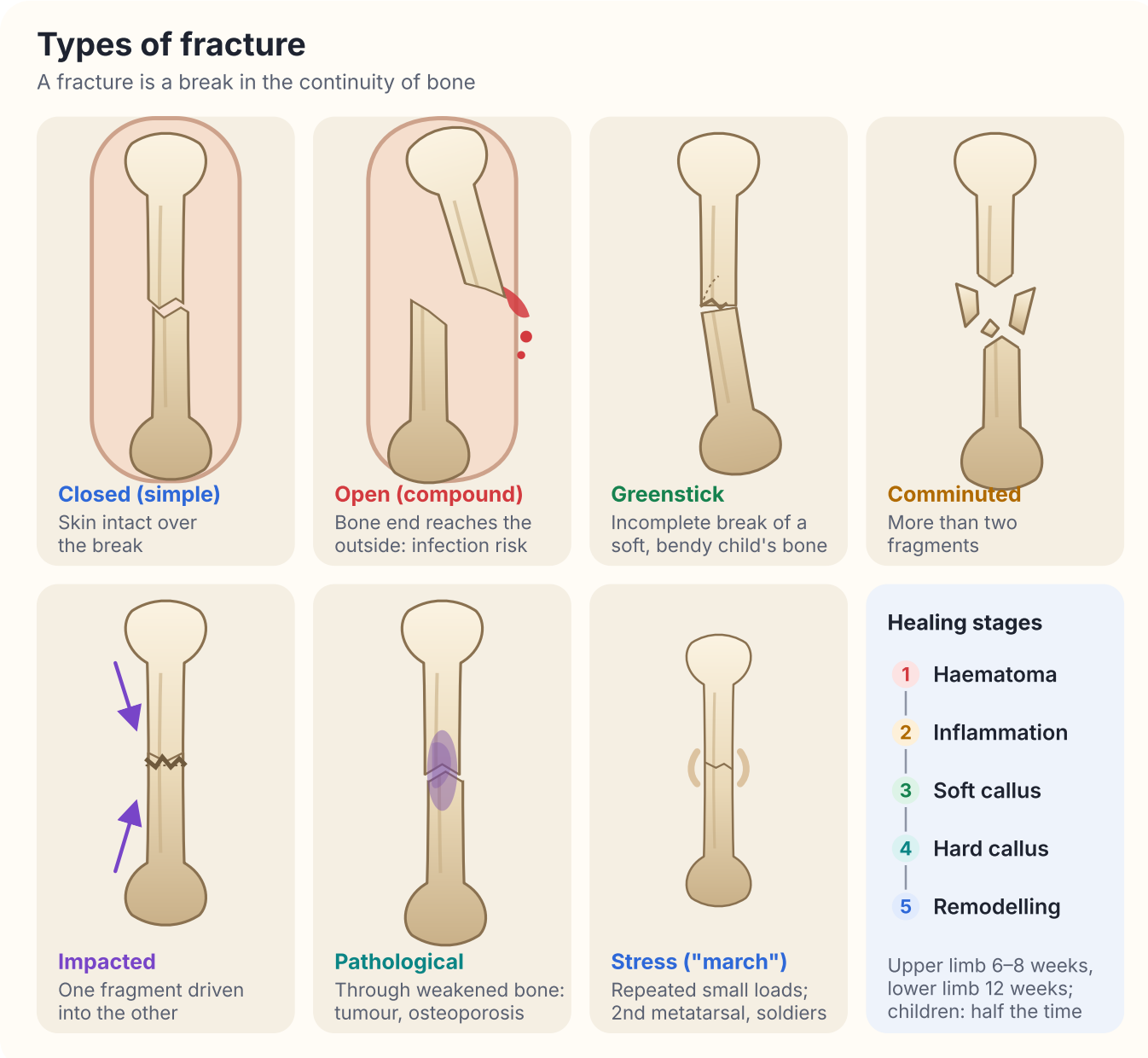
9.2 Traumatology and orthopaedics

Expect eponymous fractures, the difference between a sprain, strain and dislocation, and the X-ray signs of osteoarthritis and spondylosis.

Fractures

A fracture is a break in the continuity of bone. **Signs**: pain, swelling, deformity, abnormal mobility, crepitus, loss of function. **Closed** (skin intact) or **open/compound** (communicates with the outside, risk of infection).

Type	Meaning
Greenstick	Incomplete fracture of a soft, bendy child's bone
Comminuted	More than two fragments
Pathological	Through bone already weakened (tumour, osteoporosis)
Stress (fatigue, "march")	Repeated small loads; 2nd metatarsal in soldiers
Impacted	One fragment driven into the other

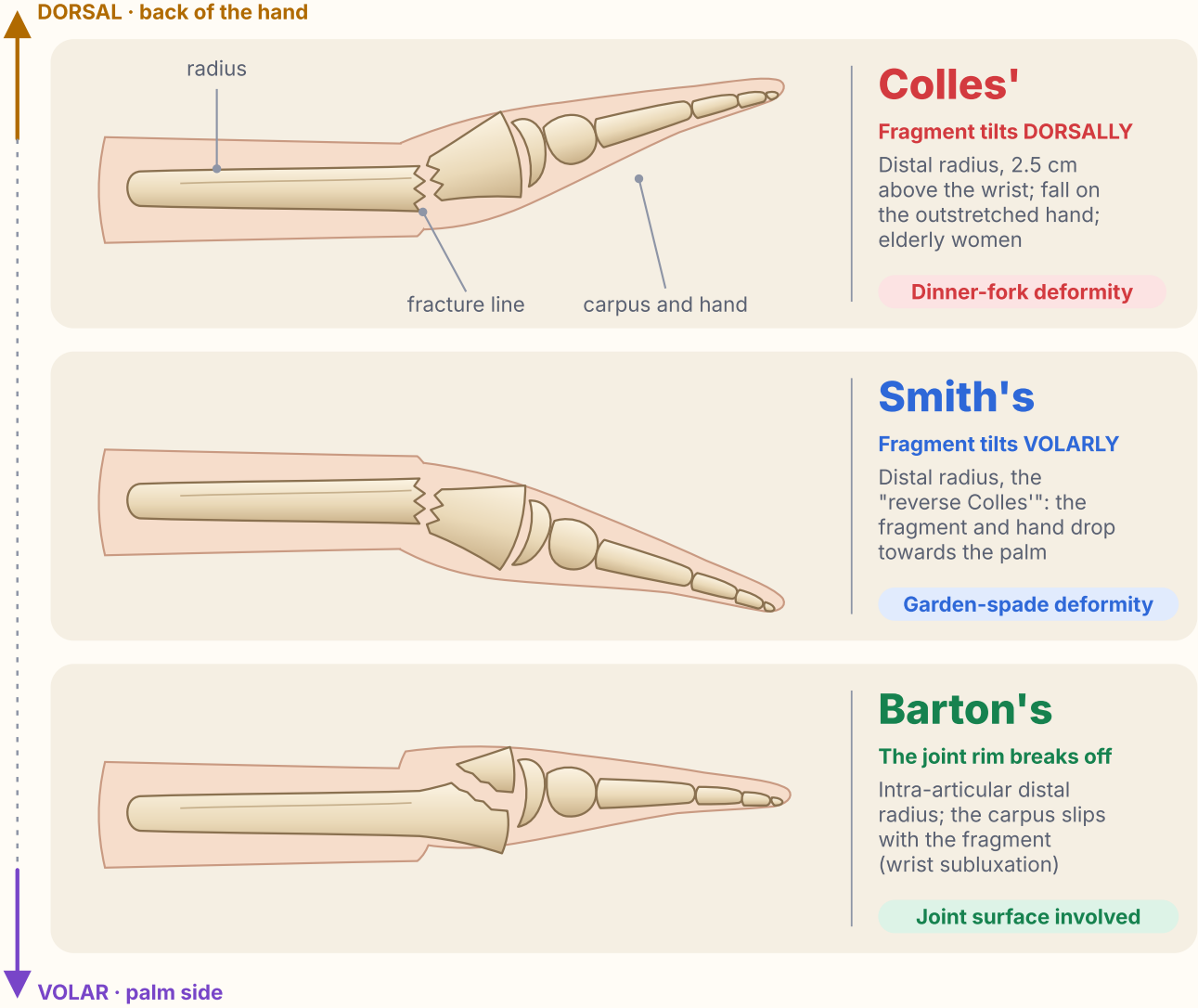


Types of fracture · how each one looks, and the five healing stages

Eponymous fracture	Bone and deformity
Colles'	Distal radius, about 2.5 cm above the wrist, **dorsal displacement** ("dinner-fork" deformity); fall on the outstretched hand, elderly women
Smith's	Distal radius, **volar** displacement ("reverse Colles'", garden-spade deformity)
Barton's	Intra-articular distal radius with subluxation of the wrist
Monteggia	Fracture of upper ulna + dislocation of the radial head (**MU**: Monteggia = Ulna)
Galeazzi	Fracture of lower radius + dislocation of the distal radio-ulnar joint
Pott's	Bimalleolar fracture of the ankle
Bennett's	Fracture-dislocation of the base of the first metacarpal (thumb)
Boxer's	Neck of the 5th metacarpal
Jones	Base of the 5th metatarsal

Wrist fractures: Colles', Smith's and Barton's

Same bone, same place, different direction · side view of the right forearm



Wrist fractures · Colles' tilts dorsally, Smith's volarly, Barton's breaks the joint rim

- **Healing stages**: haematoma → inflammation → soft callus → hard callus → remodelling.
- Upper limb bones unite in about 6–8 weeks, lower limb in 12 weeks; children heal in half the time.
- **Complications**: shock, fat embolism (long bones, 24–72 h), compartment syndrome (pain on passive stretch: the earliest sign), avascular necrosis (scaphoid, femoral head), malunion, non-union, Volkmann's ischaemic contracture (supracondylar fracture of humerus in children).

First aid: immobilise the joint above and below with a splint, do not try to straighten deformity, cover open wounds, treat for shock.

Dislocations and soft-tissue injury

Term	Meaning
Dislocation	Complete loss of contact between joint surfaces

Term	Meaning
Subluxation	Partial loss of contact
Sprain	Injury to a **ligament** (commonest: lateral ligament of the ankle, inversion injury)
Strain	Injury to a **muscle or tendon**

Shoulder is the commonest joint to dislocate, **anterior** in about 95% (axillary nerve at risk). **Hip** dislocation is usually **posterior** (dashboard injury; limb flexed, adducted, internally rotated; sciatic nerve at risk). **RICE** for sprains: Rest, Ice, Compression, Elevation.

Osteomyelitis

- Infection of bone.
- Acute haematogenous osteomyelitis is commonest in children, in the **metaphysis** of long bones (lower femur, upper tibia), caused mostly by ***Staphylococcus aureus***.
- Signs: high fever, severe local pain, refusal to move the limb.
- X-ray is normal for the first 10–14 days; MRI shows it early.
- Chronic osteomyelitis: **sequestrum** (dead bone), **involucrum** (new bone around it), discharging sinus.
- Tuberculosis of the spine is **Pott's disease** (gibbus, cold abscess, paraplegia).

Degenerative joint disease

	Osteoarthritis	Rheumatoid arthritis (for contrast)
Age	Over 50, weight-bearing joints	Young and middle-aged women
Joints	Knee, hip, DIP (**Heberden's nodes**), PIP (**Bouchard's nodes**)	Small joints of hands, symmetrical, MCP and PIP
Stiffness	Under 30 minutes, worse after use	Morning stiffness over 1 hour
X-ray	**LOSS**: Loss of joint space, Osteophytes, Subchondral sclerosis, Subchondral cysts	Erosions, osteopenia

- **Cervical spondylosis**: neck pain and stiffness radiating to the shoulders and arms, numbness and tingling (radiculopathy, commonest at C5–C6); vertigo on turning the head.
- **Lumbar spondylosis / disc prolapse**: low back pain with sciatica (L4–L5 and L5–S1), positive straight-leg-raising test.

- **Frozen shoulder** (adhesive capsulitis): painful restriction of all shoulder movements, commoner in diabetics.

Homoeopathic therapeutics

Picture	Remedy
Sprain or strain; stiffness worse on first motion, better from continued motion	**Rhus toxicodendron**
Bruised tendons and periosteum, sprained wrists and ankles, eye strain	**Ruta graveolens**
Fractures slow to unite; pricking pain at the site	**Symphytum**
Delayed union in a thin, anaemic patient	**Calcarea phosphorica**
Osteoarthritis with joint pain worse from the slightest motion, better from pressure and rest	**Bryonia alba**
Cervical spondylosis with stiffness and vertigo, worse on turning the head	**Conium**, Cocculus
Sciatica with cramping pain, better from pressure and heat; left side	**Colocynthis**
Sciatica with numbness, worse sitting, better lying	**Gnaphalium**
Osteomyelitis with chronic suppuration, fistulous openings, offensive discharge	**Silicea**; bone necrosis: **Calcarea fluorica**

Memory hook: the **shoulder** slips **forward** (anterior), the **hip** is pushed **back** (posterior, dashboard injury). In osteomyelitis the **S**equestrum is the **S**lain bone and the **I**nvolucrum is the new bone that **I**nvests it.

9.3 ENT and ophthalmology

Six conditions are named in the syllabus; for each, know the classic picture, one complication and one or two remedies.

ENT

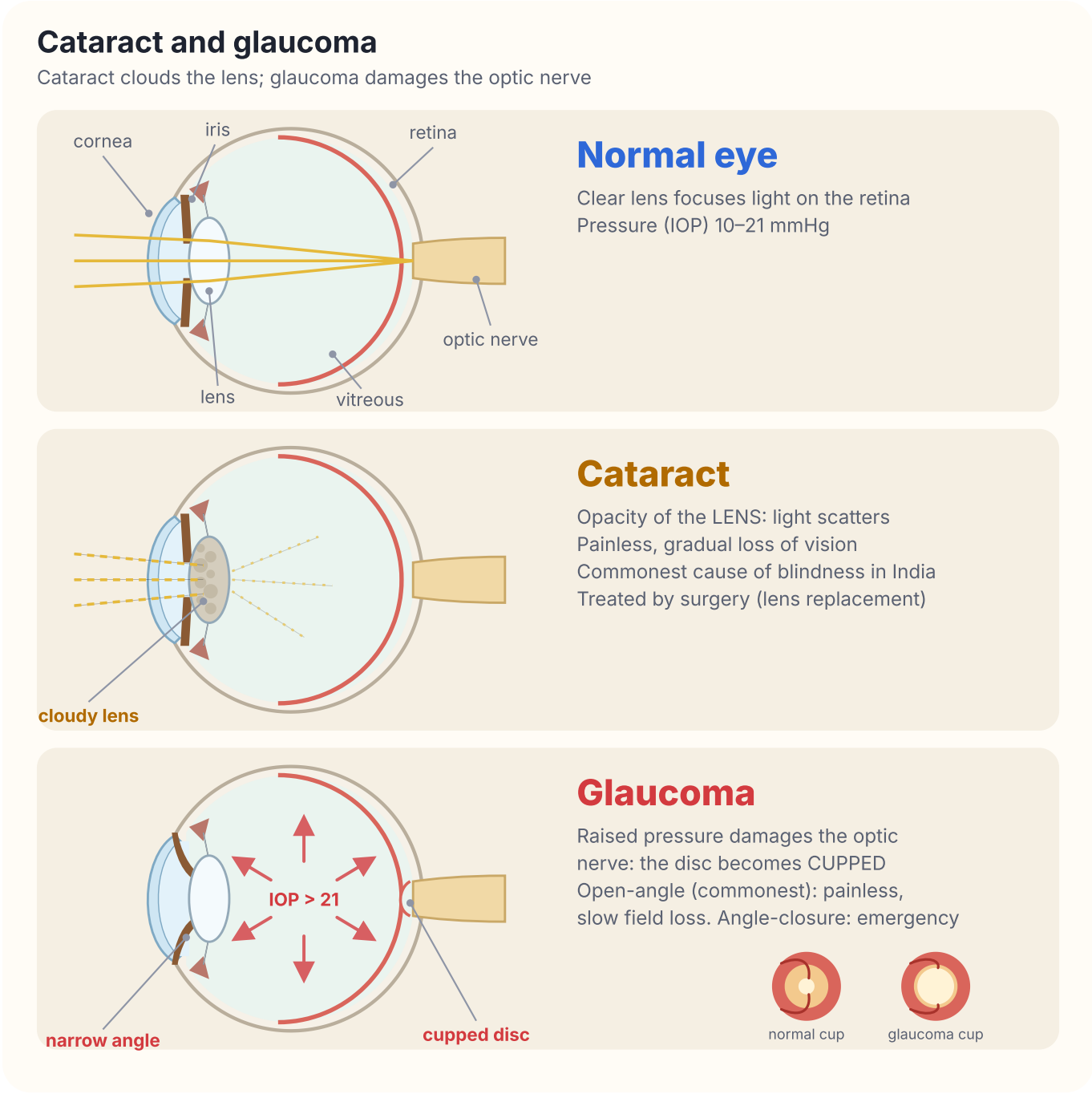
Condition	Key facts	Remedies
Sinusitis	Commonest sinus affected in adults: **maxillary** (drains poorly, opening high up); in children: ethmoid. Facial pain worse on bending forward, nasal block, purulent discharge, post-nasal drip	Kali bichromicum (thick, ropy, stringy discharge; pain at the root of the nose); Hepar sulph; Silicea (chronic)
Acute otitis media	Middle-ear infection after a cold, commonest in children (short, horizontal Eustachian tube). Earache, fever, **red, bulging drum**; if it bursts, discharge and pain relief. Chronic suppurative otitis media (CSOM): safe (tubotympanic, central perforation) or unsafe (atticoantral, cholesteatoma)	Belladonna (sudden, throbbing, red), Chamomilla (unbearable pain, angry child), Pulsatilla (bland discharge, worse warm room), Ferrum phos (first stage)
Tonsillitis	Sore throat, fever, painful swallowing, enlarged red tonsils with exudate; commonest bacterial cause **Group A beta-haemolytic streptococcus** (risk of rheumatic fever and glomerulonephritis). Peritonsillar abscess = **quinsy**	Belladonna (right side, bright red), Lachesis (left to right, worse empty swallowing, hot drinks), Lycopodium (right to left), Baryta carb (recurrent, enlarged tonsils), Hepar sulph (suppuration)
Otosclerosis (contrast)	Progressive conductive deafness in young women, worse in pregnancy; hears better in noise (paracusis Willisii)	—
Epistaxis	Commonest site: **Little's area** (Kiesselbach's plexus, anteroinferior septum); first aid: sit up, lean forward, pinch the soft nose for 10 minutes	Ferrum phos, Millefolium, Phosphorus

Hearing tests. Rinne: air conduction better than bone = positive (normal or sensorineural loss); negative in conductive loss. Weber: lateralises to the **affected** ear in conductive loss, to the **better** ear in sensorineural loss.

Eye

Condition	Key facts	Remedies
Conjunctivitis	Red eye with discharge, vision normal, pupil normal. Bacterial: sticky purulent; viral: watery, often with preauricular nodes; allergic: itching, ropy discharge (spring catarrh in boys)	Euphrasia (acrid tears, bland nasal discharge), Argentum nitricum (profuse purulent, ophthalmia neonatorum), Pulsatilla (bland yellow discharge), Apis (swollen lids, stinging)

Condition	Key facts	Remedies
Cataract	Opacity of the lens; painless, gradual loss of vision; **commonest cause of blindness in India**; senile cataract commonest; treated by surgery (lens replacement)	Calcarea fluorica, Causticum, Phosphorus, Cineraria maritima (local) in early cases
Glaucoma	Raised intraocular pressure (normal **10–21 mmHg**) damaging the optic nerve (cupping); **acute angle-closure**: sudden painful red eye, **halos round lights**, vomiting, fixed semi-dilated oval pupil — an emergency. **Open-angle** (commonest): painless, slow field loss	Belladonna, Phosphorus, Spigelia (pain in and around the eye), Osmium
Stye (hordeolum)	Painful infection of a lash follicle at the lid margin	Pulsatilla, Staphysagria (recurrent)
Chalazion	Painless swelling of a meibomian gland, away from the margin	Staphysagria, Thuja



Cataract and glaucoma · the lens clouds in one, the optic nerve cups in the other

Leukocoria (white pupil) in a child must raise suspicion of **retinoblastoma**. Vitamin A deficiency causes night blindness and Bitot's spots. A chemical splash: wash the eye at once with plenty of clean water for 15–30 minutes.

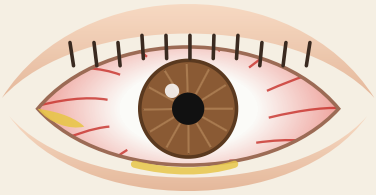
Red eye decision chart: check pain, vision, pupil and pressure to tell the three common red eyes apart.

Feature	Acute conjunctivitis	Acute angle-closure glaucoma	Acute iritis (anterior uveitis)
Pain	Grittiness, little pain	Severe, with headache and vomiting	Moderate ache, photophobia
Vision	Normal	Markedly reduced, haloes round lights	Mildly blurred

Feature	Acute conjunctivitis	Acute angle-closure glaucoma	Acute iritis (anterior uveitis)
Redness	Conjunctival, worst at fornices	Ciliary flush	Ciliary (circumcorneal) flush
Discharge	Present (mucopurulent or watery)	None, watering only	None, watering only
Cornea	Clear	Hazy, oedematous	Clear; keratic precipitates
Pupil	Normal, reacting	Mid-dilated, vertically oval, fixed	Small, irregular, sluggish
Intraocular pressure	Normal	Very high (stony hard eye)	Normal or low
Key action	Hygiene, treat at home	**Emergency referral** to save sight	Refer for slit lamp examination
Leading remedies	Euphrasia, Argentum nitricum, Pulsatilla	Spigelia, Phosphorus, Belladonna (only alongside urgent referral)	Mercurius corrosivus, Rhus toxicodendron

Three red eyes: look at the pupil

A red eye with a fixed mid-dilated pupil is glaucoma until proved otherwise



Acute conjunctivitis

PAIN
Gritty, little pain

VISION
Normal

REDNESS
Worst at the fornices

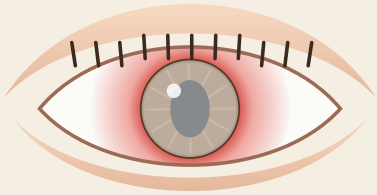
DISCHARGE
Present

CORNEA
Clear

PUPIL
Normal, reacting

PRESSURE
Normal

Hygiene, treat at home



Acute angle-closure glaucoma

PAIN
Severe, headache, vomiting

VISION
Much reduced, halos

REDNESS
Ciliary flush

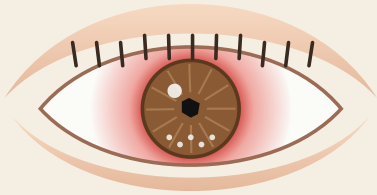
DISCHARGE
None, watering

CORNEA
Hazy, oedematous

PUPIL
Mid-dilated, oval, fixed

PRESSURE
Very high, stony hard

EMERGENCY referral



Acute iritis
(anterior uveitis)

PAIN
Moderate ache, photophobia

VISION
Mildly blurred

REDNESS
Ciliary flush

DISCHARGE
None, watering

CORNEA
Clear; keratic precipitates

PUPIL
Small, irregular, sluggish

PRESSURE
Normal or low

Refer for slit lamp

Three red eyes · the pupil and the pressure tell them apart

☆ **Exam tip:** a red eye with a **fixed mid-dilated pupil** is glaucoma until proved otherwise; a red eye with a **small pupil** points to iritis.

Memory hook: Weber **comes to** a conductive loss and **runs from** a nerve loss: the sound goes to the **affected** ear in conductive deafness and to the **better** ear in sensorineural deafness.

9.4 Surgical therapeutics and first aid

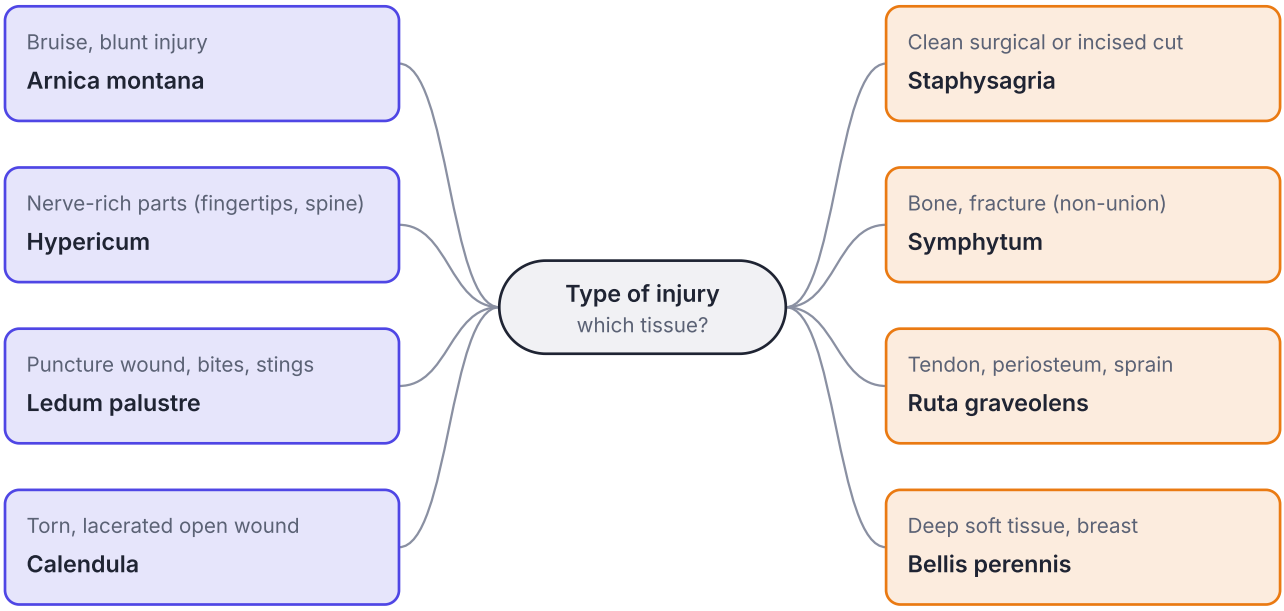
The injury remedies are among the most reliable marks in the paper: each remedy is matched to one kind of tissue or wound.

Remedies for mechanical injuries

The map below pairs each type of injury with its remedy; the table adds the keynote.

Pick the remedy by the tissue injured

Eight injury remedies; first aid and surgical referral come first



Injury → remedy map · remedy chosen by the tissue injured

Match the tissue first: a nail puncture is Ledum, a crushed fingertip is Hypericum, and a fracture slow to unite is Symphytum.

Injury	Remedy	Keynote
Blunt injury, bruises, shock of injury, concussion	**Arnica montana**	Sore, bruised feeling; says he is well; bed feels too hard
Injury to nerve-rich parts (fingertips, spine, coccyx); shooting pains along nerves	**Hypericum**	"The Arnica of the nerves"; also prevents tetanus-like pains after punctured nerve injuries
Punctured wounds, insect bites, rat bites; wound cold yet better from cold	**Ledum palustre**	Ascending pains; prevents sepsis in punctured wounds
Lacerated, open wounds; promotes healing and prevents suppuration	**Calendula** (local and internal)	"The homoeopathic antiseptic"
Clean incised (surgical) wounds; pain after operations and catheterisation	**Staphysagria**	Sense of insult or humiliation
Fractures slow to unite; injury to the eyeball by a blunt object	**Symphytum**	"Knitbone"

Injury	Remedy	Keynote
Bruised tendons, periosteum and joints; eye strain	**Ruta graveolens**	Lameness after sprain
Deep bruising of soft tissues and breasts; injury of deep pelvic organs	**Bellis perennis**	Bruised soreness of deep tissues after surgery
Haemorrhage, bright red, from injury	**Millefolium**, Arnica	—
Burns with blisters and burning pain	**Cantharis** (also Urtica urens for first-degree)	Better from cold applications
Ill effects of a fall on the head, concussion	Arnica, **Natrum sulphuricum** (mental symptoms after head injury)	—

Hook: "ARNICA for the Bruise, HYPERICUM for the Nerve, LEDUM for the Needle, CALENDULA for the Cut-and-Torn, STAPHYSAGRIA for the Scalpel, SYMPHYTUM for the Skeleton, RUTA for the Rope (tendon)."

Pre- and post-operative remedies

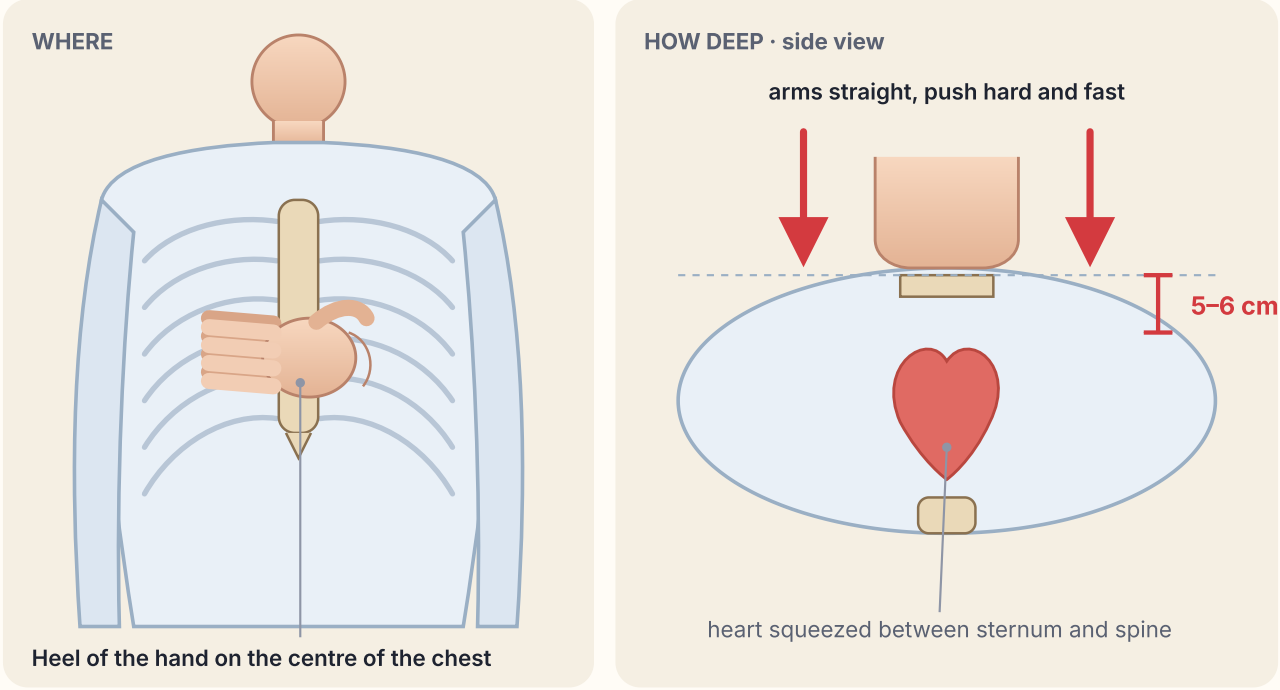
Situation	Remedy
Fear and anticipatory anxiety before surgery, trembling, diarrhoea	**Gelsemium**, Argentum nitricum
Restless fear of death before surgery	**Aconitum napellus**
Shock, bruised soreness after surgery	**Arnica**
Pain of a clean incision	**Staphysagria**
Retention of urine after surgery	**Causticum**
Flatulent distension after abdominal surgery	**Carbo vegetabilis**, Raphanus
Vomiting after anaesthesia	**Phosphorus**
Ill effects of anaesthesia, slow recovery, sleepiness	**Opium**
Collapse with cold breath and craving for air	**Carbo vegetabilis**

First aid

Basic life support (adult) — C-A-B. Check response and breathing → call for help → chest compressions **30 : 2 breaths**, rate **100–120 per minute**, depth 5–6 cm, centre of the chest; continue until help arrives. Recovery position for an unconscious patient who is breathing.

CPR: 30 compressions, 2 breaths

Where to push, how deep and how fast



Adult basic life support: C-A-B (Circulation, Airway, Breathing)

Check response and breathing → **Call** for help → **30 compressions** 100–120 per minute → **2 breaths** then repeat 30 : 2

Breathing but unconscious: recovery position. Continue CPR until help arrives.

CPR · centre of the chest, 5–6 cm deep, 100–120 a minute, 30 : 2

Emergency	First aid
Snakebite	Reassure, **immobilise the limb** like a fracture, remove rings and tight items, keep the patient still, rush to hospital for anti-snake venom. **Do not** cut, suck, apply a tight tourniquet, ice or herbs ("Do it RIGHT": Reassure, Immobilise, Get to hospital, Tell the doctor the symptoms)
Neurotoxic bites (cobra, krait)	Ptosis, difficulty swallowing, respiratory paralysis; krait bites at night while asleep, often painless

Emergency	First aid
Haemotoxic bites (Russell's viper, saw-scaled viper)	Local swelling, bleeding from gums and bite site, kidney failure; 20-minute whole-blood clotting test (20WBCT)
Poisoning (general)	Remove from the source, clear the airway, place in the recovery position, keep the container or sample, do not induce vomiting after corrosives or kerosene, shift to hospital
Corrosive or kerosene swallowed	No vomiting (risk of further burns or aspiration pneumonia)
Bleeding	Direct pressure and elevation; tourniquet only as a last resort for life-threatening limb bleeding
Fracture	Immobilise the joints above and below; do not straighten
Choking (adult)	5 back blows, then 5 abdominal thrusts (Heimlich)
Heat stroke	Move to shade, remove clothing, cool with water and fanning
Drowning	Clear airway, start CPR with rescue breaths
Dog bite	Wash the wound with soap and running water for 15 minutes; rabies vaccination

Homoeopathic support in bites and stings. Ledum (punctured bites), Apis (stinging, oedematous swelling), Lachesis (bluish, septic bite wounds), Arsenicum album (collapse, anxiety, burning). Snakebite is always treated with anti-snake venom in hospital; remedies are supportive only.

Rapid revision: one-line facts

Area	Fact
Wounds	First intention = clean incised wound, minimal scar; second intention = granulation, wide scar
Wounds	Keloid grows beyond the margins; hypertrophic scar stays within
Sinus / fistula	Sinus = blind track; fistula = link between two epithelial surfaces
Infection	Abscess = Staph. aureus, drain it; cellulitis = Strep. pyogenes, spreading
Shock	Commonest after trauma = hypovolaemic; neurogenic = low BP with slow pulse
Ulcers	TB undermined; rodent rolled; squamous everted; syphilitic punched out; varicose above the medial malleolus

Area	Fact
Burns	Blisters = superficial partial thickness; painless = full thickness; cool 20 minutes under running water
Rule of nines	Head 9, each arm 9, front trunk 18, back trunk 18, each leg 18, perineum 1
Fractures	Colles' = distal radius, dorsal displacement, dinner fork; Smith's = volar
Fractures	Monteggia = ulna + radial head; Galeazzi = radius + distal radio-ulnar joint; Pott's = ankle
Fractures	Compartment syndrome earliest sign = pain on passive stretch; scaphoid and femoral neck = avascular necrosis
Joints	Sprain = ligament; strain = muscle or tendon; shoulder dislocates anteriorly, hip posteriorly
Bone	Osteomyelitis = metaphysis, Staph. aureus; sequestrum = dead bone; involucrum = new bone
Osteoarthritis	Heberden's nodes (DIP), Bouchard's (PIP); X-ray LOSS
ENT	Maxillary sinus commonest; red bulging drum = acute otitis media; quinsy = peritonsillar abscess; Little's area = nosebleed
Eye	Cataract = lens opacity, commonest cause of blindness in India; glaucoma IOP over 21 mmHg, halos = acute angle closure
Eye	Leukocoria in a child = retinoblastoma until proven otherwise
Remedies	Arnica bruise; Hypericum nerve; Ledum puncture; Calendula torn wound; Staphysagria incision; Symphytum fracture; Ruta tendon; Bellis deep tissue
Remedies	Pre-op fear = Gelsemium, Aconite; post-op retention = Causticum; vomiting after anaesthesia = Phosphorus; flatulence = Carbo veg
First aid	CPR 30 : 2, 100–120/min, 5–6 cm; snakebite: immobilise, no cutting or sucking, hospital for ASV

One-glance recall card

Topic 9 recall card: the 15 facts asked most

Cover the small lines, say the fact from the big cue, then check

Rule of nines head 9, arm 9, trunk 18+18, leg 18, perineum 1	**Palm = 1%** patient's palm with fingers is about 1% of the body	**Cool water** burn first aid: 20 min, no ice, no butter
30 : 2 CPR, 100-120 a minute, 5-6 cm deep	**Shoulder: anterior** commonest dislocation; hip goes posterior	**Dinner fork** Colles' fracture of the distal radius
Metaphysis osteomyelitis site; Staph aureus	**Sequestrum** dead bone; involucrum is the new bone around it	**Weber** to the affected ear: conductive; to the better ear: nerve
Fixed mid-dilated red eye with this pupil is acute glaucoma	**Leukocoria** white pupil in a child: think retinoblastoma	**Arnica** bruises, blunt injury, shock of injury
Ledum punctures and stings; cold yet better cold	**Hypericum** nerve-rich parts: fingertips, spine	**Symphytum** fracture slow to unite; Staphisagria: clean cuts

Topic 9 recall card · 15 tiles

For the last ten minutes before the exam: cover the small lines of each tile and say the fact from the big cue.

Practice MCQs

Twenty questions grouped by syllabus heading; the answer key follows.

General surgical conditions

1. An abnormal communication between two epithelial surfaces is a: (a) sinus (b) fistula (c) abscess (d) ulcer
2. An ulcer with an undermined edge is typical of: (a) basal cell carcinoma (b) tuberculosis (c) syphilis (d) varicose veins
3. Burns with blisters and severe pain are: (a) superficial (first degree) (b) superficial partial thickness (c) full thickness (d) fourth degree
4. By the rule of nines, one entire lower limb of an adult is: (a) 9% (b) 18% (c) 27% (d) 36%

5. Low blood pressure with a slow pulse after spinal injury suggests: (a) hypovolaemic shock (b) neurogenic shock (c) septic shock (d) cardiogenic shock

Trauma and orthopaedics

6. Colles' fracture involves the distal radius with: (a) volar displacement (b) dorsal displacement (c) ulnar dislocation (d) scaphoid fracture
7. A sprain is an injury to a: (a) bone (b) muscle (c) ligament (d) nerve
8. Acute haematogenous osteomyelitis in children most commonly affects the: (a) epiphysis (b) metaphysis (c) diaphysis (d) flat bones
9. Heberden's nodes in osteoarthritis are found at the: (a) MCP joints (b) PIP joints (c) DIP joints (d) wrist
10. The commonest direction of shoulder dislocation is: (a) anterior (b) posterior (c) inferior (d) superior

ENT and eye

11. Peritonsillar abscess is called: (a) quinsy (b) Ludwig's angina (c) adenoiditis (d) laryngitis
12. The commonest site of nosebleed is: (a) Woodruff's area (b) Little's area (c) the turbinates (d) the sphenoid sinus
13. Sudden painful red eye with halos around lights and vomiting suggests: (a) conjunctivitis (b) acute angle-closure glaucoma (c) cataract (d) stye
14. Opacity of the lens of the eye is: (a) glaucoma (b) cataract (c) pterygium (d) keratitis
15. Tonsillitis beginning on the left and moving to the right, worse from hot drinks: (a) Lycopodium (b) Lachesis (c) Belladonna (d) Baryta carb

Surgical therapeutics and first aid

16. Injury to nerve-rich parts such as crushed fingertips calls for: (a) Arnica (b) Hypericum (c) Calendula (d) Ruta
17. Punctured wounds that are cold yet better from cold applications: (a) Ledum (b) Arnica (c) Staphysagria (d) Symphytum
18. Pain after a clean surgical incision with a sense of insult: (a) Bellis perennis (b) Staphysagria (c) Hypericum (d) Ruta
19. First aid for snakebite includes: (a) cutting and sucking the wound (b) a tight tourniquet (c) immobilising the limb and urgent transfer to hospital (d) ice packs
20. In adult CPR the ratio of chest compressions to breaths is: (a) 15 : 2 (b) 30 : 2 (c) 5 : 1 (d) 30 : 1

Answer key

Q	Answer	Why
1	b	Sinus is blind; fistula links two surfaces
2	b	TB = undermined edge
3	b	Blisters + pain = superficial partial thickness
4	b	Each lower limb 18%
5	b	Loss of sympathetic tone: low BP, slow pulse
6	b	Dinner-fork deformity
7	c	Strain = muscle or tendon
8	b	Sluggish metaphyseal blood flow
9	c	Heberden = DIP; Bouchard = PIP
10	a	About 95% anterior
11	a	Quinsy = peritonsillar abscess
12	b	Kiesselbach's plexus on the anterior septum
13	b	Emergency: raised IOP
14	b	Cataract
15	b	Lachesis: left to right, worse hot drinks
16	b	Hypericum = nerve injuries
17	a	Ledum keynote
18	b	Staphysagria = incised wounds
19	c	No cutting, sucking or tight tourniquet
20	b	30 : 2 at 100–120/min

Topic 10: Community Medicine, Forensic Medicine & Ethics

How to use this doc

- This is study material for **Core Topic 10** of the GPSC Homoeopathic Physician CBRT (Advt 06/2026, exam **1 November 2026**).
- Core carries 25 of the 60 marks across Topics 1–10, so expect **2–3 questions** from this topic.
- They are usually one-line facts: a definition, a formula, a colour code, a section of an Act, or a classical sign.

What it covers

- 10.1 Epidemiology, levels of prevention, screening, and National Health Programmes, including vaccines.
- 10.2 Demography, vital statistics, basic biostatistics, water, sanitation and biomedical waste.
- 10.3 Forensic medicine and toxicology: death and post-mortem changes, injuries, asphyxia, common poisons.
- 10.4 Medical jurisprudence and ethics for a homoeopathic physician: NCH Act 2020, professional conduct, consent, negligence, records.

Standard books followed

- K. Park, *Park's Textbook of Preventive and Social Medicine*.
- B.K. Mahajan, *Methods in Biostatistics*.
- K.S.N. Reddy, *The Essentials of Forensic Medicine and Toxicology*.
- C.K. Parikh, *Textbook of Medical Jurisprudence, Forensic Medicine and Toxicology*.
- Modi's *Medical Jurisprudence and Toxicology*.
- The National Commission for Homoeopathy Act, 2020, and the professional conduct regulations for homoeopathic practitioners.

Scope note: conventional (allopathic) drug treatment is left out. Vaccines and programme schedules are kept because they are public-health facts. Antidotes appear only as forensic toxicology facts that examiners ask.

How to revise: read each section once, look at the diagram, then do the Rapid revision list and the 20 MCQs. Mark any wrong answer and re-read only that paragraph.

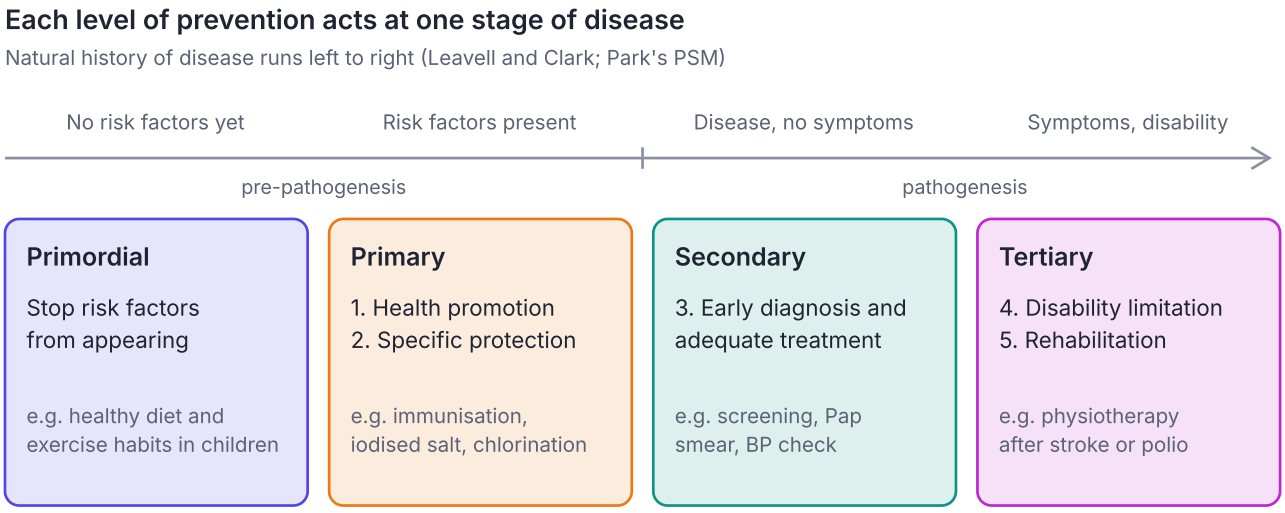
10.1 Epidemiology, prevention and National Health Programmes

Health and disease

- **WHO definition of health (1948):** "a state of complete physical, mental and social well-being and not merely an absence of disease or infirmity". Spiritual health is a proposed fourth dimension.
- **Epidemiological triad:** agent, host and environment. For non-communicable diseases a **web of causation** (multiple factors) is used instead.
- **Iceberg phenomenon:** the visible tip is clinical cases; the submerged part is sub-clinical, undiagnosed and carrier cases. Hypertension, diabetes and malnutrition show a large hidden part.
- **Epidemiology** is the study of the distribution and determinants of health states in populations and its application to their control (Last).

Levels of prevention

Prevention is matched to the stage of disease. Leavell and Clark gave **five modes of intervention**: health promotion, specific protection, early diagnosis and treatment, disability limitation, and rehabilitation.



Levels of prevention · five modes of intervention across the natural history of disease

Read the chart left to right: primordial prevention acts before any risk factor exists, and tertiary prevention acts once disease has caused disability.

Measuring disease

- **Incidence:** new cases in a period ÷ population at risk. It measures risk and is used for acute diseases.
- **Prevalence:** all existing cases (old + new) at a time ÷ total population. It is useful for chronic diseases and planning services.
- **Prevalence = Incidence × Mean duration** (P = I × D).
- **Attack rate:** an incidence rate used during an epidemic. **Secondary attack rate** = cases among contacts ÷ susceptible contacts; it measures infectiousness.
- **Incubation period:** time from infection to first symptom. **Serial interval:** gap between onset of primary and secondary cases.
- **Herd immunity:** a high proportion of immune people protects the unimmunised; it underlies mass immunisation.

Epidemics

- **Common-source, single exposure:** cases rise and fall within one incubation period (e.g. food poisoning).
- **Propagated (person to person):** gradual rise, several waves (e.g. hepatitis A, measles).
- **Endemic:** constant presence in an area. **Pandemic:** spread over many countries.
- **Sporadic:** scattered, irregular cases.

Study designs (a frequent question)

Design	Starts with	Gives	Example use
Cross-sectional	A population at one time	Prevalence	Survey of anaemia
Case-control	Cases and non-cases, looks back to exposure	**Odds ratio**	Rare disease, quick, cheap
Cohort	Exposed and non-exposed, follows forward	**Incidence, relative risk, attributable risk**	Smoking and lung cancer (Doll and Hill)
Randomised controlled trial	Random allocation to groups	Efficacy of an intervention	Gold standard for therapy

Blinding: single (subject unaware), double (subject and investigator unaware), triple (also the analyst).

Screening

Use the 2×2 table: TP, FP, FN, TN.

- **Sensitivity** = TP ÷ (TP + FN): ability to detect true cases (true positive rate).
- **Specificity** = TN ÷ (TN + FP): ability to identify true non-cases.
- **Positive predictive value** = TP ÷ (TP + FP). It **rises with prevalence**.
- **Negative predictive value** = TN ÷ (TN + FN).
- A screening test should be sensitive; a confirmatory (diagnostic) test should be specific.
- Wilson and Jungner (WHO, 1968) gave the criteria for a screening programme.

Immunisation and the cold chain

- **Active immunity:** from infection or vaccine; slow but long-lasting. **Passive:** ready-made antibodies (e.g. immunoglobulin, maternal antibodies); immediate but short.
- **Cold chain:** vaccines are kept at **+2 to +8 °C** in an ice-lined refrigerator (ILR) at the health facility.
- **Most heat-sensitive vaccine:** OPV. **Freeze-sensitive vaccines** (never freeze): Hepatitis B, pentavalent, DPT, Td, IPV.
- **Shake test** detects a frozen vaccine. **Vaccine vial monitor (VVM):** discard when the inner square matches or is darker than the outer circle.
- The full Universal Immunisation Programme schedule is in the Topic 8 doc.

Genus epidemicus (homoeopathic public health)

- In an epidemic, symptoms of several patients are pooled to form one picture, and the remedy matching it is the **genus epidemicus** (Organon §100–102).
- Hahnemann used **Belladonna** as a prophylactic in **scarlet fever** (1799). This is the classic historical example.

National Health Programmes

Programme	Key facts
National TB Elimination Programme (NTEP)	Renamed from RNTCP in 2020; aim to end TB ahead of the SDG 2030 target; Nikshay portal; DOTS strategy
National Leprosy Eradication Programme (NLEP)	Multi-drug therapy free; Sparsh awareness campaign on 30 January
National Centre for Vector Borne Diseases Control (NCVBDC)	Malaria, dengue, chikungunya, filaria, kala-azar, Japanese encephalitis
National AIDS Control Programme (NACP)	95-95-95 target; ICTC for counselling and testing

Programme	Key facts
NP-NCD (formerly NPCDCS)	Screening of hypertension, diabetes, common cancers at 30+ years
National Programme for Control of Blindness and Visual Impairment	Cataract is the leading cause of blindness
National Mental Health Programme	District Mental Health Programme; Tele-MANAS helpline 14416
Pulse Polio, Mission Indradhanush (2014)	India declared polio-free in 2014; Indradhanush covers missed children
Rashtriya Bal Swasthya Karyakram (RBSK)	Screens 0–18 years for the 4 Ds: defects at birth, diseases, deficiencies, developmental delays
Janani Suraksha Yojana (2005), JSSK (2011)	Cash for institutional delivery; free delivery and care for mother and sick newborn
Ayushman Bharat (2018)	Ayushman Arogya Mandir (health and wellness centres) and PM-JAY cover of ₹5 lakh per family per year

Health care delivery norms

Facility	Plain area population	Hilly or tribal population
Sub-centre	5,000	3,000
Primary Health Centre	30,000	20,000
Community Health Centre (30 beds, 4 specialists)	1,20,000	80,000

One **ASHA** serves about 1,000 population. The National Rural Health Mission began in 2005 and became part of the National Health Mission in 2013. **Bhore Committee (1946)** laid the foundation of primary health care in India; the **Alma-Ata Declaration (1978)** set "Health for All".

Memory hook: prevention runs **Promote, Protect** (primary), **Diagnose early** (secondary), **Limit disability, Rehabilitate** (tertiary). For screening tests: **SnNout** (a **S**ensitive test, when **N**egative, rules **out**) and **SpPin** (a **S**pecific test, when **P**ositive, rules **in**).

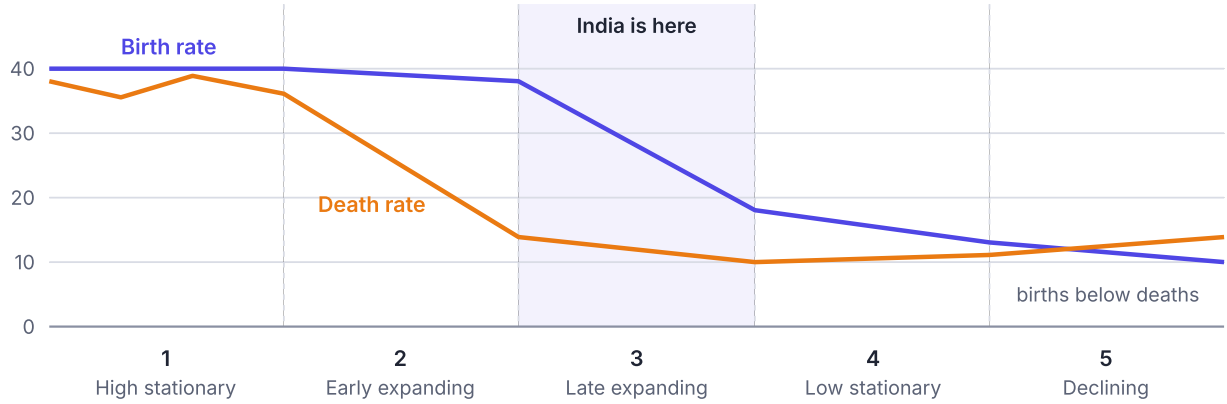
10.2 Demography, biostatistics and environmental health

Demography

Demography studies the size, composition and distribution of a population. Its three processes are **fertility, mortality and migration**.

Demographic cycle: the gap between birth and death rates drives growth

Schematic, rates per 1,000 population; India is in the late expanding stage



Stage 2 has the fastest growth: deaths fall while births stay high.

Demographic cycle · five stages of birth and death rates

The population grows fastest when the death rate has fallen but the birth rate is still high; India's birth rate is now falling, which places it in stage 3.

- **Census 2011 (latest completed census):** population 121.09 crore; sex ratio 943 females per 1,000 males; child sex ratio (0–6 years) 919; literacy 74.04%; density 382 per sq km; decadal growth 17.7%.
- Gujarat's sex ratio was 919.
- The census is conducted under the **Census Act, 1948**.

Vital statistics: definitions and denominators

Indicator	Formula (denominator is the exam trap)
Crude birth rate	Live births in a year per **1,000 mid-year population**
Crude death rate	Deaths in a year per 1,000 mid-year population
Infant mortality rate (IMR)	Deaths under 1 year per **1,000 live births**
Neonatal mortality rate	Deaths in the first 28 days per 1,000 live births
Perinatal mortality rate	Still births (28 weeks+) and deaths in the first 7 days per 1,000 **live births + still births**
Maternal mortality ratio (MMR)	Maternal deaths per **1,00,000 live births**

Indicator	Formula (denominator is the exam trap)
General fertility rate	Live births per 1,000 women aged 15–49
Total fertility rate (TFR)	Average children a woman would have in her reproductive life; India about 2.0 (NFHS-5)
Net reproduction rate (NRR)	Daughters per woman surviving to mothers' age; **NRR = 1** equals replacement level (TFR about 2.1)
Dependency ratio	(0–14 + 65 and over) ÷ 15–64 population

IMR is the best single indicator of a community's health status. Registration of births and deaths falls under the **Registration of Births and Deaths Act, 1969**: births within 21 days.

Biostatistics essentials

- **Data types:** qualitative (sex, blood group) versus quantitative; quantitative is discrete (number of children) or continuous (height, Hb).
- **Mean** is affected by extreme values; **median** is the middle value and is best for skewed data; **mode** is the most frequent value.
- **Measures of dispersion:** range, mean deviation, **standard deviation** (most used), variance (SD^2), coefficient of variation = SD ÷ mean × 100.
- **Normal curve:** bell-shaped, mean = median = mode. Mean ± 1 SD covers 68%, ± 2 SD covers 95%, ± 3 SD covers 99.7% of values.
- **Standard error of mean** = SD ÷ √n.
- **Tests:** chi-square for proportions (qualitative data); t-test for two means (small samples); z-test for large samples; ANOVA for three or more means.
- **p < 0.05** means the result is statistically significant. A **type I error** (α) is rejecting a true null hypothesis; a **type II error** (β) is accepting a false one.
- **Correlation coefficient (r)** ranges from −1 to +1; it is shown on a **scatter diagram**.
- **Diagrams:** bar chart and pie chart for qualitative data; **histogram** and frequency polygon for continuous data; **ogive** for cumulative frequency; line diagram for a trend over time.

Water

- **Safe water:** free from pathogens and harmful chemicals, pleasant to drink, usable for domestic purposes. The daily need is about 150–200 litres per person in cities.
- **Purification on a large scale:** storage → filtration → chlorination.

- **Slow sand filter:** the **vital layer (Schmutzdecke)** of algae and bacteria on the sand surface does the work. **Rapid sand filter:** needs prior coagulation with alum; cleaned by back-washing.
- **Chlorination:** the free residual chlorine must be **0.5 mg/L after 1 hour of contact**. **Break-point chlorination** is the point after which added chlorine appears as free residual chlorine.
- **Horrock's apparatus** finds the dose of bleaching powder for well water. Bleaching powder contains about **33% available chlorine**.
- **Fluoride:** desirable level 0.5–0.8 mg/L; above 1.5 mg/L causes fluorosis; correction is by the **Nalgonda technique**.
- **Hardness:** temporary hardness (bicarbonates) is removed by boiling; permanent hardness (sulphates, chlorides) by the base-exchange process.

Sanitation: excreta and refuse disposal

Faecal-oral diseases (typhoid, cholera, hepatitis A, polio, worms, diarrhoea) spread along the **F-diagram**: Faeces → Fingers, Flies, Fields, Fluids → Food. A **sanitation barrier**, the safe disposal of excreta, breaks every one of these routes at the start.

Method	Key facts
Sanitary latrine	Must not pollute the soil or water, and must keep excreta away from flies and animals; sited at least 15 m (about 50 feet) from a water source
Water-seal latrine	A trap of water seals the pan from the pit, keeping out flies and smell; the RCA (Research-cum-Action) type is suited to rural India
Pit and bore-hole latrines	Simple dry pits for places without a water supply
Septic tank	For houses without sewers: solids settle and are digested anaerobically; the effluent drains to a soak pit
Sewerage	Water carriage of sewage to a treatment plant (towns and cities)
Refuse (solid waste)	Disposed of by composting, sanitary landfill or incineration

Swachh Bharat Mission (launched 2 October 2014) aims at an open-defecation-free (ODF) India, with household toilets and solid-waste management.

Memory hook: the five Fs (faeces, fingers, flies, fields, fluids) lead to food; a latrine 15 metres from the well is the first barrier.

Air, noise and housing

- Indicators of air pollution: sulphur dioxide, smoke, grit and dust. Ventilation standard: about 3,000 cubic feet of fresh air per person per hour.
- Noise above **85 dB** risks hearing loss over long exposure.

Biomedical waste: colour codes (Bio-Medical Waste Management Rules, 2016)

Bag or container	What goes in	Final treatment
Yellow	Human anatomical waste, soiled dressings and cotton, expired or discarded medicines, chemical waste, microbiology cultures	Incineration or deep burial
Red	Contaminated recyclable plastics: tubing, bottles, IV sets, catheters, syringes **without needles**, gloves	Autoclave or microwave, shred, then recycle
White (translucent), puncture-proof	Sharps: needles, syringes with fixed needles, scalpels, blades	Autoclave, shred, sharps pit or encapsulation
Blue (cardboard box)	Glassware (broken or discarded) and metallic body implants	Disinfect or autoclave, then recycle

Biomedical waste: which colour?

Segregate at the source, by colour



Yellow

bag

WHAT GOES IN

Anatomical waste
Soiled dressings, cotton
Expired medicines
Chemical waste
Microbiology cultures

TREATMENT

Incineration or deep burial



Red

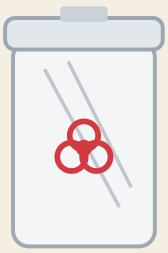
bag

WHAT GOES IN

Recyclable plastics: tubing, bottles, IV sets, catheters, gloves, syringes WITHOUT needles

TREATMENT

Autoclave or microwave, shred, recycle



White

puncture-proof, translucent

WHAT GOES IN

Sharps: needles, syringes with fixed needles, scalpels, blades

TREATMENT

Autoclave, shred, sharps pit or encapsulation



Blue

cardboard box

WHAT GOES IN

Glassware (broken or discarded)
Metal body implants

TREATMENT

Disinfect or autoclave, then recycle

Bio-Medical Waste Management Rules, 2016 · treat within 48 hours · hook: Yellow body, Red plastic, White sharps, Blue glass

Biomedical waste · Yellow body, Red plastic, White sharps, Blue glass

Waste must be treated and disposed of within **48 hours**. Bags must not be chlorinated plastic (no PVC).

Memory hook: waste bags by colour: **Yellow** for the body (anatomical and soiled waste), **Red** for recyclable plastic, **White** for sharps, **Blue** for glass. Census 2011 in two numbers: **943** women per 1,000 men and **74.04%** literacy.

10.3 Forensic medicine and toxicology

New criminal laws: from **1 July 2024** the IPC, CrPC and Indian Evidence Act were replaced by the **Bharatiya Nyaya Sanhita (BNS)**, **Bharatiya Nagarik Suraksha Sanhita (BNSS)** and **Bharatiya Sakshya Adhiniyam (BSA)**. Learn both numbers, as questions may use either.

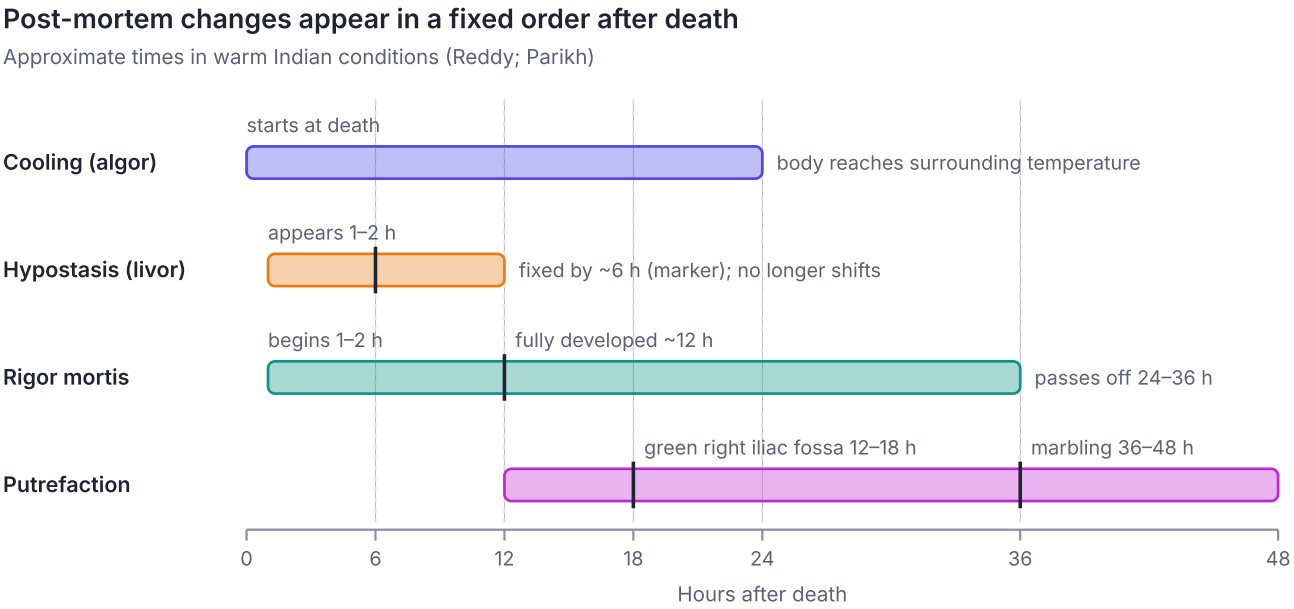
Subject	Old section	New section (2023 Acts)
Murder (punishment)	IPC 302	BNS 103

Subject	Old section	New section (2023 Acts)
Grievous hurt (definition)	IPC 320	BNS 116 (severe pain period cut from 20 to **15 days**)
Police inquest	CrPC 174	BNSS 194
Magistrate inquest (custodial death)	CrPC 176	BNSS 196
Dying declaration	Evidence Act 32	BSA 26

Death and its signs

- **Somatic death:** permanent stop of heart, lungs and brain function. **Molecular (cellular) death** follows over the next hours.
- **Brain-stem death** is recognised for organ donation under the Transplantation of Human Organs and Tissues Act, 1994.
- **Suspended animation:** apparent death (e.g. drowning, electrocution, newborns); resuscitation must be attempted.

Post-mortem changes



Post-mortem changes · approximate timeline used to estimate time since death

Estimate the time since death by reading several changes together; no single sign is reliable alone.

- **Hypostasis (livor mortis):** blood settles in dependent parts; spared at pressure points (contact flattening). It is **cherry-red in CO and cyanide** poisoning and **chocolate (brown)** in nitrite or potassium chlorate poisoning.

- **Rigor mortis:** follows **Nysten's rule**: eyelids → face → neck → trunk → upper limbs → lower limbs, and passes off in the same order. The heart (myocardium) stiffens first of all.
- **Cadaveric spasm (instantaneous rigor):** stiffening at the moment of death in a tense person, e.g. weeds clutched in drowning or a weapon gripped in suicide. It cannot be faked and is strong evidence of what happened at death.
- **Heat stiffening** (burns, above 65 °C) and **cold stiffening** are different from rigor.
- **Putrefaction:** the first external sign is **greenish discolouration over the right iliac fossa** (caecum lies close to skin). **Casper's dictum:** decay in air, water and earth is in the ratio `1:2:8`.
- **Adipocere (saponification):** body fat turns soapy in warm, moist conditions (water, damp graves). **Mummification:** drying in hot, dry air; features are preserved for identification.

Identification

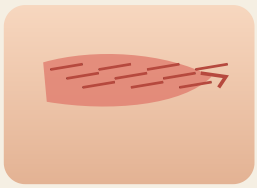
- **Locard's exchange principle:** every contact leaves a trace.
- **Fingerprints (dactylography):** Galton's system; **loops** are the most common pattern; no two persons share fingerprints.
- **Age:** from teeth (Gustafson's method) and ossification of bones; the third molar erupts at 17–25 years.
- **Sex:** pelvis is the best bone, then skull.
- **DNA profiling** is the most accurate method today.

Mechanical injuries

Injury	Key forensic feature
Abrasion	Only the epidermis is lost; shows the direction of force and the pattern of the object; heals without scar
Contusion (bruise)	Bleeding under intact skin; colour changes with age: red → blue (1–3 days) → brown → **green (5–6 days)** → **yellow (7–12 days)** → normal (about 2 weeks)
Laceration	Tearing by a blunt force; ragged margins, **tissue bridges** in the wound, bruised edges
Incised wound	Sharp-edged weapon; clean cut, longer than deep; **tailing** shows the end of the cut. **Hesitation cuts** suggest suicide; **defence wounds** on palms and forearms suggest homicide
Stab wound	Depth greater than length; shape tells single- or double-edged blade
Firearm wound	Entry: inverted edges, abrasion collar, burning and tattooing at close range. Exit: everted, larger, no collar

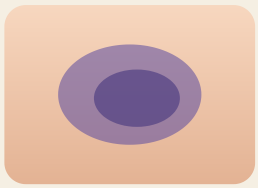
Mechanical injuries: read the wound

What each wound looks like, and how a bruise ages



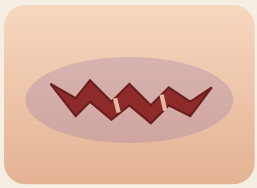
Abrasion

Only the epidermis is lost; shows direction of force; heals without a scar



Contusion (bruise)

Bleeding under intact skin; its colour dates the injury (strip below)



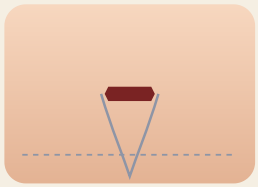
Laceration

Blunt force tears: ragged margins, tissue bridges, bruised edges



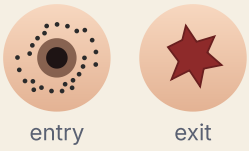
Incised wound

Sharp edge: clean cut, longer than deep; the tail shows where it ended



Stab wound

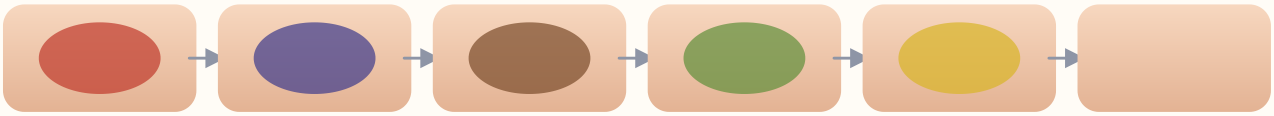
Depth greater than length; shape tells a single- or double-edged blade



Firearm wound

Entry: small, inverted, collar, tattooing at close range. Exit: larger, everted, no collar

A bruise changes colour as it ages



Red	Blue	Brown	Green	Yellow	Normal
fresh	1–3 days		5–6 days	7–12 days	≈2 weeks

Mechanical injuries · what each wound looks like, and how a bruise ages

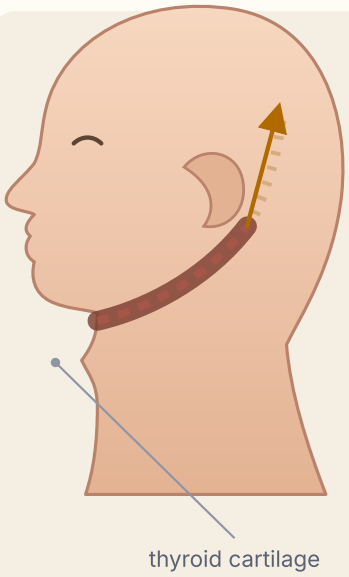
Asphyxial deaths

Feature	Hanging	Ligature strangulation
Usual manner	**Suicidal**	**Homicidal**
Ligature mark position	High, **above the thyroid cartilage**	Low, **at or below** the thyroid cartilage
Direction	**Oblique**, rising to the knot	**Horizontal**, transverse
Continuity	**Incomplete** (gap at knot)	**Complete**, encircles the neck

Feature	Hanging	Ligature strangulation
Face	Pale	Congested, petechiae
Salivary dribbling	Present; a sign of hanging while alive	Absent

Hanging vs ligature strangulation

Read the ligature mark: direction, level and continuity



Hanging

DIRECTION
Oblique, rising to the knot

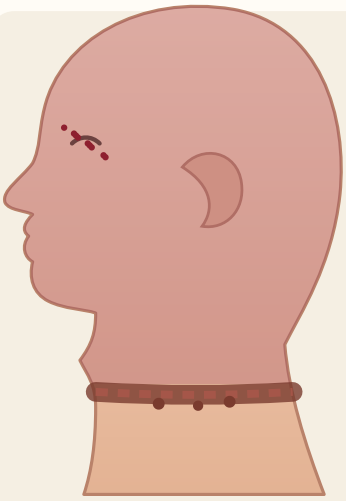
POSITION
High, above the thyroid cartilage

CONTINUITY
Incomplete (gap at the knot)

FACE
Pale

SALIVA DRIBBLING
Present: hanged while alive

USUAL MANNER
Suicidal



Ligature strangulation

DIRECTION
Horizontal, transverse

POSITION
Low, at or below thyroid cartilage

CONTINUITY
Complete, encircles the neck

FACE
Congested, petechiae

SALIVA DRIBBLING
Absent

USUAL MANNER
Homicidal

Throttling (manual strangulation): finger-nail marks and bruises; hyoid fracture is commonest here.

Hanging vs ligature strangulation · direction, level and continuity of the mark

- **Throttling (manual strangulation):** finger-nail marks and bruises on the neck; the **hyoid bone fracture** is commonest here.
- **Drowning:** fine, persistent **froth at mouth and nostrils**; weeds in the hand (cadaveric spasm); **diatoms** in bone marrow confirm death by drowning.
- **Tardieu spots:** petechial haemorrhages under pleura and pericardium in asphyxia.

Poisons: classical signs examiners ask

Many of these poisons are also homoeopathic polychrests; their toxic picture forms part of the proving.

Poison	Classical forensic facts
Organophosphate insecticides	Commonest poisoning in rural India; **pin-point pupils**, salivation, sweating, fasciculations; kerosene-like smell; low cholinesterase; atropine is the standard antidote
Datura (*Datura stramonium*)	"**Dry as a bone, red as a beet, blind as a bat, mad as a hatter, hot as a hare**"; dilated pupils; seeds resemble chilli seeds; used by road poisoners
Arsenic	Rice-water stools; **Mees' lines** on nails; raindrop pigmentation; detected in hair and nails long after; delays putrefaction; **Marsh and Reinsch tests**
Lead	**Burtonian blue line** on gums; basophilic stippling of red cells; wrist drop; colic
Mercury	Acrodynia (pink disease); Minamata disease from methyl mercury
Cyanide	Fastest-acting poison; **bitter-almond smell**; bright red hypostasis
Carbon monoxide	**Cherry-red** colour of blood and hypostasis
Opium (morphine)	Pin-point pupils, deep coma, slow breathing
Aconite	Most poisonous plant in India; tingling and numbness of mouth and limbs
Strychnine (*Nux vomica* seed)	Convulsions with **opisthotonus**, risus sardonicus; mind clear between fits
Oleander (*Nerium*, yellow oleander)	Cardiac poison; irregular slow pulse
Abrus precatorius (ratti)	Seeds made into **sui** (needles) to kill cattle
Calotropis (madar)	Milky juice used as an abortifacient and for infanticide

Snake bite

- **India's "big four":** common cobra, common krait, Russell's viper, saw-scaled viper.
- **Elapids (cobra, krait)** are **neurotoxic**: ptosis, difficulty swallowing, respiratory paralysis. Krait bites at night and is often painless.
- **Vipers** are **haemotoxic**: local swelling and bleeding. The **20-minute whole blood clotting test** shows non-clotting blood.
- First aid: reassure, immobilise the limb, no cutting or sucking, and transfer quickly for **polyvalent anti-snake venom**, which covers the big four.

Memory hook: the new laws keep their initials: **N**yaya replaces the penal code, **N**agarik **S**uraksha replaces the CrPC, **S**akshya replaces the Evidence Act (BNS, BNSS, BSA from 1 July 2024). A **hanging** mark climbs **obliquely above** the thyroid and breaks; a **strangulation** mark runs **flat and complete** at or below it.

10.4 Medical jurisprudence and ethics

Regulating homoeopathy in India

Milestone	Fact
Homoeopathy Central Council Act, 1973	Set up the Central Council of Homoeopathy (CCH) to keep the Central Register and set education standards
Department of ISM&H (1995) → Department of AYUSH (2003) → **Ministry of AYUSH (November 2014)**	Government body for Ayurveda, Yoga, Unani, Siddha, Sowa-Rigpa and Homoeopathy
National Commission for Homoeopathy (NCH) Act, 2020	Replaced the 1973 Act; the Commission was constituted in **July 2021**
Three autonomous boards under NCH	**Homoeopathy Education Board**; **Medical Assessment and Rating Board for Homoeopathy**; **Board of Ethics and Registration for Homoeopathy**
National Register	Kept by the Board of Ethics and Registration; only registered practitioners may practise homoeopathy
Central Council for Research in Homoeopathy (CCRH), 1978	Apex research body under the Ministry of AYUSH

State level: the Gujarat Board of Homoeopathic System of Medicine keeps the State Register for practitioners in Gujarat.

Cardinal principles of ethics

- **Autonomy** (respect the patient's choice), **beneficence** (do good), **non-maleficence** (do no harm, *primum non nocere*), **justice** (fairness in care).
- **International codes:** Hippocratic Oath; **Declaration of Geneva (1948)**, the modern physician's pledge; **Helsinki (1964)**, research on human subjects; **Tokyo (1975)**, doctors must not take part in torture; **Sydney (1968)**, determination of death; **Oslo (1970)**, therapeutic abortion.

- **Organon §1:** the physician's high and only mission is to restore the sick to health. §2: the ideal cure is rapid, gentle and permanent.

Duties and misconduct

- **Duties to the patient:** attend in an emergency, keep confidentiality, obtain consent, keep records, give proper prescriptions with the registration number, and refer when beyond one's competence.
- **Professional ("infamous") misconduct** includes: adultery or improper conduct with a patient; issuing false certificates; **advertising** or self-promotion; **fee-splitting (dichotomy)**; covering an unqualified person; disclosing a patient's secrets without cause; performing an illegal abortion or sex determination; practising another system of medicine without qualification.
- **Penalties:** warning or censure; temporary removal; **penal erasure** of the name from the register.
- **Cross-practice:** in *Poonam Verma v. Ashwin Patel* (1996) the Supreme Court held that a homoeopath prescribing allopathic drugs without qualification is **negligent per se**.

Consent

- **Implied consent:** the patient's conduct shows agreement (e.g. coming to the clinic, holding out an arm). It covers only a general examination.
- **Expressed consent:** oral or written; written consent is needed for any procedure with risk.
- **Informed consent:** the patient is told the nature, risks, alternatives and consequences of refusal, in a language they understand.
- **Age:** a person aged **18 years or more** can give valid consent (IPC 87). For a child **under 12** or a person of unsound mind, the guardian consents (IPC 89). In an **emergency**, the doctor may act in good faith without consent (IPC 92). The BNS sections 25–30 carry these rules forward.
- **Blanket consent** ("any treatment") is not valid.
- A **medico-legal examination** of a victim needs that person's consent; an accused may be examined at police request without consent (BNSS 51, formerly CrPC 53).

Professional negligence

- **Elements, the "4 Ds":** **D**uty of care, **D**ereliction (breach) of that duty, **D**irect causation, **D**amage.
- **Civil negligence** brings compensation; **criminal negligence** needs gross negligence or recklessness (IPC 304A, now **BNS 106**).
- **Bolam test:** a doctor is not negligent if a responsible body of fellow practitioners would have acted the same way.
- ***Jacob Mathew v. State of Punjab* (2005):** guidelines that protect doctors from frivolous criminal prosecution.

- ***Res ipsa loquitur*** ("the thing speaks for itself"): the facts prove negligence, e.g. a swab left in the abdomen or the wrong limb operated on.
- **Contributory negligence:** the patient's own carelessness (e.g. not following advice) reduces the doctor's liability.
- **Novus actus interveniens:** a new, unrelated act breaks the chain of causation.
- **Consumer protection:** *Indian Medical Association v. V.P. Shantha* (1995) brought paid medical service under consumer law. Under the **Consumer Protection Act, 2019**, claims go to the **District Commission (up to ₹50 lakh)**, **State Commission (₹50 lakh to ₹2 crore)** or **National Commission (above ₹2 crore)**.

Confidentiality and records

- **Professional secrecy:** the doctor must not disclose what the patient tells them.
- **Privileged communication:** disclosure is justified when the law or the public interest requires it, e.g. notifiable diseases, a court order, a patient's infection endangering a spouse, or a danger to others (e.g. an epileptic bus driver).
- **Records:** keep in-patient records for **at least 3 years**; keep **medico-legal records until the case is closed**. A patient (or their legal heir) may ask for copies, which should be given within **72 hours**.
- The **Charter of Patients' Rights (NHRC, 2019)** lists rights such as information, records, emergency care, informed consent, confidentiality, a second opinion, non-discrimination, and choice of where to buy medicines and tests.

Legal procedure and evidence

- **Inquest:** an inquiry into an unnatural death. **Police inquest** is the commonest in India; **magistrate inquest** is mandatory for deaths in custody, dowry deaths within 7 years of marriage and exhumation.
- **Medico-legal autopsy:** ordered by the police or magistrate; relatives' consent is not needed. A **clinical (academic) autopsy** needs the relatives' consent.
- **Dying declaration:** a statement by a dying person about the cause of death; best recorded by a magistrate, but a doctor may record it if death is imminent. The doctor certifies that the person is **conscious and of sound mind (compos mentis)**. No oath is given. A **dying deposition** is recorded by a magistrate in the presence of the accused, with cross-examination, and carries more weight.
- **Witness examination order:** examination-in-chief → **cross-examination** → re-examination. **Leading questions** are allowed only in cross-examination.
- **Perjury:** giving false evidence under oath (IPC 191, now BNS 227).
- **Summons and conduct money:** a doctor must obey a court summons; in a criminal case it cannot be refused for lack of fees.

Other Acts a homoeopath must know

Act	Key fact
Medical Termination of Pregnancy Act, 1971 (amended 2021)	Up to 20 weeks on one registered (allopathic) practitioner's opinion; 20–24 weeks on two, for special categories; beyond 24 weeks by a medical board for major foetal anomalies
PC-PNDT Act, 1994 (amended 2003)	Bans sex determination before or after conception
POCSO Act, 2012	Age of child below 18; **mandatory reporting** of child sexual abuse by any doctor
Mental Healthcare Act, 2017	A suicide attempt is presumed to be due to severe stress; it is no longer punished
Transplantation of Human Organs and Tissues Act, 1994	Recognises brain-stem death; bans commercial organ trade
Drugs and Cosmetics Act, 1940 and Rules	Homoeopathic medicines are regulated here; **Schedule M-I** sets good manufacturing practice for homoeopathic medicines
Drugs and Magic Remedies (Objectionable Advertisements) Act, 1954	Bans advertising remedies that claim to cure listed diseases

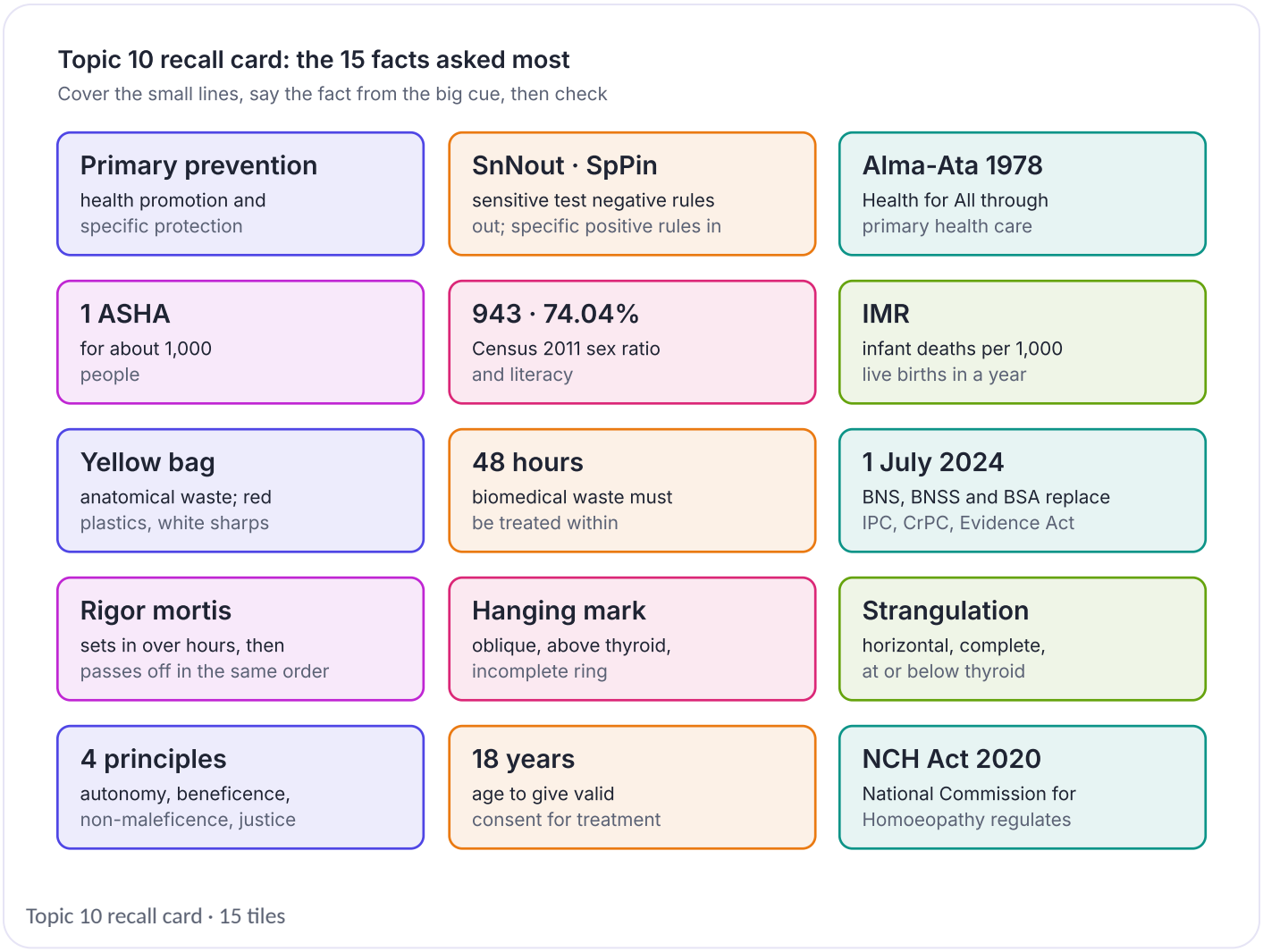
Memory hook: the four principles are "**A** **B**ig **N**ice **J**ob": **A**utonomy, **B**eneficence, **N**on-maleficence, **J**ustice. Valid consent for treatment needs a patient of **18 or over**, informed and free to refuse.

Rapid revision

1. WHO definition of health: 1948; physical, mental and social well-being.
2. Epidemiological triad: **agent, host, environment**; web of causation for non-communicable disease.
3. Five modes of intervention: health promotion, specific protection, early diagnosis and treatment, disability limitation, rehabilitation.
4. Immunisation = **primary prevention (specific protection)**; screening = **secondary**; physiotherapy = **tertiary**.
5. **P = I × D**.

6. Case-control gives **odds ratio**; cohort gives **relative risk** and incidence.
7. Sensitivity = TP/(TP+FN); specificity = TN/(TN+FP); PPV rises with prevalence.
8. Cold chain **+2 to +8 °C**; OPV most heat-sensitive; Hep B and pentavalent must not freeze; shake test for freezing.
9. Genus epidemicus: Organon §100–102; Belladonna in scarlet fever (1799).
10. RBSK screens 0–18 years for 4 Ds.
11. Sub-centre 5,000; PHC 30,000; CHC 1,20,000 (plain areas).
12. IMR per **1,000 live births**; MMR per **1,00,000 live births**.
13. Replacement level: **NRR = 1**, TFR about 2.1.
14. Census 2011 sex ratio 943; literacy 74.04%.
15. Normal curve: ±1 SD 68%, ±2 SD 95%, ±3 SD 99.7%.
16. Chi-square for proportions; t-test for two small-sample means.
17. Residual chlorine **0.5 mg/L after 1 hour**; Horrock's apparatus; bleaching powder 33% chlorine.
18. Slow sand filter: **Schmutzdecke** (vital layer).
19. Fluoride above 1.5 mg/L causes fluorosis; Nalgonda technique.
20. BMW: **yellow** anatomical/soiled; **red** recyclable plastics; **white** sharps; **blue** glass and metal implants; treat within 48 hours.
21. Rigor follows **Nysten's rule**; first sign of putrefaction is green **right iliac fossa**.
22. Casper's dictum 1 : 2 : 8 (air : water : earth).
23. Cherry-red hypostasis: **CO and cyanide**.
24. Cadaveric spasm cannot be faked: weeds in drowning, weapon in suicide.
25. Hanging: oblique, high, incomplete mark; strangulation: horizontal, low, complete.
26. Hyoid fracture commonest in **throttling**; diatom test for drowning.
27. Datura: dry, red, blind, mad, hot; arsenic: Mees' lines, rice-water stools; lead: Burtonian line.
28. Elapids neurotoxic; vipers haemotoxic; **20-minute whole blood clotting test**.
29. IPC 302 → **BNS 103**; IPC 320 → **BNS 116**; CrPC 174 → **BNSS 194**; Evidence Act 32 → **BSA 26**.
30. NCH Act 2020 replaced the HCC Act 1973; three autonomous boards.
31. Negligence: duty, dereliction, direct causation, damage; *res ipsa loquitur*; Bolam test.
32. *Poonam Verma v. Ashwin Patel* (1996): cross-practice is negligence per se.
33. Valid consent from **18 years**; guardian for a child under 12; emergency without consent in good faith.
34. Leading questions only in **cross-examination**; dying declaration has no oath.
35. Consumer Protection Act 2019: District up to ₹50 lakh, State to ₹2 crore, National above.

One-glance recall card



For the last ten minutes before the exam: cover the small lines of each tile and say the fact from the big cue.

20 practice MCQs

Cover the answer key, attempt all 20 in about 15 minutes, then check.

1. Specific protection belongs to which level of prevention? A. Primordial B. Primary C. Secondary D. Tertiary
2. Prevalence equals: A. Incidence ÷ duration B. Incidence × duration C. Incidence + duration D. Incidence − deaths
3. The odds ratio is calculated in a: A. Cohort study B. Case-control study C. Cross-sectional study D. Field trial
4. Positive predictive value of a test increases when: A. Prevalence rises B. Prevalence falls C. Sensitivity falls D. Sample size falls

5. Which vaccine is most sensitive to heat? A. Hepatitis B B. BCG C. OPV D. Td
6. Hahnemann used Belladonna as a prophylactic against: A. Cholera B. Scarlet fever C. Smallpox D. Typhus
7. Maternal mortality ratio is expressed per: A. 1,000 live births B. 1,000 women 15–49 C. 1,00,000 live births D. 1,00,000 population
8. India is in which stage of the demographic cycle? A. High stationary B. Early expanding C. Late expanding D. Low stationary
9. The best measure of central tendency for skewed data is the: A. Mean B. Median C. Mode D. Range
10. After chlorination, the free residual chlorine at 1 hour should be: A. 0.1 mg/L B. 0.2 mg/L C. 0.5 mg/L D. 2 mg/L
11. Discarded glass vials go into the: A. Yellow bag B. Red bag C. White container D. Blue cardboard box
12. The first external sign of putrefaction appears over the: A. Face B. Right iliac fossa C. Left iliac fossa D. Back
13. Cherry-red hypostasis is seen in poisoning by: A. Arsenic B. Carbon monoxide C. Opium D. Organophosphate
14. A ligature mark that is oblique, above the thyroid cartilage and incomplete suggests: A. Throttling B. Hanging C. Ligature strangulation D. Smothering
15. Mees' lines on the nails are seen in chronic poisoning by: A. Lead B. Mercury C. Arsenic D. Copper
16. The 20-minute whole blood clotting test is used in: A. Cobra bite B. Krait bite C. Viper bite D. Scorpion sting
17. The punishment for murder is now under: A. BNS 101 B. BNS 103 C. BNS 116 D. BNSS 194
18. The National Commission for Homoeopathy Act was passed in: A. 1973 B. 2014 C. 2019 D. 2020
19. "The thing speaks for itself" refers to: A. Novus actus interveniens B. Res ipsa loquitur C. Bolam test D. Contributory negligence
20. Leading questions may be asked during: A. Examination-in-chief B. Cross-examination C. Re-examination D. None of these

Answer key: 1 B · 2 B · 3 B · 4 A · 5 C · 6 B · 7 C · 8 C · 9 B · 10 C · 11 D · 12 B · 13 B · 14 B · 15 C · 16 C · 17 B · 18 D · 19 B · 20 B

Why the tricky ones: Q4, PPV depends on how common the disease is. Q8, India's birth rate is falling but still above its death rate. Q11, glass and metal implants go in the blue box, not the red bag. Q16, viper venom stops blood clotting; elapid venom is neurotoxic.